Efficient Entangled Biphoton Production and Manipulation for Quantum Applications

by

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Dissertation submitted in partial fulfillment of the
requirements for the degree of Doctor of Philosophy
in the Department of Physics
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ABSTRACT

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Abstract

The creation and manipulation of biphotons is important for many applications in quantum optics and quantum information. Topics that benefit from efficient biphoton sources range from the most fundamental quantum science experiments to the highly applied fields of quantum communication and quantum computation. Biphoton sources have long been hailed as one of the leading methods for creating entangled photon pairs for tests of Bell's inequality, creating heralded photon pairs that are used in on-demand single-photon sources and heralded measurement techniques, and for quantum communication protocols to name a few. Specifically for quantum communication, biphoton sources are commonly used for cutting edge quantum key distribution (QKD) protocols.

In the first part of the thesis, I focus on realizing an efficient biphoton source that produces high yield photon pairs. More specifically, I develop an optimized biphoton source using the nonlinear optical process of spontaneous parametric down-conversion in a second-order nonlinear crystal. I develop a formalism for predicting the two important metrics of a biphoton source: the heralding efficiency and joint count rate. I show how, from a large parameter space, one can tailor the phase matching of the nonlinear interaction to create a high quality biphoton source that produces both high heralding efficiency and high joint count rate. I achieve heralding efficiencies of 86±5% and joint count rates of 2.58±0.6 kHz per mW pump power. I show that using a collinear nondegenerate geometry allows for heralding efficiencies of up to 99.7% assuming no loss in the system. I verify the theoretical model with experimental results and find good agreement.

In the second part of the thesis I turn to manipulating the single photons born from the biphoton source for applications in creating single-photon spectrometers and time-

frequency QKD systems. The security of QKD is only guaranteed if the two parties have access to a set of states called mutually unbiased states. I create a set of these states in time and their conjugate states in the frequency basis and show that I can manipulate single photon correlations in time and frequency so that an eavesdropper can be detected if she localizes a photon to a 1 ns time interval.

Additionally, in these experiments, I stretch a single photon wavepacket of 5-ps-width to a wavepacket of 8.3-ns-width and subsequently recompress it to at least the resolution of the detectors ($\sim$ 300 ps). This demonstrates a stretch factor of $>$1600 for a single-photon pulse using a group velocity dispersive material. To my knowledge, this is the largest reported stretch factor for a single-photon wavepacket produced by a biphoton source. The ability to stretch and recompress a single photon by this amount has applications in creating high-resolution, high-efficiency, single-photon spectrometers as well as advancing time-frequency QKD systems and other temporal pulse shaping applications.

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Acknowledgements

The journey through a physics Ph.D. (or maybe any Ph.D. for that matter) is never an easy one, nor do I believe it should be. What I do believe in is the saying "It takes a village to raise a child." This is how I've always pictured my journey through physics Ph.D -land – among a village full of innovative, brilliant, hard-working and genuinely amazing people who have helped me learn, helped me grow as a physicist and as a person, taught me how to think better, work harder and above all challenged me to always keep going. I have always felt, in conversations with my fellow lab mates, professors, and graduate students that I have been walking among giants (intellectually, of course). To all of these people, I owe my deepest thanks.

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Chapter 1

Introduction

1.1 Introduction to Quantum Optics and Quantum Information

The field of quantum optics focuses on studying and manipulating various states of light, specifically nonclassical light, where the electromagnetic field of the light is quantized. That is, it is the study of light in its most fundamental form, that of waves and particles. Quantum optics as a formal field began in the 1950s with the introduction of coherent states and statistical distributions of light, and has bloomed into one of the largest fields today crossing the interdisciplinary boundaries of physics, photonics, electrical and computer engineering, and computer science. It is a natural testing bed for the fundamental properties of quantum mechanics, and additionally has founded applications in modern technologies including quantum information, quantum metrology, quantum lithography, and quantum imaging.

The early foundation of quantum optics traces its origins to the first several years of the 20th century, with Einstein's study of the photoelectric effect that indicated that light was both a particle and a wave [1]. However, most of the theory for developing a formalism for the quantum nature of light was not begun until the 1950s when Roy Glauber first discovered the coherent state of the photon using quantum field theory [2]. The discovery of the coherent state paved the way for understanding the statistical properties of light – namely that light was no longer considered only wave-like, but now instead it could contain individual photons with statistical properties that had thermal, Poisonnian, sub- or super-Poissonian statistics. This, in turn, led to the discovery of more exotic states of light, including squeezed states, where the limits of the quantum uncertainty principle dictate the state of the light.

The development of the laser in the 1960s and 1970s spurred a growing interest in how light and matter interacted on a fundamental level. The field of quantum optics bloomed after the invention of the laser because the laser allowed the possibility of studying coherent states as well as other statistical properties of light. In addition, laser light could achieve high intensity levels as well as long coherence times, which enables many other experiments in the atomic and molecular optics community.

While important topics in atomic physics were explored with the aid of the laser, the interaction of light and matter at the quantum level allowed for a new type of experimental realization: fundamental quantum science. Until this point, most intriguing questions in quantum mechanics were investigated with *Gedankenexperiments*, or thought experiments. Such thought experiments hypothesized the nature of entanglement and non-locality and what implications they had for classical physics. In 1962 Bell showed that entanglement could be proven to be explained by a quantum mechanical theory instead of a local hidden variable theory (LHV) using experimental quantities and that violate the so-called "Bell inequality" [3]. Using the newly developed laser, the first “proof” of quantum mechanics using a Bell-state violation was performed experimentally by Freedman and Clauser in Ref. [4] using polarization entangled photons emitted in an atomic cascade in calcium. Although ground-breaking at the time, this experiment made several strong assumptions to demonstrate the violation of the Bell inequality due to experimental parameters. In 1981 and 1982 Aspect demonstrated Bell’s inequality with greater statistical accuracy using a higher quality entangled photon source, made possible by a stronger laser [5]. Because of this technological improvement, the authors were able to get rid of the strong supplementary assumptions that previous experiments had to make. This is one example of how improvements in technology pushes the bounds of fundamental science experiments.

Over the last forty years after this first Bell experiment was performed to test the hypothesis of quantum mechanics and disprove a local hidden variable theory (LHV) many groups have continued to perform similar experiments to test the building blocks of quantum mechanics. Recently, the advances in high-purity, high-rate entangled photon sources and high-efficiency detectors have contributed to being able to close several loopholes in for tests of Bell's inequality [6, 7]. It is now only a matter of time before all loopholes are simultaneously closed and LHVs are disproven. This advancement could not have been made without the progress in creating high-quality entangled and correlated photon sources. These sources are relevant to fundamental quantum mechanics, and also to a variety of quantum optics and quantum information applications. In my work, I focus on creating high-efficiency, high-rate, correlated and entangled biphoton sources. In the next section I will introduce the biphoton source and discuss its relevance in many of the fields in quantum optics including fundamental questions in quantum mechanics, as well as the applications in quantum metrology, quantum lithography, and quantum information.

The field of quantum information creates and uses quantum states, primarily photons, to achieve communication and computational possibilities that are not classically feasible. This field stemmed from the joining of quantum optics and information science, and is based on using quantum states as ideal conduits of information. Quantum information typically includes the fields of quantum computation, whose goal is to use quantum states to perform computational tasks that classical computers cannot, and quantum communication, whose goal is to transmit secure information using quantum states instead of classical.

Both of these fields have greatly advanced over the past two decades thanks to the discovery and advancement of high quality quantum state sources such as entangled photon sources, the quality of fiber optic channels, the advancements in single-photon detector technology, and the ability to manipulate photonic states. In the next section I will describe how my work contributes to the field of both quantum optics and quantum information through the improvement of entangled and correlated photon sources. Additionally, in the following section I will describe my work on manipulating these single photon states in the time and frequency degrees-of-freedom. The manipulation of single photon sources in time and frequency has generated much interest in applications aiming to create single-photon spectrometers and in quantum key distribution systems aiming to create high-dimensional states. My work paves the way for future improvements in both of these technologies.

1.2 Correlated and Entangled Biphoton Sources

1.2.1 What are correlated and entangled photon sources?

A biphoton wave-packet contains strictly two photons that are correlated in any number of degrees-of-freedom and will maintain these correlations, even at great distances. Correlations or entanglement are created in any number of degrees-of-freedom including time, frequency, space, momentum, and polarization. Biphoton sources, sometimes referred to as two-photon sources, are particularly useful for the study of fundamental quantum mechanics and for numerous quantum-based applications, because they produce a pair of photons where knowing the state of one of the photons reveals information about the state of the other. In this section I define an entangled photon pair as well explain the difference between correlated and entangled photon pairs. I further detail modern applications that use these ubiquitous quantum-based sources to push the limits of technology as a motivation for my experiments characterizing a popular biphoton source. I then discuss how my work impacts these fields.

An entangled photon pair consists of two particles whose wavefunctions cannot be factored into the direct product of two single particle wavefunctions. Consider two subsystems described by orthonormal basis states $|a\rangle$ and $|b\rangle$, where $|a\rangle$ is the basis for system 1 and $|b\rangle$ is the basis vector for system 2. Now consider the state given by

$$|\Psi\rangle = \sum_{(a,b)} f(a,b)|a\rangle|b\rangle, \tag{1.1}$$

where $f(a,b)$ represents the coefficients for each $|a\rangle|b\rangle$ state. If $f(a,b)$ is separable, this means that it can be factored into $f(a,b) = g(a)\times h(b)$ where each coefficient only depends on one of the two subsystems. If this is not true, and $f(a,b) \neq g(a) \times h(b)$, $|\Psi\rangle$ is an entangled state. Furthermore, let $\hat{\rho}$ denote an entangled pure state created from the density matrix of $|\Psi\rangle$ such that

$$\hat{\rho} = |\Psi\rangle\langle\Psi| \neq \hat{\rho_1}\hat{\rho_2}, \tag{1.2}$$

where, $\hat{\rho_1} = g(a)\sum_{(a)}|a\rangle\langle a|$ and $\hat{\rho_2} = h(b)\sum_{(b)}|b\rangle\langle b|$. Again, $\hat{\rho}$ is not separable and represents an entangled density matrix. The specific entangled states of position-momentum entanglement, known as the EPR pairs were first suggested in Ref. [8] by Einstein, Podolsky, and Rosen. Bohm further investigated other entangled states such as the spin-1/2 entangled states, which evolved into the popular polarization entangled states which are frequently used in Bell-type experiments [9].

A correlated photon pair can be a separable pair state, meaning the pair state is not entangled and can be expressed as a direct product of the wavefunctions of the two photons. However, knowledge of one partner's state must always reveal the state of the other. For example, if a two-photon state were created at exactly the same moment, it might be in a state represented by, $|\Psi\rangle = |t_{1_A}\rangle|t_{1_B}\rangle$ where A and B represent each particle and t_1 is the time they were born. Measuring one photon in this pair would give information about the time it was born as well as the time its twin was born. To distinguish this from entangled states, let us write down a similar expression for a time-entangled state. In a time-entangled state, the two photons would need to be in an unfactorable state. One such state is

$$|\Psi\rangle \propto A(|t_{1_A}\rangle|t_{1_B}\rangle + |t_{2_A}\rangle|t_{2_B}\rangle), \tag{1.3}$$

where t_1 and t_2 are two different times the photon pairs could be born and A is the normalization constant. This function is inseparable, and thus entangled. However it is also a correlated state because measuring state A in time t_1 means that state B must also have been born at time t_1.

Biphoton sources, both correlated and entangled, are used in many applications. One of the most popular sources for generating these biphoton states is a nonlinear optical process called spontaneous parametric down-conversion (SPDC), which I discuss in more detail in Sec 2.3.1 and Ch. 3. Briefly, in SPDC, a single pump photon is annihilated in a nonlinear crystal and produces two daughter photons that are correlated and can be entangled in any number of degrees-of-freedom. These sources are exceptionally popular in the fields I describe in this chapter and are the main focus of much of my research with biphoton sources. The need for high-efficiency, high purity sources is further driven by the specific applications, which I discuss next.

1.2.2 Applications for biphoton sources

Fundamental Quantum Science

Fundamental quantum science aims at obtaining a deeper understanding of basic quantum phenomena including quantum entanglement, nonlocality, and quantum teleporation. The creation of entangled states of photons has driven experiments towards proving that quantum entanglement cannot be explained by LHV theories. The central quantity used to prove this is called the Bell inequality, proposed by John Bell in 1965 [3]. If the inequality is violated in an experiment, it proves that quantum mechanics cannot be explained by LHVs. This inequality has been violated in numerous experiments since it was first proposed. However, there are a set of loopholes in these experiments which when not accounted for, leave room for an LHV interpretation. Recently, progress has been made on closing these loopholes using entangled photon states, with the ultimate goal of a completely loophole-free test of Bell's inequality [6, 7]. Such an accomplishment would be marvelous for quantum physics. However, until recently, it has been unlikely due to the nature of the loopholes, specifically the detection loophole, which arises from inefficiencies in the experiments such as detector inefficiency and noise.

In more detail, the detection loophole or "fair sampling" assumption which states that a classical reason could account for the violation of the inequality if only certain pairs of photons are considered. Specifically, if entangled photon pairs are measured more efficiently than uncorrelated photon pairs, the violation could occur due to biased sampling. To close this loophole, a minimum photon detection is needed so that there is no unfair sampling. For the Clauser-Horne-Shimony-Holt (CHSH) inequality, which is experimentally simpler to realize than the Bell inequality, that minimum detection was originally 83% – which represents the overall efficiency of the whole system including collection from the source, losses in the channel and optics, and the inefficiencies in the detectors [10]. This condition was relaxed by Eberhart in [11] who showed that the minimum system efficiency for a loophole-free test is 66% with non-maximally entangled states. This value represents the minimum efficiency a system needs to violate a Bell inequality without using the fair-sampling assumption. There are several methods of increasing the efficiency of a system including simply obtaining lower loss components and using a less noisy channel. However, from a physics standpoint, a more interesting and useful way of increasing the efficiency is by understanding the emission process of the source, commonly an SPDC source, and tailoring the collection optics to maximally capture the biphotons. For an SPDC source, if a quantity called the heralding efficiency is made high, this corresponds to a higher system efficiency. The heralding efficiency is the probability that if one photon in the biphoton pair is detected, its twin is also detected. I discuss the heralding efficiency in detail in Chs 3 and 4. Also important in SPDC sources is the rate at which entangled photon events are produced and captured. One of the main reasons for moving to SPDC instead of atomic cascades is the rate of production of entangled photon pairs is orders of magnitude higher, meaning the experiments collect enough data to violate Bell's inequalities over the time scale of seconds to minutes instead of hours to days.

In this thesis, I present my work for optimizing both the biphoton count rate and the heralding efficiency, which is the probability that if one photon from a biphoton source is detected in detector A, that its twin is detected in detector B. Optimizing heralding efficiency is of utmost importance for loophole-free tests of Bell's inequality, because if the efficiency of collection from the source can be made very high, it is possible to have enough "overhead" to not only close the detection loophole but simultaneously close all of the loopholes, leading to a true loophole-free test of Bell's inequality. Currently, the state of the art systems have just enough efficiency to close the detection loophole and require a few percent more to simultaneously close all the loopholes. My work shows how to obtain a theoretical maximum of 97% heralding efficiency assuming perfect detectors and no channel loss, while simultaneously maintaining a high biphoton count rate. When combined with low-loss optics and superconducting detectors with efficiencies above 90% [12], a full loophole-free test of Bell's inequality using entangled photons is now possible. Although fundamental science applications are one area which require high-quality entangled photon sources, other quantum-based applications such as quantum metrology also benefit from high heralding efficiency and high count rates

Quantum Metrology

The main goal of quantum metrology is to make high-resolution measurements on quantum systems. These measurements should be highly sensitive in the sense that they are able to distinguish the physical parameters being measured. Because quantum mechanics sets the absolute limit on the amount of precision or uncertainty any measurement can achieve, quantum methods are therefore the best way of making the most precise measurement of a system. A few of the quantities quantum metrology experiments aim to measure are absolute quantum efficiency of detectors, absolute spectral intensity of various sources, and high-precision timing. Atomic clocks that can achieve resolution down to one second over 300 million years which is an accuracy of 16 orders of magnitude [13].

Correlated photon pairs play an important role in various experiments in quantum metrology [14]. Consider the following example of how quantum metrology utilizes a biphoton source to achieve high accuracy measurements of the quantum efficiency of a detector illustrated in Fig. 1.1. In this experiment, the quantity of interest is the absolute quantum efficiency of the detector, which is defined as the probability that a given detector will output an electrical signal (“click") when a photon is incident on the detector. One way to measure the absolute quantum efficiency is to create a pair of correlated photons with an SPDC process. These two photons exist in a biphoton state so that if one photon is present at detector A, then the other is present at detector B. In one variation of this measurement, each photon in the photon pair is sent to a different detector. When the photon arrives at detector A, it triggers an electrical signal, which is used to gate detector B. The gated pulse tells detector B to “look" in the gate window for the twin photon. In an ideal case, the twin photon is always present at detector B, so that if B does not click, it is from an inefficiency in the detector. A popular source for such experiments is the SPDC source because it can be tailored to emit a broad spectral range of photons, where the two photons are always anti-correlated in frequency to conserve energy. This broad bandwidth is necessary for wavelength-dependent quantum efficiency measurements.

While this is a common method of measuring the quantum efficiency of the detector, it is prone to errors if both photons in the photon pair are not collected 100% of the time. Said differently, if one photon is going to herald the presence of the other photon in the channel, then they both need to be present for any measurement of the detector quantum efficiency to be accurate. However, high efficiency collection of both photons in the biphoton pair can be a challenging task. This, again, returns to the notion of heralding efficiency – a measure that says if photon A has been collected into arm A, what is the probability that it's twin is present in arm B. Increasing this value increases the ability to perform measurements using correlated photon pairs.

Although this is one of the major experiments in quantum metrology using biphoton sources, there are many other experiments that also utilize them. Other applications using biphoton sources include absolute spectral radiance measurements of an uncalibrated source and polarization-mode dispersion measurements with sub-

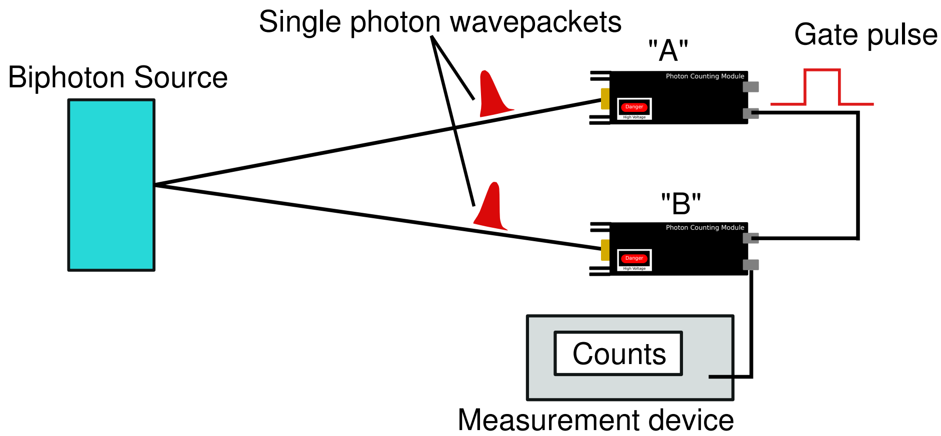

Figure 1.1: **Measuring absolute detector quantum efficiency with biphoton source** A biphoton source creates two twin photons that get sent to two single photon counting detectors (“A" and “B"). When the photon arrives at A and is measured, detector A sends a gated pulse to detector B. The gate basically tells the detector to “turn on" or look in the window of the gate pulse. Because the twin photon should arrive in this window with an appropriate time delay between the two detectors, the number of time detector B registers a photon during the gate time is a measure of its quantum efficiency.

femtosecond accuracy for various materials. Each of these experiments benefits from having a high efficiency biphoton source, meaning that the heralding efficiency is high. Additionally, they benefit from having a high biphoton rate so that signal to noise improves. In my thesis, I demonstrate how to create such a source using SPDC. I specifically show how to obtain both high heralding efficiency as well as joint count rates. These sources are invaluable for quantum metrology measurements, and by improving the sources, more accurate and higher-speed metrology measurements are possible. While quantum metrology experiments push the boundaries of measuring experimental parameters, quantum lithography and imaging use entangled photons to push the boundaries of classical lithography and imaging.

Quantum Lithography and Quantum Imaging

Quantum lithography and quantum imaging both utilize entangled photon sources to surpass current limits of classical lithography and classical imaging. In standard lithography, beams of photons write or etch patterns in photosensitive materials. One major industry that uses photolithography techniques almost exclusively is the microchip fabrication and processing industry. As the need for smaller and smaller microchips containing more components grows, methods of beating the standard diffraction limit become more important. The diffraction limit puts a bound on how closely two features can be resolved, imaged, or written on a microchip. The diffraction limit is proportional to the wavelength of light which is why deep UV rays are often used in classical photolithography.

Quantum lithography can beat these limits by using entangled states [15]. A type of entangled state called a N00N state is frequently used. N00N states act like a quasi-particle with the sum of the energies of the number of photons in the N00N state. This essentially changes the effective wavelength and sub-wavelength features can be written using this method. Again, a high-quality source of entangled biphotons is essential, with SPDC being a commonly used source of N00N states. I discuss such a high-quality source of entangled biphotons in this thesis, although I do not create N00N states.

Quantum imaging follows the same principles as quantum lithography in that the goal of the field is to be able to image features much smaller than the diffraction limit for classical light. Most of the applications of quantum imaging use entangled photons sources. An example of this is entangled two-photon microscopy, where the same idea as in lithography applies, where the two photons can impart as much energy in the sample as a single higher energy photon thus increasing the resolution. Here, the higher the entangled photon rate, the faster the images can be recorded. In addition to creating a high heralding efficiency photon source, I also discuss how to maximize entangled photon rates in the SPDC process. Quantum metrology, imaging, and lithography are all fast-growing fields requiring entangled photon sources, however, one of the biggest fields utilizing entangled photon sources is quantum information.

Quantum Information

Quantum information applications are the most prevalent users of entangled biphoton sources. Both quantum computation and quantum communication protocols depend on entangled photon sources for use in various applications. One such way is creating an on-demand single-photon source that is used in such applications as linear-optics quantum computing [16], quantum communication [17, 18], and quantum repeaters [19]. To date, heralded single photon sources have led the way in applications requiring single-photon sources [20, 21]. Single-photon sources are difficult to implement because all laser sources are inherently probabilistic. Even a laser that has been attenuated such that on average, it emits a single photon in a pulse, has some probably of emitting two photons in a pulse and also a high probably of emitting zero photons. For both quantum computation and quantum communication, this is a fundamental limitation.

Heralded single-photon sources are one way of solving this issue because detection of one of the twin photons heralds the other. Although this source is not fully "on-demand" it is highly probable that a single photon exists in the conjugate state when its twin is detected. Again, the heralding efficiency of such a source is crucial for these applications. Also important is the rate in which the photon pairs are collected.

Quantum communication, specifically quantum key distribution, is another application that relies on correlated and entangled biphoton sources. Many QKD protocols rely on the entanglement produced by SPDC sources to secure the information being sent from one party to another. The security proofs of QKD rely on the quantum nature of such sources and thus they play an important role in determining how successful a certain QKD protocol can be. Biphoton sources were shown in theory to provide provable security in an ideal QKD system, although practically, experiments have difficultly boasting such results. As the field of QKD grows, the need for higher rate, higher efficiency sources also grows. This is one of the driving factors for creating and characterizing an SPDC source – for implementation in a high rate high dimensional QKD system, which I discuss in Ch. 2.

Other quantum communication protocols benefit from high heralding efficiency as well. One-sided device-independent QKD required a minimum efficiency of 66% to be carried out. One-sided device-independent QKD is a QKD protocol which does not depend on the equipment or setup used. Instead, it simply assume “black boxes” for Alice and Bob’s measurement apparatuses and uses Bell inequalities to prove security given this minimum channel efficiency [22]. Additionally, three-way quantum communication protocols, such as the ping-pong protocol, require a minimum efficiency of 60% [23, 24].

My work

In Chs. 3 and 4 of this thesis, I focus on formulating a theoretical model to predict the heralding efficiency and biphoton count rate for SPDC in a nonlinear crystal collected into single mode fibers. Specifically, I focus my work in a Type-I system, meaning that the daughter polarizations are the same. I explore different geometries for the pump, signal, and idler wavevectors, as well as using different interaction wavelengths, focusing conditions, and filtering conditions. I show how to obtain simultaneously a high heralding efficiency and a high count rate by tailoring the collected biphoton and sin-

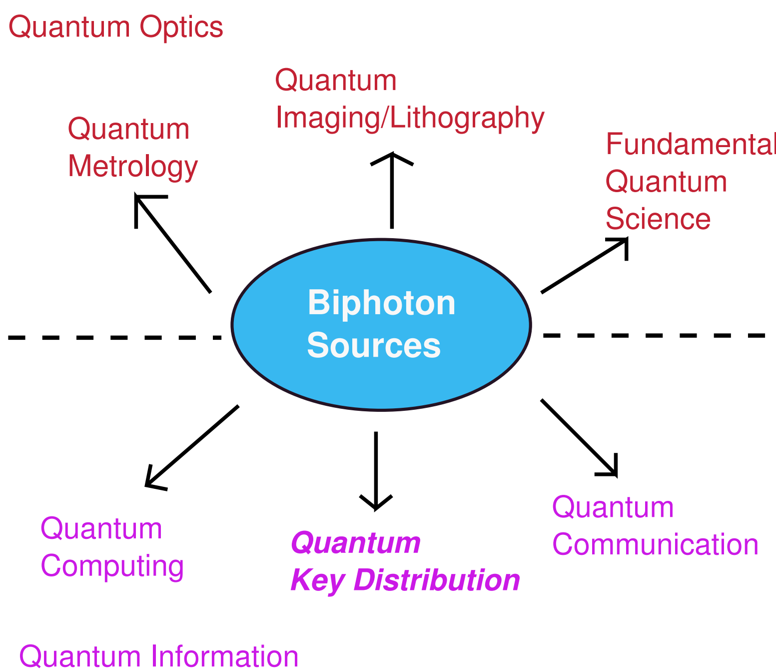

Figure 1.2: **Biphoton source applications** Biphoton sources impact many fields in quantum optics and quantum information. Although all of these applications are interesting, my original motivation for creating a biphoton source is high-speed, high-dimensional quantum key distribution (bold italics).

gle photon spectra. Achieving high heralding efficiency, high biphoton count rate, or both simultaneously is important for all of the applications mentioned above and although my work originally focused on achieving this for high-speed QKD applications, my results have a much broader impact across other fields as well. A summary of the fields that utilize biphoton sources is given in Fig. 1.2.

1.3 Manipulation of correlated single photons

Understanding how to manipulate and measure the spectrum of single photons is another useful tool for various areas of quantum information and quantum optics. Currently, this is an exciting current area of ongoing research in time-frequency high-dimensional QKD and in creating single photon spectrometers. It is important to realize that when talking about measuring the spectrum of single photons, the spectrum is obtained with multiple collections of a single-photon wavepacket. The single-photon wavepacket is a probability distribution, and when measured, it is measured in a particular state. After many such subsequent measurements, the total probability distribution of the spectrum (or of the timing distribution) is obtained. In this section, I discuss both of these applications and how my contribution to realizing temporal-spectral manipulation of single photons.

Creating single photon spectrometers is a challenging task because practical realizations tend to have high loss and low signal-to-noise. Losses arise from both gratings and photomultiplier tubes, which are the common detector choice for measuring low-light levels. Typically, the loss in such systems is so detrimental that the signal-to-noise is prohibitively low for many experiments and special lock-in amplification techniques are needed to measure the spectrum. Another method of measuring the spectrum of a single photon is by using a standard grating and then an array of single-photon counting detectors, where each detector measures a specific wavelength. This idea, while theoretically sound, is hard to implement due to the fact that, for high resolution, many more detectors are needed, which eventually becomes cost-prohibitive. This issue led Avenhaus *et al.* in Ref. [25] to use an optical material with large group velocity dispersion (GVD) to measure the spectrum of a single photon produced by a biphoton source. In this experiment, only one single photon counting detector is used and the loss through the system is quite low. The authors measure the spectrum by using time-to-frequency mapping, which turns the spectral measurement into a time-of-arrival measurement. In my work, I use the same concept of time-to-frequency mapping to manipulate the spectral-temporal correlations produced by a biphoton SPDC source. Although the original motivation for this work was to implement a time-frequency based high-dimensional QKD system, which I discuss in detail in Ch. 2, the work I have done with measuring spectral-temporal correlation using dispersive optics can readily be applied to single photon spectrometers.

Another area in which manipulating single photons is useful is in QKD. In QKD, significant work has been done on creating time-energy or time-frequency entangled states for high-dimensional protocols, which aim at making high-speed QKD a reality [26–30]. One of the challenges of making high-dimensional QKD with time bins as a basis is that, for full security, the conjugate frequency states must also be measured. I explain the reason for this in Ch. 2. Here, the goal is the ability to accurately measure the spectrum or spectral correlations of the biphoton source that is used to create the secret key.

In my work I focus on manipulating the spectrum of twin photons from a biphoton SPDC source. I study the correlations between the two photons when they are sent through a group velocity dispersive material and study how to stretch and subsequently recompress the temporal packets of the photons without destroying the correlations. I also focus on obtaining large group velocity dispersion for NIR wavelengths, instead of the more commonly-used telecommunications wavelengths. This, to my knowledge, is the first time a single photon packet from a biphoton source has been stretched and recompressed at these wavelengths. This has important implications for time-frequency QKD, single-photon spectrometers at non-telecommunications wavelengths, and potentially even applications where information is written into the phase of the photon wavepacket before compression. Here, the compression itself is one potential method of securing information against simple eavesdropping attacks.

1.4 Thesis Overview

In this thesis, I explore a mix of interesting physics including fundamental properties of nonlinear and dispersion based optics. I produce original theory and experiments on the nonlinear optical process of spontaneous parametric down-conversion, which is essential for all the applications discussed above. Furthermore, I develop and implement parts of a timing-based high dimensional QKD system and demonstrate one method that could allow for securing the information by creating access to the states in the frequency basis using dispersive optics.

In Ch. 2, I give an overview of the main components of a QKD system as a platform for studying biphoton sources and temporal-spectral manipulation of single photons. I detail the most common security attacks of such systems and explain how the quantum nature of such systems, at least of ideal systems, allows detection of an eavesdropper. Finally, I discuss the basic components of the QKD system I implement, including the biphoton source, the free-space channel, and the detection scheme. I introduce and define some important quantities of the SPDC process that are essential for creating a higher speed system.

In Ch. 3, I explore the process of spontaneous parametric down conversion in detail. Specifically, I derive expressions for the joint and the singles spectrum, which are important quantities in determining the overall count rate of the system. Beginning with the Hamiltonian for the SPDC process, I derive equations for predicting the differential spectral rate and the total spectral rate for different geometries, wavelengths and filtering conditions. In this chapter, I develop the framework which I use for predicting the heralding efficiency and total biphoton count rate.

In Ch. 4, I use the framework developed in Ch. 3 to predict the heralding efficiency and joint count rate for four different geometry/wavelength combinations. I show how joint count rates and heralding efficiency can be made high simultaneously, although previous work has always claimed a trade-off between optimization of the two quantities. I then describe my experimental setup and measurements for each combination and show good agreement between the predictions and experimental results. I discuss how these results are not only important for QKD applications, but also other applications that require high heralding efficiency.

In Ch. 5, I discuss more details of the SPDC process, specifically the shape of the emission pattern produced by two different nonlinear crystals. The emission pattern from SPDC is typically circular and can thus be used for spatial multiplexing. Interestingly I discover that elliptical patterns are actually emitted for a specific nonlinear crystal that I use in my experiments. I show that elliptical emission is due to the polarization-dependent refractive index the daughter photons experience in the crystal, and that for a different crystal where the daughter photons have a different polarization-dependent refractive index, I indeed observe circular patterns. I discuss the impact this effect has on spatial multiplexing and entanglement purity around the down-conversion ring.

In Ch. 6, I describe one implementation of a time-bin QKD system and how to secure the time bins. I use a dispersive material to transform timing information to the frequency basis and vice versa so that I have access to both mutually unbiased bases of my system. I introduce fiber Bragg gratings at the attractive choice for dispersive material in my system. I describe my experiments using these fiber Bragg gratings to stretch and recompress single photon pulses so that I can measure correlations in both the time and frequency bases. Finally, I compare experimental data with theoretical predictions.

In Ch. 7 I conclude.

In Appendices A and B, I give details of my experimental setup (Appendix A) and details on detectors and detector technology (Appendix B).

Chapter 2

Quantum key distribution: A platform for the study of creating and manipulating biphoton sources

2.1 Introduction

In Ch. 1, I introduce several applications in quantum optics and quantum information that utilize biphoton sources and benefit from improvement in their efficiency. I also discuss several applications that take advantage of single photon manipulation in the time and frequency degrees-of-freedom. Although the improvement of biphoton sources and manipulation of single photons impacts all of these fields, the major field I focus on in this thesis is QKD. My work supports the efforts of the DARPA InPho program whose goal was to design and build a high-speed, high-dimensional free-space QKD system using multiple degrees-of-freedom of the photon. This project was a large collaboration between several major groups and individuals including Paul Kwiat's group at U. of Illinois, Mile Padgett's group at U. of Glasgow and Robert Boyd's group at U. of Rochester among others. The collective goal was to build a high-speed QKD system, and the results of my project supported this work. Specifically, my experiments aim at improving various parts of a high-speed, high-dimensional QKD system. Therefore, I use QKD as a platform for discussing the various techniques of optimizing a biphoton source and manipulating single photons.

In this chapter, I discuss the basics of creating the "front end" of a high-speed, high-dimensional QKD system – specifically the source, channel and detection. Although building and implementing a full QKD system is beyond the scope of this thesis, my experiments are aimed towards the improvement of high-rate time-frequency QKD using an SPDC source.

2.2 Key Sharing - Review of BB84 and E91

The pioneering work in creating a quantum key distribution system was done in 1984 by Gilles Brassard and Charles Bennet [31]. The protocol is named for the letters of their two last names and the year – BB84. In their protocol they imagined two parties, typically called Alice and Bob, communicating bits of information encoded in the polarization states of free-space photons. Their protocol is depicted in Fig. 2.1. Alice prepares a photon in one of four possible linear polarization states: two states, horizontal (H) and vertical (V), belong to the rectilinear basis and two states, antidiagonal (A) and diagonal (D), belong to the diagonal basis. Alice has the ability to switch randomly between these bases as well as between the two states in each basis. To send information she randomly chooses a basis and a state within that basis and sends the photon to Bob.

Bob has a similar setup where he chooses to measure the incoming photon's state in either the rectilinear or diagonal basis. However, Bob's choice of basis is random and may be different than Alice's basis for any given photon. After Alice has sent a certain number of photons depending on the desired key length, Alice and Bob perform a sifting protocol in order to obtain identical keys. They communicate over a public channel their individual basis choices for each bit. Each time they choose different bases, they throw away that bit: this happens roughly half the time. If no errors occur due to eavesdropping (which will be discussed in the next section), Alice and Bob each have a copy of the same states of polarization. Since each basis has two states, they assign either a zero or a one two each of the states and end up with an identical

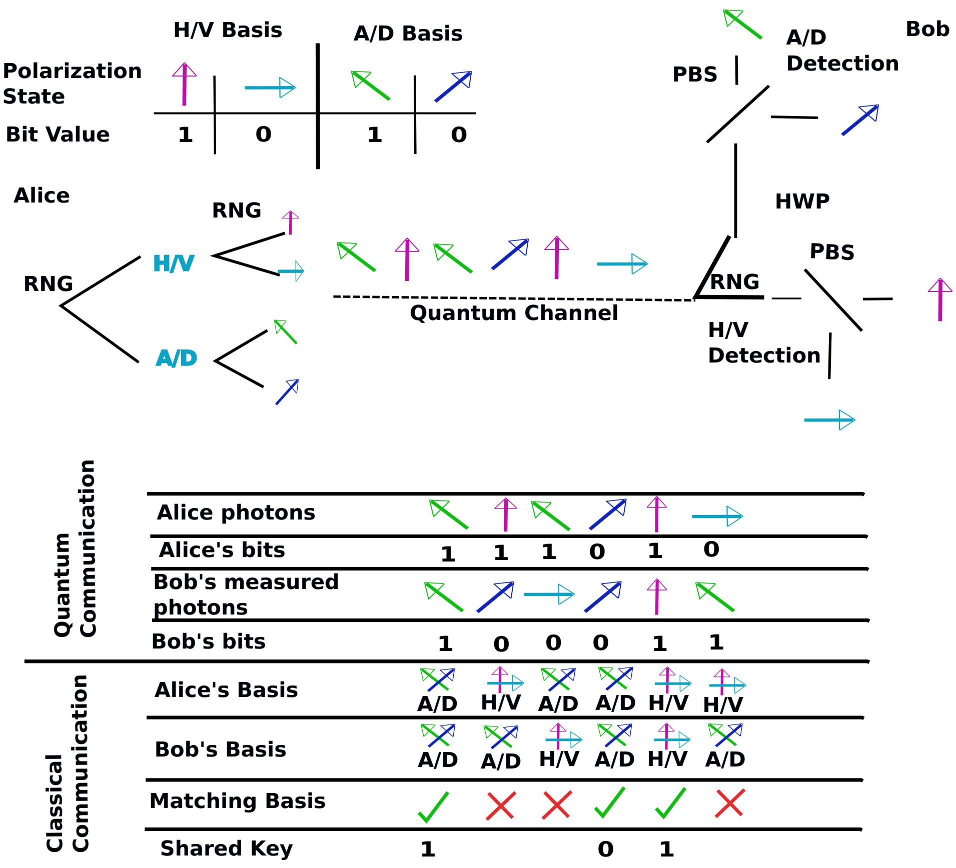

Figure 2.1: **BB84 Protocol.** Alice uses a random number generator (RNG) to choose which basis and which polarization state she sends to Bob. She chooses between the rectilinear basis which includes horizontally polarized (H) and vertically polarized (V) photon states and the diagonal basis which includes anti-diagonally polarized (A) and diagonally (D) polarized photon states. Bob then uses a RNG to send the photons through one of two detection paths. The A/D detection arm has a half-waveplate which rotates the state of polarization from the A/D basis to the H/V basis and vice versa. The photons in this arm are then sent through a polarizing beam splitter (PBS) and detected with photon detectors in each arm. The other arm for H/V detection consists of a PBS and detectors. After Alice has sent her photons, they use a classical communication channel which is insecure to compare their basis choices (but not their bits!). Every time they choose differing bases, they throw away the bit. When they choose the same basis they keep the bit. The only information exchange over this channel is the basis choice, but not the key itself. After the protocol they are left with a sifted identical key.

random binary key.

In the BB84 protocol, the sharing occurs between Alice, who creates and keeps a copy of the key, and Bob, with whom she shares the key. In contrast, there are other QKD sharing systems in which neither Alice nor Bob are in control of creating the key, but instead the key is generated and distributed to each Alice and Bob. The first proposal of this type of QKD system was published in 1991 by Artur Ekert (hence the E91 protocol) [32].

In the E91 protocol, a source produces entangled photon pairs in a spin-1/2 singlet state and one photon of each pair is distributed to Alice and Bob. Alice and Bob then make spin component measurements on their photon with analyzers oriented in three possible azimuthal angles. Each measurement will either yield a $+\hbar/2$ or $-\hbar/2$. Because the photon pair is originally in a spin singlet state, the two particles must have opposite polarizations. After the photons are sent and measured, Alice and Bob communicate to compare which orientations of the polarization analyzers they chose for each received photon over a classical public channel. They divide each measurement into two groups: one group for which they chose the same orientation and one group for which they did not. From the group where they chose different orientations of the analyzers, they reveal their results of their measurements over the classical public channel and from this, determine correlation coefficients. The correlation coefficients are made up of the joint probabilities of measuring the two photons in the same polarization basis are defined in Ref. [3]. These correlation coefficients are used to determine whether they have a pair of entangled particles. If they have entangled particles, then their results from the group where they used the same analyzer orientation form what they call the sifted key. Their sifted key is the result they obtain after they have thrown away all the times where their bases did not match each other.

Demonstrating that they have entangled particles is done by a violation of the Bell inequality [3]. In Ref. [3], a simple yet groundbreaking proposal was put forth whose premise was that certain combinations of correlation measurements on entangled particles could prove that they were quantum mechanically entangled and no local-hidden variable theory was in play. This remains one of the primary methods of demonstrating that a source is entangled. Since then, other inequalities have been imagined that are more experimentally feasible to demonstrate. One of these is the CHSH inequality, which Alice and Bob use in the E91 protocol to verify that they have entangled particles [33].

Although the key sharing techniques described above are only two of many protocols, they are both still widely used in QKD schemes today, in fact, most commercial QKD systems are based on these protocols. My key sharing experiment using the timing basis, which I describe in a Ch. 6 is based on these protocols. These protocols demonstrate the physical distribution ("key sharing") part of QKD. While this is the major objective, the key that Alice and Bob receive from these protocols is still prone to error. In the next subsection I discuss methods of how to minimize errors in the key without leaking information and how to maximize secure information content.

2.3 Major Components for a QKD system

Each QKD system consists of several major components, among these are the source, the channel, and the detection scheme. In this section, I discuss the source for my experiments, which is the nonlinear optical SPDC source I introduced in Ch. 1, as well as some of the major components of detection, although many of the details are left for Appendices A and B. In the experimental system I implement, the channel is roughly one meter of free space and does not substantially affect any of the parameters that drive the physics and so I will not discuss it here. However, it is important to realize that, in a real world setup with a free-space channel, atmospheric turbulence, and non-ideal conditions will also play into the final key rate as well as the error rate.

2.3.1 Source: SPDC

I use a nonlinear optical process to produce a pair of twin photons that are correlated in time and anticorrelated in frequency as the source for the QKD experiment. The process is called spontaneous parametric down-conversion (SPDC) and it occurs when a single pump photon (p) interacts with a nonlinear crystal in such a way that it decays into two outgoing photons called the signal (s) and idler (i). The nonlinear polarization vector that results from this three-wave mixing process is given by

$$P_a(\omega_p) = 2\epsilon_0 \sum_{b,c} \sum_{s,i} d_{abc} E_b(\omega_s) E_c(\omega_i), \tag{2.1}$$

where ω_j where $j = p, s, i$ is the frequency of the pump, signal, and idler respectively, E is the electric field and d_{abc} is the abc'th component of the nonlinear susceptibility tensor ([34], Ch. 3). SPDC must obey energy conservation so that the signal and idler frequencies sum to the pump frequency given by

$$\omega_p = \omega_s + \omega_i. \tag{2.2}$$

In *degenerate* SPDC, $\omega_s = \omega_i = \omega_p/2$, while in *nondegenerate* SPDC $\omega_s \neq \omega_i$. In addition, there are two different configurations for SPDC: *collinear*, in which the two daughter photons are emitted along the original direction of the pump, and *non-collinear*, in which the daughter photons are emitted at angles on either side of the pump (see Fig. 2.2).

While the degenerate/nondegenerate and collinear/noncollinear choices dictate

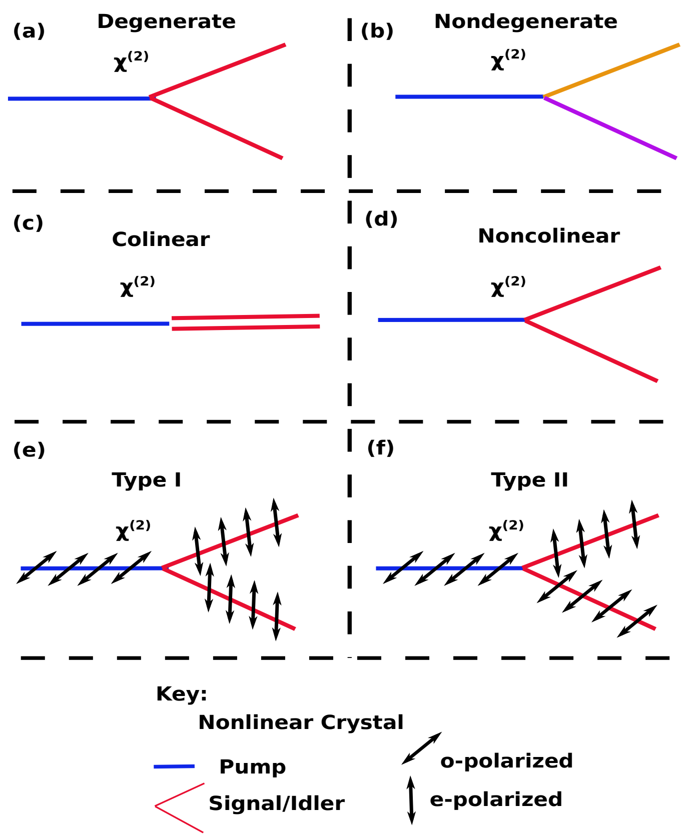

Figure 2.2: **SPDC Types** (a)(b) is a degenerate (nondegenerate) configuration where the signal and idler photons have the same (different) frequency(ies). (c) is the collinear geometry where the signal/idler photons are emitted in the same direction as the pump photon. (d) is the noncollinear case where the daughter photons are emitted on either side of the pump photon. (e) is a Type-I interaction where the polarization of the daughter photons is the same and opposite to that of the pump photon. (f) is a Type-II interaction where one of the two daughter photons has the same polarization as the pump, while the other photon does not.

the outgoing frequencies and direction respectively, the polarizations of the outgoing daughter photons are governed by the type of interaction. For daughter photons that have the same polarization as each other and opposite of the pump, the interaction is called Type-I. For daughter photons with opposite polarizations, the interaction is called Type-II. Typically, in uniaxial crystals, which are crystals with a single axis of symmetry, the polarization of the photons is referred to as ordinary (o-rays or o-polarized) when their polarization vector is perpendicular to the plane containing the optic axis of the crystal and the photon propagation vector, or extraordinary (e-rays or e-polarized) where the photon's polarization is in that plane. The geometry, wavelengths and polarization of the SPDC interaction will play an important role in tailoring the SPDC source to achieve high data rates, which I will discuss in Chs. 3, 4 and 5.

Phasematching

Energy conservation dictates which daughter frequencies are emitted in the SPDC process, but for any finite crystal length, phase-matching must also be considered. For either the collinear or noncollinear process, the photons carry momentum

$$\vec{k}_j = \frac{n_j(\omega_j, \hat{s}_j)\omega_j}{c}\hat{s}_j, \qquad (j = p, s, i) \tag{2.3}$$

where c is the speed of light in vacuum, $\vec{k}_j$ is the wavevector, $n_j(\omega_j, \hat{s}_j)$ is the frequency- and directional-dependent refractive index, and $\hat{s}_j$ is the unit vector in the direction of propagation. In collinear SPDC, the vectors all lie in the same plane and scalar quantities replace the vectors. In noncollinear SPDC, the daughter photons are emitted at angles (θ_s, θ_i) on either side of the pump beam. For details of phasematching, see

Fig. 2.3. Efficient SPDC requires phasematching (momentum conservation) so that

$$\vec{k}_p = \vec{k}_s + \vec{k}_i. \tag{2.4}$$

This equation only strictly holds for the case of an infinitely long crystal and and infinitely wide pump beam [35, 36]. Classically, the phase mismatch arrises from the chromatic dispersion that is present in most crystals. Because of this dispersion, different frequencies travel at different speeds and the difference between the interacting wavevectors does not sum to zero. Quantum mechanically, there is also a phase mismatch that arrises from the uncertainty principle given by $\Delta x \Delta p \geq \hbar/2$. This can be rewritten as $\Delta L \Delta k \geq 1/2$ where L is the crystal length. This means that the precise location and momentum of the photon that is born in the proccess cannot be known with arbitrary accuracy and so a phase mismatch arrises unless the crystal is infinite.

SPDC will still occur with a small phase mismatch which I define as

$$\vec{\Delta k} = \vec{k}_p - \vec{k}_s - \vec{k}_i. \tag{2.5}$$

When $\vec{\Delta k} = 0$, the interaction is perfectly phase matched. In general, the phase-mismatch determines the conversion efficiency of the process. This will be discussed in great detail in Ch. 4, but briefly, the output intensity of the SPDC interaction depends on a sinc-squared function with the argument of the sinc being $\Delta k L/2$. The sinc function arises from performing an integral of the interacting fields over the finite crystal length which is shown in Ch. 4. The sinc function is maximized at $\Delta k = 0$ and has its first zero at $\Delta k L/2 = \pi$, so that a phase-shift of π for the quantity $\Delta k L/2$ brings the conversion efficiency from its maximum value to zero.

Phasematching dictates the emission directions of the daughter photons. In non-

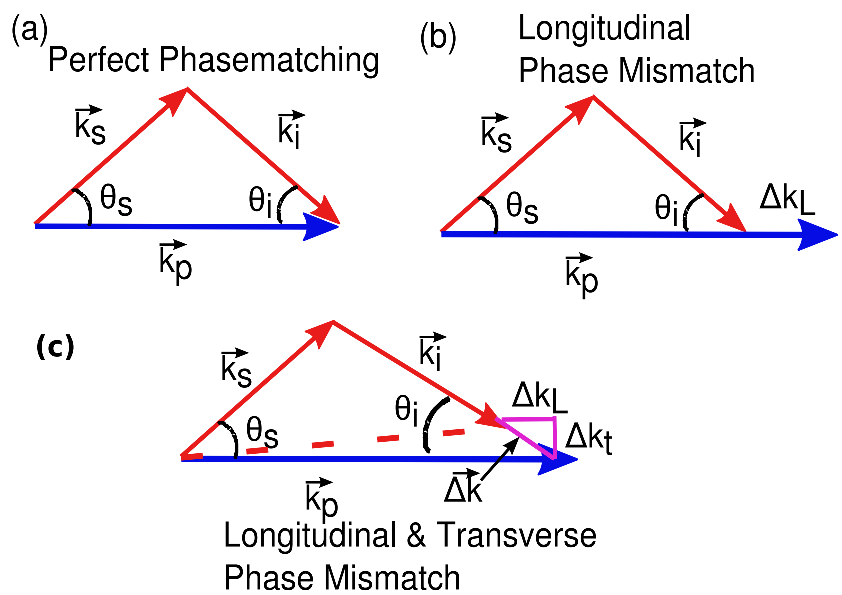

Figure 2.3: **Phase-matching** In (a) I show a diagram of perfect phase-matching where the momentum vectors of the signal and idler completely sum to the pump momentum wavevector. In (b) there is a longitudinal phase mismatch (Δk_L) and in (c) there is both a longitudinal (Δk_L) and transverse (Δk_t) mismatch

collinear SPDC, I solve for the angles θ_s and θ_i that satisfy the phasematching conditions given by the following equations in each direction (x,y,z)

$$n_p(\theta_p, \phi_p, \omega_p)\frac{\omega_p}{c}\hat{s}_p = n_s(\theta_p, \phi_p, \theta_s, \phi_s, \omega_s)\frac{\omega_s}{c}\hat{s}_s + n_i(\theta_p, \phi_p, \theta_i, \phi_i, \omega_i)\frac{\omega_i}{c}\hat{s}_i, \tag{2.6}$$

where the angles are depicted in Fig. 2.4. I choose the pump phase-matching angles (called the crystal cut angles), θ_p and ϕ_p and either ϕ_s or ϕ_i and solve for the emission angles, θ_s and θ_i. I could also do this process in reverse: select a direction of emission and then find what crystal cut angle I need to achieve this. To solve for the phasematching conditions (angles), I use a root-finding program to solve Eq. 2.5 for $\Delta k = 0$. The daughter photons are emitted at angles θ_s and θ_i at local azimuthal angles, ϕ, around the pump and will thus form a cone in three-dimensions. A transverse

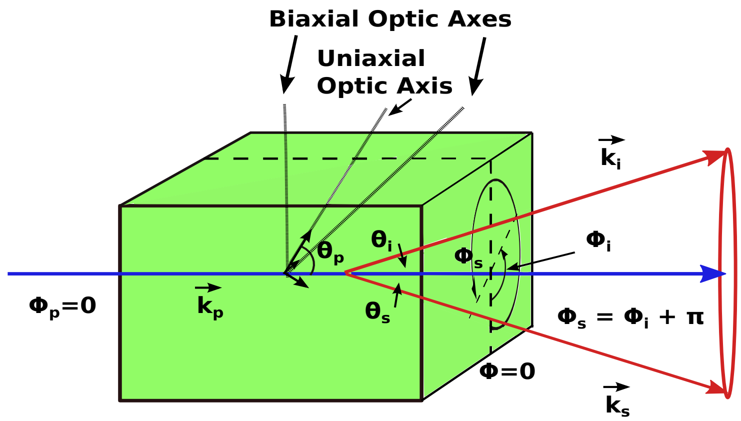

Figure 2.4: **SPDC geometry in nonlinear crystal** A pump photon enters a nonlinear crystal at angles θ_p, ϕ_p with respect to the crystal's optic axis (or axes if it is a biaxial crystal). A signal (idler) photon is emitted at angles θ_s, ϕ_s (θ_i, ϕ_i) relative to the original pump direction. They have wavevectors $\vec{k}_s$ and $\vec{k}_i$ respectively. The signal and idler photon must be emitted on opposite angles of the pump in the same plane so that $\phi_s = \phi_i + \pi$.

slice of this cone appears as a ring called the down-conversion ring. In Chs. 3 and 4, I discuss how phase mismatch affects the spectrum and provide more details for the specific cases of degenerate and nondegenerate SPDC.

Nonlinear Crystals

There are a variety of nonlinear crystals used in SPDC processes that differ for various applications and geometries. Periodically-poled crystals are typically used in collinear and Type-II geometries [37, 38]. In my experiment I use thin crystals and Type-I phase matching because I am able to entangle both the polarization degree-of-freedom as well as timing degrees-of-freedom. Polarization entanglement is one possible method of adding security to information encoded in the timing degree-of-freedom, and so to keep this possibility open, I use a setup that lends itself to polarization entanglement.

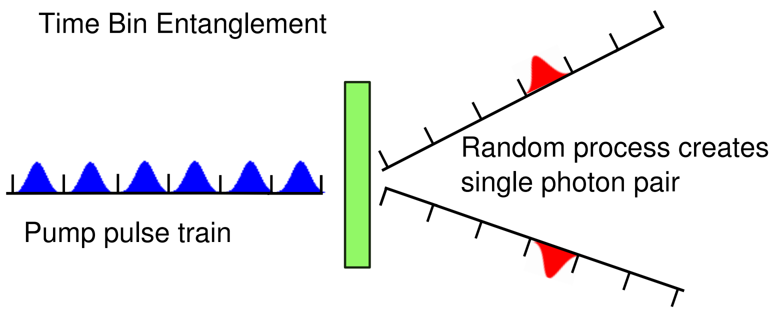

Figure 2.5: **Time bin entanglement**
Time-bin entanglement is created when SPDC photons are randomly born from a pulse train of pump photons. Because there is no information about which pulse in the pulse train produced the photon pair, the photons are entangled in time and represented by a state similar to the one in Eq. 2.7.

One of the most commonly used crystals for noncollinear setups is a crystal called beta barium borate (BBO). BBO is a uniaxial crystal, meaning it has a single axis of symmetry. This crystal is commonly used because it has a high nonlinear coefficient d_{eff}, which dictates the likelihood of getting a daughter photon pair with a pump photon. Recently, however, it was shown that a biaxial crystal, bismuth barium borate (BiBO) has an even higher d_{eff} by almost a factor of two as well as a higher damage threshold compared to BBO [39, 40]. It also has a larger phase-matching range, making it an attractive choice for my experiments.

2.3.2 Time Bin Entanglement

I encode bits of information in the timing degree-of-freedom of photons born from the SPDC process. These photons are entangled in the timing degree-of-freedom because they originate from a train of pump pulses with time between these pulses less than that of the time frame. This pump pulse train is sent into the crystal and because the process of SPDC is a random one, the daughter photons are born at random times from random pulses in the pulse train. The photons are then in a superposition of the time bins in a single frame because, until a measurement is made, they could have resulted from any one of the string of pulses. The "which-path" information here is which pulse in the pump pulse train created them, and this is not known. This is depicted in Fig. 2.5. The wavefunction expressed in the time basis is then written

$$|\psi\rangle \propto |t_1\rangle_A|t_1\rangle_B + |t_2\rangle_A|t_2\rangle_B + |t_3\rangle_A|t_3\rangle_B + ... + |t_N\rangle_A|t_N\rangle_B, \tag{2.7}$$

where 1,2,3...N are the first through N^{th} time bin, and A and B are the two photons, signal and idler. Because the particles are entangled this is not a separable wavefunction.

2.3.3 Detection

The final piece of the QKD/SPDC system is the detection of the shared key by Alice and Bob. Alice and Bob each have collection lenses that focus a part of the downconversion ring into single mode collection fibers in my experiment. Depending on the geometry and wavelength in the interaction, spectral filtering may be used to select out portions of the spectrum where correlations are the highest. This will increase a quantity called the heralding efficiency which is introduced in the next section. After passing thought the filter and lens, the photon enters a single mode fiber that is connected to a single photon counting detector.

Single-photon avalanche photodiodes (SPADs) make up the majority of single-photon detectors in the visible and near-IR wavelength range. They have high quantum efficiencies, allowing for more possible detected events, as well as fast circuits, which allow them to maintain high detection rates. The key specifications that control the overall rate of particular SPADs are the deadtime of the detector, the timing jitter, and the quantum efficiency.

The deadtime of a detector is the time in which, after an event has been registered, the detector is basically in an "off" position and not ready to receive another photon. After a photon hits the detector, an electron-hole pair is created in the p-n material of the detector. For wavelengths in the visible and NIR part of the spectrum, the primary p-n material used is silicon. The p-n material has a large reverse bias voltage across it that allows the single electron-hole pair to be accelerated, creating more electron-hole pairs. Those pairs are accelerated, creating even more pairs in an avalanche process. When enough pairs are created, the current can be registered as a pulse. It is the avalanche process that allows a single photon to create a measurable current in the device.

During the avalanche process, there are electron-hole pairs that are "left behind" in the material that do not get accelerated immediately. After the main pulse is registered, it is possible for these pairs to create another avalanche event that is not triggered by the arrival of a new photon. This is called "afterpulsing" of the detector and results in errors in the final key. To minimize these errors, the reverse bias voltage that allows the electron-hole pairs to accelerate in the material is switched off after a main event is registered. The time it takes to shut the current off, wait until the afterpulsing probability decays (see Appendix B) and switch the bias voltage back on is called deadtime – the total time in which the detector does not register any incoming photons. In order to have a high-speed system, this deadtime should be as short as possible.

Another parameter in SPADs that limits their speed is a quantity called the jitter. The jitter is intimately linked to the width of the time bins in my system because it is a measure of how accurately the detector registers the exact timing of an incoming photon. If there were a periodic train of single photons that arrived every 1 μs at the detector, the detector would measure them arriving at every $1\mu\text{s} \pm \delta t$ where δt is the jitter. In order to reliably place a single photon in one of N time bins, the time bin width must be at least the jitter time of the detector. Therefore, the ultimate speed in the high-dimensional QKD system is limited by the width of the time bins, which must be at least the temporal jitter of the detector. For my experiments, the jitter depends on which detector I use and typically $\sim$50-200 ps.

Finally, the quantum efficiency of the detector will affect the total key generation rate. The quantum efficiency of a single photon detector is the probability that, if a photon hits the active material of a SPAD, this will result in a measurable electrical output pulse. SPAD performance has recently increased with high efficiency detectors in the visible and near IR ranges boasting above $70-80\%$ quantum efficiency. High efficiency at these wavelengths as well as a strong pump laser in the UV was the driving factor for choosing the wavelengths in my system.

2.4 Security of QKD

Because my work has been supporting the goal to build an entire QKD system using time-bin entangled states, one of the issues that arises in these systems is security. Although my final goal does not include building a fully secure QKD system, I still work under the assumption that my system will need to be secured in the future and aim to make this physically possible. In this section I discuss two main eavesdropping attacks, although there are many more – these two attacks are fundamental to using the time-energy or time-frequency duality for the basis states described in Ch. 6. The security of QKD protocols is based on the ubiquitous nature of the quantum world — namely the idea that a system can and will change somehow when it is observed. There is a footprint left by anyone who attempts to observe the quantum bits while they are transferred across the quantum channel. In this section I review two types of attacks on a QKD system: the intercept-and-resend attack and the quantum nondemolition attack.

2.4.1 Intercept-and-Resend

The simplest type of attack that can be made on a QKD system, and one that is independent of imperfections of physical devices, is the intercept and resend attack. In this attack, the eavesdropper (Eve), steals the photon Alice sends to Bob, measures its state, and based on the result, prepares and sends that exact state to Bob. To examine this more closely, recall the BB84 protocol where Alice and Bob are sending and measuring photons with a certain polarization in one of two bases (H/V or A/D). Suppose Eve does an intercept and resend attack on their bit string of photons. In each case, she "steals" the photon Alice sends, and does a measurement on it. She, of course, has no idea what basis Alice prepared her photon in, and so she must randomly guess which basis to measure the photon in. After she makes the guess, she measures the state of the photon, prepares that exact state, and sends it to Bob. Bob then, as he does in BB84, randomly guesses a basis in which to measure this photon and makes a measurement. In doing so, Eve forces the quantum state of the photon to collapse when she measures it - she *changes* the quantum state. Alice and Bob can now detect this change and verify her presence.

To observe Eve's presence and thus verify security of their system, Alice and Bob compare the actual bit values of a small part of their sifted key. Since their sifted key contains only those measurements taken using the same basis, they expect to have a key that has 100% identical bits. However, consider the following: Alice sends Bob an H-polarized photon, but Eve measures that photon in the A/D basis. Expressing a horizontally polarized photon in the diagonal basis

$$|H\rangle = \frac{1}{\sqrt{2}}\left(|A\rangle + |D\rangle\right). \tag{2.8}$$

This means when Eve measures in the A/D basis, she has a 50% chance of measuring A-polarized and 50% change of measuring D-polarized. Let us suppose she measures it in the A-polarized state. She then prepares and sends an A-polarized photon. If Bob has chosen the H/V basis to measure the photon (which he and Alice would have kept for their sifted key), he has a 50% chance of measuring H-polarized and 50% chance of measuring V-polarized for the same reasoning as above. If he measures V-polarized, when he compares his bit string with Alice, they will have an error.

Eve measures the wrong bit every time she measures the opposite basis that Alice has chosen ($\sim$ 50% of the time) and Bob projects the incorrect state Eve sends in the incorrect basis into the incorrect state in the correct basis ($\sim$ 50% of the time). This means that approximately 25% of the time Alice and Bob will see an error in their sifted key. When this error rate occurs, they know Eve is present!

The security that quantum mechanics guarantees in this type of attack is based on the information being encoded in these two bases which have a special relationship to each other. In the previous example, the H/V basis and the A/D basis are examples of *mutually unbiased bases* (MUBS) which are a set of bases for which the knowledge of a state in one basis gives no information about the state if measured in the other basis. For example, knowing the state in Eq. 2.8 is H in the H/V basis gives no information about what the state (A or D) would be if measured in the A/D basis, as their probabilities of occurring are equal.

The security of QKD in the intercept-and-resend attack is based on the fact that an eavesdropper will change something about the state she is measuring if she measures in the opposite MUB from Alice and Bob. More generally, for any attack, if Alice and Bob have knowledge of both the basis they are using and its MUB, Eve will never be able to make a measurement without changing the state in one of the two MUBS and in doing so, she leaves herself open for detection. The intercept and resend attack is a simple and common attack to implement on any QKD in theory. In the next section I describe an attack where the eavesdropper is more difficult to detect called the quantum non demolition attack.

2.4.2 Quantum Nondemolition

Quantum nondemolition (QND) measurements are basically “perfect” measurements of a system. They are “back-action” evading, meaning that the outcome of a measurement of the system does not change with successive measurements if the system is measured in the same basis each time. The state of the system immediately after the measurement is in the eigenstate of the observable and will continue to evolve this way [41, 42] when measured in the same basis. The danger in QND measurements is that Eve can probe the system and gain information about the state in one basis without destroying it. However, even though she does not change the state in one basis, by the uncertainty principle, she must change the state in its conjugate basis. If Eve makes a QND measurement in, say the time basis, she gains the timing information of the photon without changing it and so may go by undetected. If she localizes a state in one basis, she must necessarily cause a change in its conjugate. Conjugate bases and MUBS are, in fact, the same. This means, that if Alice and Bob were to monitor both MUBS they would be able to detect and eavesdropper doing a QND measurement on either basis. The conjugate basis to time is frequency, and this will be important for the secure key I discuss in Ch. 6.

2.5 My approach: A High-Dimensional, High-Speed QKD Scheme Using Free-space Photons

2.5.1 Limitations of Current QKD Schemes

In most early implementations of QKD schemes, only one bit of information was encoded on the photon because the early protocols predominantly used the polarization degree-of-freedom of the photon. Although current off-the-shelf commercial QKD systems offered by companies such as idQuantique, MagiQ, and Quiessense, have moved away from the polarization degree-of-freedom and now use timing states and phase state, they still only encode 1 bit per photon because they only create two such timing or phase states. This, along with loss in the channel, is responsible for the low achievable key-rates of these commercial QKD systems which is on the order of 1000 bits per second (1 kbps).[1] This low data rate is a severe drawback of these systems in today's increasingly fast information network.

In order to make any QKD system practical by current standards, it has to be made roughly 10^6 times faster. Internet data rates run upwards of 10-100 Gbps, with this limit being pushed constantly. The goal for the DARPA project was to build a QKD system that ran at speeds greater than 1 Gbps. Although a daunting task, there are two methods I use to increase the rate of data sent in my system. First, I encode multiple bits of information on a single photon and second, I use faster detectors and sources to ensure that the rate of photon transfer is also faster. In this section, I will discuss the first point: how I encode multiple bits of information on a photon making it high-dimensional QKD (HDQKD).

[1]http://www.idquantique.com/photon-counting/clavis.html

2.5.2 High-Dimensional QKD

The question of how many bits of information can be encoded on a photon is understood by considering the number of physical states that photon can occupy. In information theory, if a device can be in one of N states, it maximally encodes

$$\#\ \text{of bits} = \log_2(\text{N}), \tag{2.9}$$

bits of information [43] when the average signal photon number is much greater than the average thermal photon number, the source is Poissonian and the average photon number is much less than 1. In this limit, Yammamoto and Haus showed that the channel capacity is given by

$$C = B\overline{n_s} \overbrace{\log_2\left(\frac{1}{\overline{\text{n}_\text{s}}}\right)}^{\#\ \text{bits per photon}}, \tag{2.10}$$

where C is the channel capacity, B is the channel bandwidth, and $\overline{n_s}$ is the average signal photon number [44]. If on average, a photon is placed in one of N different states, this gives $\overline{n_s} = 1/N$, which agrees with Eq. 2.9. Therefore, a photon that is in one of two polarization states stores one bit of information in the polarization basis. In order to increase the number of states the photon has access to, information is encoded in different degrees-of-freedom including but not limited to orbital angular momentum states [45, 46], spatial states [47, 48], and time/frequency states [26, 49, 50]. Compared to polarization, each of these other degrees-of-freedom allows the photon to occupy many possible states at once, thereby allowing more information to be encoded.

Although all of these degrees-of-freedom are potentially good candidates to use, I choose to use the timing degree-of-freedom of the photon for a few specific reasons.

First, the timing degree-of-freedom is relatively simple to create and measure. Photons are always born at a specific time and are detected at a specific time. Key sharing is implemented using the timing degree-of-freedom when both users share the same clock. No expensive custom optics or devices are needed to create this. Secondly, the timing degree-of-freedom does not decohere as much in free-space channels or in fiber-based channels as some other degrees of freedom, although over long fiber distances, dispersion can cause timing fluctuations.

My source produces two photons that are born simultaneously and therefore carry identical timing information. Although time is a continuous variable, I segment it into separate time bins that contain on average one pump pulse per time bin. On average one down-converted photon pair is born from the series of pump pulses in a segment called a time frame. A construct depicting this idea is shown in Fig. 2.6 wherein there are a number of discrete time bins of width Δt that make up a time frame T_f. When pumped with a pulse train, a pair is born randomly in one of N time bins in this frame on average. If the frame consists of two time bins, then according to Eq. 2.9 the photon would carry maximally one bit of information because it could occupy two possible time states. If the photon has N possible time bins in a frame it carries maximally $\log_2(\mathrm{N})$ bits of information from Eq. 2.9 in the noiseless limit. Increasing the number of time bins therefore increases the number of bits per photon I achieve.

The drawback to encoding information in this manner is that there is a trade-off between the number bits per photon and the total data rate. Because there is one photon (on average) per time frame, the total photon rate decreases when the time between sending photons increases, for example, for large N. Although I want to encode many bits per photon, I simultaneously need to send them at high rates. One way of achieving data rates in excess of 1 Gbps is to encode 10-bits-per-photon and send each photon at a rate of 100 MHz. The number of bits per photon corresponds

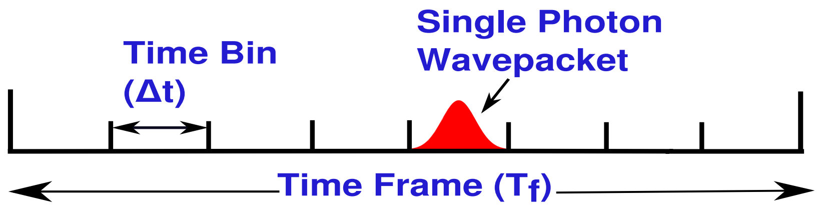

Figure 2.6: **Encoding time bits** I encode individual bits of timing information on a single photon using discrete time bins. A single photon is randomly placed in any one of N time bins which have a temporal width (Δt). There are N of these time bins that make up one time frame. A single photon placed randomly in this frame carries $\log_2(N)$ bits of information.

to 1024 time bins, while the data rate sets a maximum limit on the width of the time bin ($\Delta t \approx 10$ ps). Although my current experiment does not run at this speed due to limitations in the detectors, which I discuss later in the chapter, I have the capability to run only roughly a factor of 10-20 slower ($\Delta t \approx 100 - 200$ ps). There are, of course, trade-offs to running the system with this time bin size, which I will discuss in Ch. 6 and Appendix B.

2.5.3 High-Speed QKD

There are two main parts of my system which contribute to the high rate. The first is the efficiency of the SPDC source, meaning how well twin photons that are correlated in time are created and collected. Briefly, the source is made up of a nonlinear crystal which is pumped by a strong laser. The nonlinear optical process of spontaneous parametric down-conversion converts the pump photons into twin daughter photons born at the same time. The details of this process will be discussed in the first half of this thesis. There are two metrics I used to quantify how well I can create and collect these twin photons – the heralding efficiency and joint count rate, and these both play an important role in determining the rate of my system. I will discuss these parameters extensively in Chs. 3 and 4.

The second aspect that dictates the speed of the system is speed of the physical components in the system. The maximum speed of physical systems is limited by the slowest constituent component of the setup. In many QKD systems, mine included, the slowest component is the single-photon detectors. As part of my thesis work, I worked on developing and characterizing different types of single-photon detectors that would perform optimally for my system. I focused on minimizing a quantity called the timing jitter as well as maximizing the efficiency of the detectors. I discuss more details of the basics of single photon counting detectors in Sec. 2.3.4 and fully describe my efforts in detector characterization in Appendix B.

The experimental details are discussed fully in Appendices A and B.

Chapter 3

Joint and Singles Spectrum

3.1 Introduction

In this chapter, I derive expressions for the spectra of both the biphoton wavefunction (joint) as well as the individual single photon wavefunction (singles). The joint and singles spectrum contribute significantly to the overall heralding efficiency and joint count rate, which are the two important metrics that I aim to optimize in my system. In Sec. 3.2, I give a brief overview of heralding efficiency and joint count rate and motivate why optimizing them is important for my experiments as well as many other experiments in quantum science fields. I then discuss the formalism for predicting the joint and singles spectrum in Sec. 3.3 and derive expressions for the joint spectrum (Sec. 3.4) and singles spectrum (Sec. 3.5). In Sec. 3.6, I discuss four geometry/wavelength cases; collinear degenerate, collinear nondegenerate, noncollinear degenerate and noncollinear nondegenerate and how the spectra for each of these cases differs for a given set of parameters.

3.2 Overview of Heralding Efficiency and Joint count rate

Heralding efficiency and joint count rate (also known as biphoton rate or coincidence rate) are two of the metrics that are important in creating a high-rate source for QKD using SPDC. The heralding efficiency I consider here is the symmetric heralding efficiency given by

$$\eta = \frac{R}{\sqrt{R_s R_i}}, \tag{3.1}$$

where $R_{s(i)}$ is the rate of single detections for the signal (idler) beam and R is the joint count rate or rate of correlated photon events. These events occur when both photons of the pair are detected simultaneously in a time window called the coincidence window. Typically, the window is equal to, or shorter than, the pulse width of the single photon counting detector. Optimizing the joint rate is crucial for high-speed QKD systems because the secure key rate is proportional to R.

The heralding efficiency represents the probability of detecting one twin photon (the idler) *conditioned on* its twin (the signal) being detected. The maximum conditional probability is unity; however, if one of the photons in the pair is not detected while its twin is, the heralding efficiency decreases. This occurs due to losses in the system and detectors and inabilities to collect each pair of photons into a single mode given the phase matching conditions discussed later in this chapter. Enhancing the heralding efficiency is important in a number of systems described in Ch. 1 that are sensitive to this value. These include detection loophole-free tests of Bell's inequality ($\eta_T \geq 66\%$)[6, 7, 11], one-sided device independent QKD (DI-QKD) ($\eta_T \geq 66\%$) [22], and three party quantum communication ($\eta_T \geq 60\%$) [23, 24]. Here, η_T denotes the total efficiency of the system including the heralding efficiency from the source, efficiencies in the optics, channel and detectors. Increasing the heralding efficiency increases the overall efficiency of the system.

Both the heralding efficiency and the joint count rate depend on almost all the accessible parameters of the system including the beam waist sizes in the crystal, the crystal length, frequencies, geometry of the interaction, type of crystal, and the spectral filtering. Interestingly, there is often a trade-off between heralding efficiency and joint count rate [51].

In this thesis, I consider only the so-called Type-I interaction where all the beams are linearly polarized and the two daughter photons have identical polarization that is perpendicular to the pump polarization.

3.3 Formalism for prediction joint and singles spectral rate

The procedure and formalism for predicting the joint and singles spectrum has been well established. Here, I adapt the formalism of Ling [52], who predicts the joint spectral function and subsequently the joint spectral rate. This work does not, however, predict the singles spectrum, which is needed to obtain the heralding efficiency. In addition, the joint spectral rate that Ling *et al.* derive is valid only for Type-II interactions and nondegenerate Type-I interactions. Here, I derive expressions that are valid for Type-I degenerate and nondegenerate interactions. I adapt the approach of Bennink [51] for calculating the singles spectrum and singles count rates, although his work primarily considers collinear geometries in a periodically poled medium. I finally incorporate ideas from Mitchell [53] to model my spectral filter functions.

I calculate the joint and singles spectral functions with the interaction Hamiltonian for the SPDC process acting on a vacuum state. The interaction Hamiltonian is given by

$$\hat{H}_I = -\frac{2\epsilon_0\chi^2}{8}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} dx\,dy \int_{-L/2}^{L/2} dz\,\mathbf{E}_p^{(+)}\hat{\mathbf{E}}_s^{(-)}\hat{\mathbf{E}}_i^{(-)} + H.c., \tag{3.2}$$

where ϵ_0 is the permitivity of free-space, χ^2 is the second order nonlinearity constant, L is the length of the crystal, and $\mathbf{E}_j$ is the electric field mode operators for $j = p, s, i$ respectively. I will define these in more detail in the following section. When this

Hamiltonian acts on the vacuum state, creation operators in the Hamiltonian populate the target states. Finding the joint spectral rate involves an integral of the product of the pump, signal, and idler modes over a volume of the nonlinear optical crystal and includes a term that is dependent on the phase matching conditions for the interaction which I detail in the next section. The geometry of the interaction is shown in Fig. 3.1.

I make several assumptions that are valid for my experimental setup and allow me to obtain a closed analytic expression for the joint and singles spectrum. First, I assume the paraxial approximation holds for the interaction beams. I collect the photons into single mode fibers whose modes are given by the lowest order Gaussian beams

$$U_j(x,y) = e^{-(x_j^2+y_j^2)/W_j^2} \; (j = p,s,i), \tag{3.3}$$

where W_j is the $1/e$ radius of the field mode. The spatial modes are normalized by

$$\alpha^2 \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} |U(x,y)|^2 = 1, \tag{3.4}$$

where

$$\alpha_j = \sqrt{\frac{2}{\pi W_j^2}}. \tag{3.5}$$

I further assume that the pump is in a coherent state and that it is monochromatic so that there is a strict relationship between the frequencies of the pump, signal, and idler photon, namely, $\omega_p = \omega_s + \omega_i$. This assumption is valid for my experimental setup because the pump beam I use is a transform-limited sech-squared pulse with a temporal width of ∼5 ps. This means its bandwidth is ∼ 30 GHz at 355 nm, which corresponds to a bandwidth of 13 pm. This should be compared to the bandwidth of the down-converted light, which is between 20-60 nm. This is orders of magnitude less

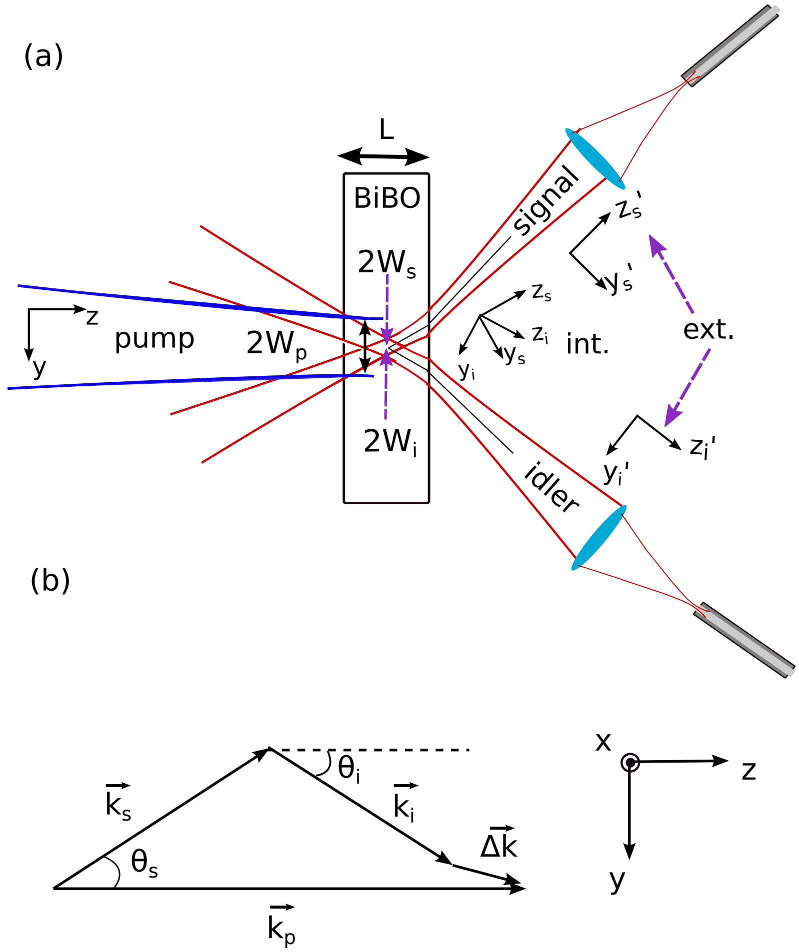

Figure 3.1: Beam interaction geometry and phasematching for noncollinear SPDC
(a) A pump beam, propagating along the z-axis is focused into a nonlinear crystal of length L to a Gaussian waist of W_p. The back-propagated fiber modes are also focused into the crystal with waists of $W_{s(i)}$. The axes for the down-converted beams $(y_{s(i)}, z_{s(i)})$ are calculated by a simple transformation in Eq. 3.10. When the beams reach the crystal-air interact they refract according to Snell's law which is indicated by the primed coordinates $(y'_{s(i)}, z'_{s(i)})$.
(b) Relationship between the wave vectors of p, s, i for noncollinear geometries. The phase mismatch is in both y and z directions and is given by $\vec{\Delta k}$.

and so I consider the pump to be monochromatic. I assume that the nonlinear crystal is thin so that the Gouy phase of the Gaussian beams are constant over the length of the crystal and therefore ignored.

Also, I assume that the transverse dimensions of the crystal are large enough that the interacting beams do not diffract. Diffraction occurs if the transverse dimensions of the crystal are similar to the pump, signal and idler waist sizes in the crystal. For both the theoretical modeling and the experimental implementation, I use pump beam sizes of 250 μm or 150 μm. For the signal/idler beams I use beams sizes of $\sim 100\,\mu$m or 50 μm. The transverse dimensions of the crystal are typically 5 $\times$ 5 mm, so the assumption of nondiffracting beams is valid. I calculate the transition rate of the interaction using Fermi's Golden rule. Finally, once I have the joint or singles spectral function, I integrate over all frequencies of the interaction to obtain a total joint or singles spectral rate.

3.4 Joint Spectrum

To calculate the joint spectrum and subsequently the joint count rate, I first compute the overlap integral of the three interacting beams in the nonlinear crystal. This overlap integral arises from the interaction Hamiltonian in Eq. (3.2) acting on the vacuum states. The mode field operators involve raising operators for the signal and idler field and an annihilation operator for the pump field. However, because the pump is strong compared to the down-converted fields, I treat the pump classically. I assume a non-depleted pump field where the pump field mode can be written as

$$\hat{\mathbf{E}}_p^{+}(\vec{r},t) = \frac{1}{2}[E_p^0 \mathbf{e}_p e^{ik_p z} U_p(x,y) e^{-i\omega_p t}], \tag{3.6}$$

where $\mathbf{e}_p$ is the polarization vector, $k_p = n_p(\omega_p, \vec{k_p})\omega_p \hat{k_p}/c$ and E_p^0 is the pump field given by

$$|E_p^0| = \alpha_p^2 \frac{2P}{\epsilon_0 n_p c} \tag{3.7}$$

where P is the pump power, ϵ_0 is the permitivity of free space, $n_p(\omega_p, \vec{k_p})$ is the direction and frequency dependent refractive index for a photon of frequency ω_p, $\hat{k_p}$ is the unit vector in the direction of propagation and c is the speed of light in vacuum. The signal and idler field modes are given by

$$\mathbf{E}^{-}_{s(i)} = \frac{-i}{2} \sum_{k_{s(i)}} \sqrt{\frac{2\hbar\omega_{s(i)}}{n^2_{s(i)}\epsilon_0}} \frac{\alpha_{s(i)}}{\sqrt{L}} \mathbf{e}_{s(i)} e^{-ik_{s(i)}z} U_{s(i)}(x,y) e^{i\omega_{s(i)}t} \hat{a}^{\dagger}_{k_{s(i)}} \tag{3.8}$$

where L is the crystal length, $\mathbf{e}_{s(i)}$ is the polarization vector for signal (idler), $k_{s(i)} = n_{s(i)}(\omega_{s(i)}, \vec{k_{s(i)}})\omega_{s(i)}\hat{k_{s(i)}}/c$ where, $n_{s(i)}(\omega_{s(i)}, \vec{k_{s(i)}})$ is the frequency and direction dependent refractive index for a photon of frequency $\omega_{s(i)}$. The wavevectors for the signal and idler can be expressed in the coordinate system of the signal and idler or in that of the pump. The two coordinate systems are related by the simple transformation

$$\begin{aligned} y_s &= cos\theta_s y + \sin\theta_s z \\ z_s &= -sin\theta_s y + \cos\theta_s z \\ y_i &= cos\theta_i y - \sin\theta_i z \\ z_i &= sin\theta_i y + \cos\theta_i z, \end{aligned} \tag{3.9}$$

depicted in Fig. 3.1. The signal and idler experience refraction at the crystal-air interface giving rise to the primed coordinate system in Fig. 3.1. I calculate the exterior angles (θ_s, θ_i) using Snell's law where,

$$n_{s(i)}(\omega_{s(i)}\vec{k_{s(i)}}) \sin\theta'_{s(i)} = \sin\theta_{s(i)}, \tag{3.10}$$

where $n_{s(i)}(\omega_{s(i)}\vec{k_{s(i)}})$ is the refractive index in the crystal and is generally a function of frequency and propagation direction inside the crystal, and $\theta'_{s(i)}$ is the external emission angle. For biaxial crystals, such as the BiBO crystal I use, the refractive index is either angle-dependent or angle-independent over certain frequency ranges. In my case, for a pump wavelength in the UV part of the spectrum and the daughter photons in the red/NIR part of the spectrum, the daughter photons experience an angle-dependent refractive index. The impact of this on the spatial emission pattern for the whole down-conversion ring will be discussed in Ch. 5. For the purposes of this chapter, I find that the joint and singles spectra do not change significantly inside versus outside the crystal. In this chapter, I calculate the joint and singles spectra inside the crystal.

Using the field modes in Eq. 3.6 and 3.8 the interaction Hamiltonian becomes

$$H_I = -\frac{2\epsilon_0\chi^2}{8}\int_{-\infty}^{\infty} dxdy \int_{-L/2}^{L/2} dz \sum_{k_s,k_i} \frac{\hbar\sqrt{\omega_s\omega_i}}{n_s n_i}\frac{\alpha_s\alpha_i E_p^0}{L} * e^{-i\Delta\omega t} \tag{3.11}$$

$$\times U_p(x,y)U_s(x,y)U_i(x,y)e^{i(k_p-k_s-k_i)z}\hat{a}^\dagger_{k_s}\hat{a}^\dagger_{k_i}. \tag{3.12}$$

To calculate the SPDC interaction rate, I use the interaction Hamiltonian acting on the vacuum state to populate a final desired state. If the initial vacuum state for the signal and idler is $|0_{k_s}, 0_{k_i}\rangle$, the Hamiltonian acts on this state so that,

$$|1_{k_s}, 1_{k_i}\rangle = \hat{a}^\dagger_{k_s}\hat{a}^\dagger_{k_i}|0_{k_s}, 0_{k_i}\rangle. \tag{3.13}$$

The rate of the transition from the vacuum state to the single photon pair state involves an integral over the geometrical part of the Hamiltonian, otherwise known as the overlap integral. This is given by

$$\Phi(\Delta k) = \int dz \int dxdy U_p(x,y)U_s(x,y)U_i(x,y)e^{(k_p z - k_s z_s - k_i z_i)}. \tag{3.14}$$

From the transformations in Eq. 3.10, I rewrite this as

$$\Phi(\Delta k) = \int dz \int dxdy U_p(x,y)U_s(x,y)U_i(x,y)e^{i[k_p z - k_s(-\sin\theta_s y + \cos\theta_s z) - k_i(-\sin\theta_i y + \cos\theta_i z)]}. \tag{3.15}$$

Using the signal/idler coordinate system, I rewrite $k_{s(i)}$ in terms of the pump coordinate system as

$$\begin{aligned} k_{s_y} &= -\sin\theta_s k_s, \\ k_{s_z} &= \cos\theta_s k_s, \\ k_{i_y} &= \sin\theta_i k_i, \\ k_{i_z} &= \cos\theta_i k_i, \end{aligned} \tag{3.16}$$

so that

$$\Phi(\Delta k) = \int dz \int dxdy U_p(x,y)U_s(x,y)U_i(x,y)e^{i(k_p z - k_{s_y} y - k_{s_z} z - k_{i_y} y - k_{i_z} z)} \tag{3.17}$$

$$= \int dz \int dxdy U_p(x,y)U_s(x,y)U_i(x,y)e^{i(\Delta\vec{k}\cdot\vec{r})}, \tag{3.18}$$

where $\Delta\vec{k}\cdot\vec{r} = \Delta k_y y + \Delta k_z z$. I assume that the modes all lie in the y-z plane so that $\Delta k_x = 0$. Writing out the integral fully, I find

$$\begin{aligned} \Phi(\Delta k) = \int dz \int dxdy e^{i\Delta\vec{k}\cdot\vec{r}} \mathrm{Exp}\left[\frac{-(\mathrm{x}^2+\mathrm{y}^2)}{\mathrm{W_p^2}}\right] \\ \times \mathrm{Exp}\left[\frac{-(\mathrm{x}^2+(\mathrm{y}\cos\theta_\mathrm{s}+\mathrm{z}\sin\theta_\mathrm{s})^2)}{\mathrm{W_s^2}}\right] \mathrm{Exp}\left[\frac{-(\mathrm{x}^2+(\mathrm{y}\cos\theta_\mathrm{i}-\mathrm{z}\sin\theta_\mathrm{i})^2)}{\mathrm{W_i^2}}\right]. \end{aligned} \tag{3.19}$$

I expand the terms in square brackets and rearrange terms to obtain,

$$\Phi(\Delta k) = \int dz \int dxdy e^{i\vec{\Delta k}\cdot\vec{r}}$$
$$\mathrm{Exp}\left[-\left[\left(\frac{1}{\mathrm{W_p^2}}+\frac{1}{\mathrm{W_s^2}}+\frac{1}{\mathrm{W_i^2}}\right)\mathrm{x}^2+\left(\frac{1}{\mathrm{W_p^2}}+\frac{\cos^2\theta_\mathrm{s}}{\mathrm{W_s^2}}+\frac{\cos^2\theta_\mathrm{i}}{\mathrm{W_i^2}}\right)\mathrm{y}^2\right.\right.$$
$$\left.\left.+\left(\frac{\sin^2\theta_s}{W_s^2}+\frac{\sin^2\theta_i}{W_i^2}\right)z^2+\left(\frac{2\cos\theta_s\sin\theta_s}{W_s^2}-\frac{2\cos\theta_i\sin\theta_i}{W_i^2}\right)yz\right]\right]. \tag{3.20}$$

Then I define

$$\begin{aligned}
A &= \frac{1}{W_p^2}+\frac{1}{W_s^2}+\frac{1}{W_i^2}, \\
C &= \frac{1}{W_p^2}+\frac{\cos^2\theta_s}{W_s^2}+\frac{\cos^2\theta_i}{W_i^2}, \\
D &= \frac{2\cos\theta_s\sin\theta_s}{W_s^2}-\frac{2\cos\theta_i\sin\theta_i}{W_i^2}, \\
&= \frac{\sin 2\theta_s}{W_s^2}-\frac{\sin 2\theta_i}{W_i^2}, \\
F &= \frac{\sin^2\theta_s}{W_s^2}+\frac{\sin^2\theta_i}{W_i^2},
\end{aligned} \tag{3.21}$$

I rewrite the overlap integral as

$$\begin{aligned}
\Phi(\Delta k) &= \int dz \int dxdy e^{i\vec{\Delta k}\cdot\vec{r}} e^{-\left[Ax^2+Cy^2+Fz^2+Dyz\right]}, \\
&= \int dz \int dxdy e^{i(\Delta k_y\cdot y+\Delta k_z\cdot z)} e^{-\left[Ax^2+Cy^2+Fz^2+Dyz\right]}, \\
&= \int dz e^{i\Delta k_z\cdot z} e^{-Fz^2} \int dxdy e^{i(\Delta k_y\cdot y)} e^{-\left[Ax^2+Cy^2+Dyz\right]}.
\end{aligned} \tag{3.22}$$

Solving the integral in the x-direction yields

$$\int_{-\infty}^{\infty} dx e^{-Ax^2} = \sqrt{\frac{\pi}{A}}. \tag{3.23}$$

The y-integral is given by

$$\int_{-\infty}^{\infty} dx e^{-Cy^2+i\Delta k_y y-Dzy} = e^{\frac{-\Delta k_y^2-2i\Delta k_y Dz+Dz^2}{4C}}\sqrt{\frac{\pi}{C}}. \tag{3.24}$$

I am finally left with

$$\Phi(\Delta k) = \frac{\pi}{\sqrt{AC}} e^{-\frac{\Delta k_x^2}{4A}} e^{-\frac{\Delta k_y^2}{4C}} \int dz e^{iKz} e^{-Hz^2}, \tag{3.25}$$

where

$$H = F - \frac{D^2}{4C}, \tag{3.26}$$

and

$$K = -\Delta k_y \frac{D}{2C} + \Delta k_z. \tag{3.27}$$

In order to simplify the integral in Eq. 3.25, I make an approximation of a thin crystal so that walk-off due to noncollinear mode propagation is minimal. Physically this means that the crystal is so thin that even with the finite angles (θ_s, θ_i) of the signal and idler beams, the interaction volume is small enough that the beams do not physically separate by much over the length of the crystal. In a thick crystal, this is not true as further propagation in the material leads to greater separation of the beams. For the spot sizes I consider as well as the crystal thickness, I have found making the thin crystal assumption changes the count rates by only 2% and the heralding efficiency $< 0.4\%$, which is negligible.

The thin crystal assumption is similar to making the emission angles (θ_s, θ_i) very small, meaning that the quantity $H << 1$ so the real part of the exponential can be ignored. This leaves,

$$\Phi(\Delta k) = \frac{\pi}{\sqrt{AC}} e^{-\frac{\Delta k_y^2}{4C}} \int_{-L/2}^{L/2} dz e^{iKz}, \tag{3.28}$$

so that the final expression is given by

$$\Phi(\vec{k}) = \frac{\pi}{\sqrt{AC}} e^{-\frac{\Delta k_y^2}{4C}} L\mathrm{Sinc}[\Delta k_z L/2]. \tag{3.29}$$

Obtaining the transition rate into the biphoton mode is calculated from Fermi's golden rule [52] as

$$\frac{dR}{d\omega_s} = \frac{\eta_s \eta_i P d_{\mathrm{eff}}^2 \alpha_s^2 \alpha_i^2 \alpha_p^2 \omega_s \omega_i}{\pi \epsilon_0 c^3 n_s n_i n_p} |\Phi(\Delta\vec{k})|^2, \tag{3.30}$$

where $\eta_{s(i)}$ is the overall efficiency of the signal (idler) path, which includes all the losses. This is the differential joint spectral rate as a function of signal frequency. The fact that the left-hand-side of Eq. 3.30 is a differential with respect to only the signal frequency reflects the assumption I made that the pump beam is monochromatic and thus, $|d\omega_s| = |d\omega_i|$.

I calculate the total joint spectral rate R by integrating the differential joint rate over all frequencies with a filter transmission function under the integral. I assume a perfectly flat-topped, unit-efficiency filter with bandwidth $\Delta\omega_f$. The transmission functions I assume are simple step functions and the total rate is given by,

$$\begin{aligned} R &= \frac{\eta_s \eta_i P d_{\mathrm{eff}}^2 \alpha_s^2 \alpha_i^2 \alpha_p^2 \omega_s \omega_i}{\pi \epsilon_0 c^3 n_s n_i n_p} \\ &\times \int_{-\infty}^{\infty} T_s(\omega_s) T_i(\omega_p - \omega_s) |\Phi(\Delta\vec{k})|^2 \, d\omega_s, \end{aligned} \tag{3.31}$$

where I have denoted the filter transmission functions as $T_{s(i)}$.

In this section, I derived an expression for the differential joint spectral rate and the total joint spectral rate which are essential for determining the heralding efficiency and for the total key rate of the system. The other piece needed to calculate the heralding efficiency is the singles spectrum and the total singles rate, which I determine in the next section.

3.5 Singles Spectrum

I derive an expression for the singles spectrum in a similar manner to that of the joint spectrum. The main difference is that when calculating the singles spectrum for say the signal (idler) photon, I calculate the joint spectral rate with emission into any direction for the idler (signal). This is because, for the signal photon's singles rate, the type of mode, or direction of idler emission should not affect the rate of that signal photon and vice versa. In order for one photon, say the idler, not to affect the signal singles rate, all its higher order modes must be summed over so that all dependence on the idler drops out. Formally, I do this by using an entire set of transverse higher-order modes. This is possible using any orthonormal set of modes and here I use the Hermite-Gauss modes. The field modes are then given by

$$U_{l,m}(x,y,z) = \alpha^{(n,m)} G_n\left[\frac{\sqrt{2}x}{W(z)}\right] G_m\left[\frac{\sqrt{2}y}{W(z)}\right] e^{\left[-ikz-ik\frac{x^2+y^2}{2R(z)}+i(n+m+1)\zeta(z)\right]}, \tag{3.32}$$

where the normalization parameter is given by

$$\alpha^{(n,m)} = \sqrt{\frac{2}{2^{n+m}n!m!\pi W_s^2}}, \tag{3.33}$$

and

$$G_n(u) = H_n(u)e^{\left(\frac{-u^2}{2}\right)}, \tag{3.34}$$

with the Hermite functions obeying the recurrence relationship

$$H_{n+1}(u) = 2uH_n(u) - 2lH_{n-1}(u). \tag{3.35}$$

I have previously shown that the Gouy phase can be ignored so that the Hermite-Gauss modes simplify to,

$$U_{l,m}(x,y,z) = \alpha^{(n,m)} G_n \left[\frac{\sqrt{2}x}{W(z)}\right] G_m \left[\frac{\sqrt{2}y}{W(z)}\right] e^{-ikz}. \tag{3.36}$$

I calculate the singles rate by substituting the higher-order field modes into either the signal or idler mode and perform a summation over all modes (n,m). For example, the signal singles spectral rate is determined by,

$$\frac{dR_s}{d\omega_s} = \sum_{n,m=0}^{\infty} \frac{\eta_s \eta_i P d_{\text{eff}}^2 (\alpha_i^{(n,m)})^2 \alpha_s^2 \alpha_p^2 \omega_s \omega_i}{\pi \epsilon_0 c^3 n_s n_i n_p} |\Phi_s^{(n,m)}(\Delta\vec{k})|^2. \tag{3.37}$$

where $\Phi_s^{(n,m)}(\Delta\vec{k})$ is the generalized mode overlap function for order (n,m).

I calculate the total singles rate for the signal photon in a similar manner to that of the joint spectrum. I take the integral over all differential signal frequencies with the filter transmission functions included. This total rate is given by

$$\begin{aligned} R_s &= \sum_{n,m=0}^{\infty} \frac{\eta_s P d_{\text{eff}}^2 (\alpha_s^{(n,m)})^2 \alpha_i^2 \alpha_p^2 \omega_s \omega_i}{\pi \epsilon_0 c^3 n_s n_i n_p} \\ &\times \int_{-\infty}^{\infty} T_s(\omega_s) |\Phi_s^{(n,m)}(\Delta\vec{k})|^2 \, d\omega_s. \end{aligned} \tag{3.38}$$

Here, I have only given an expression for the signal photon, the idler is just as easily calculated by making the switch $s \longrightarrow i$. Although the analytic expression for the singles requires an infinite sum, I find good convergence when I truncate the sums after 10 modes (n,m >10). For the plots I show in the following section, the sums have been truncated at $n = m = 12$.

In general, the analytic solutions for each mode are lengthy and get very complicated for higher order mode numbers. It is instructive, however, to understand the contributions in some of these higher order modes, so here I write out the first two lowest order modes after $|\Phi_s^{(0,0)}(\Delta\vec{k})|$. I find that the odd-numbered modes in the x-direction (n-odd) are all zero due to the fact that there is no phase mismatch in the x-direction ($\Delta k_x = 0$). Therefore, the first two higher order modes are $|\Phi_s^{(0,1)}(\Delta\vec{k})|$ and $|\Phi_s^{(2,0)}(\Delta\vec{k})|$. These modes are given by,

$$\begin{aligned}\Phi_s^{(0,1)}(\Delta\vec{k}) = & \frac{i\pi\sqrt{2}e^{-\Delta k_y^2/(4C)}}{W_i A^{3/2}\sqrt{C}} \\ & \times \left\{\cos\theta_i \Delta k_y L \,\mathrm{sinc}\left[\frac{\Delta k_z L}{2}\right]\right. \\ & + (\cos\theta_i D + \sin\theta_i 2C) \\ & \left.\times \frac{L}{\Delta k_z}\left(\cos\left[\frac{\Delta k_z L}{2}\right] - 2\,\mathrm{sinc}\left[\frac{\Delta k_z L}{2}\right]\right)\right\},\end{aligned} \tag{3.39}$$

and

$$\begin{aligned}\Phi_s^{(2,0)}(\Delta\vec{k}) = & \frac{2\pi}{\sqrt{AC}}\left(\frac{2}{AW_i^2} - 1\right) \\ & \times e^{-\Delta k_y^2/(4C)} L\mathrm{sinc}\left[\frac{\Delta k_z L}{2}\right],\end{aligned} \tag{3.40}$$

where

$$D = \frac{\sin(2\theta_\mathrm{s})}{W_s^2} - \frac{\sin(2\theta_\mathrm{i})}{W_i^2}. \tag{3.41}$$

In the next sections, I refer back to these expressions to demonstrate similarities and differences that arise in the joint versus singles spectrum.

3.6 Joint and Singles Spectra for Various SPDC geometries

In this section, I discuss the joint and singles spectra calculated in the previous two sections for four different geometries: collinear degenerate, noncollinear degenerate, noncollinear degenerate and noncollinear nondegenerate. Collinear geometries have the signal and idler copropagating with the pump, while noncollinear geometries have the signal and idler emitted on either side of the pump as in Fig. 3.1. Degenerate wavelengths are at twice the wavelength (half the frequency) of the pump, while non-degenerate ones simply satisfy $\omega_p = \omega_s + \omega_i$ with $\omega_s \neq \omega_i$. I will begin with a more intuitive discussion of the phase mismatch parameter and how it plays an important role in the joint and singles spectra. I will then briefly overview the parameter I choose for making specific plots of the joint and singles spectrum. In Sec. 3.6.3-3.6.6 I discuss the joint and singles spectrum for each of the four geometry/wavelength cases.

3.6.1 Phase Mismatch

The phase mismatch in both the y- and z-directions plays an important role in determining both the joint and singles spectra as I have outlined in the previous two sections. The mismatches are given formally by the equations,

$$\Delta k_z = k_p - k_s \cos\theta_s - k_i \cos\theta_i \tag{3.42}$$

$$\Delta k_y = k_s \sin\theta_s - k_i \sin\theta_i . \tag{3.43}$$

Before delving into the details of the spectra, it is important to gain a physical understanding of how the phase mismatch behaves and affects the rates differently for the four different geometries listed above. To do this, I consider a Taylor-series expansion of the wavevector magnitudes as a function of frequency and I truncate the series after second order. From this approach I have,

$$k_{s(i)}(\omega) \simeq k_{s(i)0} + \frac{n_{g,s(i)}}{c}(\omega - \omega_{s(i)0}) + \frac{1}{2}k''_{s(i)}(\omega - \omega_{s(i)0})^2, \tag{3.44}$$

where $k_{s(i)0} = k|_{\omega=\omega_{s(i)0}}$ is the wavevector magnitude, $n_{g,s(i)} = c\,\partial k_{s(i)}/\partial \omega_{s(i)}|_{\omega=\omega_{s(i)0}}$ is the group index, and $k''_{s(i)} = (1/c)(\partial n_{g,s(i)}/\partial \omega_{s(i)}|_{\omega=\omega_{s(i)0}})$ is the group velocity dispersion. I evaluate all of these expressions at the carrier frequency $\omega_{s(i)}$ of the signal (idler) beam. These frequencies are chosen such that I obtain perfect phase matching for them ($\Delta\vec{k} = 0$). I angle-tune the crystal so that the angle between the pump wavevector and the crystal optics axes θ_p is phase matched for the target modes and I obtain,

$$\begin{aligned} k_p - k_{s0}\cos\theta_s - k_{i0}\cos\theta_i &= 0 & (3.45)\\ k_{s0}\sin\theta_s - k_{i0}\sin\theta_i &= 0. & (3.46)\end{aligned}$$

Substituting the expanded $k_{s(i)}$ into 3.42 and 3.43, I find

$$\begin{aligned} \Delta k_z &= \overbrace{k_p - k_{s0}\cos\theta_s - k_{i0}\cos\theta_i}^{0} - \left(\frac{n_{g,s}}{c}(\omega - \omega_{s0}) + \frac{1}{2}k''_s(\omega - \omega_{s0})^2\right)\cos\theta_s \\ &- \left(\frac{n_{g,i}}{c}(\omega - \omega_{i0}) + \frac{1}{2}k''_i(\omega - \omega_{i0})^2\right)\cos\theta_i, & (3.47)\\ \Delta k_y &= \overbrace{k_{s0}\sin\theta_s - k_{i0}\sin\theta_i}^{0} + \left(\frac{n_{g,s}}{c}(\omega - \omega_{s0}) - \frac{1}{2}k''_s(\omega - \omega_{s0})^2\right)\sin\theta_s \\ &- \left(\frac{n_{g,i}}{c}(\omega - \omega_{i0}) + \frac{1}{2}k''_i(\omega - \omega_{i0})^2\right)\sin\theta_i. & (3.48)\end{aligned}$$

For the case of frequency-degenerate SPDC, $\omega_{s0} = \omega_{i0}$, $\omega - \omega_{s0} \equiv \Delta\omega_s = -\Delta\omega_i = -(\omega - \omega_{i0})$. In this case, the emission angles are also equal ($\theta_s = \theta_i \equiv \theta$), as well as the group indices, ($n_{g,s} = n_{g,i} \equiv n_g$) and group velocity dispersion parameters ($k_s'' = k_i'' \equiv k''$). In this case, 3.47 and 3.48 reduce to,

$$\Delta k_z = -k'' \cos\theta\,(\omega_s - \omega_p/2)^2 \tag{3.49}$$

$$\Delta k_y = 2n_g \sin\theta(\omega_s - \omega_p/2)/c. \tag{3.50}$$

The longitudinal phase mismatch (Δk_z) has a quadratic frequency dependence and no linearly dependent term. The transverse phase mismatch (Δk_y) has only a linearly dependent term. For frequency nondegenerate cases, both transverse and longitudinal components of the phase mismatch have linear and quadratic terms. When I take the limit of $\theta = 0$ for collinear geometries, the transverse component drops out entirely. This behavior of the phase mismatch functions will dominate the shapes of the spectrum as I detail in the next sections.

3.6.2 Parameters

For the remainder of this chapter, I discuss the specific joint and singles spectra for the four geometries previously mentioned. For all of the modeling, I use the following assumptions and parameters as these will most closely represent my experimental setup. The base configuration assumes a 600-μm-long BiBO crystal pumped by a modelocked 355-nm-wavelength laser with a ∼5-ps-long pulse duration that has an average power of $P = 1$ mW. The spectral width of the pump light is 0.013 nm (sech^2 pulse shape), which is much less than the spectral width of the down-converted light ($\sim 10-80$ nm). Hence the assumption of a monochromatic pump beam is applicable [54]. I use two different focusing conditions: “loose” focusing where $W_p = 250$ μm and $W_s = W_i = 100$ μm and “tight” focusing where $W_p = 150$ μm and $W_s = W_i = 50$ μm. For the degenerate configuration I consider wavelengths at twice the pump wavelength (710 nm)

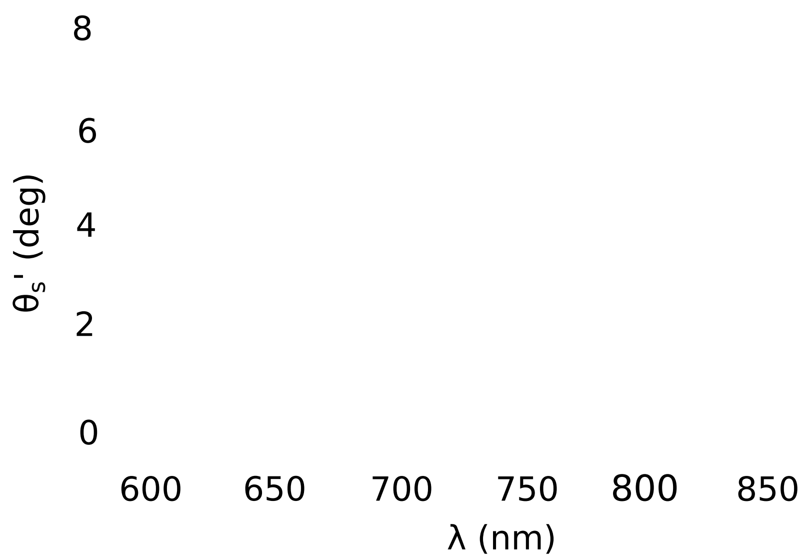

Figure 3.2: **Noncollinear opening angle versus wavelength** Exterior opening angle, θ_s' as a function of wavelength for crystal tilt angles: $\theta_p = 141.9°$ (blue line), $\theta_p = 142.2°$ (maroon dashed line), and $\theta_p = 143.0°$ (gold line). For certain values of θ_p, the angle versus wavelength curves intersect a horizontal line, which denotes a single angle, at two wavelengths. Both of these wavelengths will be collected into the fiber mode. The vertical dashed lines represent a conjugate pair of wavelengths. These show that the two angles in the same direction are similar to, but not exactly frequency conjugates for noncollinear nondegenerate geometries. For collinear and noncollinear degenerate geometries, the two wavelengths that are phase matched for the same direction are frequency conjugates of each other.

and for nondegenerate wavelengths, I use a signal wavelength of 850 nm and an idler wavelength of 609.6 nm. I assume that the signal and idler path have no loss in the ideal case so that $\eta_{s(i)} = 1$.

I use the biaxial crystal BiBO that can be phase matched in the wavelength range of the pump and angle-tuned to adjust the emission angles for specific wavelengths. In this crystal, the pump experiences an angle-independent refractive index, similar to an o-ray in a uniaxial crystal and the daughter photons experience an angle-dependent refractive index.

One interesting result of phase matching for nondegenerate wavelengths is that in a single direction more than one wavelength is phase matched for the process. Figure 3.2 shows the external angle of emission θ_s' that is phase matched for the process as a function of wavelength. The horizontal line represents one exterior angle and clearly the angle curve crosses this line twice, meaning that two wavelengths are emitted in the same direction and they are both phase matched. It is interesting to note that for noncollinear nondegenerate geometries, these wavelengths are not frequency conjugates of each other. For collinear geometries and for degenerate wavelengths the two wavelengths that are phase matched for the process become frequency degenerate. Because in the noncollinear nondegenerate case, these frequencies are not conjugate, care must be taken to filter one peak out by using a low- or high-pass filter because uncorrelated photon events registered as hits will degrade the heralding efficiency.

3.6.3 Collinear Degenerate

The first case I consider is the collinear degenerate case where $\theta_s = \theta_i = 0$ and the daughter photons have the same frequency. This configuration is not experimentally practical because of the difficulty in separating the signal and idler beam with low loss. Because I use Type-I, the polarizations of the signal and idler are the same as well as the direction and frequency. It is possible to put a 50:50 beamsplitter in the path and detect signal and idler photons and observe correlations. Because of the random nature of the beamsplitter however, the heralding efficiency is maximally limited to 50%, making this an impractical setup for applications aiming for high heralding efficiency. Despite its impracticality, I present the collinear degenerate case here because of the interesting physics underlying the interaction.

The phase mismatch in the transverse direction is zero ($\Delta k_y \sim 0$) as discussed previously, leaving only the phase mismatch in the longitudinal direction to dictate the shape of the spectrum. In Fig. 3.3 (a), I plot the longitudinal phase mismatch (Δk_z) as a function of frequency around the degenerate frequency. The degenerate wavelength is 710 nm corresponding to a carrier frequency of $\omega_{s0} = 2.65 \times 10^{15} s^{-1}$. It is clear from this figure that the quadratic dependence dictates the joint spectral rate (Fig. 3.3 (b)). The spectrum is broadened from the broad longitudinal phase mismatch parameter. Looking at the overlap function in Eq. 3.29, the term that dominates is the sinc function instead of the exponential, which goes to 1 for $\Delta k_y = 0$. The result of this is that the spectrum is highly sensitive to the crystal length and much less sensitive to the waist parameters in C and thus the focusing conditions. The width of the joint spectrum can be approximated from the condition $\Delta k_z L \sim 2\pi$.

The higher-order modes contribute very little to the overall singles spectrum. For the collinear case, $D = 0$, so that $\Phi_s^{(0,1)} \sim 0$. The higher-order mode $\Phi_s^{(2,0)}$ has the same shape as the joint mode with the frequency dependent sinc function dominating. Therefore, I expect the joint and singles spectra to have identical frequency dependencies and indeed this is shown in Fig. 3.3 (b). The coefficient in Eq. 3.40 can be made small with appropriate focusing conditions given by

$$\left| \frac{2}{AW_s^2} - 1 \right| = \frac{1}{1 + 2W_p^2/W_s^2} \ll 1, \tag{3.51}$$

when $W_p \gg W_i$, assuming that $W_s = W_i$, which is known to maximize R [52]. When this coefficient is small, the joint and singles spectra become almost identical (Fig. 3.3 (b)). This situation gives rise to very high heralding efficiency but only if it is possible to separate the signal and idler photon experimentally.

Maximizing the joint count rate in a given bandwidth is equivalent to maximizing the area under the joint spectral rate curve. This is optimized with different focusing

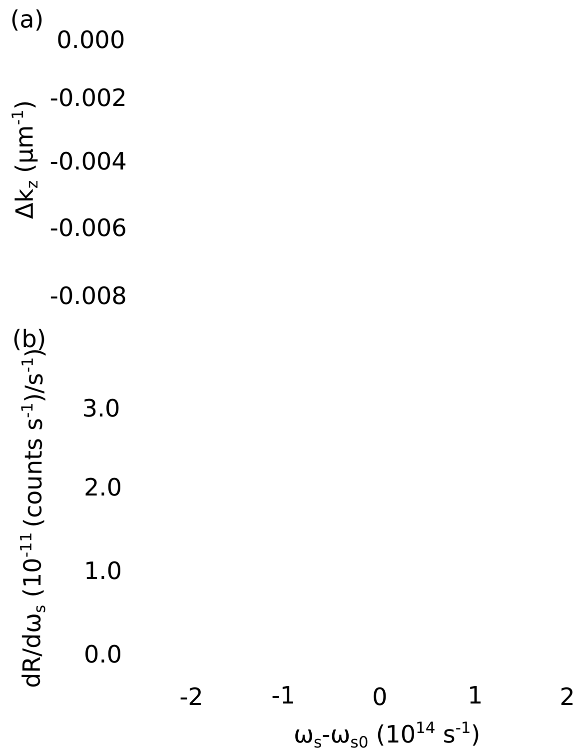

Figure 3.3: **Collinear Degenerate** (a) The longitudinal phase mismatch as a function of frequency where the carrier frequency is $\omega_{s0} = 2.65 \times 10^{15}\ s^{-1}$, for the collinear degenerate case. (b) The joint spectral rate (purple, dashed) and the singles spectral rate (blue, solid) are essentially identical. Here, $\theta_p = 142°$.

conditions than those that would maximize heralding efficiency as has been pointed out by a number of papers [51, 52, 55]. Typically, making the pump beam smaller maximizes the joint count rate. However, I find that heralding efficiency can still be kept high while maintaining high joint count rates by an appropriate focusing ratio of pump, signal, and idler beams as I will detail later.

3.6.4 Noncollinear Degenerate

In the noncollinear case, I angle tune the crystal slightly to obtain spatial separation of the signal and idler. This results in the down-converted light being emitted in a small

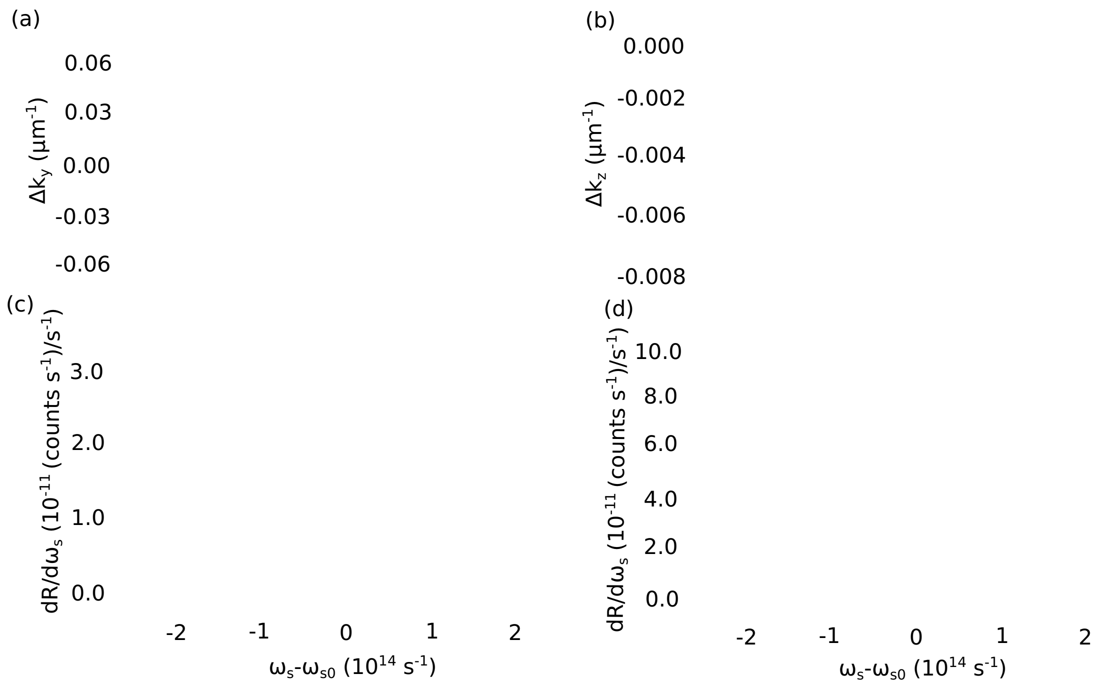

Figure 3.4: **Noncollinear Degenerate** (a) The transverse and (b) longitudinal phase mismatch as a function of frequency where the carrier frequency is $\omega_{s0} = 2.65 \times 10^{15}$ s^{-1}, for the noncollinear degenerate case. The joint spectral rate (purple, dashed) and singles spectral rate (blue, solid) for (c) "loose" and (d) "tight" focusing conditions. The green vertical lines at frequency offsets of $\pm 3.73 \times 10^{13} s^{-1}$ correspond to a full bandwidth of $\sim$20 nm for a degenerate wavelength of 710 nm. Here, $\theta_p = 141.9°$, $\theta_s = \theta_i = 1.64°$, corresponding to $\theta_s' = \theta_i' = 3.04°$.

cone around the original pump direction. Although in this chapter and in Ch. 4, I assume a single collection plane, the down-converted light is allowed in any azimuthal direction around the pump, which I will address in more detail in Ch. 5. The external angle of emission is $\sim 3 - 5°$ in a typical experiment. The signal and idler beams are then located on opposite sides of the pump.

The phase mismatch is now nonzero in both longitudinal and transverse directions, effectively changing the spectrum drastically compared to the collinear case. As seen in Fig. 3.4 (b), the longitudinal phase mismatch is essentially the same as that for the collinear case. However the transverse phase mismatch (Fig. 3.4 (a)) is now a linear function of frequency. The effect this has on the spectrum is now that the exponential term in Eq. 3.29 dominates and so the width of the joint spectrum is approximately, $\Delta k_y \sim 2\sqrt{C}$ where the parameter C depends on the ratio of the beam waists. Due to this fact, the spectrum is now strongly dependent on the focusing condition more than the crystal length. This is depicted in Fig. 3.4 by comparing panels (c) and (d), which show the spectra for the "loose" (c) and "tight" (d) focusing conditions. Clearly, the joint spectrum widens with the tighter focusing conditions. In addition, because of the strong dependence on Δk_y for this case, the shape of the joint spectrum is Gaussian-like due to the exponential term in Eq. 3.29.

The singles spectra are substantially different from the joint spectra for this case due to the contribution of the higher-order terms. Physically, this translates into a higher probability that if a single photon is in one single mode state, that its twin is detected in a higher-order mode state. This is seen by examining Fig. 3.4 (c) and (d), where the singles spectrum is indeed much broader and differs in shape from the Gaussian-like joint spectra. Examining the higher-order modes, I note particularly that $\Phi_s^{(0,1)}$ does not vanish and the term in the second line of Eq. 3.39 is linear in Δk_y which results in a broadening of the spectrum.

Interestingly, the different focusing conditions shown in Fig. 3.4 (c) and (d) do not change the shape of the singles spectrum significantly, although the tighter focusing conditions do give rise to much higher joint and singles count rates (almost a factor of 3), as discussed in Ch 4.

3.6.5 Collinear Nondegenerate

I now consider the collinear nondegenerate SPDC case where different signal and idler frequencies are both emitted along the same direction as the pump. In this case the

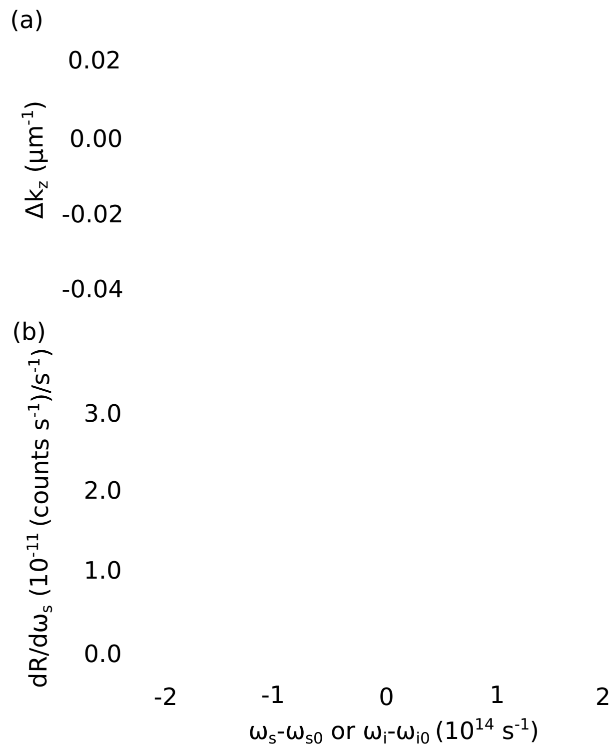

Figure 3.5: **Collinear Nondegenerate** (a) Longitudinal phase mismatch (Δk_z) as a function of frequency. (b) The joint spectral rate (purple, dashed) and the singles spectral rates (green, solid) are essentially identical. Here, $\omega_{s0} = 3.09 \times 10^{15}\ s^{-1}$ (corresponding to a wavelength of 850 nm), $\omega_{i0} = 2.22 \times 10^{15}\ s^{-1}$ (corresponding to a wavelength of 609.6 nm), and $\theta_p = 143.22°$.

use of spectral filtering (highpass and lowpass filters) aids in separating the signal and idler beam from each other and from the pump. This case is similar to the collinear degenerate case in the fact that there is no transverse component of the phase mismatch ($\Delta k_y \simeq 0$). The longitudinal component has an approximately linear dependence on the frequency. In contrast to the quadratic longitudinal dependence of the previous two cases, the roughly linear variation in Δk_z results in a much narrower spectrum, as seen in Fig. 3.5 (b). The shape of the spectrum is again dictated by the sinc function and thus is more sensitive to the crystal length instead of focusing conditions.

In Sec 3.6.3, I discussed the reasons for the similarity between the singles and joint spectrum for the collinear degenerate case. The arguments presented there also hold for the collinear nondegenerate case where some higher order modes such as $\Phi_s^{(0,1)}$ vanish and the non-vanishing ones such as $\Phi_s^{(2,0)}$ have the same spectral shape as the joint and can be made small. Therefore as Fig. 3.5 (b) shows, the joint and singles curves are very similar and I expect to get high heralding efficiency from this case as I will discuss in the following chapter.

3.6.6 Noncollinear Nondegenerate

The final system I consider is the noncollinear nondegenerate geometry, where two different frequencies are emitted on either side of the pump beam. As discussed in Sec 3.6.2, there are two different wavelength solutions to phase matching in a certain direction. Here I only consider the signal at 850 nm and idler at 609.6 nm. Again, both the transverse and longitudinal phase mismatch play a role in determining the joint and singles spectrum. The longitudinal phase mismatch shown in Fig. 3.6 is similar to the collinear nondegenerate case. The transverse phase mismatch dominates and is a linear function of frequency (Fig. 3.6 (a)). In this situation, the phase mismatch in the longitudinal direction slightly dictates the spectrum so that it is weakly dependent on the focusing conditions. In Fig. 3.6 (c) and (d) I show the difference in spectra for the "loose" (c) and "tight" (d) focusing conditions where there is a slight broadening for the tighter focusing condition. Interestingly, the singles spectrum for this case is not significantly affected by the higher order modes so that the joint spectrum and singles spectrum are similar, but not identical.

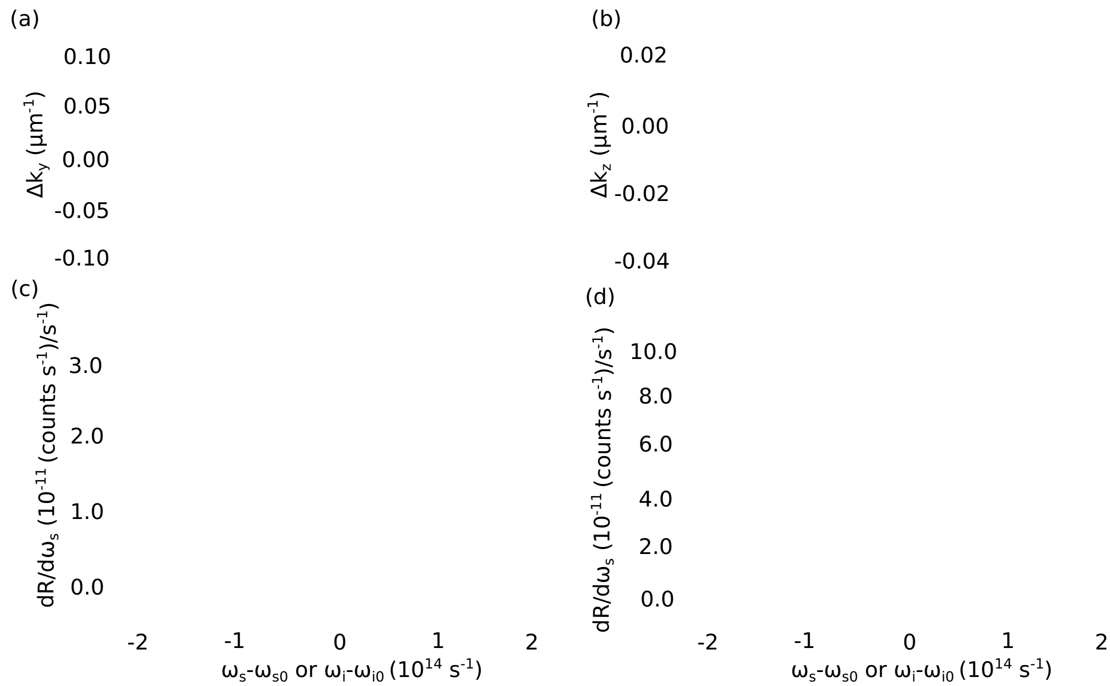

Figure 3.6: **Noncollinear Nondegenerate** (a) The transverse and (b) longitudinal phase mismatch as a function of frequency where the carrier frequency is $\omega_{s0} = 2.65 \times 10^{15}\ s^{-1}$, for the noncollinear nondegenerate case. The joint spectral rate (blue, dashed) and the singles spectral rate for the signal (gold, solid) and idler (purple, dotted) for the (b) "loose" and (c) "tight" focusing conditions. Carrier frequencies are the same as those given in the caption to Fig. 3.5, $\theta_p = 142.44°$, $\theta_s = 3.05°$, $\theta_s' = 5.62°$, $\theta_i = 2.17°$, and $\theta_i' = 4.02°$. Green vertical lines at frequency offsets of $\pm 3.73 \times 10^{13} s^{-1}$ correspond to a full bandwidth of ∼20 nm for a central wavelength of 710 nm.

3.7 Conclusions

In this chapter, I derive both the joint and singles spectrum for the four different cases under the given conditions and assumptions. I show how the phase mismatch can be interpreted physically with a Taylor expansion and then further showed the importance of the phase mismatch on the spectrum. I discuss how different cases can depend more strongly on the longitudinal or transverse phase mismatch and how this dictates both the shape of their spectra and which physical parameters they are more sensitive to. This dependence is important for being able to maximize both heralding efficiency and joint count rate, which I discuss in the following chapter.

Chapter 4

Heralding Efficiency and Joint Count Rate

4.1 Introduction

In the previous chapter, I derived expressions for the joint and singles spectra, which dictate the biphoton joint count rate and heralding efficiency. I plotted the different joint and singles spectra for each of the four geometries and, from this, made some qualitative predictions about the heralding efficiency. In this chapter, I will discuss the detailed results for the heralding efficiency and joint count rate for the "loose" and "tight" focusing conditions discussed in Ch. 3, as well as various filtering conditions. In Sec. 4.3, I discuss my experimental setup and specify how I measure the heralding efficiency, joint count rate, and singles spectrum. I compare my experimental results to my theoretical predictions and show good agreement between the two. The results in this chapter are published in the IEEE Journal of Selected Topics in Quantum Electronics special issue on Quantum Communication and Cryptography [56].

4.2 Heralding Efficiency and Joint Count Rate

The heralding efficiency defined in Eq. 3.1 is the joint biphoton count rate divided by the square root of the product of the singles count rate for the signal and the idler. The heralding efficiency is determined by taking the ratio of the areas under the joint spectrum and the singles spectrum. Therefore high heralding efficiency is achievable if the joint and singles spectra for a given interaction are similar to each other. Filtering out specific parts of the joint and singles spectra can make the two more similar, leading to a higher heralding efficiency. The inherent trade-off of this is the decrease in total joint count rate as I shown in the following sections. The total joint count rate (singles count rate) is calculated by the area under the joint (singles) spectrum.

4.2.1 Collinear Degenerate

For the collinear degenerate case, the joint and singles spectrum shown in Fig. 3.3 are remarkably similar; their differences cannot be identified on the plot. From this I expect the heralding efficiency to be quite high. In Fig. 4.1 (a) and (c) I plot the heralding efficiency and joint count rate as a function of filter bandwidth for both focusing conditions. Figure 4.1 (a) shows that extremely high ($\sim 99.5\%$) heralding efficiency is achievable in this case due to the joint and singles spectra being identical over the whole bandwidth. The difference between the two focusing conditions here is slight, although it is worth noting that, for a slightly looser focusing condition of $W_p = 300$ μm and $W_s = W_i = 100$ μm, joint and singles spectra curves lay directly on top of each other (not shown). This indicates that, for a pump:signal:idler ratio of 3:1:1, the heralding efficiency is higher than for a pump:signal:idler of 2.5:1:1 shown by the solid blue line in Fig. 4.1 (a). Due to the fact that the joint and singles spectra have the same shape, no filtering is needed to achieve high heralding efficiency.

The joint count rate is shown in Fig. 4.1 (c) and is strongly dependent on the filter bandwidth, as expected. A broader bandwidth filter allows more of the spectrum to be collected into the single mode fiber resulting in higher count rates. Interestingly, in Fig. 4.1(c), the count rate increases linearly over a large bandwidth before it tapers off, as opposed to Fig. 4.1 (d) and Fig. 4.2 (c) and (d). This long linear dependence of the rate on filter bandwidth is due to the broad, almost-flat-topped nature of the joint spectrum for this specific case. All three other cases have joint spectra that are much

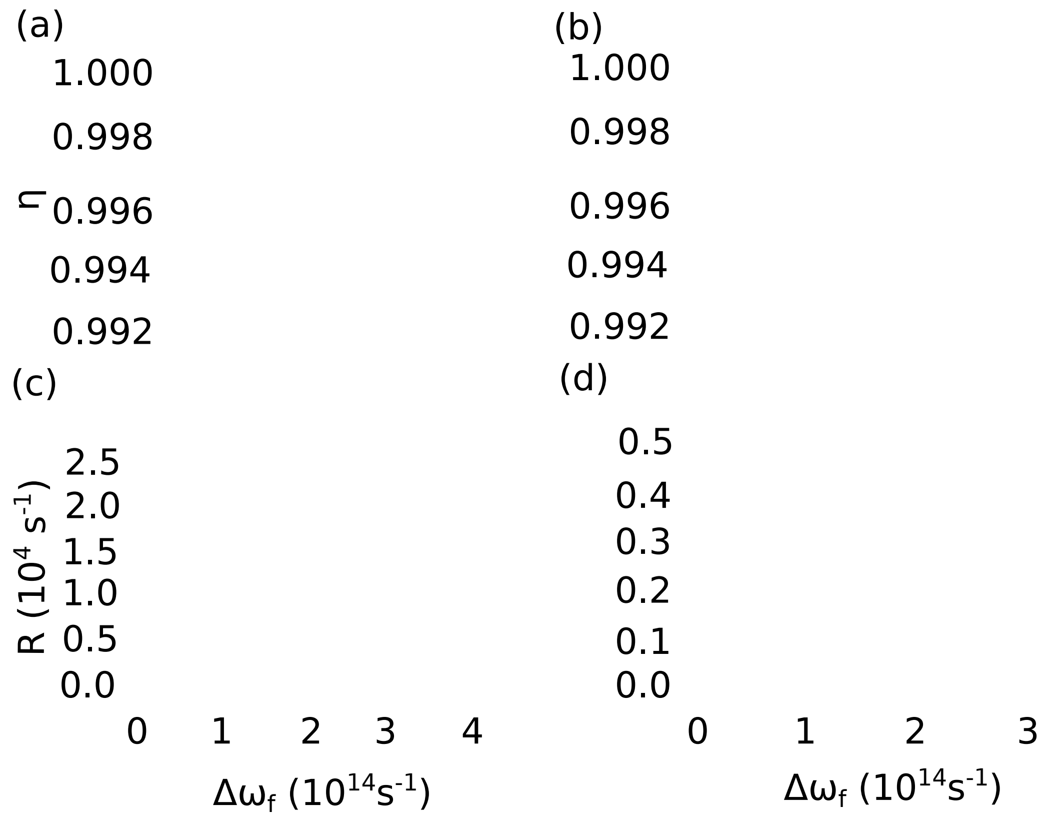

Figure 4.1: **Collinear heralding efficiency and joint count rate** Purple (dashed) lines denote "tight" focusing condition with waist parameters in the crystal of $W_p = 150\ \mu$m, $W_s = W_i = 50\ \mu$m while blue (solid) lines denote the "loose" focusing condition and have waists of $W_p = 300\,\mu$m, $W_s = W_i = 100\ \mu$m. Vertical green lines at 7.46×10^{13} s^{-1} denote a ∼20 nm filter at 710 nm. (a) and (b) are plots of the heralding efficiency as a function of filter frequency for the two sets of waist parameters for (a) collinear degenerate and (b) collinear nondegenerate. The heralding efficiency is very high and constant over a large filter bandwidth because the singles and joint spectra are roughly identical. (c) and (d) show joint count rates for (c) collinear degenerate and (d) collinear nondegenerate.

narrower with Gaussian-like profiles.

While high heralding efficiency is theoretically achievable for this case, I do not account for the 50% loss that results from not being able to practically separate the twin photons. If this is factored in, the highest achievable heralding efficiency for the loosely focused case is 49.725% and for the tightly focused case itis 49.875 %.

4.2.2 Collinear Nondegenerate

The collinear nondegenerate case, shown in Fig. 4.1 (b) and (d) also has high heralding efficiency, which was predicted in Ch. 3 because of the similarities between the joint and singles spectrum (Fig. 3.5 (b)). As in the collinear degenerate case, the tighter focusing condition improves the heralding efficiency by a very small amount, ($\sim$ 0.3%) so that high heralding efficiency is achievable for either focusing parameter. In this case, as in the collinear degenerate case, filtering does not increase the heralding efficiency and comes at a cost of limiting the joint spectral rate.

The joint count rates for this configuration are much lower than for the collinear degenerate case due to the narrower spectrum. For the "loose" focusing case, I obtain rates of $R = 19.5$ kHz/s/mW, whereas for the "tight" focusing case I obtain a rate of $R = 56.8$kHz/s/mW. Thus, high heralding and high count rate are achievable for the more tightly focused case using the collinear nondegenerate scheme.

4.2.3 Noncollinear Degenerate

For the noncollinear degenerate case, the joint spectrum is much narrower than the singles spectrum leading to a low heralding efficiency if no spectral filtering is present. This is shown in Fig. 4.2 (a) for large bandwidths. This case is very sensitive to the focusing conditions with the "tighter" focusing condition leading to a greater than 20% increase in heralding efficiency with no spectral filtering.

Examining Fig. 3.4 (c) and (d), I find that the joint spectrum for the "tight" focusing case is much broader, and therefore more similar to the singles spectrum, leading to the increased heralding efficiency. The "loose" focusing case has lower heralding efficiency due to the narrow joint spectrum compared to the much broader singles spectrum. This is reasonable because, in this configuration, the spectrum is more dependent on Δk_y

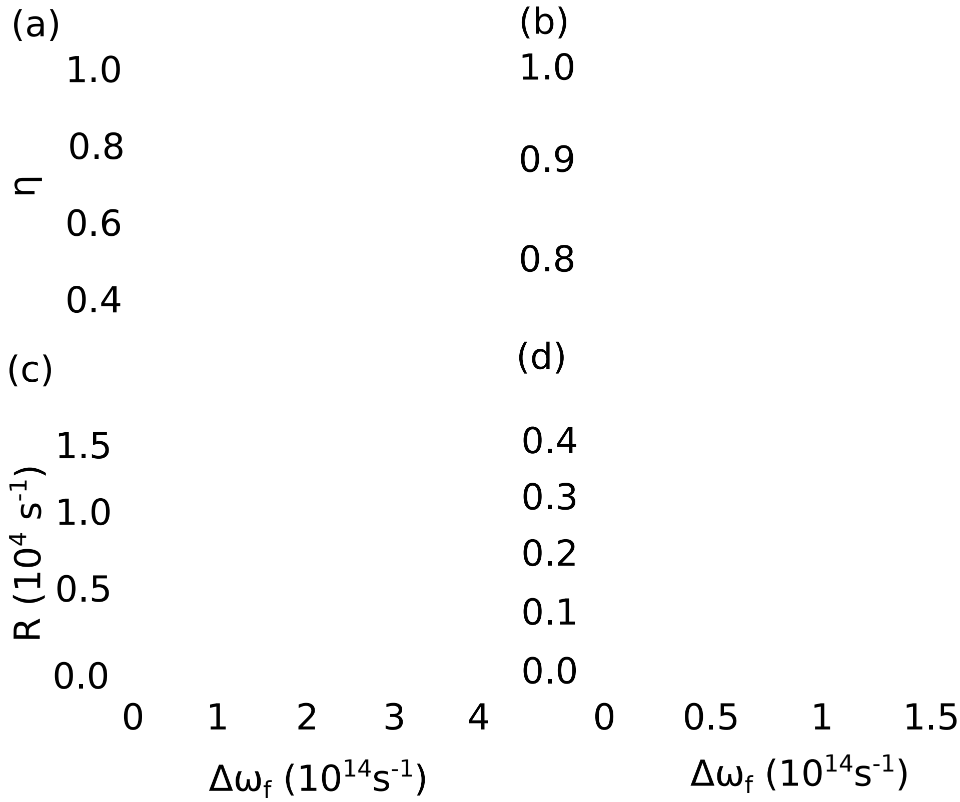

Figure 4.2: **Noncollinear heralding efficiency and joint count rate** Purple (dashed) lines denote "tight" focusing condition with waist parameters in the crystal of $W_p = 150\mu$m, $W_s = W_i = 50\,\mu$m while blue (solid) lines denote the "loose" focusing condition and have waists of $W_p = 300\,\mu$m, $W_s = W_i = 100\,\mu$m. Vertical green lines at 7.46×10^{13} s^{-1} denote a $\sim$20 nm filter at 710 nm. (a) and (b) are plots of the heralding efficiency as a function of filter frequency for the two sets of waist parameters for (a) noncollinear degenerate and (b) noncollinear nondegenerate. The heralding efficiency is low for large filter frequencies due to the dissimilarity of the joint and singles spectra. As the filter gets narrower, the part of the spectra selected out gets more similar and thus the heralding efficiency increases. (c) and (d) show joint count rates for (c) noncollinear degenerate and (d) noncollinear nondegenerate. As the filter gets narrower, the count rate allowed to pass through the filter gets lower.

and hence more sensitive to the waist parameters.

In the case of noncollinear degenerate SPDC spectral filtering is needed to increase the heralding efficiency. Spectral filtering essentially selects the part of the spectrum where the joint and the singles spectrum are more similar, and in doing so, increases the heralding efficiency. The trade-off is a lower joint count rate depicted in Fig. 4.2 (c). As the bandwidth of the filter gets narrower, the heralding efficiency gets higher and the joint count rate gets lower. In this case, it is beneficial to use the "tighter" focusing parameters because they give both higher heralding efficiency as well as higher joint count rates.

4.2.4 Noncollinear Nondegenerate

For the noncollinear nondegenerate case, the singles and joint spectra are much more similar (Fig. 3.6 (c) and (d)) compared to the noncollinear degenerate case, and therefore the heralding efficiency is higher (Fig. 4.2 (b)) even without spectral filtering. The "tight" focusing case has, again, a higher heralding efficiency compared to the "loose" focusing case, but the difference is smaller, with only $\sim 10\%$ difference between the two cases. Heralding efficiencies of $\sim 85\%$ are achievable with tighter focusing and no spectral filtering, while spectral filtering can increase this number up to close to 100% for very narrow bandwidths. Even narrower band filters are needed for this case because the spectrum is much narrower compared to the noncollinear degenerate case.

Tighter focusing is also more advantageous for the joint count rates as well, although, for both focusing conditions, the joint count rate is lower than the noncollinear degenerate case due to the narrower spectrum.

4.3 Experiment

Having explored the effect of the joint and singles spectra on the joint count rate and heralding efficiency in the previous section, I now describe the experiments I perform on measuring the heralding efficiency and joint count rate for the configurations and discuss their agreement with the model given here. I begin with a description of the experimental setup (Sec. 4.3.1) and then detail my results for each case in Sec. 4.3.2. Finally, I compare my findings to other similar experiments and theory in the literature in Sec. 4.3.3.

4.3.1 Experimental Setup

A high-power modelocked UV laser (Coherent Paladin, 4 W, 5-ps-long pulse duration, 120 MHz repetition rate) with a 355-nm-wavelength is focused into a 600-μm-long BiBO crystal (Newlight Photonics). The crystal is anti-reflection coated on both faces for the pump wavelength (355 nm) and the degenerate down-conversion wavelength (710 nm). The down-converted light from the crystal is collected with a well-corrected achromatic lens (Schaefter and Kirchhoff GmbH, 60FC-T-0-M20l-02). The lens couples the light into a single mode fiber. I use anti-reflection (AR) coated fibers that are AR-coated for the degenerate wavelength (Oz Optics custom fiber) and fibers that are not AR-coated for the nondegenerate wavelengths (Thorlabs custom fiber). The schematic setup for noncollinear SPDC is shown in Fig. 4.3.

Alignment of the system involves back-propagation using a fiber pigtailed alignment laser (Thorlabs LP705-SF15 for $\sim$ 703-708 nm, Edmund Optics # 56-112 for 850 nm). The back propagated light gets focused by the lens into the crystal. I remove the crystal and detect the light with a scanning knife edge beam profiler (Thorlabs, BP209-VIS) which allows me to measure the spot size ($1/e^2$ intensity) of the light in the crystal. By changing the focal length and lens-to-fiber distance, I adjust the beam to the desired size.

I collect single photons exiting the crystal into a single mode fiber and detect them with silicon avalanche photodiodes operated in Geiger mode (Perkin-Elmer/Excelitas, SPCM - AQRH). The quantum efficiency of these detectors is peaked around 710 nm

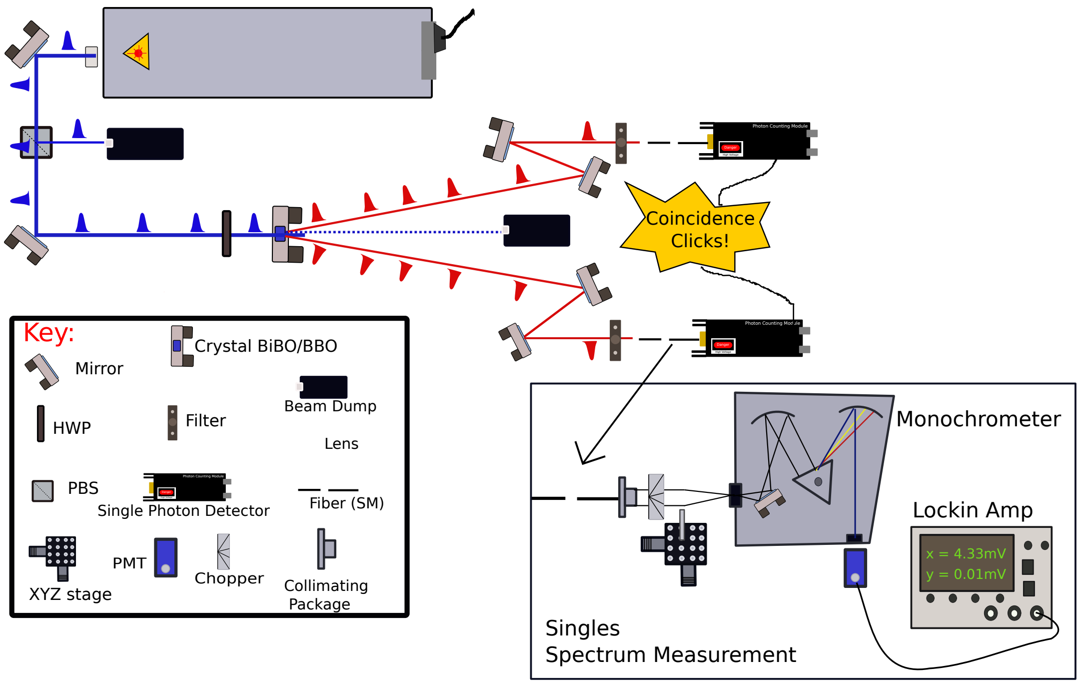

Figure 4.3: Noncollinear experimental setup Experimental setup for the noncollinear geometry. UV pump laser pumps a BiBO crystal and produces daughter photons on either side of the pump beam. Signal and idler pairs are collected via achromatic lenses into fibers and subsequently detected by single photon counting detectors and coincident counting electronics. Single photon spectra is collected into a fiber then focused into a triple grating monochrometer and read out with a PMT and lockin amplifier. For full details, see text.

with an efficiency of $\sim 73\%$. The detectors output electrical TTL pulses which are sent to a custom-programmed field-programmable gate array (FPGA) for both singles and coincidence counting. The coincidence time window is adjustable by shortening the incoming pulses. For the experiments presented here, there was no shortening of the incoming pulse width of 19 ns.

To measure the singles spectra, I couple light from the fibers into a triple grating monochrometer (Newport, Cornerstone 260 1/4 m), and a photomultiplier tube (Hammamatsu, H6780) followed by a transimpedance amplifier and a lock-in amplifier (SRS 850 DSP). I calibrate the system's spectral response using a high-pressure tungsten halogen lamp (Ocean Optics, HL2000).

4.3.2 Experimental Results

Noncollinear Degenerate

Due to the fact that the heralding efficiency is strongly dependent on the singles spectrum, I measure the singles spectral differential rate as a function of wavelength for the signal beam in the noncollinear degenerate case and plot it in Fig. 4.4. I observe a very broad emission spectrum that agrees quite well in bandwidth and overall shape with the theoretical predictions. Because the emission is so broadband, in order to obtain higher heralding efficiency, I use a 23-nm-bandwidth filter (Semrock TBP 704/13) with a 99% efficiency that is close to an ideal top-hat filter. I angle-tune this filter so that the center wavelength of the filter is centered at the degenerate wavelength.

Using the single photon counting setup previously described, I measure a heralding efficiency of $\eta = 43 \pm 0.5\%$ with $W_p = 250 \pm 5$ μm, $W_s = W_i = 100 \pm 5$ μm and without accidental counts factored in. I measure accidental coincidence counts by putting a very long delay line in one arm so that there is no possible way for an actual coincident event to contribute, and measure the coincidence counts that are produced by random processes like dark counts and afterpulsing. For this setup, I measure an accidental heralding efficiency of 0.5% which should be subtracted from η. When comparing this number to my theoretical predictions, I correct for the transmission/detection losses which I estimate to be $\eta_s = \eta_i = 52.0 \pm 3.3\%$. Different factors contribute differently to the loss and I list them here as follows: $62.5 \pm 2.5\%$ for the detection efficiency, $84 \pm 2\%$ for the non-ideal behavior of the spectral filters and minimal loss on the AR-coated fibers of $< 1\%$. For details of detector efficiency versus wavelength and other important detection parameters see Appendix A2. The non-ideal behaviour of

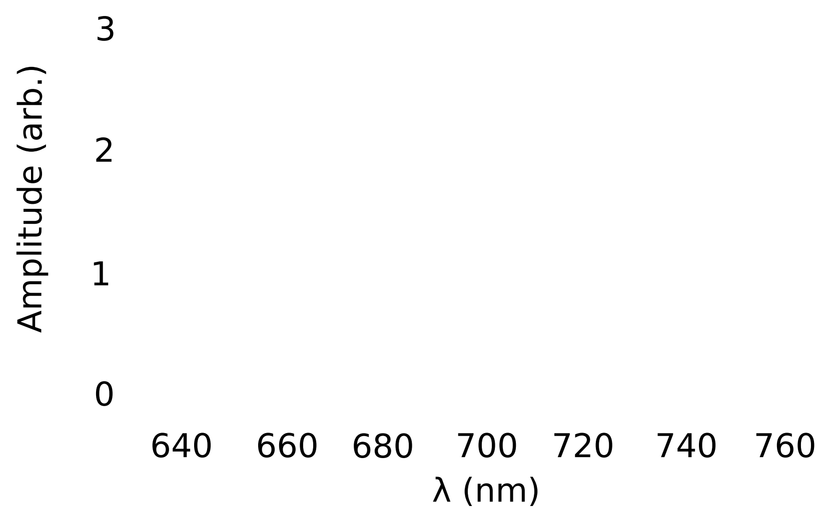

Figure 4.4: **Experimental Singles Spectrum for Noncollinear Degenerate SPDC** Signal singles spectral rate for noncollinear degenerate SPDC with $\theta_s' = 3.04°$. The experimental data is shown by the blue solid line and the predictions are shown by the red dashed line. I use a least-square fit to choose the vertical scaling of the theoretical predictions. The green cross-bars denote errors in each direction. In these plots, the error in wavelength is due to systematic error from calibration of the monochrometer while the error in the amplitude is dominated by statistical error from fluctuations in power and PMT gain, and Shot noise in the electronics.

the spectral filters is due to them not having infintiely sharp edges and perfect flat-tops with 100% efficiency. Typically, when using bandpass filters that are not centered at the nominal wavelength used which is 710 nm in this case, I either tilt the filter if the center wavelength is higher or heat-treat the filter if the center wavelength is lower. Both of these processes create unwanted ripples and less-sharp edges in the filter. To calcualte the spectral efficiency I calculate the overlap integral of the transmission of joint spectral field of the filters, and divide this by the square-root of the product of the transmission field curve for each of the individual filters. This results in the $84 \pm 2\%$ that I call the non-ideal behaviour for the filter. Using these values, I obtain a corrected heralding efficiency $\eta_{correct} = 81.7 \pm 5.5\%$, with accidentals subtracted. I compare this to the theoretical prediction of $84.8 \pm 1.3\%$ and find that the agreement is excellent and well within the assigned errors.

I measure the total joint count rate R by adjusting the average pump power to $\sim$ 100 mW so that I am well below the saturation rate of the detector, but also so that my signal rate is significantly higher than the noise, or background counts. I then scale the experimental results to reflect a power of 1 mW to compare with my theoretical predictions. I find a rate of $R = 982 \pm 20$ kHz, which I correct for by dividing by the product, $\eta_s \eta_i$ to arrive at $R_{correct} = 3.6 \pm 0.5$ kHz. My theoretical prediction for the joint count rate is $R = 2.3 \pm 0.7$ kHZ so that the agreement is nearly within the error of my measurements.

Noncollinear Nondegenerate

For the case of noncollinear nondegenerate SPDC, I use a signal (idler) wavelength of 850 nm (609.6 nm) and the similar waist parameters as above ($W_p = 250 \pm 5$ μm, $W_s = 120 \pm 5$ μm, $W_i = 100 \pm 5$ μm). In this case, I do not need narrow spectral filtering to obtain high heralding efficiency, however, I do use a low-pass filter in the idler arm to block out the 869.3 nm emission that is also phase matched for this process and a long-pass filter in the signal arm to block out the 600 nm emission arising for the same reason (see Sec. 3.6.2 and Fig. 3.2 and associated discussion).

I obtain a heralding efficiency of $\eta = 29.4 \pm 0.6\%$, with accidental coincidences contributing 0.4%. I estimate a total loss in the signal arm of $\eta_s = 34.9 \pm 2.3\%$, with contributions of $38 \pm 2.5\%$ from the detector efficiency and $8 \pm 0.2\%$ from the Fresnel reflections from the fibers. The Fresnel reflections are from the fibers not having an AR coating on them. Here, there is no inefficiency from filters because I use only a highpass or low pass filter because the emission is narrow. In the idler arm I estimate an efficiency of $\eta_i = 53.3 \pm 2.3\%$ with contributions of $58 \pm 2.5\%$ from the detector and $8 \pm 0.2\%$ from the Fresnel reflections from the fibers. Using these values I find

$\eta_{correct} = 68.1 \pm 4.3\%$, which agrees well with my predicted value of $73.2 \pm 1.2\%$.

I measure $R = 410 \pm 12$ Hz, which I divide by $\eta_s \eta_i$ to find $R_{correct} = 2.2 \pm 0.3$ kHz. This result compares favorably with my predicted joint count rate of 1.4 ± 0.4 kHz. There is an additional systematic error included here from flucuations in the power meter head. This error affects the joint count rate but not the heralding efficiency. The measured power is 100 mW $\pm$ 20 mW which is factored into the error in $R_{correct}$.

Collinear Nondegenerate

In the collinear nondegenerate case, I use a different configuration from the non-collinear SPDC setup previously discussed. I use a single achromatic lens to couple both wavelengths (850 nm and 609.6 nm) into a single mode fiber. I filter out the copropagating pump light with a high-power high-pass filter (Semrock, Di02-R405-25x36) that has high transmission ($> 95\%$) at both signal and idler wavelengths and a pump suppression $> 10^6$. I then guide both daughter photons back into free-space where I split them on a dichroic mirror to separate the signal and idler beams and couple each into a multi-mode fiber. Because the achromat is not perfectly correct the back-propagated waists I measure are not identical at the crystal. I measure them to be $W_s = 120 \pm 10$ μm and $W_i = 95 \pm 10$ μm.

I measure the singles spectrum in the same manner as the noncollinear degenerate case and plot the results against the theoretical predictions in Fig. 4.5. The spectrum is much narrower than the noncollinear degenerate case as expected and is in good agreement with the theory. The dual peaks represent the two solutions to the phase matching equations. In the collinear case, the two solutions are frequency conjugates of each other so that the peaks should be at 850 nm and 609.6 nm. The agreement between the predicted spectrum in the relative heights, spectral shape and features is excellent.

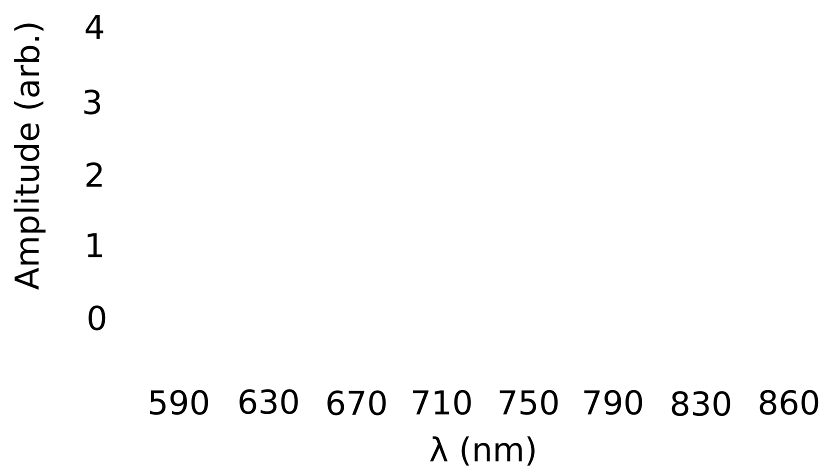

Figure 4.5: **Experimental Singles Spectrum for Collinear SPDC** Signal singles spectral rate for collinear nondegenerate SPDC with $\theta_s' = 3.04°$. The experimental data is shown by the blue solid line and the predictions are shown by the red dashed line. For discussion of sources of error see Fig. 4.4 caption. Here, the spectra for the collinear nondegenerate case is much narrower than the degenerate case and matches well with the theoretical prediction.

I measure a heralding efficiency of $\eta = 33.5 \pm 0.5\%$, which is corrected to yield $\eta_{correct} = 86 \pm 5\%$. Here, I estimate the signal channel efficiency as $\eta_s = 28.9 \pm 1.8\%$ with a detector quantum efficiency of $38 \pm 2.5\%$, a combined coupling loss of $77.5\% \pm 1\%$ from coupling the beam into free-space and back to fiber, and $98\% \pm 1\%$ from the filter. I couple the beam into freespace and back into fiber with standard collimating packages nominally optimized for 780 nm. The slight focusing that occurs for 609.6 nm and defocusing for 850 nm causes the coupling loss. The idler efficiency is $\eta_i = 52.7 \pm 2.3\%$ with a detector quantum efficiency of $58 \pm 2.5\%$, a combined coupling loss of $92.8\% \pm 0.7\%$, and $98\% \pm 1\%$ from the filter. This result is considerably lower than the predicted value of $94.8 \pm 0.9\%$, where I have accounted for the different signal and idler waists in the theoretical prediction.

One potential reason for a lower measured heralding efficiency is that there is a shift in the location of the waists of the signal and idler beams of ~ 40 mm due to the non-perfect achromats. I measure this total shift using three lasers at different wavelengths (633 nm (JDS Uniphase), 705 nm (Thorlabs LP705-SF15), 850 nm (Edmund Optics # 56-112)) propagating through the achromat package. I focus one wavelength (for example the 850 nm) to a spot size of ~ 100 μm. I then replace the 850 nm laser with the 710 nm laser and the 633 nm laser. I measure where the waists are of each of these wavelengths using a beam profiler (Thorlabs BP209-VIS) and find the 633 nm one is almost 40 mm from where the waist of the 850 nm one is. This is the total shift from the location of a waist at ~ 609 nm to the waist at ~ 850 nm. Optimally, one would put each waist roughly 20 mm from the spot of collection in the crystal. I estimate the loss that arises from coupling the diverging beam at a shifted location from its minimum waist. For the case given above I assume the beam sizes are given, but that the waists are 20 mm away from where the beam sizes are measured in the crystal. Here then I obtain the minimum waists are $W_{s_{min}} = 110$ μm and $W_{i_{min}} = 83$ μm. I then calculate the overlap integral of the target Gaussian mode, where the minimum waists are $W_s = 120 \pm 10$ μm and $W_i = 95 \pm 10$ μm and the realistic case where the beam sizes are shifted by 20 mm. This integral represents mode matching for the two Gaussian beams, the target (modeled) mode and the actual (experimental) mode. It is given by

$$\eta_{overlap} = \frac{\left| \int E_1^* E_2 dA \right|^2}{\int |E_1|^2 dA \int |E_2|^2 dA}, \tag{4.1}$$

where $E_{1(2)}$ are the electric fields, modeled here as Gaussians, for the target and actual mode respectively and the integral is over the two-dimensional area of the beam. I find that the signal mode has an efficiency of 95% and the idler mode has an efficiency of 92.7%. When I factor this inefficiency into the expressions for η_s and η_i, I find that the corrected heralding efficiency is increased to $\sim 92\%$, which is well within error of the

Table 4.1: SPDC Data and Theory

	Herlading Efficiency (%)		JCR (kHz)	
	Exp	Theory	Exp	Theory
Noncollinear Degnerate	81.7±5.5	84.8±1.3	3.6± 0.5	2.3±0.7
Noncollinear Nondegenerate	68.1±4.3	73.2±1.2	1.35±0.38	2.2±0.3
Collinear Degenerate (Loose)	86±5	94.8±0.9	2.57±0.84	1.86±0.5
Collinear Degenerate (Tight)	82.6±4	89±2	5.84±0.8	5.02±1.4

theoretical value. This non-ideality could be avoided if the signal and idler where to be spatially separated, say on a dichroic mirror, and then separately coupled into two different achromatic lenses and single-mode fibers.

I measure $R = 392 \pm 12$ Hz, which I correct to $R_{correct} = 2.57 \pm 0.84$ kHz, while the predicted joint count rate is 1.86 ± 0.5 kHz. For this configuration I also decrease all the mode waist sizes to test the "tight" focusing condition. Here, the waist parameters in the crystal are $W_p = 150 \pm 5$ μm, $W_s = 67 \pm 5$ μm and $W_s = 47 \pm 5$ μm. I find that $\eta = 32.2 \pm 0.6\%$, (corrected to $\eta_{correct} = 82.5 \pm 4\%$), which is only slightly higher, but nearly within error of the theoretical prediction of $89 \pm 2\%$. Finally, I measure a joint count rate of $R = 890 \pm 45$ Hz, which is corrected to $R_{correct} = 5.84 \pm 0.8$ kHz. This rate is within error of the predicted value of 5.02 ± 1.4 kHz. I summarize the experimental results in Table 4.1.

4.3.3 Comparison to literature

I compare my theoretical predictions and experimental findings above to previous work whose aim is to optimize heralding efficiency and joint count rates in single mode optical fibers. Midgall *et al.* ([57] Fig. 4) show that in both noncollinear and collinear regimes, the joint count rate decreases as the pump waist decreases which agrees with my predictions for pump waist scaling. They also show that the heralding efficiency increases when there is a larger pump:signal waist, which is also in agreement with my findings. A trade-off exists between obtaining high heralding efficiency by making the pump large compared to the signal and idler, and sacrificing high joint count rates which increase with smaller pump waists. This trade-off is also in agreement with Bennink [51]. Bennink finds that as the pump waist is focused tightly, the joint count rate increases, but this comes with a decrease in heralding efficiency. Although Bennink's work is done mostly in a thick periodically-poled crystal for collinear geometries, I generally agree with his results. Recently, experimental work by Dixon *et al.* [55] has confirmed predictions by Bennink. Baek and Kim [58] show that in Type-I collinear SPDC both the joint and singles spectrum have a broad bandwidth, while for the noncollinear configuration the singles spectrum is still broad, while the joint spectrum gets much narrower. This agrees well with my findings. Finally, Carrasco *et al.* [59] show that, in both collinear and noncollinear configurations, the singles spectrum can be broadened by decreasing the size of the pump waist. In addition they show that the joint spectrum can also be broadened by tighter focusing of the pump beam. The results shown in Fig. 1 in Ref. [59] agree very well with my theoretical predictions.

4.4 Conclusions

In this chapter, I show how the heralding efficiency and joint count rate scale as a function of bandwidth for the different geometry/wavelength configurations. I predict the scaling of heralding efficiency with different filtering and waist parameters from the joint and singles spectrum introduced in Ch. 3. Most notably, I show that in the collinear nondegenerate case, very high heralding efficiencies can be achieved without the need for narrowband spectral filtering. In addition, the joint counts rates for this case can be increased by decreasing the size of the waist, while keeping a ratio of $\sim$ 3:1:1 for pump:signal:idler. For the noncollinear cases, spectral filtering must be used to obtain high heralding efficiencies. The trade-off with this is that the joint count rates are not as high due to spectral filtering. Furthermore, for the noncollinear nondegenerate case, the spectral filtering must be made narrower in order to obtain high heralding efficiency due to the narrowed spectrum.

I further demonstrate good agreement between my experimental results for the heralding efficiency and joint count rates for the noncollinear degenerate, noncollinear nondegenerate and collinear nondegenerate cases. In addition, the predicted singles spectra and my experimental data for the singles spectra for both the noncollinear degenerate and the collinear nondegenerate cases agree very well. Comparisons with other literature aiming to optimize heralding efficiency and joint count rate are in good agreement with my work.

The high heralding efficiency and high biphoton count rates I demonstrate here are important steps for each of the applications discussed in Ch. 1. All of these applications benefit from increasing the heralding efficiency and biphoton count rate and my work has important implication for the broad range of quantum optics experiments requiring both.

Chapter 5

Elliptical Emission Pattern in SPDC in BiBO

5.1 Introduction

One major goal of current QKD systems is to achieve high data rates. This can be accomplished in an SPDC source by high photon pair production and collection. Traditionally, a single pair of photons is collected from either side of the pump and subsequently analyzed and potentially used for generating the quantum key in a non-collinear SPDC configuration. The polarization degree-of-freedom has been the major focus of many of these experiments, although the pairs can be entangled in any number of degrees-of-freedom. One idea to achieve higher key rates is to collect entangled photon pairs from around the entire down-conversion ring. As these pairs are emitted in any azimuthal direction around the pump beam, the potential for spatial multiplexing is quite high.

In this chapter, I investigate the properties of the spatial emission pattern for SPDC in two different types of crystals: bismuth triborate (BiBO) which is a biaxial crystal and beta barium borate (BBO), a uniaxial crystal. The key difference between these two crystals is that in BBO, the degenerate daughter photons in the visible to infrared part of the spectrum (500 nm- $\sim$2.5 μm) experience an angle-independent refractive index while in BiBO they experience an angle-dependent refractive index. This difference leads to an interesting difference in their spatial patterns, namely that BiBO produces elliptical patterns while BBO produces circular ones. The work I present in this chapter is under review in JOSA B [60].

5.2 Polarization Entanglement for Type-I Noncollinear SPDC

Polarization entanglement in Type-I SPDC was first introduced in [61] where the authors use a double-crystal scheme to achieve the entanglement. Each crystal produces a pair of identically polarized photons, but the second crystal is rotated with its optic axis 90° to the first, making the polarization of the photons in the second crystal perpendicular to the first. If the crystals are thin, the rings from the crystal overlap to create polarization entanglement around the entire ring. For example, pumping the first crystal with V polarized photons produces two H polarized photons after the down-conversion process, (typically written as $V \rightarrow HH$) while pumping the second crystal with H, produces two V polarized photons ($H \rightarrow VV$). When the pump is polarized 45° to the H or V axis it is in an equal linear combination of H and V. Then, the state of the daughter photons is the polarization entangled state

$$|\Phi\rangle_{s,i} = \alpha(|H_s H_i\rangle + e^{i\phi}|V_s V_i\rangle), \tag{5.1}$$

where $s(i)$ denotes a signal (idler) photon and $e^{i\phi}$ is a arbitrary phase. The advantage of this method is that it enables spatial multiplexing, for example, where sets of detectors could be placed at any pair of opposing points around the ring. I depict the idea of spatial multiplexing around the ring in Fig. 5.1.

In order for this scheme to succeed, the down-conversion rings from each of the two crystals must overlap at many spatial points around the ring. There are two main factors that influence this overlap: the length of the crystal and the emission pattern of SPDC. In order for the rings to overlap at many spatial points around the down-conversion rings, the crystals must be thin. If they are thin, then the shift in origin

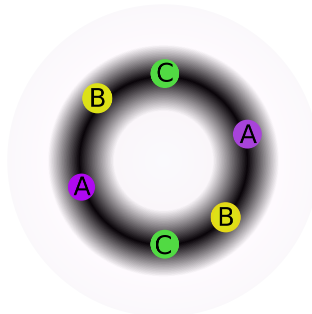

Figure 5.1: **Spatial Multiplexing** The dark circle illustrates the emission pattern of the down-conversion ring. Points A, B, and C represent pairs of daughter photons that each have a different azimuthal angles around the ring. Spatial multiplexing increases count rates by collecting multiple pairs of photons around the whole ring.

from one ring to another will be negligible and the rings will be approximately the same size in any plane perpendicular to the pump beam (see Fig. 5.2).

In addition to the crystals being thin so that the rings overlap, the SPDC emission pattern also needs to be circular in order for the rings to overlap at many points around the ring (see Fig. 5.2 (a)). If the rings are elliptically-shaped (Fig. 5.2 (b)), the patterns at most overlap at four points, but can also simply overlap at two points. Thus, thin crystals and nearly circular emission patterns are important for enabling spatial multiplexing ability.

There are two commonly-used crystals for this process – beta barium borate (BBO) and bismuth triborate (BiBO). BBO has been the more commonly used crystal for SPDC applications with its use dating back to 1992 [62], while BiBO has been a much more recently developed crystal. BiBO has the advantage of having a higher nonlinear coefficient, ($d_{\text{eff}} \sim 1-2$ pm/V in BBO across the phase-matching range versus $d_{\text{eff}} \sim 1.5-4$ pm/V in BiBO across the phase matching range for daughter photons from 600 nm-3.1 μm [63]), making it a popular choice for high-brightness applications such as high-rate QKD. In the next section, I discuss how to predict the emission pattern for these two

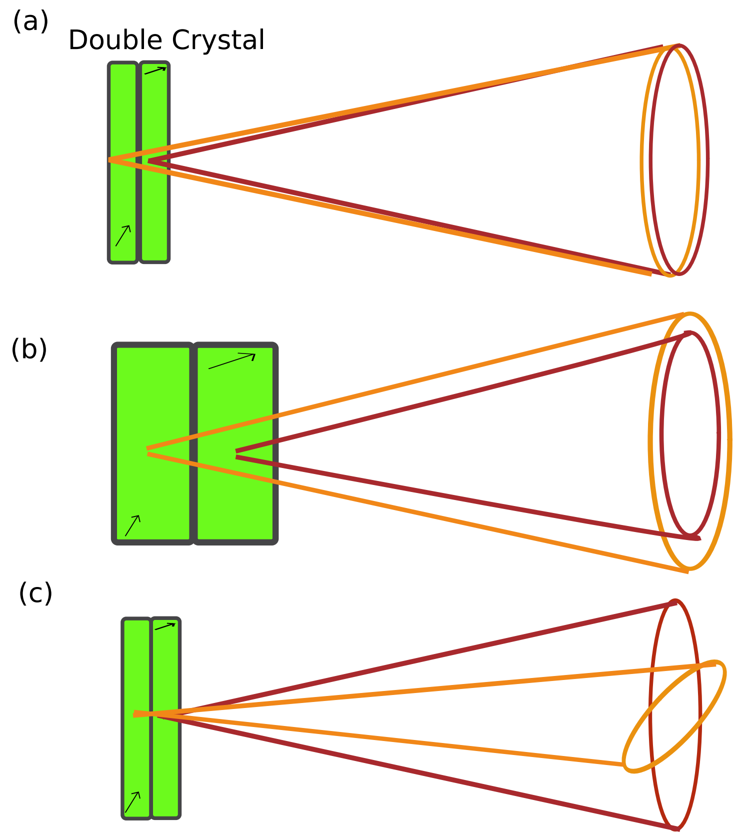

Figure 5.2: **Double Crystal Geometry** (a) shows a double crystal in the Type-I geometry where each crystal emits a pair of photons with the same polarization. The crystals are rotated with optical axes, denoted by arrows, 90° to each other so each crystal produces pairs with opposite polarization to the other crystal. Here, colours denote opposite polarizations. In (b) the thick crystals produce pairs of opposite polarizations, but the rings do not overlap in a plane because the origin of the down-converted photons are too far displaced from each other. In (c), the emitted patterns are elliptical so that they only overlap at maximally four points around the ring.

crystals and explain how both circular and elliptical patterns arise in different crystals.

5.3 Phasematching for Type-I Interactions in BiBO vs BBO

I predict the emission pattern for SPDC by finding the phase matching directions for the emitted photons using the energy and momentum conditions. In general, the momentum condition is given by

$$\vec{k}_p = \vec{k}_s + \vec{k}_i, \tag{5.2}$$

where

$$\vec{k}_j = \frac{n_j(\omega_j, \theta_j)\omega_j}{c}\hat{s}_j, \tag{5.3}$$

for $j = p, s, i$ where $n_j(\omega_j, \theta_j)$ is the frequency- and directional-dependent refractive index, ω_j is the frequency of the photon, $\hat{s}_j$ is the unit vector in the propagation direction and c is the speed of light in vacuum. In this thesis, I closely follow the geometry, conventions and notations in Ref. [36, 64].

Figure 5.3 shows a typical geometry of an interaction. An incident pump beam (p) with wavevector k_p makes an angle θ_p, ϕ_p with the optic axis of the crystal. Uniaxial crystals have a single axis of symmetry, denoted by “OA-U” in Fig. 5.3 whereas biaxial crystals have two optic axes denoted by “OA-B”. Figure 5.3 shows the configuration where $\phi_p = 0$ although I use both $\phi_p = 0$ and $\phi_p = 90°$ as possible configurations. The signal (idler) photon is emitted with wavevector $\vec{k}_{s(i)}$ at local angles $\theta_{s(i)}, \phi_{s(i)}$ with respect to the pump beam vector. The change in coordinates from the pump beam to signal (idler) coordinates is a linear transformation given in Ref. [36] as

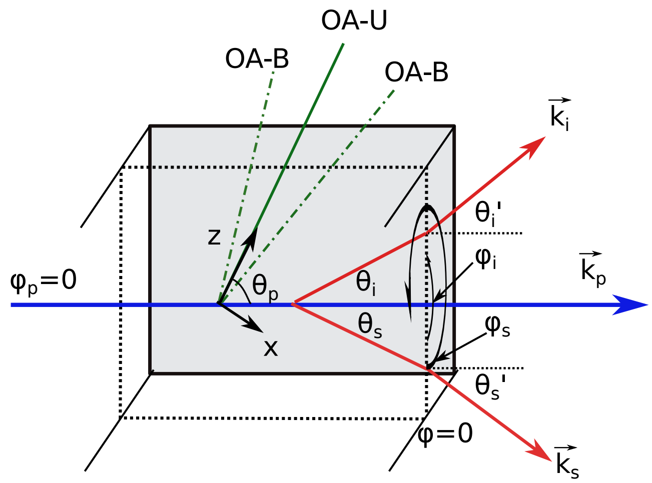

Figure 5.3: **Interaction Geometry** Pump beam (solid blue line) enters a crystal at $\phi_p = 0$. The optic axis for a uniaxial crystal points along the z-direction, while the biaxial optic axes are in that same plane on equal angles on either side of the z-axis. The pump beam makes an angle θ_p with the optic z axis. Signal (idler) beams are emitted at angles $\theta_{s(i)}$ on either side of the pump and at local azimuthal angles, $\phi_{s(i)}$. Signal (idler) beams refract outside the crystal with new angles given by Snell's law.

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \cos\theta_p \cos\phi_p & -\sin\phi_p & \sin\theta_p \cos\phi_p \\ \cos\theta_p \sin\phi_p & \cos\phi_p & \sin\theta_p \sin\phi_p \\ -\sin\theta_p & 0 & \cos\theta_p \end{pmatrix} \begin{pmatrix} x'' \\ y'' \\ z'' \end{pmatrix}.$$

When the daughter photons emerge at the crystal-air interface, they refract according to Snell's law

$$n_{s(i)}(\omega_{s(i)}, \theta_{s(i)}) \sin\theta_{s(i)} = \sin\theta'_{s(i)}, \tag{5.4}$$

where their new direction is given by the primed angles $\theta'_{s(i)}$.

The phase matching conditions contain the following parameters: the direction of propagation for pump, signal, and idler $(\theta_p, \phi_p, \theta_s, \phi_s, \theta_i, \phi_i)$, the frequencies of the interaction $(\omega_p, \omega_s, \omega_i)$, and the frequency and angle dependent refractive indices, $(n_p(\omega_p, \theta_p, \phi_p), n_s(\omega_s, \theta_s, \phi_s), n_i(\omega_i, \theta_i, \phi_i))$. Typically all the frequencies and refractive indices are known through the Sellmeier equations or chosen for a given interaction. In addition, momentum conservation implies that $\phi_s = \phi_i + \pi$ and either ϕ_s or ϕ_i is chosen. The unknown quantities are then $\theta_p, \phi_p, \theta_s, \theta_i$. Either the pump-crystal angles (θ_p, ϕ_p) or the emission angles (θ_s, θ_i) must be known in order to solve for the other two. I find either set of angles by solving

$$(\Delta k_x)^2 + (\Delta k_y)^2 + (\Delta k_z)^2 = 0, \tag{5.5}$$

and

$$\Delta k_z = 0, \tag{5.6}$$

where for example, $\Delta k_z = k_{p_z} - k_{s_z} - k_{i_z}$ and $\vec{\Delta k}$ is called the phase mismatch. Here, knowing either the pump-crystal phase match angles, or the emission angles is a choice that is easily experimentally implemented. Either I set collection optics at a chosen angle on either side of the pump beam, which sets θ_s, θ_i, and then angle-tune the crystal until I observe maximum counts into those chosen directions. Or, I choose to set the crystal-pump angle and move my collection optics to collect the specific emission angles the crystal-pump angles (θ_p, ϕ_p) produce.

Calculating the phase mismatch involves knowing the angle and frequency dependent refractive index for each photon in the interaction. In BBO, the refractive indices depend on the polarization direction of the photons which is typically either perpendicular to the plane containing the optic axis and photon propagation direction and the photon is called "ordinarily polarized" (o-polarized) or it is in the plane of the optic axis and propagation direction and called "extraordinarily polarized" (e-polarized). An o-polarized photon experiences an angle-independent refractive index, meaning that its refractive index does not change with its direction of propagation, while an e-polarized photon experiences an angle-dependent refractive index that does change with the propagation direction.

Biaxial crystals, such as BiBO, have two optic axes and are different from their uniaxial counterparts in the sense that they do not have a certain polarization direction that is always angle-independent or angle-dependent. Instead, for a given crystal, a certain range of wavelengths has a polarization direction that is angle-dependent and one that is angle-independent. In these crystals, the photons are called “fast” or “slow” instead of “o" or “e" and this nomenclature has to do with having a smaller (fast) or larger (slow) refractive index.

I determine the fast (slow) indices by solving for the half-widths of the minor (major) axes of the optical indicatrix ellipse. The optical indicatrix is given by Fresnel’s equation of wave normals and represented by

$$\frac{s_x^2}{n^{-2}(\hat{s},\omega)-n_x^{-2}}+\frac{s_y^2}{n^{-2}(\hat{s},\omega)-n_y^{-2}}+\frac{s_z^2}{n^{-2}(\hat{s},\omega)-n_z^{-2}}=0, \tag{5.7}$$

where, $n_{x,y,z}$ are the refractive indices in the principle directions in the crystal at a given vacuum frequency and $n(\hat{s},\omega)$ is the total refractive index in a given unit direction $\hat{s}$. I solve Eq. 5.7 in the manner of Ref. [36] and obtain two solutions, one for that fast polarization direction and one for the slow. The fast and slow polarization can switch depending on the wavelength range. I use the fact that BiBO is a negative biaxial crystal by the convention given in Ref. [64] so that $n_x < n_y < n_z$.

For BiBO, a pump photon in the blue part of the spectrum is a “fast” photon and experiences an angle-independent refractive index, while the daughter photon in the NIR part of the spectrum are “slow” photons and experience an angle-dependent refractive index. However, in BBO, the pump photon travels as an e-polarized photon in the crystal, experiencing an angle-dependent refractive index, while the daughter photons are o-polarized photons and experience and angle-independent refractive index.

This difference is the main reason the down-conversion rings produced by BiBO are elliptical while those in BBO are circular. Namely, the daughter photons in BBO experience the same refractive index in any azimuthal angle (ϕ_s, ϕ_i) into which they are emitted. Thus, the local emission angles (θ_s, θ_i) are a constant function of the azimuthal angle and make a circular emission pattern in a plane transverse to the pump. In BiBO, the daughter photons (at NIR wavelengths) are angle-dependent, meaning that as the local azimuthal angle (ϕ_s, ϕ_i) changes, so does the emission angle (θ_s, θ_i) which gives rise to elliptical instead of circular emission patterns.

Figure 5.4 shows a plot of the exterior opening emission angle for the signal, θ_s', versus its local azimuthal angle, ϕ_s. For BBO, the curve is a straight horizontal line which gives rise to a circular emission pattern. For BiBO, the curve is periodic with 0 and π having larger values than $\pi/2$ and $3\pi/2$, meaning that there is a minor axis and a major axis which quantify the ellipse. In the following section I discuss the details of the experimental setup and present experimental data for elliptical rings in BiBO and circular rings in BBO.

5.4 Elliptical Rings: Experimental Results

I image the spatial emission patterns from BBO and BiBO with the experimental setup shown in Fig. 5.5. A 405-nm-wavelength, continuous wave (CW) laser pumps either a BiBO or BBO crystal. Each crystal is thin – a requirement for polarization entanglement.

I perform this experiment with two different cuts of BiBO crystal. Both crystal sets are cut for degenerate down-conversion from $\lambda_p = 405$ nm to $\lambda_s = \lambda_i = 810$ nm with an opening emission angle $\theta_s \sim 3°$. One set of crystals is cut with $\theta_p = 151.7°, \phi_p = 90°$ with $d_{\mathrm{eff}} = 3.2$ and the other with $\theta_p = 51°, \phi_p = 0°$ with $d_{\mathrm{eff}} = 1.5$. The BBO crystal is

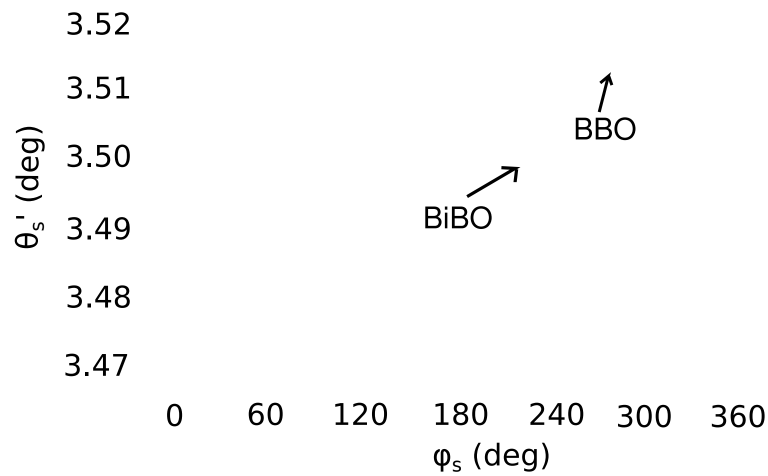

Figure 5.4: **Emission Angle versus Azimuthal Angle** Exterior emission angle for the signal photon plotted as a function of the local azimuthal angle for BBO (dashed, red line) and BiBO (solid, blue line). Here, the pump wavelength is 405 nm and the degenerate wavelengths for signal and idler are 810 nm. The pump phase matching angles for BiBO are $\theta_p = 151.563°$ and $\phi_p = 90°$, while for BBO are $\theta_p = 29.392°$.

cut at $\theta_p = 29.3°$ with $d_{eff} = 1.4$. I can make small adjustments to this angle by tilting the crystal in order to achieve other opening angles larger or smaller than $\sim 3°$.

Imaging this system and measuring the eccentricity requires a well-calibrated optical imaging system shown in Fig. 5.5. I perform the following steps to ensure that the system is well aligned. First, I removed the 40 cm focal length lens, leaving the laser, steering mirror, alignment irises, and the camera-lens system. I then place a 2 inch flat mirror against the camera lens to ensure that the laser beam is sent directly back through the apertures, meaning the camera lens is perpendicular to the pump beam. In the next step, I use a machined-milled target ring to measure the eccentricity in the imaging system. This target ring is a flat piece of metal with a 20-mm-diameter ring bored into the surface. The diameter tolerance is ~ 25 μm. The machined ring has an additional cut-out hole in the center so that it can be aligned with the laser beam path. I align this target and make small adjustments to the camera position and tilt to

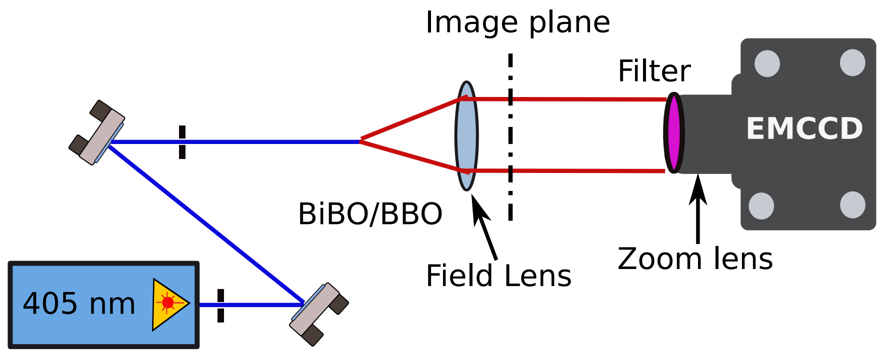

Figure 5.5: **Experimental Setup for Elliptical Ring Imaging** A 405 nm CW laser (Omicron, LDM405.120.CWA.L.WS, $<$ 0.02nm bandwidth FWHM) pumps either a BiBO or BBO crystal (Newlight Photonics). The down-converted light exits the crystal and propagates 12.4 cm before it passes through a 40 cm focal length lens (Thorlab LAC726B). I image the ring $\sim$ 10 cm after this lens by placing a zoom lens (Navitar Zoom 7000E) on an EMCCD (Andor iXon$^{\text{EM}}$). The magnification of this imaging system is 8.6. Each pixel on the camera chip is 24μm $\times$ 24 μm, the sensitive area of the chip is 3 mm $\times$ 3mm and I cool the chip down to -70° C. I put a 10 nm bandpass optical filter (Andover 810FS10-50) before the camera lens to select only nearly degenerate wavelengths.

minimize eccentricity. The eccentricity is mainly due to imperfections in alignment in the optical system. The dominant contribution to the eccentricity is tilt in the optical system, but smaller contributions arise from astigmatism and coma as well. Finally, I remove the target and replace the crystal and check back reflections with the mirror on camera lens again to ensure the pump beam is going straight back through the crystal. Finally, I add the 40 cm focal length lens back into the system and check for back-reflections.

I collect data via a computer-camera interface into a data file that contains the brightness of each pixel in the image. I collect images for each crystal and fit the observed emission pattern with an elliptical function in the two transverse dimensions with a Gaussian profile in the longitudinal dimension. The free parameters of my model include the intensity (height of the Gaussian peak), background counts (offset of the Gaussian from 0), major/minor axes for ellipse, width of Gaussian and location of the center point. I calculate the eccentricity of the ellipse by

$$\epsilon = \sqrt{1 - \left(\frac{a}{b}\right)^2}, \tag{5.8}$$

where b(a) is the major (minor) axis of the ellipse.

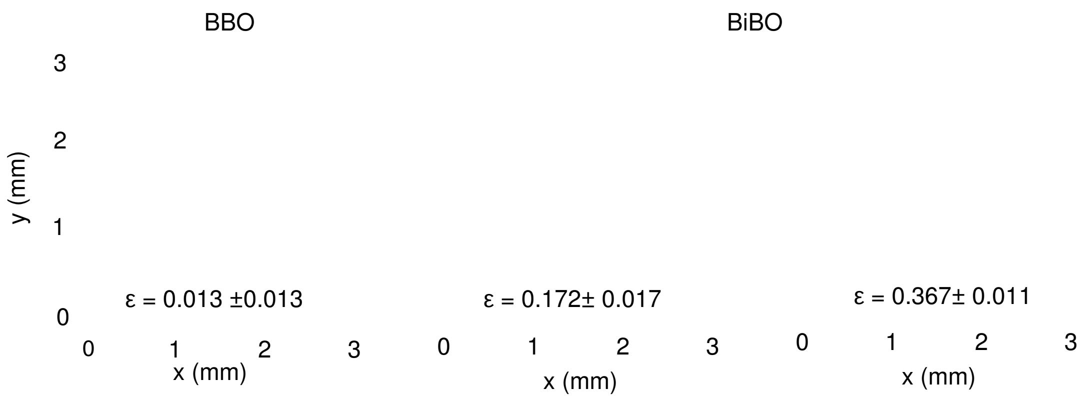

Figure 5.6: **Emission Pattern for BBO and BiBO** I plot the emission pattern from the camera versus the two transverse dimensions (x and y) where y is in the direction perpendicular to the optical table. The patterns from the camera have been scaled up by the magnification and converted from pixels into cm. The red, solid lines are ellipses with the major and minor axes taken from the fit parameters. (a) Down-conversion ring from BBO crystal is essentially circular with an eccentricity of ∼0.013. This ring is thinner than the other rings because the spread of emissions angles that are phase matched for the process is smaller, hence a thinner ring. (b) Down-conversion ring BiBO crystal cut at phase matching angle of ($\theta_p = 151.7^\circ, \phi_p = 90^\circ$) has a higher eccentricity with the major axis in the x-direction. (c) Down-conversion ring in BiBO crystal cut at phase matching angle of ($\theta_p = 51^\circ, \phi_p = 0^\circ$) has a large eccentricity.

I show the spatial emission pattern for each crystal as well as the associated eccentricity in Fig. 5.6. Here, Fig. 5.6 (a) shows the emission pattern for BBO which has a very low eccentricity of $\epsilon = 0.013 \pm 0.013$, which to within the associated error is 0 and essentially circular. I comment on sources of error at the end of this section. Figure 5.6 (b) and (c) shown the emission pattern for BiBO where there is clearly a higher eccentricity. In Fig. 5.6 (b), the eccentricity is $\epsilon = 0.172 \pm 0.019$ which is greater than that in BBO and corresponds to the pump phase matching angle of $\theta_p = 151.7°, \phi_p = 90°$ while Fig. 5.6 (c) has a much greater eccentricity of $\epsilon = 0.367 \pm 0.012$ that is easily seen by eye and corresponds to a phase matching angle of $\theta_p = 51°, \phi_p = 0°$. The reason for this difference will be explored more in the following section.

Table 5.1 shows detailed values for the data taken and plotted above in Fig. 5.6. In addition to the experimental eccentricity, I also calculate the major and minor axes in the first column. I calculate the major and minor axes by measuring the exterior angles at different azimuthal angles, $\theta_s'(\phi_s = 0°), \theta_s'(\phi_s = 90°)$. In the second column I list the theoretical values for the eccentricity. I calculate these by using one exterior angle (either $\theta_s'(\phi_s = 0°)$ or $\theta_s'(\phi_s = 90°)$), and calculating the pump cut angles used to produce this exterior angle at either $\phi_s = 0$ or $\phi_s = 90°$ using the phase matching procedure outlined in Sec. 5.3. Using this pump tilt angle, I know the emission angle at every local azimuthal angle around the entire ring and thus the eccentricity. Using Snell's law I propagate the angles for major and minor axes ($\phi_s = 0$ and $\phi_s = 90°$) outside of the crystal and calculate the eccentricity by,

$$\epsilon = \sqrt{1 - \left(\frac{\tan[\theta_s'(\phi_s = 0°)]}{\tan[\theta_s'(\phi_s = 90°)]} \right)^{\pm 2}}. \tag{5.9}$$

The final column in Table 5.1 shows the error for each of the measurements. The statistical error arises from the fitting process and is quite small. There is also a systematic error that is most likely due to tilt in the system and is the dominant contributor to the error. I measure the systematic error by taking multiple measurements over several days using the same alignment procedure outlined above.

Table 5.1: Experimental Data

Experiment			Theory	Error	
Crystal Type	$\theta_s'(\phi_s = 0)/\theta_s'(\phi_s = 90)$ (deg)	ϵ	ϵ	Statistical	Systematic
BBO	4.12358/4.12382	0.013	0	0.011	0.007
BiBO (ϕ_p = $90°$)	4.0294/4.10893	0.172	0.166	0.001	0.019
BiBO ($\phi_p = 0°$)	4.05449/4.31798	0.360	0.361	0.001	0.012

5.5 Elliptical Rings: Theoretical Model

In this section I present a theoretical model for estimating the eccentricity inside BiBO crystals. The eccentricity arising inside the crystal is the dominating factor that contributes to the eccentricity outside the crystal, because the ratio of emission angles outside the crystal is calculated from the internal emission angles and adds minimal eccentricity. The emission angles for the experiments I describe are small ($\theta_s, \theta_i << 1$), and therefore a small-angle approximation around the collinear case ($\theta_s = \theta_i = 0$) is a natural method for obtaining an approximate analytic solution for the eccentricity. Following this approach, I derive an expression for the eccentricity that fosters a more intuitive understanding of the parameters that affect it most. I begin with the phase matching conditions in the y and z directions in the pump coordinate axes given by

$$n_s(\omega_s, \theta_s)\cos(\theta_s) + n_i(\omega_i, \theta_i)\cos(\theta_i) - 2n_p(\omega_p, \theta_p) = 0 \tag{5.10}$$

$$n_s(\omega_s, \theta_s)\sin(\theta_s) + n_i(\omega_i, \theta_i)\sin(\theta_i) = 0. \tag{5.11}$$

I expand $\sin\theta_{s(i)}$ and $\cos\theta_{s(i)}$ to second order in $\theta_{s(i)}$ and also expand $n_{s(i)}(\omega_{s(i)}, \theta_{s(i)})$ because of its θ dependence. I write $n_{s(i)}(\omega_{s(i)}, \theta_{s(i)})$ as $n_{s(i)}$ for simplicity so that the expanded form is given by,

$$n_{s(i)} \approx \tilde{n}_s + \frac{\partial \tilde{n}_s}{\partial \theta_p}\delta\theta_p + \frac{\partial \tilde{n}_s}{\partial \theta_s}\theta_{s(i)} + \frac{1}{2}\frac{\partial^2 \tilde{n}_s}{\partial \theta_p^2}\delta\theta_p^2 \pm \frac{\partial^2 \tilde{n}_s}{\partial \theta_p \partial \theta_s}\delta\theta_p\theta_{s(i)} + \frac{1}{2}\frac{\partial^2 \tilde{n}_s}{\partial \theta_s^2}\theta_{s(i)}^2, \quad (5.12)$$

where θ_p is the collinear degenerate phase matching angle, and $\delta\theta_p$ represents small changes around that angle. The notation $\tilde{n}_s$ describes the refractive index for the collinear degenerate case, and I replace $\tilde{n}_i$ with $\tilde{n}_s$ everywhere because for the collinear degenerate case, they are equal. The expansion of the refractive index of the pump is

$$n_p \approx \tilde{n}_p + \frac{\partial \tilde{n}_p}{\partial \theta_p}\delta\theta_p + \frac{1}{2}\frac{\partial^2 \tilde{n}_p}{\partial \theta_p^2}\delta\theta_p^2. \quad (5.13)$$

Substituting these approximations into Eqs. 5.10 and 5.11 and using the expansions for sine and cosine, I find an expression based on $\theta_{s(i)}$ and $\delta\theta_{s(i)}$. In the noncollinear case, the eccentricity in terms of opening angles is,

$$\epsilon = \sqrt{1 - \frac{\theta_s(\phi_s = 0°) + \theta_s(\phi_s = 180°)}{2\theta_s(\phi_s = 90°)}}, \quad (5.14)$$

so that when I use the collinear case with expansion of the angles I am able to write the eccentricity as,

$$\epsilon \approx \sqrt{1 - \frac{\delta\theta_s(\phi_s = 0°) + \delta\theta_s(\phi_s = 180°)}{2\theta_s(\phi_s = 90°)}} \quad (5.15)$$

where I have used,

$$\theta_{s,(i)} = \theta_s(\phi_s = 90°) + \delta\theta_{s,(i)}. \quad (5.16)$$

I solve for $\delta\theta_s(\phi_s = 0°)$ and $\delta\theta_s(\phi_s = 180°)$ and after some algebra arrive at

$$\delta\theta_s = \frac{1}{2}\left[(-2\theta_s^{90} + e) + \sqrt{(2\theta_s^{90} + e)^2 - 2e'''}\right], \tag{5.17}$$

where

$$\begin{aligned}
a &= \frac{\partial \tilde{n}_s}{\partial \theta_s} + \frac{\partial^2 \tilde{n}_s}{\partial \theta_p \partial \theta_s}\delta\theta_p, \qquad (5.18)\\
b &= -\frac{1}{2}\left(\tilde{n}_s + \frac{\partial \tilde{n}_s}{\partial \theta_p}\delta\theta_p + \frac{1}{2}\frac{\partial^2 \tilde{n}_s}{\partial \theta_p^2}\delta\theta_p^2 - \frac{\partial^2 \tilde{n}_s}{\partial \theta_s^2}\right),\\
\alpha &= \tilde{n}_s + \frac{\partial \tilde{n}_s}{\partial \theta_p}\delta\theta_p + \frac{1}{2}\frac{\partial^2 \tilde{n}_s}{\partial \theta_p^2}\delta\theta_p^2,\\
e &= 2\left(\frac{\partial \tilde{n}_s}{\partial \theta_p}\delta\theta_p + \frac{1}{2}\frac{\partial^2 \tilde{n}_s}{\partial \theta_p^2}\delta\theta_p^2\right),\\
e''' &= \frac{e(\alpha - 2a\theta_s^{90})}{b\alpha - a^2} + \frac{a^2 e^2}{(b\alpha - a^2)^2} + 2(\theta_s^{90})^2,
\end{aligned}$$

and θ_s^{90} denotes the opening angle at $\phi_s = 90°$.

It is instructive to go one further step to solve for where the eccentricity is minimized. For applications that require multiplexing around the down-conversion ring, care must be taken to make sure the rings are indistinguishable and so a small eccentricity is important. When doing the minimization, I make a further approximation that $\delta\theta_p$ is very small and further examine the the magnitude of each specific term in the expression for various values of λ, ϕ_s, and small $\delta\theta_p$. Doing this I find that to minimize the eccentricity I must solve

$$\frac{e\alpha}{b\alpha - a^2} + (2\theta_s^{90})^2 = 0, \tag{5.19}$$

with the new approximated quantities,

$$\begin{aligned}
a &\approx \frac{\partial \tilde{n}_s}{\partial \theta_s}, \\
b &\approx -\frac{1}{2}\left(\tilde{n}_s - \frac{\partial^2 \tilde{n}_s}{\partial \theta_s^2}\right), \\
\alpha &\approx \tilde{n}_s, \\
e &\approx 2\frac{\partial \tilde{n}_s}{\partial \theta_p}\delta\theta_p, \\
(\theta_s^{90})^2 &\approx -\frac{e^{90}}{2b^{90}}.
\end{aligned} \tag{5.20}$$

In final form, I obtain

$$\left(\frac{\partial^2 \tilde{n}_s}{\partial \theta_s^2}\right)^0 - \left(\frac{\partial^2 \tilde{n}_s}{\partial \theta_s^2}\right)^{90} - \frac{2}{\tilde{n}_s}\left[\left(\frac{\partial \tilde{n}_s}{\partial \theta_s}\right)^0\right]^2 = 0. \tag{5.21}$$

This expression allows me to calculate the wavelength at which the eccentricity is a minimum simply from knowing the first and second order derivatives of the refractive index at various angles. This result agrees very well with my numerical calculations for the eccentricity and the wavelength at which it is a minimum. I plot each term in Eq. 5.21 versus wavelength in Fig. 5.7 for the crystal cut at $\theta_p = 152.077°, \phi_p = 90°$. From this plot, it is clear that the point where blue and red curves intersect will be where the eccentricity is minimized because these two terms cancel in Eq. 5.21. On either side of this intersection, the eccentricity increases. The final term in Eq. 5.21 does not contribute significantly to the eccentricity.

In addition to showing the wavelengths where minimum eccentricity occurs, this plot also shows the quantities that are the dominant factors of the eccentricity, namely the different in the first two terms of 5.21. For the crystal cut of $\theta_p = 51°, \phi_p = 0°$, the difference between these two terms is never small. This is what causes the eccentricity

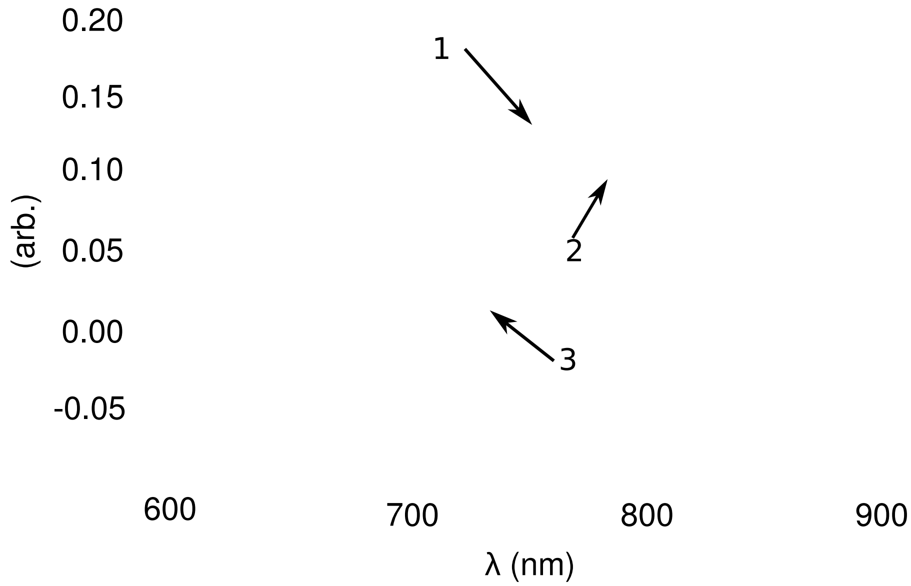

Figure 5.7: Refractive index derivatives versus wavelength Refractive index derivatives versus wavelength for each term in Eq. 5.21, "1" (blue, short-dash) $(\partial^2 \tilde{n}_s/\partial \theta_s^2)^{\phi_s=0}$ "2" (red, long-dash) $(\partial^2 \tilde{n}_s/\partial \theta_s^2)^{\phi_s=90^\circ}$, "3" (green,solid) $2/\tilde{n}_s * [(\partial \tilde{n}_s/\partial \theta_s)^{\phi_s=0}]^2$. I calculate these expressions using $\theta_p = 152.077^\circ, \phi_p = 90^\circ$.

to be so much larger for this crystal cut.

In Fig. 5.8, I plot the eccentricity as a function of wavelength for several different changes in opening angle $\delta\theta_p$. This plot shows there is a clear wavelength for each opening angle that minimizes the eccentricity. Additionally, I plot the data on the same curve and show it agrees very well within the experimental error. For experiments aiming to minimize the eccentricity for multiplexing ability, this plot shows what daughter photon wavelengths are best along with the given crystal cut.

5.5.1 Spatial Walk-off

In the calculation above, I have assumed that birefringent walk-off is negligible and ignored possible effects from it. In this section, I justify this assumption.

Birefringent walk-off occurs in a medium such as a crystal when, for a particular

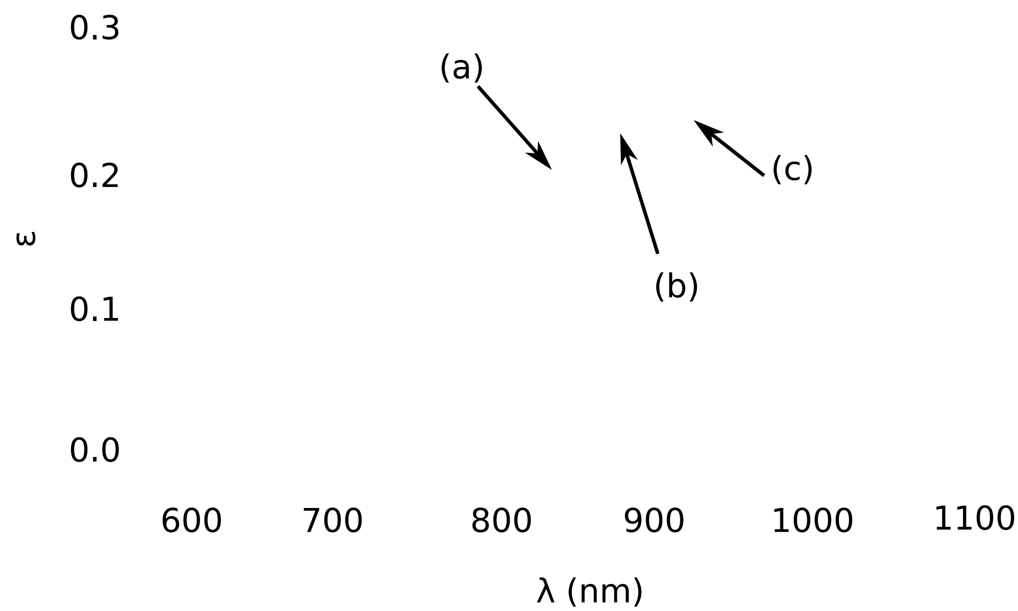

Figure 5.8: **Eccentricity and Wavelength Relationship** Plot of eccentricity versus wavelength for three different values of θ_p, (a) (blue, short-dash) $\theta_p = 152.071°$ (b) (green,solid) $\theta_p = 151.378°$, (c) (red, long-dash) $\theta_p = 149.21°$. For all three plots there is a wavelength that minimized the eccentricity. I show my experimental data point for a pump wavelength of 405 nm and a degenerate down-converted wavelength of 810 nm (purple) with associated error bars.

polarization, the momentum vector and the Poynting vector are different. This physically means that the momentum vector and energy vector do not travel in the same direction. Because I care about the intensity pattern of the light, I want to know the direction of the energy (Poynting) vector and in the theory presented above, I have used phase matching and the momentum vector to dictate the direction of the light.

Walk-off occurs in both uniaxial and biaxial crystals. While calculating the walk-off angle, which is the angle between the Poynting and momentum vector, for a uniaxial crystal is straightforward, it is more complicated in the biaxial case. I calculate the walk-off angle for BiBO using the method in Ref. [64] and find the contribution to the eccentricity is negligible.

In BiBO, the daughter photons experience an angle-dependent refractive index and have Poynting vector walk-off. Using Eqs. (20)-(22) in Ref. [64] I calculate both the unit Poynting vector, $\hat{N}$, and the unit momentum vector ($\hat{k}$). I use this to find the walk-off angles as a function of local azimuthal angle (ϕ_s) around the down-conversion ring. The walk-off angles I calculate for $\theta_p = 151.56°, \phi_p = 90°$ are 3.19° and 3.51° for $\phi_s = 0$ and $\phi_s = 180°$ respectively and 3.36° for $\phi_s = 90°$ and $\phi_s = 270°$. I project the normalized vectors onto the crystal exit face and compare the eccentricity and find that the Poynting vectors map out a ring with 0.1680 eccentricity while the momentum vectors map out a ring with eccentricity value of 0.1685 – a 0.3% difference, which is negligible.

In free-space, the Poynting and momentum vectors must be parallel, so the trajectories of these vectors to the image plane are identical. The only difference between the Poynting vector and momentum vector occurs in the crystal. In the last paragraph I showed that the eccentricity of the ring as it is about to exit the crystal is negligible. This is because I use a thin crystal so that the major/minor axes on the crystal face are calculated by

$$a = L\tan(\theta_s(\phi_s = 0°))/2, \tag{5.22}$$

or

$$b = L\tan(\theta_s(\phi_s = 90°))/2, \tag{5.23}$$

where L is the length of the crystal, in our case 800 μm. The photon path length outside the crystal is much larger than the one inside the crystal. Because the Poynting vector and momentum vector are parallel in free-space, the dominant contribution to the eccentricity comes from the paths taken outside the crystal. Essentially, the overall impact of the walk-off from intracrystal angles is very small due to the fact that the crystal length is small compared with the crystal - lens distance.

5.6 Impact on Entanglement and Joint Count Rate

The potential drawback of an elliptical emission pattern for applications in high-rate QKD, is that it may decrease the entanglement quality around the ring and/or lower the joint photon count rate. Because the rings are elliptical they no longer perfectly spatially overlap around the entire ring. This leads to fewer usable detection locations without suffering a loss of count rate or entanglement purity, thereby decreasing multiplexing ability. One could imagine putting detectors in between the two rings where they do not overlap, but in addition to suffering from a loss of count rate, the spectrum at these points could be different from each ring, leading to a degradation in entanglement quality. In this section, I look at each of these effects: I calculate the joint count rates at locations of perfectly overlapping versus non-overlapping rings and I also calculate the spectrum at these locations to estimate entanglement quality.

Using the methods presented in Chs. 3 and 4, I calculate the joint spectrum collected into single mode fibers for the biphoton wavefunction as well as the singles spectrum for both signal and idler photons. I determine these quantities at the points shown in Fig. 5.9 (a). Here, I show an example of the emission pattern from each ring for each crystal cut. I have overlapped the rings maximally at point "A" and minimally at point "B" in order to calculate the worst possible scenario.

Figure 5.9 (b) shows the joint spectrum for each crystal in the pair at point "B". The solid lines represent the joint spectrum for each crystal cut at $\theta_p = 151.7^\circ, \phi_p = 90^\circ$ while the dashed lines represent the spectrum for the $\theta_p = 51^\circ, \phi_p = 0^\circ$ crystal cut. Clearly, for the former, the joint spectral rate is higher compared to the rings with higher eccentricity. This is simply because the rings sit further apart and in order to

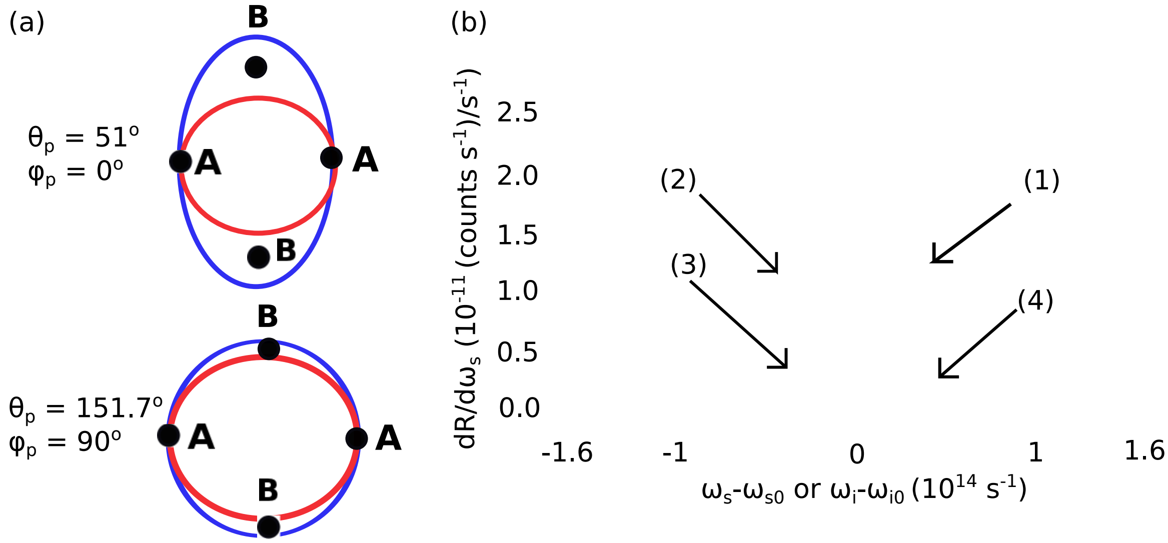

Figure 5.9: Entanglement Quality and Count Rate for Elliptical Rings (a) Points A and B represent the points of maximal and minimal overlap respectively. Red inner ring lines and blue outer ring lines illustrate the center of the emission patterns from each crystal in the crystal pair, where I have ignored the finite width of the actual ring. The ellipticity of these rings has been exaggerated for illustration purposes. Top figure depicts crystal cut at $\theta_p = 51°, \phi_p = 0$ and bottom figure depicts crystal cut at $\theta_p = 151.7°, \phi_p = 90°$(b) Joint spectra for each crystal cut at location B for each crystal cut. The y axis on the plot is the differential rate per frequency and the x axis is an angular frequency shift around the degenerate frequency. Solid lines (green (2) and gold (1)) represent the joint spectral for each crystal when the crystals are cut at $\theta_p = 151.7°, \phi_p = 90°$ while the dashed lines (blue (3) and purple (4)) represent the joint spectra from each crystal when the crystals are cut at $\theta_p = 51°, \phi_p = 0$. The pump wavelength for this simulation is 405 nm and the degenerate wavelengths 810 nm. The vertical dashed green lines in the plot indicate a 20 nm bandwidth for reference.

collect evenly from both crystals, so as not to destroy entanglement purity, the collection optic must be placed between the optimal collection location for each ring. Integrating over the joint spectra into a certain bandwidth gives the total joint count rate which is significantly higher for the crystal cut at $\theta_p = 151.7°, \phi_p = 90°$.

I calculate the entanglement purity by calculating the overlap integral of the joint spectra for each crystal. The overlap integral of the spectra is one method of determining the entanglement purity. The joint spectrum for each crystal at point “B” is shown in Fig. 5.9 (b). The difference between the two joint spectra of each crystal is minimal, leading to a high overlap integral. For point “A” I calculate the overlap integral to be 100%, because the rings completely overlap here and spectra are identical, while at point “B” I calculate the overlap integral to be $99.4 - 99.97\%$ for each of the two crystal cuts. This means that there is no significant degradation in entanglement quality in either crystal cut.

Because there is little degradation in the entanglement, both crystal cuts are feasible candidates for polarization entanglement. However, if count rates are a factor in the experiments, the crystal cut at $\theta_p = 151.7^\circ, \phi_p = 90^\circ$ is a better choice for maximizing counts around the entire down-conversion ring for multiplexing applications. For applications aiming at ultra-high entanglement purity ($> 99.9\%$), the spatial locations are limited to those points where the rings can be made to completely overlap.

5.7 Conclusions

In this chapter, I present experimental findings showing the SPDC process in BiBO pumped in the blue part of the spectrum produces elliptical emission patterns while those in BBO do not. The physical reason for this is that the daughter photons in BiBO experience an angle-dependent refractive index so that photons at different azimuthal angles are emitted at different angles with respect to the pump. In BBO, the daughter photons experience an angle-independent refractive index and are emitted at the same angle with respect to the pump around all azimuthal angles.

Additionally, I show that, by making some approximations, an analytic solutions can be found to calculate the eccentricity. Minimizing this quantity reveals that there is a wavelength for which the eccentricity is very nearly zero. This analytic solution allows me to pinpoint the quantity that dominated in making one crystal cut significantly more elliptical than the other. My experimental data and theoretical predictions agree well with each other. Finally, I discuss the potential impact of elliptical rings on applications that require high polarization entanglement. I find that, although entanglement quality around the ring suffered very little impact, the total count rates decrease significantly for rings with higher eccentricity.

Chapter 6

Temporal and Spectral Manipulation of Single Photons

6.1 Overview

In this chapter, I discuss the temporal and spectral manipulation of single-photon wavepackets. Manipulation of the spectrum and temporal profile of single photons is interesting for many applications including realizing single-photon spectrometers, time-lensing applications, and high-dimensional QKD using time-frequency encoding. In discussing the spectrum and temporal profiles of a single-photon wavepacket, I am referring to the probability distributions of the wavepacket. Over many measurements of a single-photon wavepacket, these distribution appear, and these will be the types of measurements I perform in this chapter.

I begin this chapter with a brief motivation for studying the spectral-temporal characteristics of single photons. Specifically, I focus on spectral-temporal correlations in the single-photon wavepackets produced by an SPDC source. These spectral-temporal correlations are important for single photon spectrometers, which I introduce and discuss in Sec. 6.2 and for securing high-dimensional QKD setups, which I introduce in Sec. 6.3. I investigate how security protocols rely on access to all the spectral information as well as the temporal information for these systems and show how the same temporal-to-spectral mapping used in single-photon spectrometers can be implemented in a QKD system using a dispersive medium. In Sec. 6.4, I discuss dispersive material possibilities for implementing temporal-to-spectral mapping. Of these possibilities I work with fiber Bragg gratings (FBGs), which I discuss in Sec. 6.5. In Sec. 6.6 and 6.7, I describe two potential high-dimensional time-bin QKD systems using FBGs and my experimental implementation of the key parts of this system. In Sec. 6.8, I discuss theoretical work and in Sec. 6.9, I conclude.

6.2 Single-photon spectrometer

With increasing interest from the quantum optics and quantum information communities to be able to measure various degrees-of-freedom of a single-photon wavepacket, comes the need to be able to measure a single photon's spectrum. This measurement is challenging because of the typical loss in commercial spectrometers. Commercial spectrometers typically consist of a grating element that angularly disperses the incoming light based on its wavelength and then an array of detectors (typically a CCD array) measures the amount of light dispersed into certain angles. Single-photon spectrometers use single-photon counting detectors instead of CCDs, so, depending on the quality of the single-photon detector, creating arrays of them can be cost ineffective. The other issue is that typical gratings are lossy, especially if the polarization of the incoming light is not aligned with the high-efficiency polarization direction of the grating. This high loss means that large averaging must be done to maintain a good signal-to-noise ratio and this is not always practical depending on the type of experiment. For example, in the data I present in Ch. 4 for measuring the spectrum, I used very long time constants for the lock-in amplifier and averaged over multiple runs. This requires ample time and stability – something not all experiments have.

In Ref. [65], the authors introduce a novel technique for building a single-photon spectrometer using a group velocity dispersive material. Such a material has the ability to map frequency components to different times. Group velocity dispersive materials (GVD) materials rearrange the phases of an incoming pulse, and add a chirp, which essentially delays certain frequency components more than others. The effect a GVD material has on an input pulse is that the different frequency components of the input pulse get mapped to different times in the output pulse. Ref. [65] realized that combining a GVD material with a single-photon counting detector with good timing resolution could make a single-photon spectrometer. The spectrometer works by taking the single photon input packet and sending it through a GVD material; in their case, a dispersion-compensating fiber. This delays certain frequency components with respect to time. The photon is then time-tagged. By knowing the dispersion parameter of the material and the time the photon arrived, relative to a short-input wavepacket duration, the authors reconstruct each frequency from the pulse.

The timing resolution of the detectors and the amount of stretch achievable by the GVD material control the resolution of the measurement. In the experiments I present, I demonstrate a very large stretch, up to 16.6 ns from an original 5 ps pulse. I also use very fast detectors, which have resolutions of ~ 50 ps for an individual detector. This setup that I demonstrate could be used for high-resolution spectral measurement of SPDC light. In the next section, I discuss another application using SPDC and GVD materials – a time-frequency QKD system.

6.3 High-dimensional time-frequency QKD

The security of time-bin QKD is based on Alice and Bob being able to measure both the timing states, and the equivalent states in the conjugate basis. In Ch. 2, I introduce the term *mutually unbiased bases,* (MUB) which I define as two or more conjugate bases where the state in one basis is an equal superposition of states in the other basis or bases. Security is based on the fact that if an eavesdropper attempts a measurement in either MUB, the state in the other MUB will change based on the uncertainty principle. If information about the states in both bases are known or measurable by Alice and Bob, any eavesdropping attack could, in theory be detected.

In a polarization basis, such as those used in BB84, it is simple to see that the H/V and A/D basis are mutually unbiased because the state, say $|V\rangle$ in the H/V basis is an equal superposition of A and D states, namely, $|V\rangle = 1/\sqrt{2}(|A\rangle \pm |D\rangle)$. In this work, I am interested in finding only one mutually unbiased basis for the timing states in my system. The mutually unbiased basis must satisfy

$$|\langle e_j|f_k\rangle|^2 = \frac{1}{d}, \tag{6.1}$$

where $j,k \in 1...d$ and $(|e_1\rangle...|e_d\rangle)$ is a basis orthonormal to $(|f_1\rangle...|f_d\rangle)$ [66] where d is the dimension of the system.

I use discrete time bins as my basis for encoding information. These time-bins are entangled in the timing degree-of-freedom from the SPDC source as discussed in Ch. 2. Using a pump source with a pump pulse in every time bin I obtain the following state on the output of the SPDC process

$$|\Psi\rangle_t = \frac{1}{N}\bigg(|t_{1_A}t_{1_B}\rangle + |t_{2_A}t_{2_B}\rangle + |t_{3_A}t_{3_B}\rangle...|t_{N_A}t_{N_B}\rangle\bigg), \tag{6.2}$$

where N is the number of time bins and N must be equivalent to the coherence length of the pump laser, and the subscripts A (B) stand for Alice's (Bob's) photon.

It is important to discuss the validity of creating discrete time states from the truly continuous variable, time. Unlike other work where a CW laser is used to create twin photons and time bins are created in post-processing [67–69], I used localized pulses in time that are smaller than the time bins. In this way, the system is already discrete in the sense pulses are localized in time, and creating time bins is a natural extension. I therefore work with discrete-variable QKD (DV-QKD) versus continuous-variable QKD (CV-QKD). Both DV-QKD and CV-QKD have various security proofs showing that one can bound the amount of information that Eve obtains even against collective attacks. On the DV-QKD side, Refs. [67, 68] have shown security by discretizing the time-energy basis, although it remains an open question if discretizing time states in this manner is valid.

Because I use discrete time bins for my system, I need a way, at least in theory, to create an equivalent conjugate basis for time. There are several options for this, and I will discuss two obvious choices: phase-states and frequency states.

6.3.1 Phase-state as a MUB

One important work in showing security for time-bins is Ref. [70, 71] by using Franson interferometers. Although the authors use a simple system to prove the security, in practice, this sort of system would be challenging to scale up to large numbers of time bins. However, the security proof paves the way for other methods trying to create time-frequency or time-phase mutually unbiased bases.

The QKD system described in Refs. [71] is based on securing time bins with phase states created by an interferometer. They label the normalized timing states by

$$|k\rangle = \frac{|\tilde{k}\rangle}{\sqrt{T}}, \tag{6.3}$$

where

$$|\tilde{k}\rangle = \int_{kT}^{(k+1)T} \hat{a}^{\dagger}(t')|\text{vac}\rangle \text{dt}', \tag{6.4}$$

where T is the width of the time bin, k represents the kth time bin, and $\hat{a}^{\dagger}(t')$ is the creation operator which acts on the vacuum state at a time t'. They then write down a set of states in a basis that is mutually unbiased with respect to time. They call these states phase states and they are given by

$$|\phi_n\rangle = \frac{1}{\sqrt{M}} \sum_{k=0}^{M-1} e^{\frac{2\pi ink}{M}} |k\rangle, \tag{6.5}$$

where $n = 0, 1, 2...M-1$. Here, M is the number of time bins that make up each time frame. Measuring the phase states $|\phi_n\rangle$ can be done with a Franson interferometer. This interferometer consists of two Mach-Zehnder interferometers with different delay paths and subsequent timing detection with single photon detectors. Two photons from an entangled source are separated and each travel through one of the Mach-Zehnders. Depending on the temporal difference between the two delay arms of the Mach-Zehnders they will either interfere or not, creating something similar to the Hong-Ou-Mandel dip as the time delay between the two interferometers is varied. The Franson interferometer is therefore a way to measure the temporal coherence, which Eve would disturb if she tries to measure the timing information of the photon.

In order to fully secure the time bins, one Franson interferometer would be needed for each time bin. This makes the setup particularly challenging for increasing the number of time bins in the system, because Alice and Bob would need to align and balance a large number (M) of interferometers. Another suggestion by the same authors is to use a single Franson interferometer for security, where a slightly different setup is used and presented in Ref. [70]. The authors note that a single Franson interferometer is not sufficient to build a full MUB, and cannot therefore be proven to fully secure all of the bits. However, this approach does make convincing arguments as to how any disturbance made by the eavesdropper should be able to be detected in the setup proposed.

6.3.2 Measuring frequency states

Another natural choice for a mutually unbiased basis with respect to time is the frequency basis. It is well known that time and frequency are conjugates of each other in the Fourier transform sense. The time-energy uncertainty principle can be thought of as the time-frequency uncertainty principle, where, localizing a single photon in time, necessarily means that one could not measure a well-defined spectra simultaneously. Supposing that Eve makes a time-localizing measurement in the time basis, she would necessarily change the spectrum, or the state in the frequency basis. If Alice and Bob have a method to measure the full state in each of these bases, they are able to detect an eavesdropper.

In order to detect the spectrum with enough resolution, Alice and Bob need to be able to measure the smallest changes in the spectrum that Eve could make. For example, imagine a system which secures 10 bits per photon per time frame, meaning it has 1024 individual time bins of each 100 ps so that the total frame time was 102.4 ns. Suppose Eve were to try to localize the photon to one half of this frame such that she makes a QND measurement on either the first of the second half of the frame. The question is: how much does this measurement change the spectrum? Localizing the pulse to a 51.2 ns time is equivalent to changing the spectrum by roughly 1/51.2 ns. This corresponds to a frequency change of $\sim$ 19 MHz, which, at 700 nm is a wavelength change of $\sim$ 0.03 pm. No commercial spectrometer has a resolution that comes within several order of magnitude of this value, especially that also has single photon sensitivity. This is the motivation behind creating time-to-frequency states with a dispersive material.

6.3.3 SPDC QKD with a dispersive material

Using an SPDC source I create pairs of time-energy correlated photon pairs that are born at identical times (in the same time bin) and distributed to Alice and Bob who both have single-photon counting detectors and electronics to time stamp the detected photons. These photons are also time-bin entangled because I place a pump pulse with equal power in each of the time bins that pumps the crystal. As describe in Ch. 2, the SPDC photons are born in one of these time bins, are time-bin entangled. In order to illustrate how a dispersive material, specifically a fiber Bragg grating in this case, can be used to secure these time bins, I describe a potential QKD system using dispersive materials in Fig. 6.1. I devote Sec. 6.5 to discussing the details of a fiber Bragg grating. For this section it can be simply understood as a material with group velocity dispersion which chirps the phase and stretches the temporal wavepacket.

Both Alice and Bob have the following setup under their control: they each have a switch with which they can randomly alternate between measuring the direct timing of the photon or sending the photon through a dispersive material ($\pm$ FBG) and then measuring its time of arrival. Alice and Bob's dispersive material has the same GVD parameter, which I denote $\pm$ FBG. Without any eavesdropper present, Alice and Bob expect to detect correlated hits in their detectors when they choose the same path, either timing or FBG/frequency. When Alice and Bob choose the frequency/FBG path, their anti-correlated frequency photons undergo opposite chirps. Because the frequencies are anti-correlated from the source, they will be registered as coincidence counts when propagated through equal and oppositely dispersive materials. In the protocol, just as in BB84, Alice and Bob discard the bits whether they chose different bases.

When an eavesdropper is present, she can choose to make a few different types of measurements, depending on which basis she thinks Alice and Bob have chosen. First

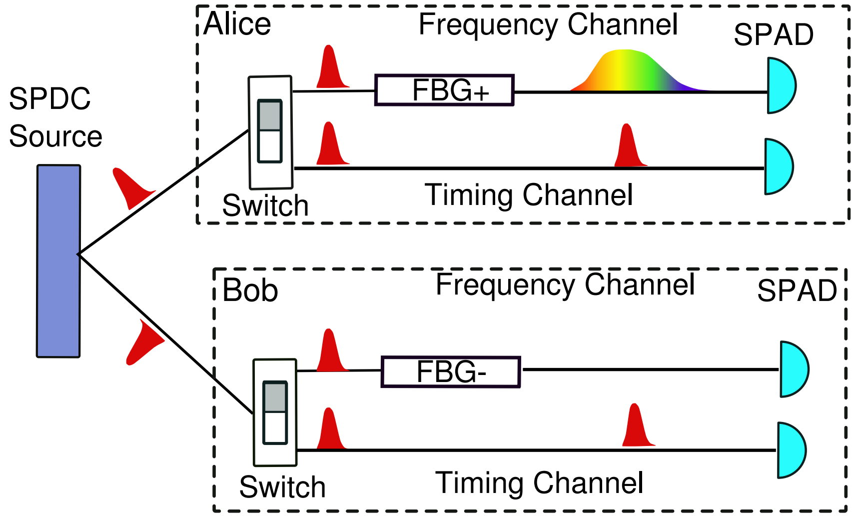

Figure 6.1: **QKD using time and frequency bases** An SPDC source produces twin photons that are time-bin entangled and one photon is sent to each Alice and Bob. Both Alice and Bob have random switches which they use to direct the photon directly into the timing channel or the frequency channel. In the timing channel, the photon is measured by a single-photon detector and its time-of-arrival is time-stamped. In the frequency arm, the single photon experiences either a positive or negative group velocity dispersion denoted by +FBG or -FBG for Alice and Bob respectively. The single photon pulses are stretched in time and their time-of-arrival detected by a single-photon detector.

I hypothesize for the case where Eve is doing an intercept and resend attack and she chooses a basis at random to measure the bit in – for example the timing basis. She measures the temporal state and resends that same state. If Alice and Bob measure in the timing basis as well, they will not detect her. However, if they have first sent it through a FBG, its spectral information is mapped to temporal information. When Eve localizes the pulse in, say Bob's channel in time and resends the timing state after the FBG, there will be decreased correlation between Alice and Bob. This is because Alice can still measure that bit in any one of the N bins it was sent in equally, while it is in a definite time bin for Bob, and one that Eve had a 1/N chance of guessing correctly. This will inform Alice and Bob that an eavesdropper is present in the system.

If Eve chooses to do the frequency measurement she uses the same system as Alice and Bob have with a GVD material to map frequency to time. She then resends the photon in the same basis – essentially, she creates another pulse and sends it through a GVD material (possibly a FBG). If Alice and Bob have chosen to check for an eavesdropper in the timing basis, and Eve has intercepted and resent Bob's photon through a GVD material, Alice and Bob will again get decreased correlation because after the FBG, the pulse has an equal probability of being in any of the time bins.

In the same way, if Eve does a QND measurement on either timing or frequency, she necessarily causes a small change in the other basis. If she only tries to partially localize the pulse in time, this changes the spectrum in some way or vice versa. A change in spectrum in one arm will cause decreased correlations between Alice and Bob when they are checking that basis.

To estimate the amount of dispersion needed to stretch a single pulse localized in a time bin of 100 ps to the length of the time frame (1024 bins equates to $\sim$100 ns), I calculate the stretch factor from Eq. 6.16. The length of the time bin is limited by the resolution of the detectors. I calculate the amount of dispersion needed to create such a stretch from Eq. 6.16 as $\sim 10^6$ ns/nm-km, which is larger than any material dispersion or current dispersive technology. However, the goal of my experiment is to demonstrate a proof-of-principle method towards securing time bins with frequency states. To accomplish this, I will describe a system in which I secure 4 bins, with the ability to secure up to 160 bins with 100 ps bin width or 32 bins with 500 ps bin width.

Due to the type of dispersive material I choose to use in the experiments, which I discuss in detail in Sec. 6.4, a modified QKD setup is required. The setup described above illustrated how eavesdropper are able to be detected with a GVD material to accomplish time-to-frequency mapping. The GVD material I use only works at one specific wavelength (850 nm) and so a modified scheme described in the next section is required. The general principle is the same.

6.3.4 Alternative QKD setup using GVD materials

In the modified QKD scheme, both FBGs are in one arm centered at 850 nm, while the other arm is 609.6 nm. Because of the theory and experiment presented in Chs. 3 and 4, I know I can achieve high heralding efficiency with a nondengenerate collinear SPDC scheme. Additionally, the singles spectra of the nondegenerate light is narrower, resulting in higher counts over the narrower bandwidth of the FBG. I show the modified setup for QKD using time bins and a GVD material shown in Fig. 6.2.

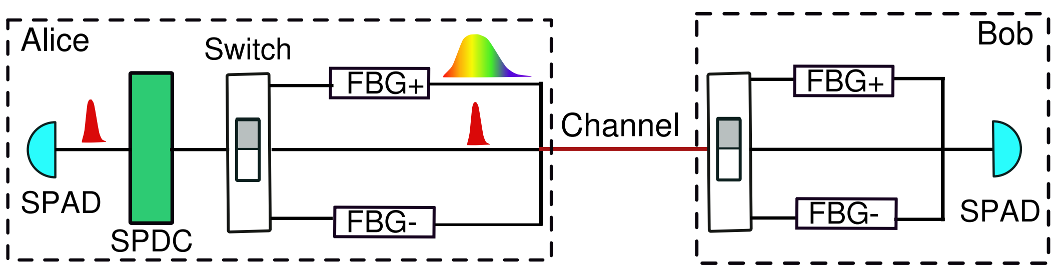

Figure 6.2: **QKD using time and frequency bases** In this QKD scheme, Alice has an SPDC source that creates twin photons. She keeps one photon and measures its timing. She then sends the other photon through one of three possibilities chosen at random. The three possibilities are the timing basis in which she does nothing to the photon, the +FBG basis where she sends the photon through a +FBG (positive GVD material) and the -FBG basis where she send the photon through a -FBG (negative GVD material). Bob similarly randomly chooses to measure the photon after sending it through the same three choices. Bob measures the photon's time-of-arrival with single-photon detectors and time-tagging equipment.

There are several key differences to this setup compared to Fig. 6.1, although the overall concept of security is the same. In this setup, Alice has the source of entangled photons. She keeps the photon from the 609.6 nm arm and detects it only in the time basis. She sends the 850 nm photon through one of three channels – the timing channel, in which she does nothing to it, the +GVD channel in which she sends it through a positive dispersive fiber Bragg grating (FBG), or the -GVD channel in which she sends it through a negative dispersive FBG. She switches between these bases at random with, for example, and electro-optic modulator. At the other end of the channel, Bob receives the photon and also randomly sends it through the same three bases. If Alice and Bob both choose the timing basis or oppositely dispersive bases they observe correlations in their single photon detectors.

An eavesdropper will be able to be detected in the same way as in Sec. 6.3.3 by destroying the correlations between Alice and Bob when they choose the correct set of bases. Eve's attack in this setup is on the channel from Alice to Bob because Alice is assumed to have the source and setup to switch between bases under her control.

In this setup, as opposed to the one in Fig. 6.1, four FBGs instead of two are used to create and detect correlations between Alice and Bob. Alice and Bob each need two FBGs with different dispersive signs to accomplish this protocol. In my experiments, I design a system that tests each arm in this setup individually, that is, I do not incorporate any switch to change between bases. Rather, I create a single channel with different combinations of FBGs and timing bases. That is, I measure a single channel with either, two plus FBGs (+/+), two minus FBGs (-/-), a plus and minus FBG (+/- or -/+), or zero FBGs (timing basis). My measurements aim to verify that I can stretch the incoming pulse to the desired width, thereby securing a certain number of bins when I use FBGs with the same basis and further verifying that using the plus/minus bases or the timing basis that I regain correlations that are only limited by detector resolution.

6.4 Dispersive Materials

In this section I first discuss some formalism related to dispersive materials. I then introduce how dispersive materials can be used in a QKD system to secure time bins and give an example of such a system.

6.4.1 Formalism for dispersive materials

A material that has group velocity dispersion is one in which the group velocity of a pulse is frequency dependent. This means that different frequency components undergo different time delays in this material, which typically leads to pulse spreading. The dispersion coefficient is related to the frequency-dependent refractive index by

$$D_\lambda = \frac{\lambda_0}{c}\frac{d^2 n}{d\lambda_0^2}, \tag{6.6}$$

where D_λ is the dispersion coefficient, λ_0 is the vacuum wavelength, n is the frequency dependent refractive index and c is the speed of light in vacuum. The dispersion coefficient can also be expressed with frequency derivatives, which may be more familiar as

$$\beta'' = 2\pi D_\nu = \frac{d^2}{d\omega^2}(\frac{\omega n(\omega)}{c}). \tag{6.7}$$

The dispersion parameter is often quoted in either ns/nm-km or ns^2/km where the conversion between the two units is

$$D_\lambda = -\frac{2\pi c}{\lambda^2}\beta'' \tag{6.8}$$

where D_λ is in units of ns/nm-km and β'' is in units of ns^2/km.

It is instructive to analyze the effect of a dispersive material on an incoming pulse in both the frequency and the time domains. To do this, I use the formalism in Saleh and Teich [72] for a transform-limited (TL) pulse. A transform-limited pulse is one that satisfies the minimum time-bandwidth product and has a constant phase. I express the amplitude, intensity, and phase of pulses in the time domain as $A(t)$, $I(t)$, $\phi(t)$ respectively, while in the frequency domain I use $V(\nu)$ or $V(\omega)$ as the spectral amplitude and $S(\nu)$ or $S(\omega)$ as the spectral intensity and $\Phi(\nu)$ or $\Phi(\omega)$ as the spectral phase. I assume an input Gaussian pulse in the temporal domain, which is a good assumption for the form of the laser and SPDC pulses. The temporal pulse envelope is given by

$$A(t) = A_0 e^{-t^2/\tau_1}, \tag{6.9}$$

where A_0 is the pulse amplitude and τ is related to the width of the pulse. The intensity is then

$$I(t) = |A(t)|^2 = I_0 e^{-2t^2/\tau_1}, \tag{6.10}$$

with I_0 being the intensity amplitude. The pulse in the frequency domain is calculated by the Fourier transform. Here, the Fourier transform is

$$F(\nu) = \int_{-\infty}^{\infty} f(t) e^{-i2\pi\nu t)} dt. \tag{6.11}$$

The spectral amplitude is given by

$$A(\nu) \propto e^{-\pi^2\tau^2\nu^2} \tag{6.12}$$

and the spectral intensity is

$$S(\nu) \propto e^{2-\pi^2\tau^2(\nu-\nu_0)^2}, \tag{6.13}$$

where $\Delta\nu = \nu - \nu_0$, and $\Delta\nu_{FWHM}\Delta t_{FWHM} = 0.44$ for Gaussian TL pulses. When this pulse enters a dispersive material, its spectral profile is multiplied by the transfer function in frequency-space. The transfer function for a dispersive material is a phase filter with a quadratic frequency dependence. It is typically called a chirp filter and has the envelope transfer function

$$H_e(f) = e^{-ib\pi^2 f^2}, \tag{6.14}$$

where $f = \nu - \nu_0$ and b is called the chirp parameter or the chirp coefficient. The chirp of a pulse is the time derivative of the instantaneous frequency – or how much the pulse is frequency modulated. Here, b is related to the GVD parameter by the equation,

$$b = 2\beta'' z, \tag{6.15}$$

where z is the length of the dispersive material. Multiplying the spectral field amplitude by the transfer function and performing an inverse Fourier transform to transform back into the time domain results in a pulse that is broadened with new width

$$\tau_2 = \tau_1 \sqrt{1 + \frac{b^2}{t_1^2}}, \tag{6.16}$$

a new chirp parameter of b/τ_1 as opposed to an originally unchirped pulse, and a reduced amplitude. It is clear from Eq. 6.16 that the greater the GVD, the more the pulse stretches. A dispersive material therefore chirps and stretches an input pulse by an amount related to the GVD parameter. It maps different frequencies to different output times and can therefore be used to detect correlations in time and frequency in SPDC as I discuss next.

6.4.2 Dispersive Material Possibilities

The ideal dispersive material has large GVD, which equates to more stretch, as well as high efficiency over the output bandwidth of the SPDC light, and the ability to exhibit both positive (normal) GVD and negative (anomalous) GVD. I investigate several possible candidates for this material including optical fibers, grating pairs, virtually imaged-phased arrays, volume holographic gratings, and fiber Bragg gratings. All of these devices exhibit dispersion, and each have drawbacks and benefits. In this section I discuss some of the factors that led to the choice in fiber Bragg gratings for the time-bin experiments.

Optical fiber has long been known to be a dispersive (GVD) material, in fact, this is commonly a negative effect in the telecommunications industry. Dispersion compensating fiber has been made to undo the effects of dispersion over long distances of regular fiber. For standard optical telecom fiber at 1550 nm, the dispersion is $\sim$ 20 ps/(nm-km), however standard optical fiber is lossy at 710 nm with $\sim$ 4 dB/km loss. This system would also be limited in scalability because, in order to secure more time bins, one would need more fiber and suffer more loss. Dispersion compensating fiber typically has a higher dispersion coefficient, although at 710 nm, is it $\sim$ -0.11 ns/nm-km. Although this is fairly large, the other issue is obtaining the same amount of positive dispersion. Typically, dispersion compensating fibers are negative dispersive materials, and for my experiment I need equal and opposite dispersion.

Grating pairs are also a promising candidate for a GVD material [73, 74]. In a grating, each wavelength is reflected by a different amount, leading to angular dispersion or spatial separation of the wavelengths. A grating pair uses two such angular dispersions to create a temporal dispersion. The two gratings are placed such that the angular spread in wavelengths from the first grating get directed back to the same direction by the second grating. There is a path length difference between shorter and longer wavelengths after they get recombined, which is equivalent to temporal dispersion. The drawbacks of this technology are that gratings are typically lossy, with only small amounts getting reflected into the correct order. Additionally, the temporal dispersion is severely limited by the angular dispersion such that long path lengths or very steep angles are needed to get any significant dispersion. For the stretch factor I need, the dispersion from a grating pair is too small for the available physical space.

A virtually-imaged phase-array (VIPA) is a tilted Fabry-Perot etalon with two highly reflective mirrors and an input window [75]. An input beam coupled into the VIPA, reflects off each window multiple times, each time transmitting a small amount of power. When the reflections are in phase, they add constructively, creating the image of a series of sources that have been delayed by different paths. When the light from these images is focused down through a lens into the Fourier plane and reflected back through the grating, it was shown that large dispersion occurs [75–77]. Although large dispersion is predicted, in practicality, I was only able to couple minimal light back through the VIPA after reflection in the Fourier plane. Such high loss is not advantageous for the QKD system.

A volume holographic grating (VHG) is a transmission grating which is similar to the standard reflection grating [78, 79]. However, in these gratings the phase of light is modulated by the periodic refractive index of the material. Each wavelength is emitted at a different angle, similar to reflection gratings. In VHGs, the periodic modulation index of the material dictates the amount of dispersion on the output. Although another promising candidate for the GVD material, these devices have a dispersion parameter of $\sim$ 10-100 ps/nm, lower than the desired dispersion.

The final candidate for creating large positive and negative dispersion is the chirped fiber Bragg grating (FBG), which I discuss in the next section.

6.5 Introduction to Fiber Bragg Gratings and Chirped Fiber Bragg Gratings

In this section, I discuss regular fiber Bragg gratings and introduce chirped fiber Bragg gratings. In my experiments, I only use chirped fiber Bragg gratings, although to understand the physical mechanism underlying the chirped fiber Bragg grating, it is useful to first understand the regular fiber Bragg grating. As a note, throughout this work, I refer to chirped fiber Bragg gratings (the ones I use) as FBGs. Whenever I refer to the regular fiber Bragg grating, which only occurs in this section, I call it a fiber Bragg grating.

6.5.1 How FBGs work

Fiber Bragg gratings are devices in which a periodic grating is written into the core of an optical fiber with the purpose of manipulating the light that enters into it. This writing is done by exposing the fiber to UV light thus changing the refractive index of the fiber [80]. Regular fiber Bragg gratings have periodic modulation of the refractive index and act to reflect a specific wavelength of light. These devices are commonly used as very narrowband filters. In more detail, at each period of the grating, some light is reflected backwards. For a certain wavelength, the reflected light is Bragg-matched, meaning that all the small reflections constructively add to form a large amount of reflected light. For all the wavelengths that are not matched to this conditions, the small amounts that are reflected do not add constructively and most of that light is transmitted through the grating. The Bragg condition is given by

$$\lambda_B = 2n_{eff}\Gamma, \tag{6.17}$$

where λ_B is the reflected wavelength, n_{eff} is the effective refractive index and Γ is the period of the grating.

Chirped fiber Bragg gratings (FBGs) have a linear frequency ramp, which multiplies the periodic modulation of the refractive index. I show this difference in Fig. 6.3 (b). This ramp produces periods with smaller separation at one end of the grating and periods with larger separation at the other, with a linear transition.

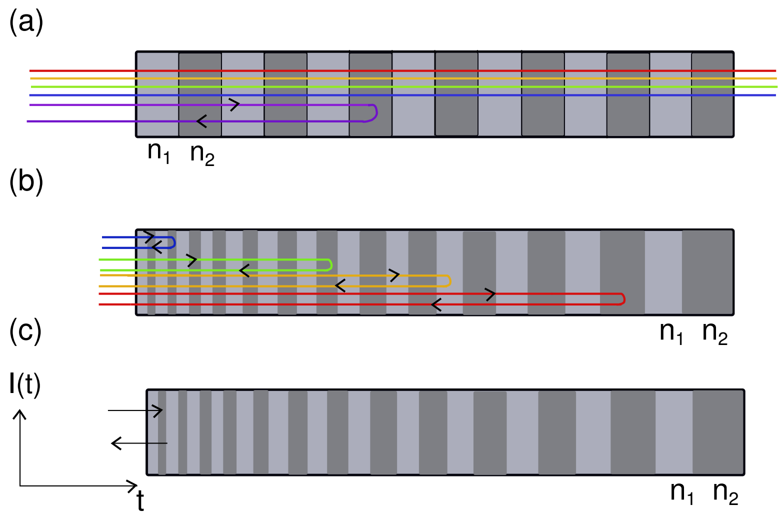

Figure 6.3: **Fiber Bragg Grating Cartoon** (a) shows a fiber Bragg grating with a periodic modulation in the refractive indices (n_1 and n_2). This modulation reflects only the wavelength that is Bragg-matched given by the condition in Eq. 6.17. (b) is another grating with periodic modulation of the refractive index, although the periods increase linearly with distance from the edge of the grating. This linear modulation causes shorter wavelengths to be reflected where the grating periods are shortest (at the first edge of the grating) and longer wavelengths to be reflected where the periods are longer (at the farther edge). In (c) I show an intensity pulse with various frequency components enter the FBG. The output result is a stretched pulse with a linear chirp, meaning that different wavelengths appear at different temporal locations.

In general, each grating period is Bragg-matched for a different wavelength. For example, the first several periods may be Bragg-matched for wavelengths in the blue part of the spectrum, while periods towards the end of the Bragg grating that have larger periods are Bragg-matched to reflect longer wavelengths depicted in Fig 6.3 (b).

Different wavelength travel different path lengths because of this effect. Therefore, if a pulse with a range of wavelengths were sent into a chirped fiber Bragg grating, the output pulse would have wavelengths that were separated in time. For example, blue wavelengths reflect off of the beginning parts of the grating and travel the shortest distance and emerge first (temporally) while redder wavelengths travel to the end of the grating before getting reflected and emerge at a later time.

This chirping, or delaying of different wavelengths with respect to time is a property of any dispersive material and very important in my QKD setup. An output pulse that has been reflected through the FBG will be stretched and chirped as discussed in Sec. 6.3.1. The amount of stretch and the output versus input power are both important characteristics of the grating which I will discuss in the next subsection.

6.5.2 Characteristics and Limitations

The two important properties of the FBGs for use in the experiment are the chirp parameter, or chirp rate, and the reflectivity curve. In the ideal case, the reflectivity would be 100%, with a completely flattop profile resembling a box function over the bandwidth of the grating. The FBGs I use are only relatively flat across the bandwidth, with almost a linear dependent of the reflectivity on wavelength. The reflectivity is also quite low– 25-40% depending on the grating. Each grating has a different exact reflectivity versus wavelength curve, with the negative chirp FBG having an overall higher reflectivity. I plot the reflectivity versus wavelength in Fig. 6.4.

The lower reflectivity simply results in not as many counts being able to pass through the grating, which affects the overall rate. Ideally, these FBGs would have a higher reflectivity, although for use in a proof-of-principle experiment such as the one I present, their reflectivity is adequate. The bigger issue with the reflectivity of

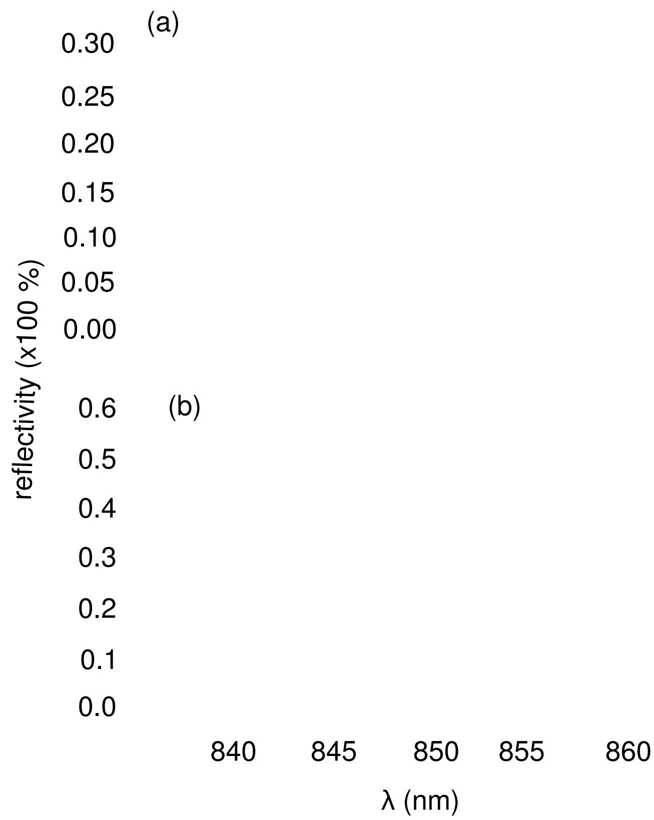

Figure 6.4: **Fiber Bragg Grating Reflectivity vs Wavelength** In (a) I show the reflectivity of two different FBGs with the same chirp parameter of -830 ps/nm while in (b) I show the reflectivity for two positively chirped FBG with chirp parameters of +830 ps/nm. This data was taken by Teraxion for the individual FBGs I use in the experiment.

these FBGs is the amount in which they are not perfectly flat-topped. Because this is the case, certain wavelengths/times are more likely to be present than others. Full security would require a perfectly even distribution among the time and frequency bins, which is only accomplished through a flat-topped FBG.

The second parameter I discuss is the chirp rate or dispersion parameter. The chirp rate dictates how much a pulse is stretched as discussed in Sec. 6.3. It is important to have the chirp rates of the FBGs of both positive and negative chirps have the same absolute value so that the pulse is stretched the same amount in either FBG. The plus and minus signs on the chirp rate simply dictate which wavelengths are delayed more with respect to others. For example, the positive FBG may have bluer wavelengths emerging before red and the negative, vice versa. The chirp rates for the Teraxion FBGs I use are $\pm$ 830 ps/nm over a bandwidth of 10 nm. This results in an 8.3 ns stretch factor if the entering pulse is greater than 10 nm, as is my case. This is a very large chirp parameter – much larger than most dispersive materials, especially fibers, posses. This is because unlike a standard off-the-shelf FBG, Teraxion custom makes their FBGs to have some of the largest dispersion available. This large dispersion was the driving reason behind using this specific technology.

The main limitation of the FBG is the center wavelength of its passband. Typically, FBGs are made at telecom wavelength (1550 nm) and fabricating them at other wavelengths requires custom runs. There is a limit to how short in wavelength a company can make a grating. This is dictated by the type of setup they have and the minimum period Γ they can resolve with their system. The FBGs I purchased had the advantage of having the largest dispersion, however the trade-off was that 850 nm was the minimum wavelength the company could fabricate.

6.6 Experimental Setup

I create an experimental setup with a pump pulse in each of four time bins. To do this, I increase the repetition rate of the Paladin laser which produces a pulse every 8.3 ns. This repetition rate means that individual bins would be 8.33 ns apart, which is very long compared to the 100-200 ps in the original scheme. To multiply the repetition rate of the pump, I multiplex pulses using the setup in Fig. 6.5 where I use two polarizing beam splitters, and two mirrors to multiply and recombine the pulse. In each repetition rate multiplier (RRM), the pump photons' polarization is rotated to an equal mixture of horizontal and vertical. These photons then get split with equal probability on the first PBS. They then get recombined on the second PBS, although the reflected photon takes a longer path and is therefore delayed with respect to the first photon. The two mirrors sit on a translation stage to make fine adjustments to the time delay between pulses. I use two RRMs – the first creating a delay of $\sim$1 ns and the second creating a delay of $\sim$500 ps which creates four pulses each spaced 500 ps apart. I choose the bin width in the experiment to be $\Delta t = 500$ ps. Although this is greater than the 100-200 ps in the original design, the two detectors limit the resolution to a 250-300 ps FWHM pulse. I choose the time bin to be twice this width to decrease the error in overlapping pulses in two adjacent bins.

I create two stages that duplicate the laser pulse and shift it so that I create four pulses each located 500 ps from the previous pulse. Although the ultimate goal is to have 16 such pulses in the 8.3 ns, or if smaller bins of 250 ps were feasible, to have 32 pulses, my goal was to create a proof-of-principle experiment with optimization details that were straightforward to accomplish at a later time. The four pulses I create mean that 2 bits/photon are achievable per time frame.

In the experimental setup I use the same configuration as in Fig. A.3 for collection of collinear nondegenerate SPDC. The 609.6 and 850 nm photons are split on a dichroic mirror and travel into two different collimating packages. The 609.6 nm path is coupled directly into a single photon counting detector (MPD PDF Module) which is the pigtailed detector described in more detail in Appendix B. The 850 nm photon takes one of several possible paths. If I measure the timing degree-of-freedom, I couple the 850 nm photon directly into a single mode fiber and single photon counting detector (SAP500 with MPD electronics). Otherwise I couple into the FBG setup shown in Fig. 6.5. This FBG setup has two FBGs setup in one of the 4 possible configurations (+/+, -/-, +/-, -,+). Because the FBGs are reflective, I use 50/50 beamsplitters to couple

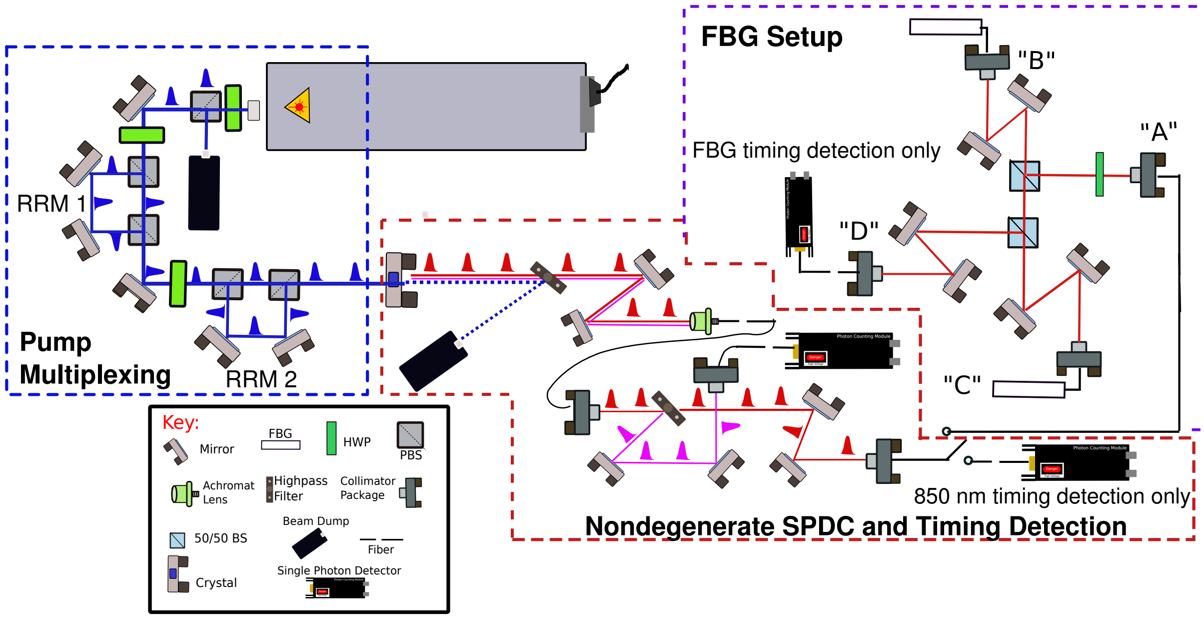

Figure 6.5: **Fiber Bragg Grating Experimental Setup** I increase the repetition rate of the pulses using two repetition rate multiplying (RRM) stages consisting of a half-waveplate (HWP) (Thorlabs WPH05M-355) two high-power polarizing beam splitters (PBS) (Precision Photonics/Advanced Thin Films PBS1005-TY), and two mirrors (Thorlabs NB1-K08). The SPDC setup for collinear nondenegerate production and collected is the same as in Fig. A.3. The 609.6 nm photon is directly coupled into a single-mode fiber pigtailed detector (MPD PDF Module). The 850 nm photon can take one of several paths. In the timing path, the 850 nm photon is coupled directly into a single-photon-counting detector (Laser Components SAP500/MPD package). For measuring in the frequency basis, the 850 nm photon is coupled into the FBG setup. The 850 nm photon is coupled into a single-mode fiber and then back out into free space via a collimating package. The photon is sent through a 50/50 beam splitter where there is a 50% probability that this photon gets directed through two mirrors and into the first FBG in arm “B". The FBG is a reflective device so the photons travel back the same path as they entered. The photon coupled through the grating then has a 25% chance of getting coupled into the second grating in arm “C" because it passes through two 50/50 beamsplitters. The photons reflected from this arm travel back through the second 50/50 beam splitter and have a 50% chance of getting coupled into the multimode fiber (Oz Optics Custom Fiber) in path “D". This photon is then sent to the SAP500.

reflected light from one fiber into the other. The 50/50 beamsplitter method has the disadvantage of being high-loss, but the advantage of being polarization insensitive. In Appendix A I discuss another method of coupling light into and out of the FBGs that is polarization sensitive and although more light is coupled through the system, it comes at a cost of having a wavelength dependent reflectivity, essentially reducing the total stretch of the system.

There are significant losses through the FBG setup. There is a 25% throughput from the first beam splitter through the first FBG and into the second beam splitter, and a 25% throughput from the second beam splitter into the second FBG and into the MMF arm (arm “D") for a total throughput from the beam splitters of 6.25%. The FBG’s have reflectivites ranging from 25-60%, with an average of around 30-35%. Additionally, coupling into the first FBG from the collimating package in arm “A" and from the first FBG to the second FBG is roughly 65-75% throughput for each while coupling into the MMF in arm “D" is roughly 90% throughput. The overall efficiency of the system is quite low, around 0.2-0.4%.

In the next section, I present my experimental results for each combination of FBG and for the timing basis using coincidence measurements. The coincidence measurements are taken with a time tagger (CAEN V1190), where the electrical pulse from the detector is time-tagged and a file written to the computer. I use Matlab code to analyze the timing coincidence measurements from each different experimental setup.

6.7 FBG Results

Here, I present the results of the experiments using coincidence measurements analyzed through time-tagged photons produced from a nondenegerate SPDC and sent through either a timing channel or a dispersive element channel consisting of two

FBGs. After I collect time-tagged data, I calculate the coincidence and singles rate for each channel, as well as the accidental coincidence rates. Most importantly, I produce a plot of the coincidence counts versus time. To plot this, I create a histogram of the time differences between detector A (Alice) and detector B (Bob) as a function of bin width, usually 100-200 ps. When the time differences between hits in each detector fall within the resolution of the two detectors, this indicates very high coincidence probability at this temporal location.

To illustrate this, consider the following system with infinitely small resolution, meaning that the detector response is infinitely sharp. SPDC creates time-correlated photon pairs, with a very small time difference equal to the coherence time of the SPDC effect ($\sim$ 50 fs) so that I expect the time differences between detection events in detectors A and B to all be zero if there is no path length difference between the detectors. If the system were indeed infinity sensitive, I would observe a very narrow function at zero time delay with the width being equal to the coherence time. When there is a finite delay in one arm, this shifts the delta function to the time difference between paths A and B. Now suppose the system has jitter in it due to the detector electronics and material thickness, and therefore has a finite resolution. In this case, instead of a delta-like function in one arm, I observe a Gaussian-like function centered at the time delay between the two paths, with the width of the Gaussian being the convolution of the jitter in each arm. However, if the photons are not correlated, or are correlated over a larger window, as they would be if they traveled through a dispersive material, then I expect to see a broad distribution of time differences between the two paths. Finally, if a photon in one arm travels through a dispersive material that stretches it so that it could be in any one of the time slots within an 8.3 ns window equally, then I expect to observe a top-hat like box with a width of 8.3 ns. Keeping this in mind, I now present my experimental data.

Timing Coincidences from SPDC

Measuring timing coincidences from the SPDC source can easily be done using only the SPDC collection part of the setup. I directly measure the timing of the 609.6 nm and 850 nm photons after they are coupled into their respective single mode fibers. For this set of experiments, I pump the down-conversion crystal with roughly 3 W of pump power, and observe singles rates in channel A of 1.38 MHz and channel B of 1.58 MHz. Channel A, which is the 609.6 nm channel, has an MPD PDF detector with ~ 50 ps jitter and $\sim 43-44\%$ quantum efficiency, while Channel B has an SAP500 detector with ~ 180 ps of jitter and a quantum efficiency of $\sim 36-38\%$. I plot a histogram of time differences between channels A and B in Fig. 6.6.

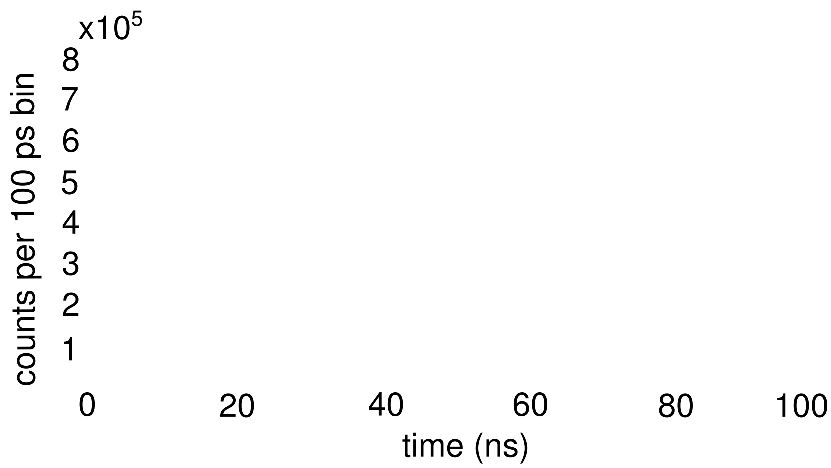

Figure 6.6: **Timing Coincidences with SPDC** Histogram of coincidence counts per 100 ps time bin versus time in ns. The sharp peak corresponds to the coincidence counts events with the offset due to the path length difference in the channels. Each small set of peaks occurs at the repetition rate of the laser (every 8.3 ns) and these counts are due to accidental coincidence counts. Within each set of peaks there are seven smaller peaks which arise from performing an autocorrelation on four separate time bins.

Here, it is clear that the very sharp Gaussian-like function represents the fact that all coincident events fall within a very narrow window that is governed by the jitter of the detectors which is $\sim$250-300 ps. The finite offset at 4-5 ns represents the path length difference between the two channels. The small peaks after the main peak occur at the repetition rate of the laser and are due to accidental counts. Accidental counts are counts which are not from correlated photon events, but show up as coincident counts in the detectors. They are mainly due to dark counts entering each detector in the coincident time window.

Single FBG Coincidences

I measure the coincidence counts between arms A and B using a single FBG in the 850 nm arm. Here, I expect the coincidence counts to be spread out over the stretch time of the FBG. Because the FBG is physically stretching the 850 nm pulse out in time, the coincidence window should match that total stretch time. The FBGs I use have 830 ps per nm of bandwidth stretch over their 10 nm bandwidth and should thus stretch the input pulse by 8.3 ns. I plot the histogram of coincidence counts in Fig. 6.6. In the single FBG experiment I obtain singles count rates of 1.51 MHz in the 609 arm and 35-36 kHz in the 850 nm arm.

In Fig. 6.7 (a), I plot the histogram of time differences between the two channels over a period of 100 ns. The sharp peaks at $\sim$ 18 ns and $\sim$ 57 ns are reflections off of the FBG when the light does not couple into it. These reflections occur when the incoming light is not polarized in the optimal direction to couple into the FBG. It reflects without any stretch and appears in the histogram as a sharp peak with an offset related to the extra path length it propagates through. The extra path length is the pigtailed end of the fiber connected to the FBG. The stretched peak centered at 37 ns and shown in more detail in Fig. 6.7 (b) is the light that couples into the FBG and gets chirped and stretched from $\sim$ 34 ns to $\sim$ 41 ns corresponding to approximately the 8 ns of stretch I predict. The shape of this peak is dictated by the shape of the FBG

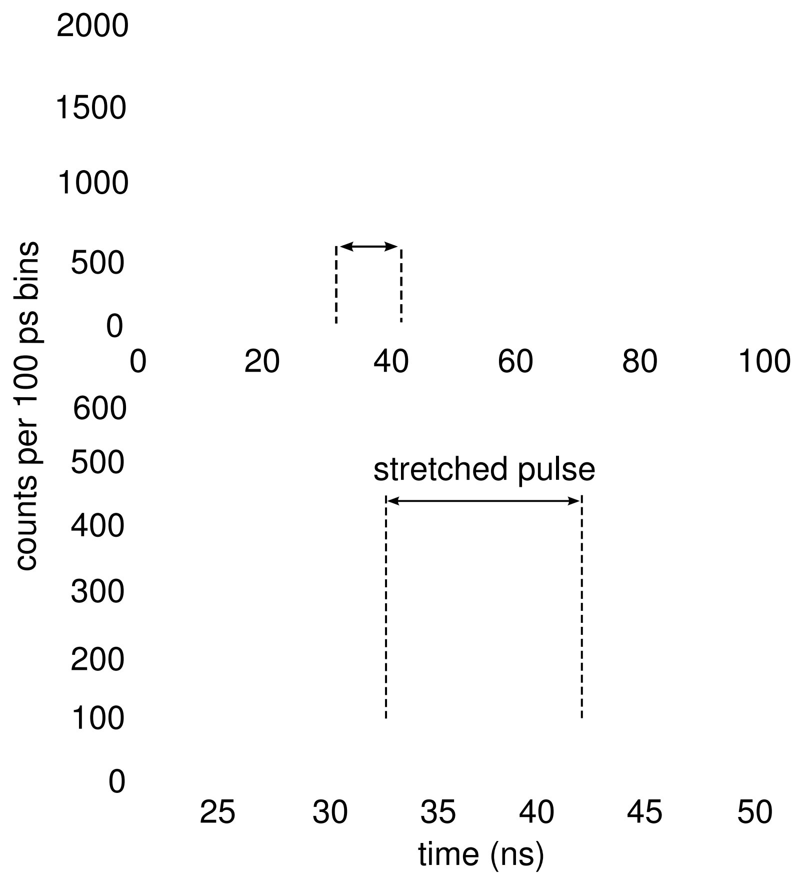

Figure 6.7: **Timing Coincidences with single FBG** Histogram of coincidence counts per 100 ps time bin versus time in ns for a single FBG in the 850 nm arm. The feature of interest is the pulse spanning 33 ns - 41 ns. The counts that fall outside this pulse are accidental counts and noise in the system.

reflectivity spectrum and the shape of the down-converted spectrum. In Sec. 6.8.1. I model a single pulse traveling through the FBG and obtain a similar profile.

Double FBG Coincidences

The QKD protocol I present requires using combinations of FBGs in both Alice and Bob's channels. For example, Alice chooses to either send the photon through the

+FBG, - FBG, or through neither. Bob can subsequently send the incoming photon through either FBG or neither. If Alice and Bob use oppositely dispersive FBGs, the pulse is stretched and recompressed back to its original width. In the experimental system, I cannot resolve the original pulse width of $\sim$ 5 ps with the detectors – the resolution is set at the 250-300 ps introduced previously for the timing coincidences for SPDC photons. The singles counts for the 850 nm channel with two FBGs is $\sim$ 4-6 kHz.

In Fig. 6.8, I show a histogram of the time differences of the coincidence counts as a function of time for a pulse that is sent through a +FBG and a -FBG where each FBG is in either arm "B" or "C" of Fig. 6.5 and a multimode fiber collects the light reflected from both FBGs in arm "D". The result is a sharp peak delayed by the appropriate amount of time for traveling through the extra path length of both FBGs as well as the free-space and additional fiber collection paths ($\sim$ 60 ns). This peak is sharp because the pulse has been stretched and recompressed down to *at least* the resolution of the detectors. Although the reflections from not entering either FBG or entering just one of the FBGs are very small, they are still present in the plot. At $\sim$ 40 ns the coincidence counts do not drop to the background between the two peaks from the accidental counts. Additionally, around 20 ns there is a sharper peak towards the right side of the group of accidental peaks. These are most likely due to the relfections off the fronts of the two FBGs.

I also plot the histogram of the time differences of coincidence counts for the case where I use two FBGs with the same dispersion, magnitude and sign. In Fig. 6.9 I show experimental data using two minus FBGs. In the plot there are three distinct peaks; the first is at $\sim$ 20 ns and corresponds to light that is not coupled into the first FBG grating and simple reflects back without getting stretched. The second peak is light that is either coupled into the first FBG and reflected off of the second or light that is

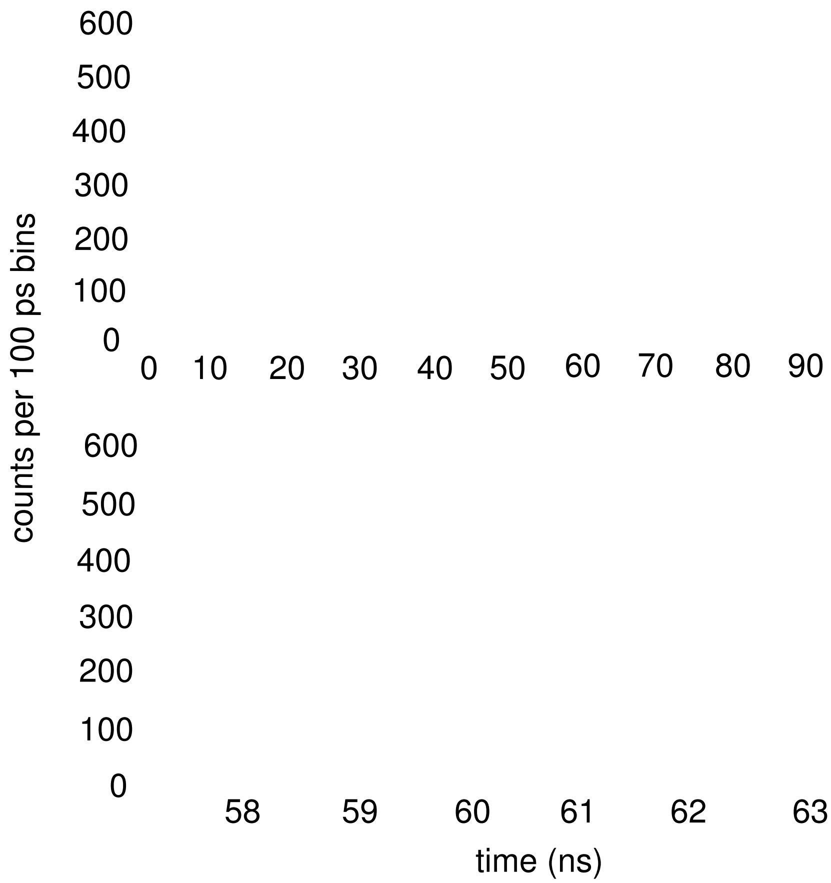

Figure 6.8: **Timing Coincidences with +/- FBG setup** Histogram of coincidence counts per 100 ps time bin versus time in ns for a +FBG followed by a -FBG in the 850 nm arm. The pulse gets stretched by the first FBG and subsequently recompressed by the second FBG because they have equal and oppositely signed dispersion or chirp parameters.

reflected off the first grating and coupled into the second. In both these processes, the light is stretched only by a single grating and so I expect the width of the peak to be ~ 8.3 ns, which it is. The third peak represent lights that has been coupled into both FBGs. Because they have the same sign and chirp parameters, light that is coupled into both should be stretched by ~ 16.6 ns. I find the width of the peak is ~ 16 ns, which compares favorably to the expected value. In the next section, I simulate pulses of light traveling through a FBG and show how the experimental data matches the expected shape and stretch given in the theory.

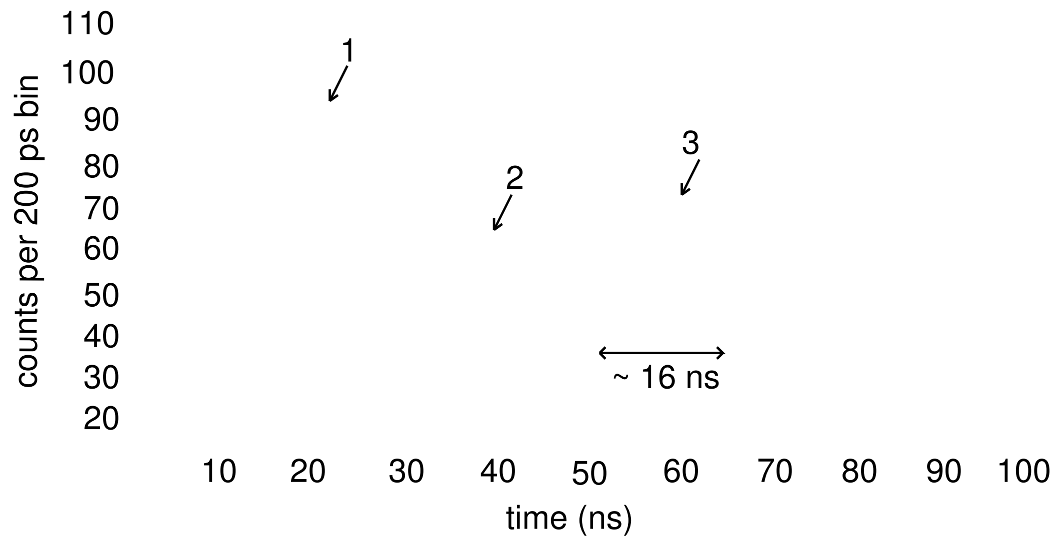

Figure 6.9: **Timing Coincidences with -/- FBG setup** Histogram of coincidence counts per 200 ps time bin versus time in ns for two -FBGs in the 850 nm arm. Peak “1” corresponds to a reflection off the front of the grating without coupling into it, while peak “2” is light that couples into one of the two gratings and peak “3” is light that couples into both gratings.

6.8 Modeling an FBG

In this section, I discuss the simulation of how a single pulse or pulse train of photons interacts and gets modified by the FBG. Fist, I simulate what occurs in the single pulse case and show how this is very similar to the data I present. Then I model a continuous pulse train, which is similar to the physical setup in the sense that the laser is a continuous pulse train instead of a single pulse. I show how the output in the pulse train case is different from the single pulse case. Finally I show a case for a set of four pulses, which most closely models my system. Throughout the first part of this section, I assume a transform-limited (also called bandwidth-limited) input pulse. In the final part of the section, I simulate how using a non-transform limited pulse changes the outcome.

6.8.1 Single pulse through an FBG

The simplest way to model the experimental system is to assume a single pulse interacting with the FBG. While this is not physically what occurs in the experiment, it is an instructive exercise to understand how the electric field of the pulse gets chirped and stretched by the FBG. I model the input pulse as a transform-limited (TL) pulse with a bandwidth of 20 nm. The bandwidth and transform limit dictate the temporal width of the intensity for a Gaussian pulse defined as

$$\Delta\tau_{FWHM}\Delta\nu_{FWHM} = 0.44, \tag{6.18}$$

where $\Delta\tau_{FWHM}$ is the full-width at half-max of the intensity for the temporal pulse envelope and $\Delta\nu_{FWHM}$ is the full-width at half-max of the spectral intensity in frequency units of s^{-1}. A 20-nm-bandwidth at 850 nm corresponds to a frequency and angular frequency difference of 16,408 GHz and 52,178 rad/ns respectively. The FWHM of the temporal pulse for 20-nm-bandwidth is $\sim$ 53 fs.

The process to calculate the output from a dispersive material was introduced previously and I outline it here briefly. I calculate the electric field from the input intensity using the fact that a TL pulse has a constant phase which can be ignored. I Fourier transform the electric field of the pulse to obtain the spectral field in the frequency domain. The spectral phase is also constant. I then multiply the spectral field by the GVD transfer function which I assume for simplicity has unit transmission (or reflectivity in this case), a 10-nm-bandpass, and a chirp parameter of 0.000318 ns^2. The factor I multiply the spectral field by is given by

$$e^{\frac{\pm i * 0.000318\omega^2}{2}}\text{HeavisidePi}[\omega/26000], \tag{6.19}$$

where HeavisidePi[ω/26000] is a unit box function with a width of 52,000 rad/ns which equates to the 10 nm bandwidth of the FBG at 850 nm and the $\pm$ denotes either positive or negative dispersion.

Using this process, I plot the intensity versus time for the pulse in Fig. 6.10 (a), the spectral intensity as a function of angular frequency in Fig. 6.10 (b), and the spectral intensity after the pulse travels through the FBG in Fig. 6.10 (c). I plot the output temporal pulse in Fig. 6.11. In Fig. 6.11 (a) I use a sharp box-like top-hat filter in the frequency domain to represent the edges of the FBG passband (see Eq. 6.19). When I multiply this function by the spectrum and inverse Fourier transform it back into the temporal domain, Gibbs-type oscillations occur over the whole bandwidth. These arise from numerical issues of approximating a box function with a Fourier series expansion of sines and cosines. In Fig. 6.11 (b), I replace the box function with a super-Gaussian function of the form $e^{-(\omega/BW)^{10}}$, where BW represents the bandwidth of the filter and ω the angular frequency. The output from the FBG is a single pulse that is stretched to a width of $\sim$ 8.3 ns with a profile that is similarly shaped to the FBG profile of an ideal flat-topped filter. In reality, the filter profile is not flat-topped and I use data provided by Teraxion, the manufacturer of the FBGs, to more accurately model the FBG reflectivity. I plot the single pulse output spectrum in Fig. 6.12. Figure 6.12 (a) shows the output data of a single -FBG filter and (b) shows the output data for a single +FBG filter. Figure 6.12 (a) compares favorably with Fig. 6.12, which plots the actual data for the output of a pulse that travels through a single -FBG.

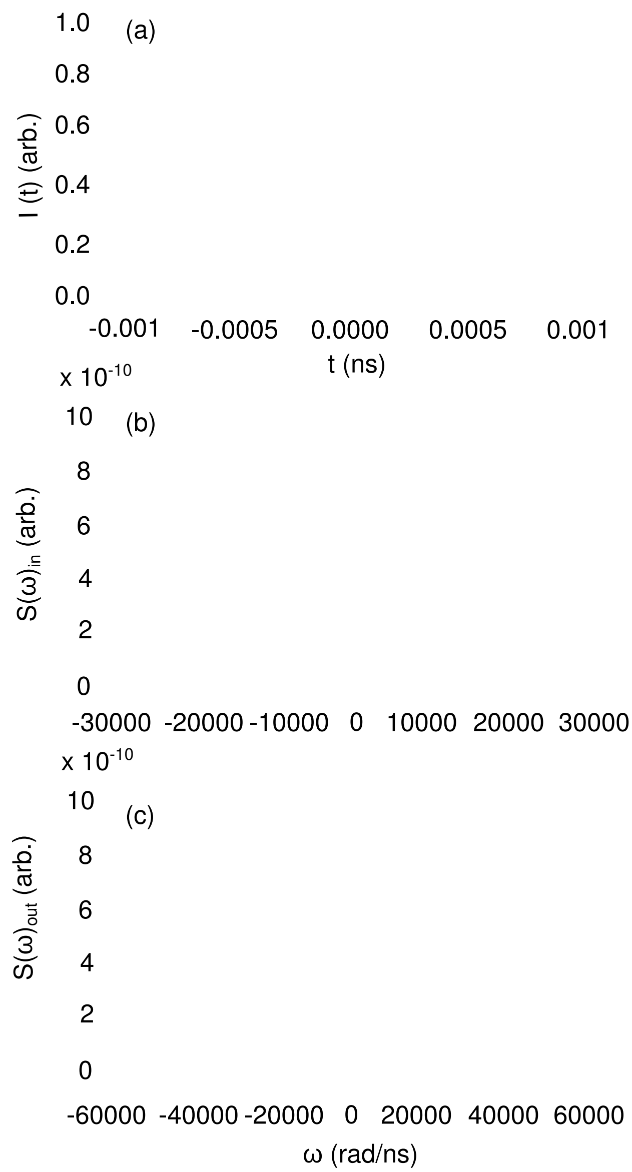

Figure 6.10: **Single pulse temporal and spectral input** (a) Intensity versus time in ns for input temporal pulse input with pulse width of 0.000318 ns. (b) The input spectrum ($\sim$ 20 nm), which is the absolute valued squared of the Fourier transform of the input field. (c) The output spectral intensity obtained from putting the input spectral field through the FBG and taking the absolute value squared. The sharp lines denote where the spectrum is truncated by the bandpass of the FBG.

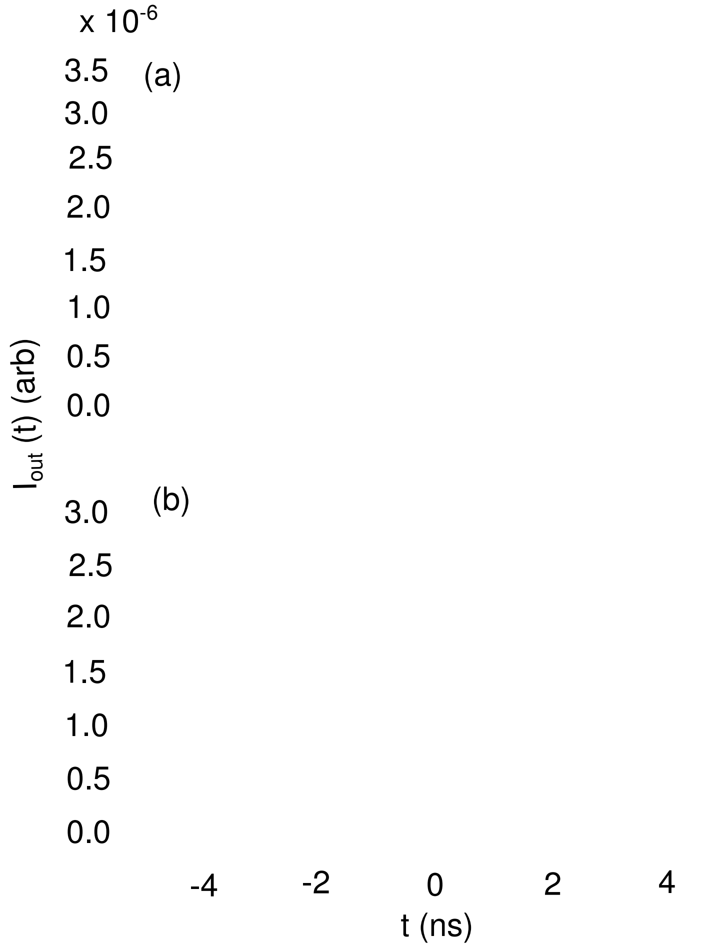

Figure 6.11: **Single pulse temporal output** The temporal output intensity versus time in ns for a single pulse that traveled through an FBG with a chirp parameter of 830 ps/nm. (a) shows the output with a sharp filter function (top-hat filter) and (b) shows the output using a super-Gaussian filter described in the text

6.8.2 Pulse train through an FBG

The next step is modeling a pulse train of TL pulses through the FBG, where the pulse repetition rate is that of the laser – 120 MHz or one pulse every $\sim$ 8.33 ns. I model a pulse train by convolving the Gaussian function for a single pulse from the previous section with a comb function. The convolution of these two is a sum of Gaussian given

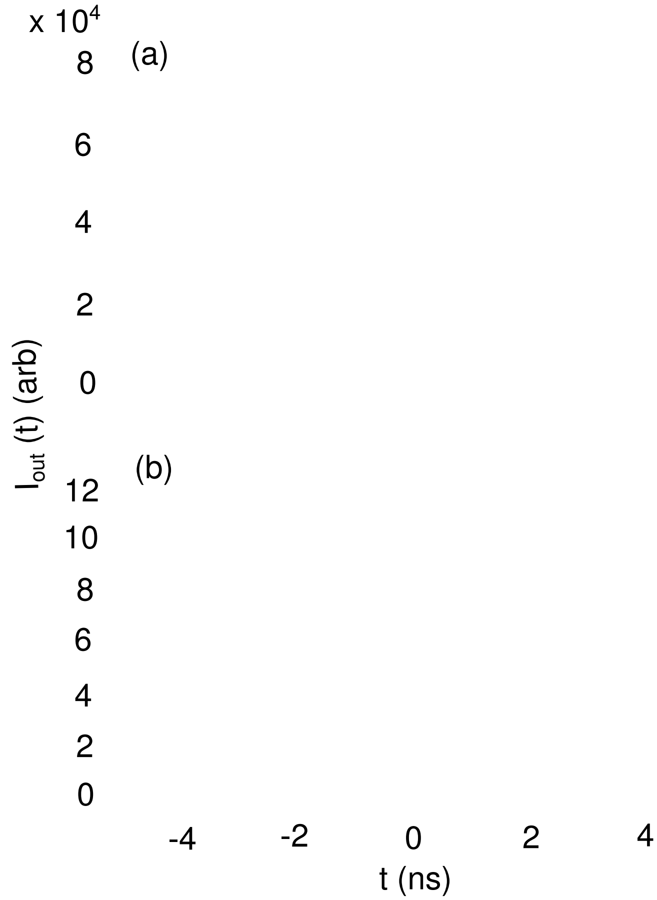

Figure 6.12: **Single pulse temporal output** Scaled single pulse output intensity versus time in ns using the reflectivity data provided by Teraxion. The difference between this figure and Fig. 6.11 is only that the reflectivity here takes on realistic values while in Fig. 6.11 it is assumed to be unity. I scale the values of the y-axis for simpler plotting so direct values should not be compared with Fig. 6.11.

by

$$\sum_{n=1}^{\infty} e^{\frac{-(t-8.33n)^2}{\Delta\tau^2}}, \tag{6.20}$$

where $\Delta\tau = 0.000318$ ns. A plot of the time series for a pulse train is given in Fig. 6.13 (a). The spectrum for a pulse train is easily calculated using the convolution property of Fourier transforms. A convolution of two functions in one basis is simply a multiplication of their Fourier transforms in the Fourier basis. The Fourier transform of a comb in time is a comb in frequency so that the spectral intensity is a comb function multiplied by the Gaussian plotted in Fig. 6.10 (b). I plot the comb and envelope separately in Fig. 6.13 (b) and (c). The FBG multiplies the spectrum in the frequency domain. When I inverse Fourier transform into the time domain I obtain Fig. 6.14. Again, in Fig. 6.14 (a) I use a top-hat filter function and in Fig. 6.14 (b) I use a super-Gaussian filter. Each pulse in the comb is stretched by ~ 8.3 ns so that the new time comb is the original single pulse temporal profile convolved with the comb function.

6.8.3 Four pulses through an FBG

In the actual time-bin QKD experiment, I create four time bins of width $\Delta t = 500$ ps with equal probability of having a pump photon occur in each time bin. This means that the down-converted light is time-bin entangled, with an equal probability of finding an SPDC photon in any of the four time bins. I now simulate the scenario of having four pulses interacting with the FBG. The results are a combination of the single pulse case and the pulse train case in the sense that the input spectrum has an envelope with fast oscillations without in. The GVD material stretches each pulse, and because they are slightly offset from each other, the temporal output profile is four-broadened pulses offset by 500 ps each. I plot the input temporal pulse series and a zoom-in of the input spectrum in Fig. 6.15.

In Fig. 6.16 I plot the output time series for four pulses. Figure 6.16 (a) is for the case where I use a box or “HeavisidePi” function for the filter bandwidth, while in Fig. 6.16 (b) I use a super-Gaussian as described previously. In both of these figures each input pulse is stretched and slightly offset from the previous pulse. However, there also appear interference fringes in each of these figures. In Fig. 6.16 (a) the oscillations are from the Gibbs phenomena and are very rapid, while in Fig. 6.16 (b) there are

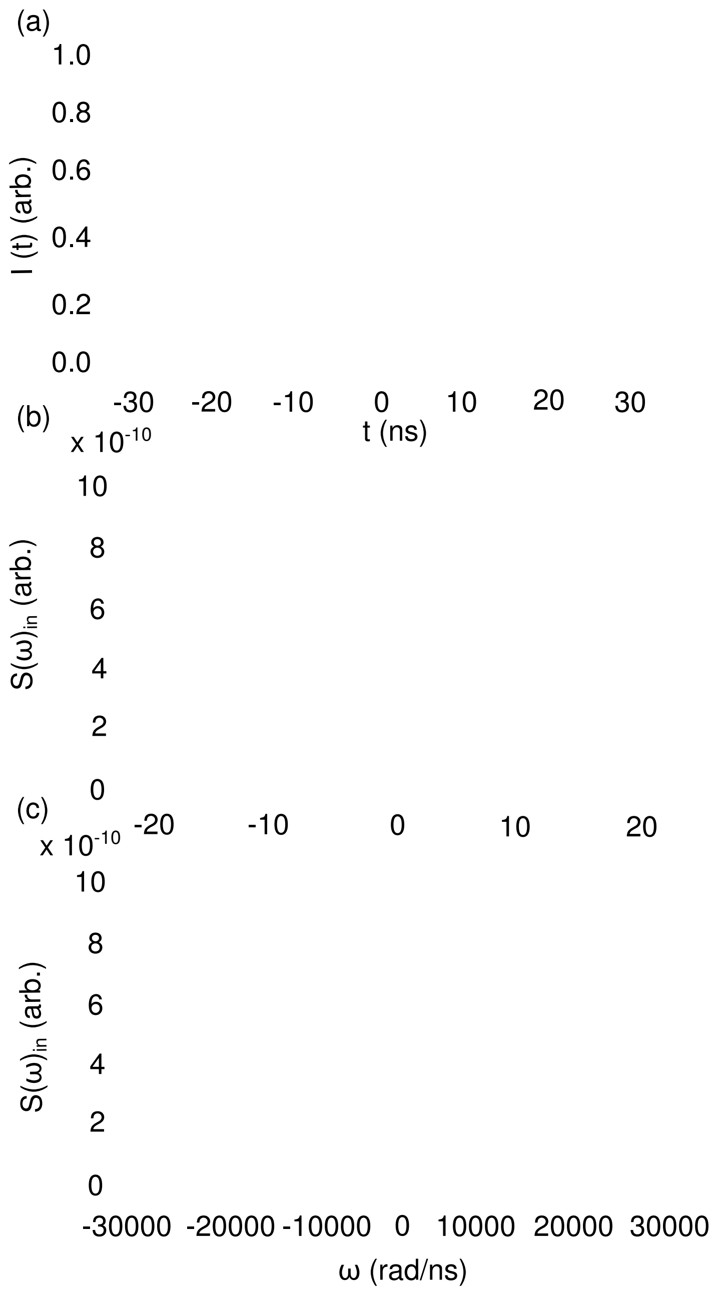

Figure 6.13: **Pulse train input** Input intensity versus time in ns (a) and spectrum (b) and (c) for pulse train with pulses every $\sim$ 8.33 ns. The pulses have the same width of 0.000318 ns. The input spectrum is made up of fast oscillations shown in (b) and a broad envelope (c), which is the same as Fig. 6.10 (b).

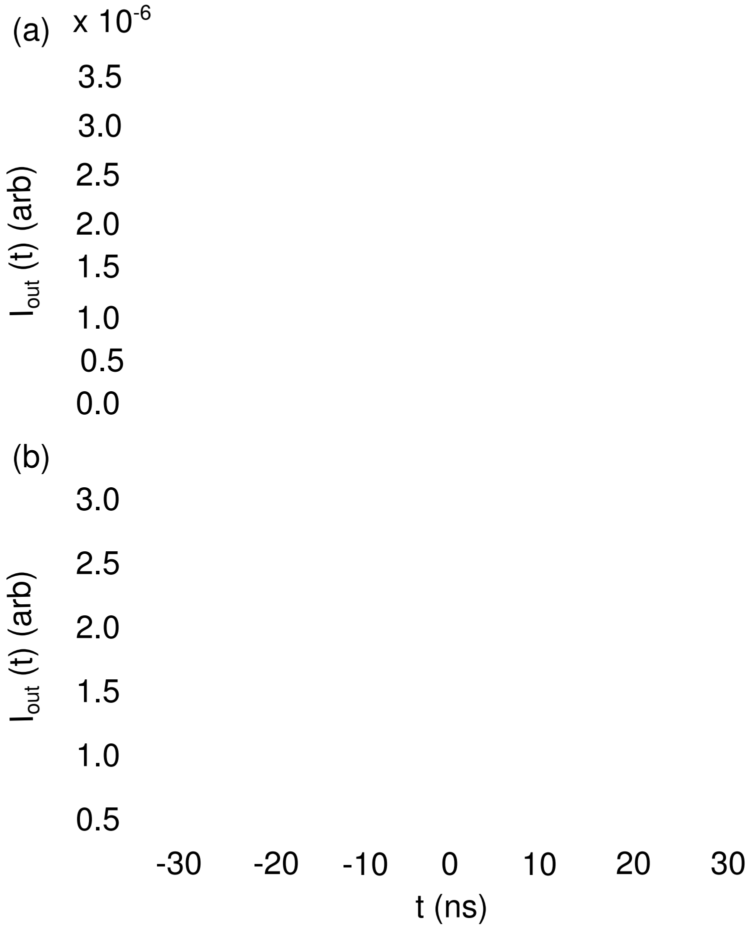

Figure 6.14: **Pulse train temporal output** The output intensity versus time in ns for a pulse train. (a) assumes a flat-topped box filter while (b) assumes the super-Gaussian filter. The output is simply a pulse train of the stretched single pulses in Fig. 6.11.

no oscillations from any numerical approximations and the fringes appear because the four separate stretched pulses now interfere.

I do not observe this interference in the experiment because each of the pump pulses has orthogonal polarization to the previous pulse and the subsequent pulse because of how the repetition rate multiplication works. This means interference between pulses should not occur. Furthermore, the coupling optics and fibers I use in the experimental setup are not polarization maintaining so that the polarization of any of

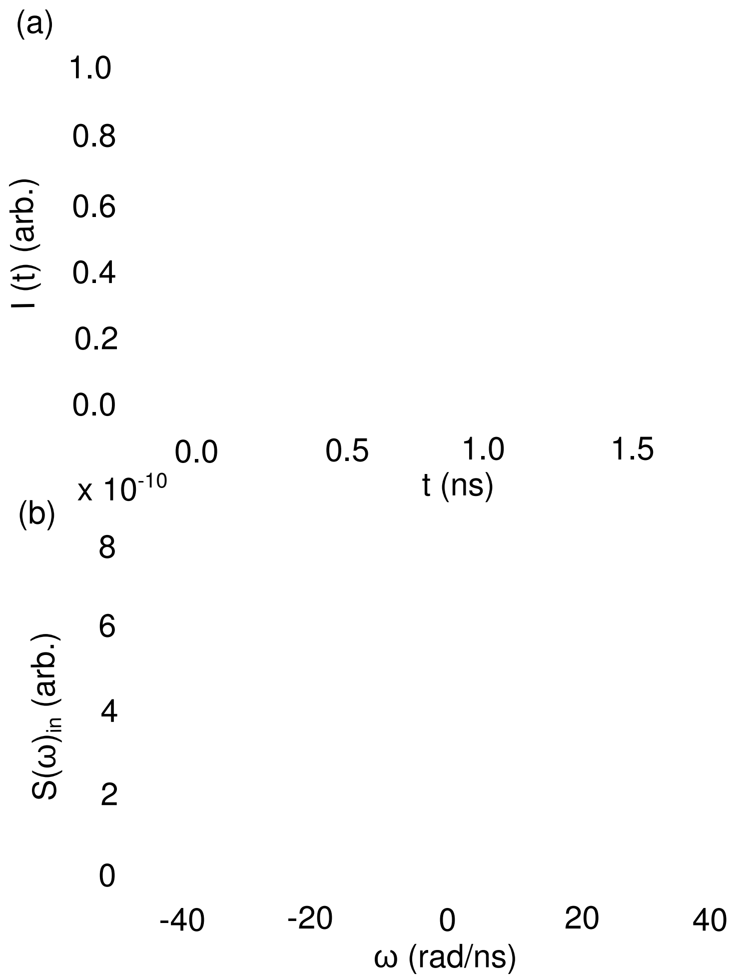

Figure 6.15: **Four pulse input** Four pulse input intensity versus time and spectrum versus angular frequency in rad/ns. The four pulses are separated by 500 ps and are each the same width of 0.000318 ns. The input spectrum has fast oscillations underneath the envelope shown in (b) and the same envelope depicted in Fig. 6.13 (c).

the photons is mixed. Subsequently, I would not expect there to interference between pulses with random linear polarizations. As a next step, I could include the polarization dependence in the model and observe how this changes the interference, however, this would only loosely model the physical system.

In this section, I describe the experimental realization of part of a proposed time-frequency QKD system. In the next section, I turn to some heuristic arguments concerning the security for the proposed system. A full security analysis is beyond the scope of this thesis, but it is none-the-less, interesting to make some arguments about proposed security for the system.

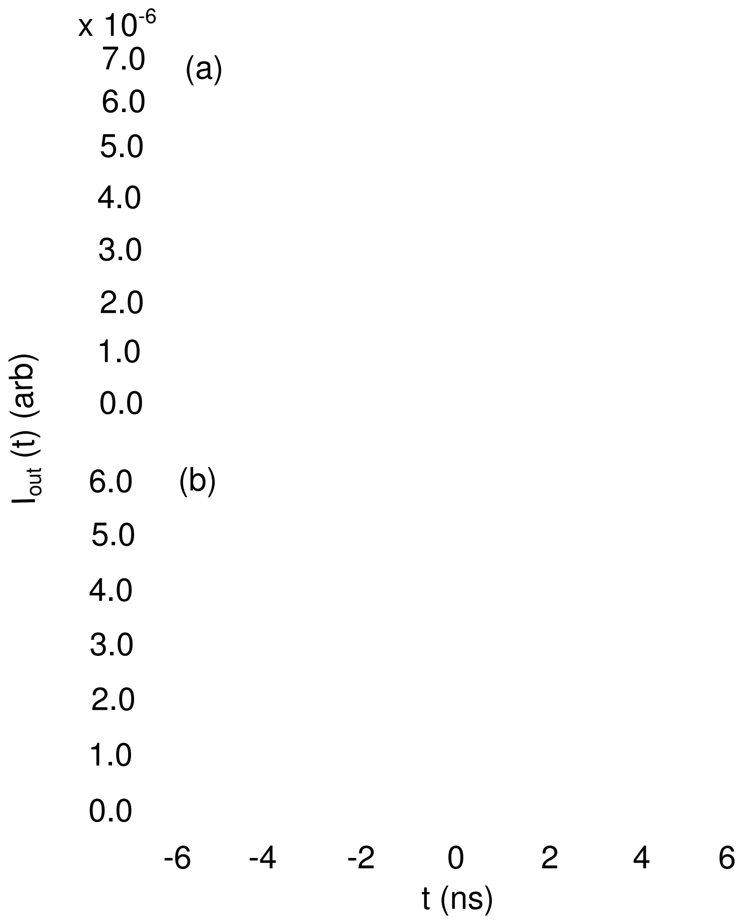

Figure 6.16: **Four pulse output** Four pulse output intensity versus time in ns. (a) shows the case where the flat-topped box function is used as the filter function while (b) is for a super-Gaussian case for the filter.

6.8.4 Security for time-bin QKD using FBGs

I introduced the notion of security against general intercept-and-resend attacks and quantum non-demolition (QND) attacks in Sec. 2.4. The security against QND attacks particularly relies on the fact that, when Alice and Bob send and receive their bits in complimentary bases (+/- GVD), and Eve measures in a different basis (timing), she will cause a change in the spectrum and thus the correlations of the output. Here, I outline a typical scenario in which Eve is detected by Alice and Bob and comment on the situations where this case is valid.

Alice creates and sends a photon through a +FBG and over the channel to Bob in Fig. 6.2. Bob randomly chooses to send the photon through the -FBG before detecting its time of arrival with his single-photon counting detector. When Alice sends the photon with an equal probability of being in any one of four time bins, and it passes through a combination of a +/- FBG, the output is identical to the input. If Bob has an infinitely sharp resolution on his detectors, every time he collects a photon, it is a coincident count with Alice's and he records which bin it arrives in. When Alice and Bob compare a segment of the key, they expect to see an equal probability for the photon to arrive in any of the four time bins.

If Eve chooses to perform a QND measurement on the photon's timing information, she must choose how many bins she will localize the photon to. For example, Eve could localize the photon to a single bin and if she guesses correctly, she gains all of the information in that particular frame (for the 4-time bin case, this is 2 bits). However, this measurement causes a larger change in the spectrum, which leads to a high probability of detection. On the other hand, Eve could localize the photon to two or three out of the four time bins, and gain less information, but have a smaller probability of being detected. Suppose Eve decides to made a QND timing measurement on the photon and localize it to half of the frame (two out of the four time bins). Further suppose that these two time bins are consecutive – and they are either the first two or second two in the frame. When Eve localizes the photon to two time bins, she limits the spectrum sent by Alice. The two pulses are still stretched and recompressed in the same manner. However, instead of measuring an even distribution of four pulses in four time bins, Bob measures a pulse in either of the two time bins. When Alice and Bob compare coincidence counts and the values of the bins they are in, they are able to detect that their coincident counts are all in the same two bins (the first two or the second two) as opposed to four. The change allows them to detect the eavesdropper.

6.8.5 Non-transform-limited pulses in an FBG

The simulations in the previous sections assume that the down-converted single photon pulses are transform-limited. The transform limit as described in Eq. 6.18 is the minimum time-bandwidth product for a Gaussian pulse. Physically, the minimum time-bandwidth product represents the minimum value of the time-energy or time-frequency uncertainty principle for a specific pulse shape. Transform limited pulses are unique because they have a constant phase across their whole spectrum. Pulses that are chirped, as previously discussed have a phase that varies quadratically with frequency and do not satisfy the minimum time-bandwidth product.

To date, the literature on using an SPDC source to create a time-frequency QKD system, assumes a non-transform limited pulse, although in general, this is not a good physical description of the system. The pump laser is typically mode-locked and transform-limited, but the SPDC daughter photons are not. In the model thus far I consider only the bandwidth to be set and I calculate the daughter photon's temporal width using the transform limit equation. Physically, however, the daughter photons are born with the same temporal width as the pump pulse ($\sim$ 5 ps) and with the bandwidth dictated by phase matching (for the case I consider here, $\sim$ 23 nm). The time-bandwidth product for SPDC is much larger than that time-bandwidth product for a transform limited pulse and therefore the phase is not a constant.

The phase of the single photon wavepacket is dictated by the type of interaction, in this case a random one. The daughter photons arise from random fluctuations in the vacuum and so have randomized phases which are similar to a thermal source. To model this, I create a color-correlated random phase that varies between -2π and 2π with a Gaussian envelope function imprinted by the pump pulse. I follow the process outlined in Ref. [81] to create a color-correlated random amplitude and phase.

Briefly, I calculate the color-correlated random phase as a function of time by determining

$$\phi_{rand}(t) = \phi_{rand}(t-1) * EE - (4D(1-EE^2)\log[\mathrm{rand}(0,1)])^{1/2} \mathrm{e}^{2\pi \mathrm{i} \mathrm{rand}(0,1)}, \quad (6.21)$$

where $EE = \lambda * \Delta t$, where Δt is the time step, λ is the inverse correlation time, D is a normalization constant, and rand$(0,1)$ is a pseudo-random number between 0 and 1. In the experiment the pulse envelope is $\sim$ 5 ps FWHM and I create time steps with Δt = 5 fs. The transform-limited temporal pulse for the down-converted light is 53 fs so that there are at least 10 sampling points for the transform-limited pulse. I use $\lambda \sim 20,000\ s^{-1}$ which gives the correct bandwidth of $\sim$ 20 nm = $\sim$ 50,000 rad/ns. I use a super-Gaussian function for the filter of the FBG as I did before with the function $e^{(-\omega/13000)^{12}}$ to give the correct 10 nm bandwidth for the filter. I choose $D = 0.01$.

I outline the process as follows. First, I create a vector of time steps with each time step being 5 fs from -5.24 ns to 5.24 ns to cover the entire range of the stretched pulse. I then create the random electric field vector (both amplitude and phase) for each of these time steps drawing from the random numbers governed by Eq. 6.21.I multiply these random numbers by the pulse field envelope, which is a Gaussian function that corresponds to a Gaussian intensity of width 5 ps at FWHM. This envelope and the random field make up the total input field. I Fourier transform this field to obtain the spectrum and multiply it by the FBG function. I then inverse Fourier transform it back into the time domain to obtain the stretched or stretched and recompressed pulse. Because the field is random, I repeat this process many times and average over them to get the physically realizable system. Typically I take 100,000 to 200,000 averages over the entire set of points.

I plot the input pulse, output spectrum and output pulse after the GVD material in Fig. 6.17. Figure 6.17 (a) is the 5-ps-long input pulse while Fig. 6.17 (b) is the output spectrum after the FBG. The FBG filters part of the spectrum to $\sim$ 10 nm. I then Fourier transform this into the time domain and plot it in Fig. 6.17 (c). Clearly the pulse is stretched by the correct amount ($\sim$ 8.3 ns). When I use a plus and minus FBG I regain the original 5 ps pulse.

From these simulations I conclude that the non-transform limited pulse behaves the same way in the GVD material as the transform limited pulse. It is both stretched and subsequently recompressed to the input pulse time, although the detectors I use do not have enough resolution to measure this. Although the non-transform limited pulse behaves in the same way that the transform limited one does, there are still important questions as to how many temporal modes are able to be secured in this system and if the nature of non-transform limited pulses changes the set of measurements one can make to characterize the whole system. I have shown that the single pulse spectrum does not change, but it is still interesting to ask what factors do change when using non-transform limited pulses, and how this affects the security of a high-dimensional time-frequency quantum key distribution system. In the next section I discuss some

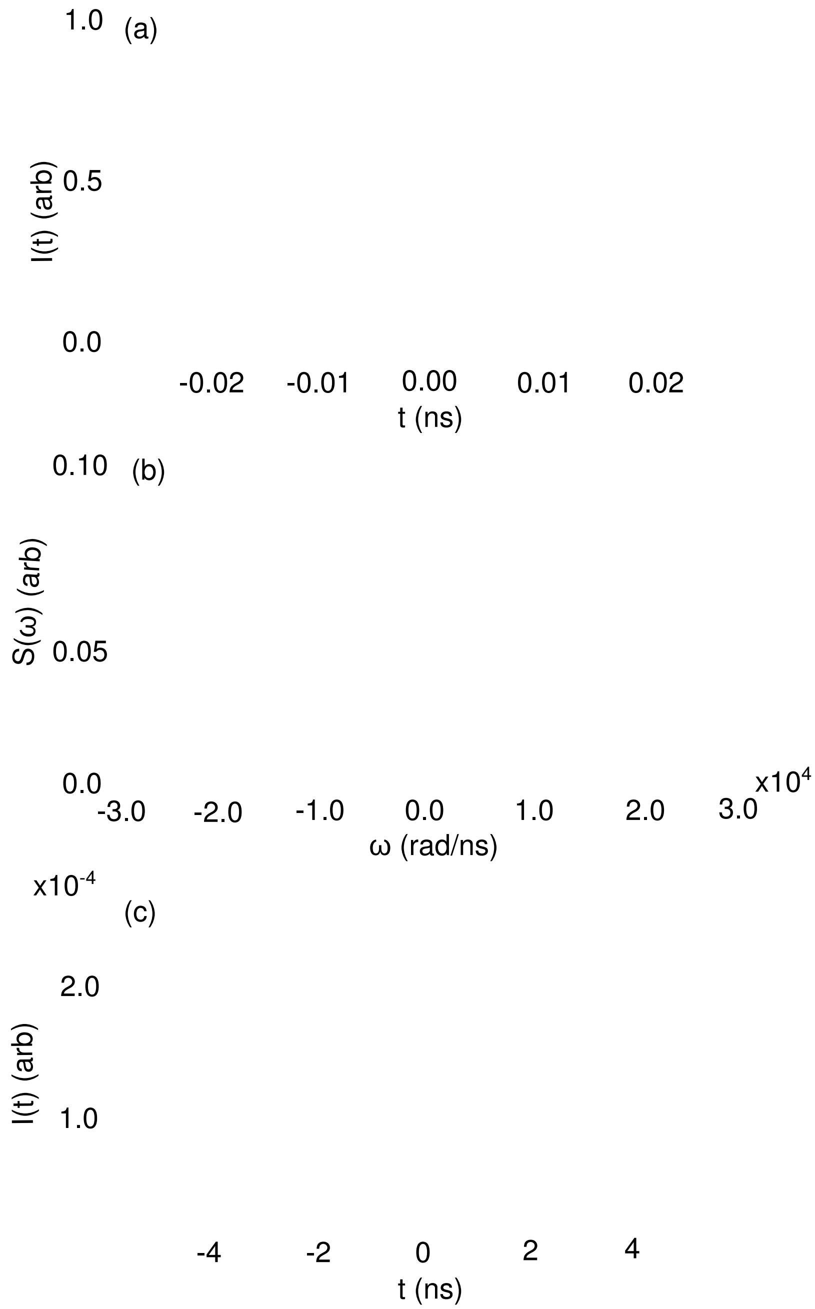

Figure 6.17: **Non-transform limited pulse with GVD** (a) is the original pulse intensity versus time with width of 5 ps. (b) is the output spectrum versus angular frequency after the FBG with a super-Gaussian filter and (c) is the output pulse in the time domain after the FBG.

possible experiments that could answer some of these questions.

6.8.6 High-dimensional Bell states and security in time-frequency QKD using non-transform limited pulses

There are several interesting questions to ask regarding the time-bin/time-frequency system presented in this chapter in terms of its entanglement properties, the security of time bins, and how to create a Bell state measurement using such a system. In this section I discuss how to measure the entanglement of a time-bin entangled state and the potential issues with using non-transform limited pulses.

The key questions are: are the time-bin states I have presented here really entangled with or without using non-transform limited pulses and what measurement could be done to show their entanglement? I assume time-bin entanglement in the QKD system discussed in this thesis due to the fact that a pair of photons is born at some time with no information about which pump pulse in a given frame produced it. However, to measure entanglement in such a system, I need to perform a Bell-state measurement (BSM) on the system to verify the entanglement.

In the typical Bell state measurement (BSM) using the polarization degree-of-freedom, coincidence count rates are measured on a polarization entangled pair after each photon is sent through a polarizer that rotates its polarization by a certain amount. More specifically, for each photon, the states $|H\rangle, |V\rangle, |H\rangle + |V\rangle, |H\rangle - |V\rangle$ are created with the polarizers and then coincidence rates are measured for different orientations of each of the two polarizers. To achieve this measurement in the time-bin case, I first take the simple system of two time bins $|t_1\rangle$ and $|t_2\rangle$. The time-bin entangled state created from the SPDC process is $\propto (|t_1 t_1\rangle + |t_2 t_2\rangle)$. The time-bin entanglement is measured by placing two Franson interferometers (one in each arm) and measuring the interference. The visibility of the interference is a measure of how strongly entangled the system is. The Franson interferometer acts like the polarizer in the standard BSM, by "rotating" individual time states into superpositions of time states $|t_1\rangle \pm |t_2\rangle$. To perform a Bell state measurement on a higher dimensional time-bin system, where more than two time bins are entangled, requires stacking multiple Franson interferometers as proposed in [71].

The issue with using non-transform limited pulses is that the phase varies randomly across the pulse widths. This can reduce the visibility by making the photons distinguishable at the beam-splitter in the Franson interferometer. The two pulses are then only indistinguishable over a time that is roughly their coherence time, which for the SPDC interaction presented here, is ~ 50 fs. Therefore non-transform limited pulses will have reduced visibility in Franson-type measurements of entanglement. Despite the potentially reduced entanglement, there is still an open question of whether these non-transform limited photons can still be used to secure timing information and what subset of measurements on the whole time-frequency state space is needed to achieve this security.

6.9 Summary

In this chapter, I describe how to manipulate the temporal and spectral components of a single-photon wavepacket. I discuss several applications that use time-to-frequency mapping, achievable with group velocity dispersive materials, to accomplish single-photon measurements. I then describe one type of time-frequency QKD setup which uses a time basis and a frequency basis to check for eavesdropper. I discuss how to implement the frequency basis correlation measurement with a GVD material and discuss specific GVD possibilities. I show how fiber Bragg gratings can be made to have large GVD parameters which is ideal for stretching single photon wavepackets by large amounts. I detail the experiments I perform in which I stretch and recompress a single-photon pulse. This marks the first time, to my knowledge, twin photons produced by SPDC have been stretched by such a large amount and subsequently recompressed. I also show that the FBGs can be used in series to create even larger stretches to achieve better resolution for a single-photon spectrometer or span more time bins in a time-bin/frequency QKD setup. Finally, I simulate how different types of pulses sequences (single pulse, pulse train and four pulses) interact with an FBG and compare the output to the experimental data and find good agreement.

Chapter 7

Conclusions

7.1 Overview of Thesis

In this thesis, I discuss three major projects that contribute to the fields of quantum optics and quantum information. I demonstrate original research in creating and optimizing a biphoton source used in numerous applications, discover different emissions patterns produced by commonly used crystals in SPDC, and achieve a large stretch and recompression of a single photon wave packet for time-to-frequency measurements. Although these three projects span different areas, they all have the common theme of creating and manipulating entangled biphoton pairs. Among the many applications that utilize this is quantum key distribution. I show how these three projects can all be used to aid in creating a high-speed, high-dimensional QKD system that pushes the limited of current QKD technology.

I review my three projects, the original physics I produce, and their possible applications in Figs. 7.1, 7.2 and 7.3. I create an entangled biphoton source in both the polarization and time-energy basis using SPDC in a nonlinear crystal. I use two main metrics to characterize this source; the heralding efficiency and the joint count rate. Many applications require one or both of these metrics to be high and I show how to optimize them individually or simultaneously. I calculate the heralding efficiency for collinear and noncollinear geometries, degenerate and nondegenerate wavelengths, narrow and broad filtering conditions, and tight and loose focusing conditions. I find the optimal combination of these parameters to achieve high heralding efficiency and high joint count rates that are integral to quantum communication, quantum metrology, quantum lithography, quantum imaging, and quantum computing. A summary of this project is shown in Fig. 7.1.

Project: Biphoton source using SPDC Chs 3 & 4

Publication: Guilbert, H.E. and Gauthier, D.J., IEEE J. Sel. Top. Quant. Electron., ***21*** *(3):1-10, 2015.*

Summary

· Created a polarization and time-energy entangled biphoton source
· Achieved high heralding efficiency and joint count rates
· Tested heralding efficiency and joint count rates for different geometries, wavelengths and filtering conditions

New Physics

· Derived analytic expressions for heralding efficiency and joint count rates for different geometries, wavelengths & filtering for Type-I SPDC
· Demonstrated that both heralding efficiency and high joint count rates can be simultaneously achvieved
· Verified theoretical predictions with experimental data

Applications

· Quantum information, quantum computation, quantum imaging, quantum metrology, quantum lithography, fundamental quantum science

Figure 7.1: Summary of SPDC Biphoton Source

Using the SPDC source, I characterize the emission pattern produced by two commonly used nonlinear crystals, BBO and BiBO. I demonstrate experimentally that the emission patterns produced in BBO are circular, while those produced in BiBO in the NIR wavelength range are elliptical. I hypothesize that this is due to the fact that the daughter photons in BiBO experience an angle-independent refractive index at NIR wavelengths, while those in BBO experience an angle-independent refractive index. I derive approximate analytic expressions for the eccentricity produced by degenerate SPDC in BiBO by using a Taylor-expansion around the collinear case for small angles. I show that the eccentricity depends on how the refractive index changes with emission angles and find a wavelength that minimizes the eccentricity in BiBO. I further show, when using two crystals oriented orthogonally to each other to produce polarization entangled photon pairs, that the joint spectrum around the ring is similar everywhere. This means entanglement purity is only minimally reduced at locations where elliptical rings produced by two orthogonally oriented crystals do not overlap. This work has implications for spatial multiplexing around the down-conversion ring. Namely, applications attempting to maximize count rates by spatial multiplexing around the ring need indistinguishability at every collection point around the ring. I show that eccentricity in the emission patterns provides a method of distinguishing between the two crystals, but not to the extent that it compromises entanglement purity. A summary of this project is depicted in Fig. 7.2.

Project: Elliptical Rings in SPDC — **Ch. 5**

Publication: Guilbert et al. Submitted to JOSAB. ArXiv:1503.04676

Summary

· Imaged emission patterns produced from Type-I SPDC in BBO and BiBO
· Showed BiBO produced elliptical rings while BBO produced circular rings
· Derived approximate expressions for eccentricity and showed certain wavelegnths minimize it
· Showed how entaglement purity is affected at different locations around ring

New Physics

· Determined physical reason for elliptical rings due to angle-dependent refractive index experienced by daughter photons
· Experimentally demonstrated large eccentricity for BiBO and zero eccentricity for BBO
· Derived analytic approximation for eccentricity in BiBO and found wavelength where eccentricity is nearly zero
· Used theory from SPDC project to approximate entanglement purity around the ring

Applications

· Quantum communication and fundamental quantum science

Figure 7.2: Summary of Elliptical Rings from SPDC in BBO and BiBO

Finally, I demonstrate stretch and recompression of a single photon wave packet using a material with group velocity dispersion. Here, I create a time-bin entangled source with four time bins of 500 ps width and send four pulses localized in the four time bins through positive and negative GVD materials. I show that fiber Bragg gratings are excellent choices because of their large group velocity dispersion. I demonstrate that input down-converted pulses can be stretched up to 16 ns with two such FBGs. They can also be stretched 8.3 ns and recompressed to at least the resolution of the detectors. I demonstrate how the FBGs create a method of measuring correlations in the frequency using only one single-photon counting detectors. By using a single-photon detector and time-tagging the arrival of the single photon, I create a spectrometer that maps frequency spectral components to the time basis. This has important implications for single-photon spectrometers. Additionally, I develop theory for predicting the expected spectrum using four time-bin entangled pulses and how the GVD material maps this spectrum to the timing basis. I also develop heuristic arguments for the security of a QKD system using the time-frequency basis and show that, if the GVD material stretches the incoming time bins over the entire frame, then POVM or QND eavesdropping attacks are known to Alice and Bob.

7.2 Future Directions

7.2.1 High(er) Heralding Efficiency

In this thesis, I present methods and data showing how high heralding efficiency is achievable in SPDC sources using collinear nondegenerate geometries. Theoretically, heralding efficiencies higher than 97% are achievable under the right conditions assuming perfect detectors and no channel loss. First, I expect that changing the current setup so that I separately couple the 609 nm and 850 nm light into two different achro-

Project: Time-to-frequnecy measuriements using GVD Ch 6

Summary

- Stretched and recompressed single photon wavepacket with original width of 5 ps to final width of 8 ns using large GVD
- Showed theoretical agreement of total stretch using spectral-temporal fourier transform methods
- Developed extension of single pulse theory to 4 pulses to show how spectral infromation is mapped to temporal information

New Physics

- Demonstrated largest stretch and recompression of single photon wavepacket
- Developed arguements for being able to secure time-bin information using frequency as the mutually unbiased basis for four-pulse case
- Modelled QND eavesdropper attack and showed eavesdropper can be detected
- Extended theory to non-transform limited pulses

Applications

- Single photon spectrometers, quantum key distribution

Figure 7.3: Summary of time-to-frequency mapping experiments using GVD

mats would make a slight increase in the heralding efficiency. I would also focus tighter to increase signals counts while keeping the 3:1:1 ratio of pump:signal:idler waist.

Additionally, small technical changes could be made to increase the overall system efficiency, namely getting AR-coated fibers for the exact nondegenerate wavelengths as well as custom couplers so that $> 95\%$ of the light that is coupled into the asphere, make it to the detector. This issue would also be aided by having separate couplers. Although separate asphere couplers to couple the nondegenerate light would increase the heralding efficiency, it would be necessary to use a very high quality dichroic or filter in the beam to prevent leaking of one wavelength into the other port and vice versa.

The greatest way to change the experimental heralding efficiency of the system is to use superconducting nanowire detectors that are optimized for both wavelengths

(609 nm and 850 nm). These detectors claim close to 100 % quantum efficiency and have very low dark counts. With these detectors, the collinear SPDC system would be able to achieve heralding efficiencies of nearly 90%. This large increase in system efficiency would allow for significantly higher key rates than the one I can currently run.

Creating such a high heralded source is useful in any number of applications and one natural broader extension of this work is to use this heralded source to create a full high-speed QKD system or a high quality heralded single photon source for any number of applications. This thesis provides insight into the fundamental principle behind making such a high quality source for use in numerous applications.

7.2.2 Securing Time Bins

In the experiment for securing time bins with fiber Bragg gratings, several aspects could be improved upon. However, in working with these specific gratings, it seems that they are highly sensitive to movement and do not give as much stretch as would be needed. Maximally, the gratings should have produced 8.3 ns of stretch each, 16.6 ns combined which would theoretically be able to secure up to 5 bits per photon given 500 ps bins. One way of improving this number is by decreasing the bin size with faster detectors. Another possibility is making fiber Bragg gratings with more stretch.

Further improvements in the FBG reflectivity would also aid in data speed. Although this method was originally meant to be used for security checking in the original scheme and would thus not affect the actual key rate, in the implementation that I described, the key rate is dependent on the rate of photons through the FBGs, which is quite poor. Higher reflectivity would produce a higher key rate. Additionally, other losses in the setup, for example, the PBSes and coupling components could be reduced by using a custom low-insertion loss fiber circulator.

A larger picture goal is to create and implement a full time-bin or time-frequency QKD setup using FBGs. Ideally, if the FBGs could be made at the degenerate wavelengths, the system I propose in Fig. 6.1 is optimal due to the fact that the key rate without security checking does not rely on the loss in the FBGs. Doing a full security analysis of this system including the total secure rate, bounding the information leaked to an eavesdropper and analyzing the quantum bit error rate would be very useful. I have built and characterized many of the components for integral parts of this system, but it was beyond the scope of this work to do the full QKD system and security analysis. This would however be a nature extension of the work I present here.

Appendix A

Overview of Experimental Systems

The purpose of this appendix is to provide more detail for the major experiments that I present in the main chapters of this thesis (Chs. 3-6). I will discuss several of the experimental setups in more detail and additionally review a few potential pitfalls of each experiment and how to trouble-shoot them. Although some of the material here is repeated in the main chapters, my aim is to write a comprehensive overview as a guide so that the reader can not only understand each of my experiments, but also be able to replicate them easily.

A.1 The SPDC experiments

In this section, I discuss my experimental work on SPDC covered in Chs. 3 and 4 in this thesis. The goal of the SPDC setup is to obtain the highest heralding efficiency and simultaneously the highest joint count rate using various geometries, wavelengths, and filtering conditions. I begin my discussion by reviewing the details of the pump laser source and optics used to create the ideal Gaussian beam with correct focused waist size. I then discuss details about the collection of the signal and idler photons, including the specific collection optics and procedures on how to optimize the heralding efficiency. In the final part of the section, I examine some potential issues working with this system, namely the issue of thermal lensing that arises from using a high-powered pump laser.

A.1.1 The pump source and focus into the crystal

The pump laser

I use a Coherent Paladin Compact 4 W, pulsed, yttrium vanadate (Nd:YVO_4) laser at 355-nm-wavelength that produces 5-ps-pulses at a rate of 120 MHz. Yttrium vanadate (often called "vanadate") is similar to more the commonly used Nd: YAG crystals and emits at the same wavelength – 1064 nm. One advantage of using vanadate over YAG is that it has a higher pump absorption and gain, which results in it being better suited for passive mode locking and producing high repetition rates. Mode locking refers to the mechanism used inside the laser resonator to create short (in this case, on the order of ps) pulses that are emitted in a pulse train. The mechanism used in this laser is passive mode locking, where the back reflector in the cavity is replaced with a saturable Bragg reflector. Under high powers, this reflector acts as a high-reflectivity ($99-99.5\%$) mirror, while at lower powers, it reflects less (95%).

Ultraviolet (UV) light is produced via two stages from the Nd:YVO_4 laser which outputs 1064 nm. Light from the Nd:YVO_4 crystal laser is frequency doubled to 532 nm using second harmonic generation. The infrared light at 1064 nm and the green light at 532 nm are then mixed via sum-frequency generation to produce 355 nm.

The average power (P_{avg}) of 4W translates to a peak power P_{peak} of ~ 6.7 kW using

$$P_{\mathrm{peak}} = P_{\mathrm{avg}} * t_{\mathrm{rep}}/t_{\mathrm{peak}}, \tag{A.1}$$

where $t_{\mathrm{rep}} = 1/120$ MHz $= 8.33$ ns is the time between pulses and $t_{\mathrm{peak}} = 5$ ps is the width of the pulse. This is a relatively high peak power in the pulse, and caution should be taken when putting anything in the beam path.

The laser beam is a lowest order Gaussian $TEM_{0,0}$ mode and is collimated to a 1-

mm-diameter $1/e^2$ intensity beam with a very small divergence angle of $< 500\mu$m. I measure the beam diameter approximately 1-2 meters away from the source to be $\sim$ 1.1-1.15 mm and verify this small divergence. I use a beam profiler (Thorlabs BP209-VIS) or a wavefront sensor (Thorlabs WFS150-5C) to measure beam sizes and wavefronts.

Alignment and setup

The first, and possibly one of the most important parts of the original pump setup, is implementing a power splitter to control how much power is sent through the setup. The power splitter consists of a half waveplate (HWP) (Thorlabs WPH05M-355), polarizing beamsplitter (PBS) (Precision Photonics/ATF PBS1005-TY) and beam dump (Thorlabs BT600) shown in Fig. A.1. I place the power splitter $\sim$ 10 cm after the laser opening. Because the output polarization of the laser is linear, the HWP rotates its polarization while the subsequent PBS separates each component of the polarization into two directions: the reflected part enters the beam dump, while the transmitted part proceeds through the setup. The power splitter allows me to control how much power flows through the rest of the setup.

After the power splitter, I use a series of mirrors – six in total – although minimally only two are necessary to steer the pump beam into the crystal. The reason I use six is for ease in switching between different focusing cases, which I discuss next. The first two mirrors (M1 and M2) after the laser are on high quality mounts (Thorlabs POLARIS-K1) and are stationary (see Fig. A.1). The next mirror (M3) is on a flipper mount, while the fourth and fifth mirrors (M4, M5) are on a translation stage (Thorlabs TB0606) and the sixth is again on a flipper mount (M6). When the flipper mounts are down, meaning only M1 and M2 are used, and all lenses are removed from the setup, I obtain a collimated beam with a short path to the crystal. Because I use various sets of lenses in order to focus the pump beam into the crystal, and because the heralding efficiency is very sensitive to pump and signal/idler overlap, I find that having extra mirrors is useful for alignment and control. I implement this control over the pump beam instead of the crystal because moving the crystal results in having to move and realign all the collection optics "down-stream." Instead, I design a system where I control all degrees of freedom of the pump and the collection optics separately and leave the crystal stationary.

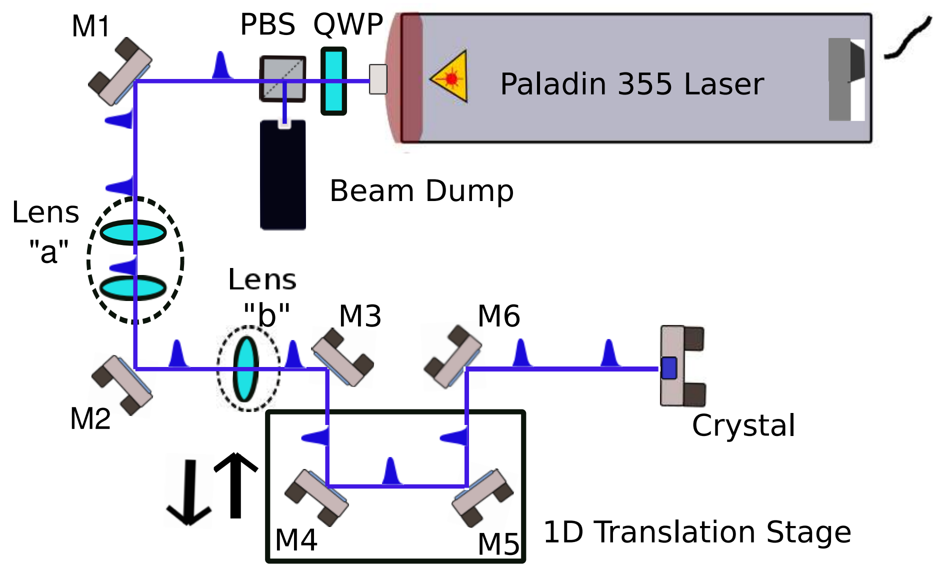

Figure A.1: **Pump beam and optics setup** 355 nm pulses at 120 MHz are sent through a power splitter, and subsequent beam steering and shaping optics. Six mirrors (M1-M6) steer the beam and control the adjustment of the focus into the crystal. (a) shows the set of two mirrors of the detuned telescope that act as a long focal length lens. These are placed further away from the crystal while the single lens with short focal length (750 mm) is placed closer to the crystal, but still farther away than the translation stage. I move this translation stage back and forth and use a beam profiler to ensure that the location of the waist is in the crystal.

In this thesis, I discuss the important role that focusing of both the pump and daughter photons play in obtaining high rates and high heralding efficiency. In the experiments I present, I use two different focusing conditions as well as additionally measuring heralding efficiency for a collimated pump, signal and idler beam. Collimated beams do not result in high count rates or heralding efficiency as I have previously discussed. The two focusing conditions I examine in this thesis I call the "tight" and "loose" focusing conditions; they have Gaussian field waist parameters of $W_p = 150$ μm, $W_s = W_i = 50$ μm and $W_p = 250$ μm, $W_s = W_i = 100$ μm respectively.

To create the tight focusing conditions, I place a 750-mm focal length lens in the collimated pump beam. Using Gaussian beam optics in the paraxial approximation and assuming a well collimated beam entering the crystal, I obtain [72]

$$W_0 \approx \frac{2}{\pi}\frac{\lambda f}{D}, \tag{A.2}$$

where W_0 is the focused waist radius, f is the focal length of the lens, D is the diameter of the incoming beam to the lens, and λ is the wavelength of the light in air (close to vacuum wavelength). Using Eq. A.2, and the fact that the pump laser outputs a 1 mm diameter beam, I calculate that a 750 mm lens focuses the 355 nm light to approximately 170 μm. I then verify this prediction with a measurement using the beam profiler.

To create the loose focusing conditions, I need a much longer focal length lens. Using Eq. A.2, the focal length lens needed to create a 250-300 μm waist is 1.1-1.5 m. These long-focal-length lens are not standard off-the-shelf parts because most lenses are made with focal lengths of 1 m or shorter. Effectively realizing a long focal length lens is easily implemented by moving one lens from a two-lens telescope system either closer or further away from the other lens. For this setup, I detune a 1-to-1 magnification telescope made up of two 100 mm focal length lenses. I separate the lenses by slightly more than the 200 mm separation used for the 1-to-1 telescope. I calculate the exact amount to separate them by solving the ABCD matrices for two thin lenses using Gaussian beam optics found in Ref. [72] (Ch. 3). I find that, in the thin-lens approximation, I obtain a 274 μm waist located 1.2 m away from the second telescope lens when I place the lenses 208 mm apart. I obtain a 323 μm waist 1.2 meters away when with a 205.8 mm focal length lens. I aim for somewhere between these two waist sizes and use a beam profiler to measure the exact waist size.

A.1.2 Collection Optics

On the other side of the crystal are the optics that collect the emitted down-converted daughter photons. Here, the lenses play the most important role in achieving high heralding efficiency. I have found, through trial and error, that single element, non-achromatically corrected lenses negatively affect the heralding efficiency. The first set of lenses I use are aspheres from Thorlabs (Thorlabs C280TMD-B). I use the 18.4 mm focal length lenses to create a beam waist of $\sim$ 100 μm in the crystal. I find, with these lenses, that although I achieve fairly high heralding efficiency, it falls below the theoretical value I expect for the heralding efficiency. I attribute this to the fact that the degenerate spectrum is quite broad and in an aspheric lens, the different wavelengths get focused to different spot sizes. To effectively collect more of the broadband spectrum, I switch from using aspheres to using achromatically-corrected lenses. These lenses are actually multi-element lenses that are designed to correct for achromatic aberrations caused by different wavelengths of light focusing at different locations, as would be the case for a single-element lens. Using these lenses increases the heralding efficiency, because over the bandwidth of the down-converted light, more of that bandwidth is collected into the fiber by the lens.

In this setup the lenses are focusing an expanding beam from the crystal down onto a single mode fiber. Another method of collection, is to place collimating aspheres closer to the crystal and collimating the beam before sending it down the table and using a second lens to focus into the fiber. This is the method that Paul Kwiat's group at University of Illinois uses for collection into single-mode fibers.

I compare the one-lens versus two-lens system to ascertain whether, given the same error in lens adjustment, one setup is more sensitive to this error than the other, resulting in a larger error and lower heralding efficiency. My system uses one lens, while Kwiat's system uses two. I find that, with the same error in lens position, the one versus two lens system does not affect the heralding efficiency significantly. Here I estimate the heralding efficiency by an overlap mode-matching integral for the three beams. I introduce imperfect focusing in the model by adjusting both waist size at the fiber as well as radius of curvature of the beam. I calculate the radius of curvature and beam waist using Gaussian beam optics, by calculating the shift in the beam focus for a system with a slightly different focal length using ray optics.

I outline the problem as follows: a single-lens system and a double-lens system shown in Fig. A.2 both accomplish the same goal of focusing rays from a point on one side of the lens or lenses to point on the other side. If the lens or lens-system has a slightly different focal length ($\Delta f \ll f$), then there is a shift in the position of the focus. Given the same focal length error for each system, I calculate which system has the greater shift in focal position, as this causes the different spot sizes and radius of curvature at the collection optics.

For any lens system in the ray optics picture,

$$\frac{1}{f} = \frac{1}{d_1} + \frac{1}{d_2} \tag{A.3}$$

where f is the focal length of the lens and $d_{1,2}$ are the distances on either side of the lens given in Fig. A.2. For a single lens system, if there is a slight change in the focal

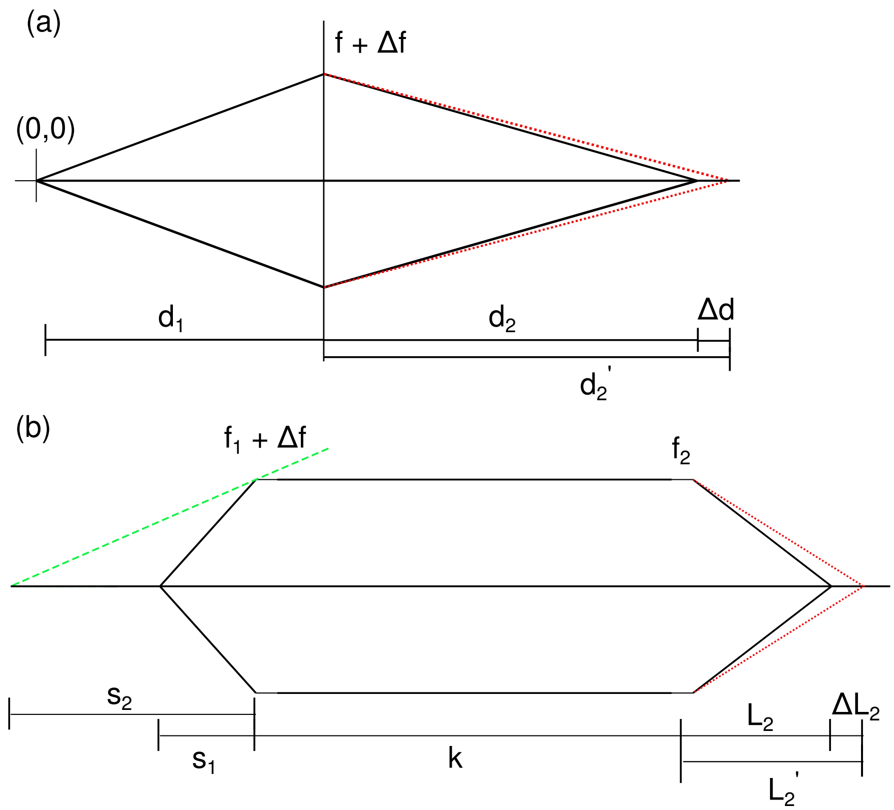

Figure A.2: **Single and Double Lens Setup** In (a) a single lens focuses rays a distance d_1 away from a lens with focal length f to a point a distance d_2 away from the lens on the other side. Introducing a focal length error $(f + \Delta f)$ causes the rays to focus at a different point, d_2' instead of d_2. In (b) two lenses focus a point a distance s_1 away from a lens with focal length f_1 by first collimating with the f_1 lens and then focusing with an additional lens with focal length f_2. The rays focus at a distance L_2 on the right side of lens f_2. Again, introducing a focal length error in the first lens $(f_1 + \Delta f)$ causes the focal point to change to a diastase L_2'.

length, Eq. A.3 becomes

$$\frac{1}{f + \Delta f} = \frac{1}{d_1} + \frac{1}{d_2'} \tag{A.4}$$

where d_2' is the new distance to the focal point. If $\Delta f << f$, I can use a Taylor expansion to simplify Eq. A.4 to find an expression for $\Delta d = d_2' - d_2$. I find

$$\frac{1}{d_1} + \frac{1}{d_2'} \approx \frac{1}{f}\left(1 - \frac{\Delta f}{f}\right), \tag{A.5}$$

which simplifies to

$$\frac{1}{d_2'} - \frac{1}{d_2} = \frac{d_2 - d_2'}{d_2' d_2} \approx -\frac{\Delta f}{f^2}. \tag{A.6}$$

Using $d_2 \approx d_2'$ I find,

$$d_2 - d_2' \approx \frac{d_2^2}{f}\frac{\Delta f}{f}. \tag{A.7}$$

The magnification (M) of the system is $M = -d_2/d_1$ so that

$$\frac{d_2 - d_2'}{d_2} \approx \frac{d_1 M}{f}\frac{\Delta f}{f} \approx -M\frac{\Delta f}{f}, \tag{A.8}$$

because $d_1 \approx f$.

For the double-lens case shown in Fig. A.2 (b), two lenses separated by a distance k focus a diverging beam to a spot at a total distance of $s_1 + k + L_2$ away from the origin using two lenses of focal lengths f_1 and f_2. For this analysis I will constrain the system to having a change in focal length for only the first lens $(f_1 + \Delta f_1)$, although further analysis suggests that the dispersion or change in the second lens does not contribute to the overall shift as much as the first lens for systems with large magnification $(M > 10)$. Let s_2 be the location of the virtual image from the first lens with focal length error $(f_1 + \Delta f_1)$ and let $s_1 = f_1$. Then

$$\frac{1}{s_1} + \frac{1}{s_2} = \frac{1}{f_1} + \frac{1}{s_2} = \frac{1}{f + \Delta f}. \tag{A.9}$$

Again, using the first-order term of the Taylor expression, Eq. A.9 simplifies to,

$$\frac{1}{s_2} \approx -\frac{\Delta f_1}{f_1^2}. \tag{A.10}$$

I then consider a source at distance $k - s_2$ from the second lens that focuses to a new point, L_2' instead of L_2. The goal is to find this new distance. I assume that $L_2 = f_2$ and solve

$$\frac{1}{-s_2+k}+\frac{1}{L_2'}=\frac{1}{f_2}=\frac{1}{L_2} \tag{A.11}$$

for an expression containing $\Delta L_2 = L_2' - L_2$ I find,

$$\frac{L_2-L_2'}{L_2L_2'}=\frac{1}{-s_2+k}=\frac{1}{s_2}\left(1+\frac{k}{s_2}\right)\approx\frac{\Delta f_1}{f_1}\left(1-\frac{k\Delta f_1}{f_1}\right). \tag{A.12}$$

If Δf_1 is small compared to the focal length (< 10 μm for a several cm focal length lens), then Eq. A.12 simplifies to

$$\frac{L_2-L_2'}{L_2}\approx L_2\frac{\Delta f_1}{f_1^2}\approx -M\frac{\Delta f_1}{f_1}. \tag{A.13}$$

This equation should be compared to Eq. A.8 and I find that they have the same dependence on magnification and focal length shift. This focal length shift is what determines the Gaussian waist size and radius of curvature at the collection fiber, and because both cases have the same dependence, neither one is better or worse at coupling the broadband spectrum.

The only parameter that I find that affects the overlap integral in a significant way is the overall magnification of the system. I find that the shift in the waist location resulting from an imperfectly aligned system simply depends on the longitudinal magnification. For systems with larger magnification (meaning the ratio between the fiber mode waist and the collection waist in the crystal) there is more shift for a given error, resulting in a lower overlap integral and lower heralding efficiency.

In Fig. A.3 I show the details of the system I use to collect light from both the noncollinear (Fig. A.3 (a)) and collinear (Fig. A.3 (b)) setups into single mode fiber. The only difference between the noncollinear degenerate versus nondegenerate case are use of the bandpass filters in the degenerate case.

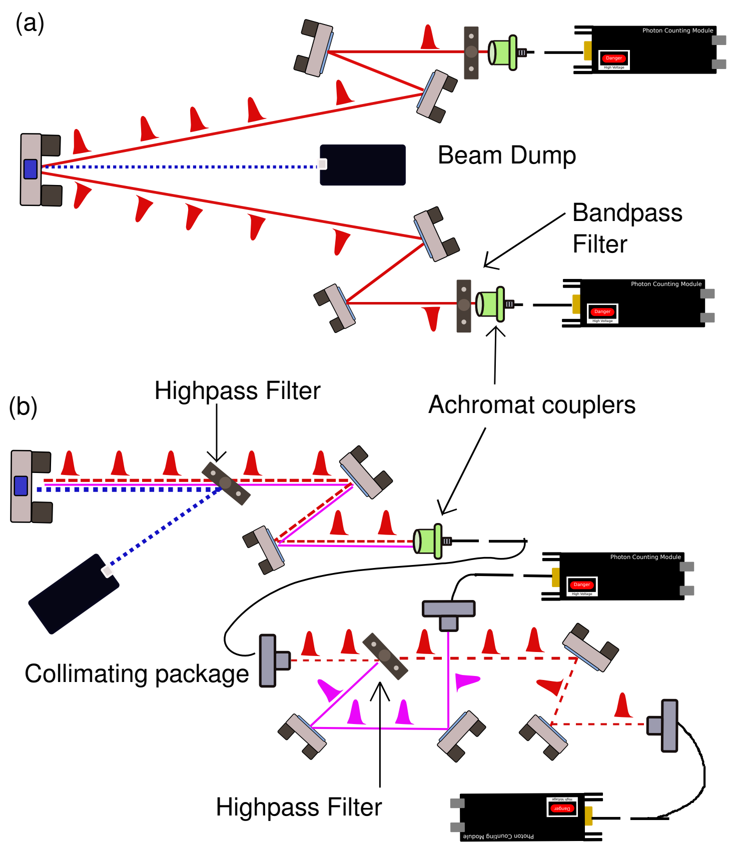

Figure A.3: **Collection setup for noncollinear and collinear SPDC** (a) is my experimental setup for noncollinear SPDC. SPDC light is emitted into a large range of angles, but I collect only from a specific angle ($\sim 3°$) and place the first set of mirrors to collect around this angle. The two mirrors on each side of the pump are for steering the beam into the achromat packages (Schaefter and Kirchhoff GmbH, 60FC-T-0-M201-02). The light is then coupled into custom coated single mode fibers (Oz Optics) and sent to detectors. The signal from the detectors is sent to the FPGA and analyzed by Labview (not shown). The collinear SPDC setup is shown in (b) where the main difference is that now all the photons (signal, idler and pump) are emitted along the same direction. The pump is split off from the signal and idler by a high-pass filter (Semrock Di02-R405-25x36). This filter has a high extinction factor of 10^6 meaning that this is the amount of pump power that leaks in the straight through path. The two wavelengths for nondegenerate light, 609.6 nm (pink, solid) and 850 nm (red, dashed) are coupled with separate steering mirrors into collimation packages (Thorlabs F220FC-780) and subsequently into multimode fibers (Oz Optics custom fiber).

A.1.3 Procedure for Maximizing HE

The method I use to obtain high singles counts as well as high heralding efficiency for noncollinear degenerate or nondegenerate setup is the following:

1. Place first collection mirror to reflect light exiting the crystal at a given exterior emission angle. For example, the desired exterior emission angle in these experiments is $\sim 3^{\circ}$ for degenerate wavelengths. I place the collection fiber roughly 1 m away from the crystal, meaning that I place the first mirror 60-65 cm away from the crystal along the pump direction and 3.5-4 cm on either side of the pump beam.

2. Place the second mirror and lens-to-fiber setup (can be an asphere lens and mount or custom achromat package) in the collection beam path.

3. Back propagate laser at chosen wavelength (either 710 nm, 850 nm or 633 nm and measure signal and idler spot with the beam profiler. Adjust the achromat packages to tune the lens-fiber distance to obtain the desired spot size.

4. Turn on the pump laser to low power and roughly overlap the pump and back-propagated laser spots "by eye" in the crystal. Use necessary laser goggles rated for the correct OD while performing this procedure. The pump scatters in the blue part of the spectrum and is easy to see. Repeat for both sides.

5. Remove the back-propagating laser and replace it with the collection fiber and single photon counting detector. At this stage, the if the beams are overlapped in the crystal fairly well, the location of the maximum singles counts should be well-matched into the fiber mode well and easy to locate. If the beams are not overlapping well, more steering of the beam is needed. Repeat for each side.

6. At this point, with high singles counts in both collection fibers, it is still very unlikely to have high heralding efficiency. This is due to the fact that high singles counts occur at many azimuthal locations around the downconversion ring, but heralding efficiency is increased only by collecting conjugate pairs (See Fig. A.4)

7. To obtain high heralding efficiency, walk/steer the mirrors on one side of the collection optics while maintaining high singles counts in that fiber. The goal is to move one set of collection optics to the spot where the conjugate photon of the other collected spot is located.

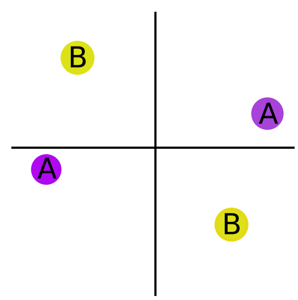

Figure A.4: **Conjugate pairs on the down-conversion ring** The ring shown represented the down-conversion ring, with the bright ring representing the location of highest singles counts. On either side of the bright ring the singles counts drop off so that the intensity cross-section is a Gaussian profile. The set of points "A" represent a conjugate pair of photons placed on either side of the pump while the set of points labeled "B" represents a separate set of conjugate points. It is possible, and often the case, to be collecting high singles counts in each detector, but to be at non-conjugate places on either side of the ring, for example, being at point "A" on the left hand side of the vertical line and point "B" on the right hand side.

This procedure is fairly general and only a few small changes are necessary for the collinear case. In the collinear case, instead of the beams being on opposite sides of the pump, they are collected along the same axis. Alignment in this case is much easier and is done by overlapping the back propagated signal/idler beam with the pump at two different spots on either side of the crystal. In my experience, obtaining heralding efficiency is much easier in the collinear case.

A.1.4 Electronics and Photon Counting

The bulk of the electronics setup for the SPDC system was built by a previous undergraduate in the group, Yu-Po Wong, and current graduate student, Nicholas Haynes, who programmed a field-programmable-gate array (FPGA) (Altera DE2-115) with Quartus II software to act as a coincidence counting device adapted from Ref. [82]. The TTL signals from the detectors are sent through resistors to ensure that 3.3-3.5 V pulses enter the FPGA. The FPGA is programmed to count hits in each channel per time for the singles rate. To count coincidences the FPGA "looks" for a hit in either channel then looks in the other channel in over a set time window (controllable by the user) for a hit. If it registers a hit in the other channel, the FPGA counts this as a coincidence count. The FPGA sends the coincidence and individual hits via an RS232 cable to a computer where a Labview program reads in the data and displays the singles count rate in each channel as well as the coincidence count rate and the heralding efficiency (expressed as the ratio of coincidences to the square root of the singles in A times singles in B). The final piece in the experiment that I have not described are the single photon counting detectors. I discuss these extensively in Appendix A2.

A.1.5 Potential Issues

The most puzzling and intriguing issue that occurs with this SPDC setup is that turning the pump power up to its maximum value does not produce the highest singles counts as one would expect. In theory, the singles rate is linearly dependent on the power so that as the power goes up, so should the singles counts. What I observe as I turn the power up is a roughly linear increase at first, then a tapering off where increasing power does not change the count rate, then finally, after a certain point, increasing power actually decreases the count rate. I trace this issue back to thermal lensing in the various optics in the pump beam path – namely the lenses and PBSes.

Thermal lensing occurs when part of a material heats up and its refractive index changes in response to this temperature change. The material may also change shape and surface curvature. Flat faces of optics undergoing thermal lensing will curve, effectively creating a lens. Although the lens in the pump beam is already focusing the beam, the distortion from the heat causes it and the PBSes to have a different effective focal length. This new focal length changes with power and creates a much larger pump beam in the crystal. Due to the fact that the singles counts vary linearly with the power but are inversely proportional to the pump waist squared, the dominant factor in determining singles counts is the pump waist, so as this increases, the count rates decrease from their expected value.

I verify this change in pump waist by using a pick-off wedge plate (CVI/Melles Griot LW-3 1037 UV), with high quality coating rated to withstand the power of the laser. The front face of the beam picks off $\sim$ 4% of the power and sends it to a beam profiler to record the waist size. I place the beam profiler in the focus of the pick-off beam, and observe, for low powers, the expected waist size. However, at high powers the waist size increases, which verifies the predictions that the increase in pump waist size causes a decrease in the singles counts. I measure the increase in the waist size at the crystal for the lowest power ($\sim$ 150-200 mW) and the highest power (3.4-4 W) and find that the $1/e^2$ waist diameter increases from 360 μm to 450-460 μm. This change is caused by a combination of 5 PBSes and one lens all changing with the power.

A.2 Collecting the spectrum

I collect the spectrum of the down-converted light using a monochromater and lock-in amplification technique because of very low-light-levels of SPDC emission. The experimental setup and details are shown in Fig. A.5. Maximally, I collect 2-5 MHz counts rates into the entire bandwidth of the SPDC corresponding to a power of 0.55-1.5 pW at 710 nm into the entire bandwidth. The issue with obtaining a spectrum is that when taking small parts of the spectrum (say a 0.5 nm piece) only a fraction of the total counts go into this band, and losses in the devices allow and even smaller fraction to propagate through the monochromater, grating, PMT, and get turned into a signal.

There are several methods I use to obtain a better signal from both the optics and the electronics portions of the experiment. On the optics side, I ensure that maximal light enters the monochromater by focusing the light with a short focal length lens into the opening slit. The micrometer-driven slits on the monochromater are on both the input and output ports and are adjustable so that more light can propagate through the system. The trade-off with opening these slits and allowing more light through is a decrease in resolution. Nominally, the monochromater has a resolution of 0.1 nm using the minimum width of the slits. The slits are adjustable from 4 μm to 3 mm. In my experiments I typically adjust the slits to between 10-30 μm which I find is the optimum trade-off between resolution and power throughput. I use a 25.4 mm focal length lens to focus the incoming beam to a spot size diameter of 10-15 μm. Additionally, as describe in Fig. A.5, I use a HWP to optimize the correct polarization for the gratings maximal reflectivity, run the detector as the highest gain, and run the TIA at the highest gain.

Both the grating in the monochromater as well as the PMT have wavelength de-

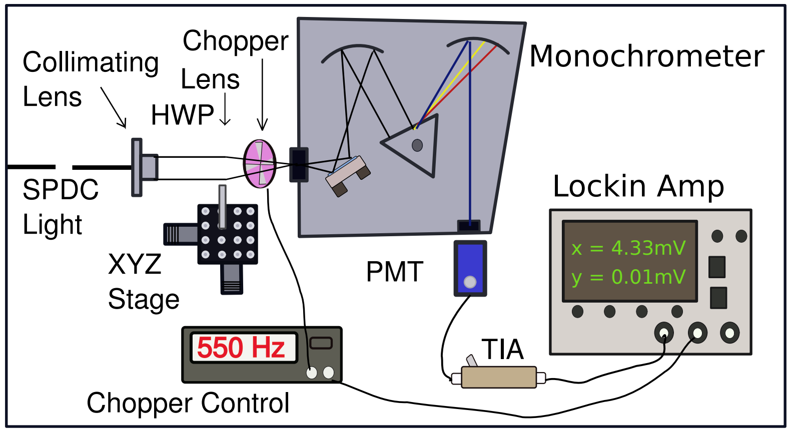

Figure A.5: **Setup for spectrum collection** 850 nm light from the SPDC source is coupled into a single mode fiber described in Fig. A.3. The light exits the fiber and travels through a collimating package (Thorlabs FC220-B) which roughly collimates the beam to ~ 1.8 mm. The beam passes through a halfwaveplate (HWP) to rotate its polarization. The gratings in the monochromater have different efficiency curves depending on the polarization of light sent in and the efficiency difference is quite high (up to 30% for 800 nm). In order to optimize the amount of throughput light, I control which polarization enters the monochromater and gratings. I place a chopper (Stanford Research Systems SR540) after the HWP to chop the light at a specific frequency that can be locked to with the lockin amplifier. I run the chopper at a frequency of 530-550 Hz. The final element before the monochromater is a lens ($f = 25.4$ mm, Thorlabs LA1951-B). The light then passes through a monochromater (Oriel Instruments Cornerstone 260 1/4 m) which is controlled with Labview software from the company or with an RS232 port. The light exits the monochromater and enters the PMT detector (Hammamatsu, H6780-20) which has an external gain voltage that controls the sensitivity. In the experiments I run the gain at the maximum (0.95-1 V) in order to get the highest sensitivity possible. The electrical current from the output of the PMT is then directed to a transimpedance amplifier (TIA). This device converts current to voltage while amplifying the signal by using a low noise operational-amplifier (op-amp). I set the TIA output to a gain of 10^8 which is the maximal setting for chopper frequencies of 550 Hz. The output of the TIA is then sent to the lock-in amplifier (SRS 850 Lockin Amplifier). The output of the chopper controller is also sent to the lock-in amplifier input locking frequency. I integrate over long time constants ($\tau_c = 10 - 30$s) so that the signal-to-noise ratio increases. Finally, I control both the monochromater and data collection from the lock-in amplifier with a Python script that I modify from online code. This python code controls the stepping of the wavelength of the monochromater and subsequent control and collection of data from the monochromater.

pendent loss and this must be corrected for in order to obtain an accurate spectrum. Additionally, different wavelengths of light will focus to different spot sizes before entering the monochromater creating another wavelength dependent loss. To correct for these, I calibrate the system with a tungsten halogen lamp (Ocean Optics HL-2000) for which the power spectral density of the light is very well known [1]. This is a broadband source covering a spectrum from $\sim$ 360 - 2400 nm. To calibrate the system, I send the light from the tungsten halogen lamp along a multi-mode fiber through the system. Because the amount of light produced by the tungsten halogen source can saturate the PMT, especially at high gain, I attenuate the light by intentionally butt-coupling poorly to a single mode fiber.

A.2.1 Calibration

In Fig. A.6, I show the expected spectrum of the tungsten halogen source with the resulting output data curve from the system. The difference between the two increases as the wavelength increases. I fit an interpolating function to the experimental data and use the ratio of the calibration curve to the interpolating function of the data as a scaling factor with which to multiply my experimental spectra by. The scaling factor is a function of wavelength so that when I multiply it by the data, I obtain wavelength-corrected data.

The error in the measurement is due primarily to statistical error. I originally measure this by taking ten full data sets, averaging them, and finding the standard deviation. However, due to the length of time needed to acquire each set of data (several hours), the system drifts significantly over the ten data sets. Instead, I choose several wavelengths and perform multiple measurements at these wavelengths and calculate the average and standard deviation to find the error.

[1] http://oceanoptics.com/product/hl-2000-family/

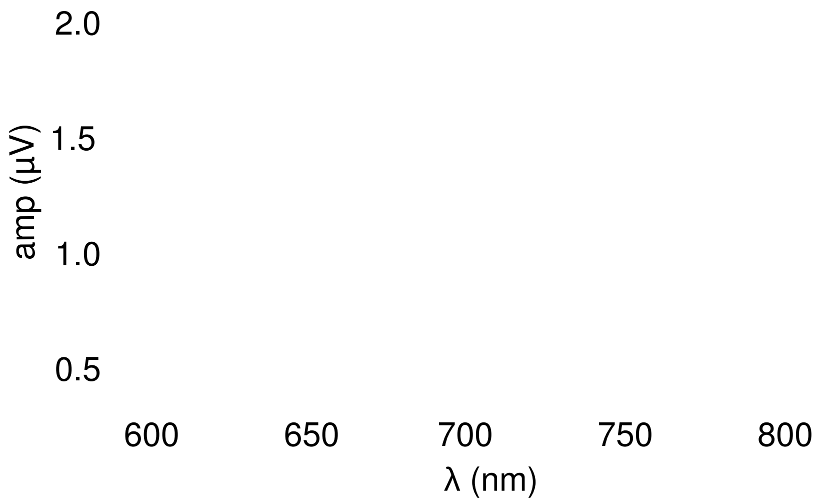

Figure A.6: **Tungsten halogen spectra** The purple, dashed curve is a calibration curve given by Ocean Optics for the tungsten halogen lamp (for details see http://oceanoptics.com/product/hl-2000-family/). The blue, solid curve is the output spectrum of the light passing through the monochromater and PMT setup. I fit a smooth interpolating function to the raw data values.

A.3 FBG setup

A.3.1 Pump pulse multiplexing

To create a time-bin-entangled source, I need to have a pump pulse in each time bin I entangle. Because this is a proof-of-principle experiment, I choose to entangle four time bins, which maximally encodes two bits of information. Each two time bins requires a multiplying stage which creates two pulses from a single pulse. In this thesis, I refer to these as repetition (rep) rate multipliers (RRMs). These multiplying stages split the incoming pump pulse, send it through two paths of different lengths, and recombine the paths resulting in two pulses with temporal spacing proportional to the path length difference of the two arms. The difference in length of these paths is dictated by the desired separation of each pulse, which I choose to be 500 ps. The reason for this choice is that each time bin needs to be much wider than the combined jitter of the detectors ($\sim$ 250 ps FWHM) for minimal error.

There are two different ways to obtain four pulses spaced 500 ps apart. In the first way, the first RRM has a delay of 500 ps in one arm, creating pulse 1 at $t = 0$ and pulse 2 at $t = 500$ ps. The second stage then has a delay arm of 1 ns so that the two pulses that travel the straight through path in this second stage are at $t = 0$ and $t = 500$ ps and the pulses that split into the longer arm are at $t = 1$ ns and $t = 1.5$ ns. The second way to achieve the same result is implementing the longer delay arm (1 ns) arm first and creating pulses 1 and 3 (at 0 and 1 ns) and then sending it through a 500 ps long-path RRM and creating pulses 2 and 4. I use the latter method because, because this gives me the option of creating two pulses separated by 1 ns (when I block the second RRM) which is helpful when troubleshooting how much resolution the single photon counting detectors have.

Figure A.7 shows the experimental setup for obtaining the four pulses spaced 500 ps apart. Each RRM has a HWP in front of it to direct power into both arms. Ideally the pulses entering each RRM are either diagonally or anti-diagonally polarized so that there is an equal probability for the photons to travel through either path. The overall power in each pulse is reduced to 1/4 of its original power, but count rates remain the same because there are more pulses each delivering less power.

Alignment of the system begins with the straight through path. Typically I align one RRM at a time, starting with the one closest to the laser. After the straight through path of each RRM is optimized, meaning that it is centered on the crystal for maximum singles counts, I use apertures and/or a laser card to overlap the delay-arm beam with the straight through path beam. I then repeat this for the second RRM. This coarse alignment gets the beams roughly to where they need to be, but fine adjustment is still necessary. Typically, over the course of 1-3 days, depending on lab conditions,

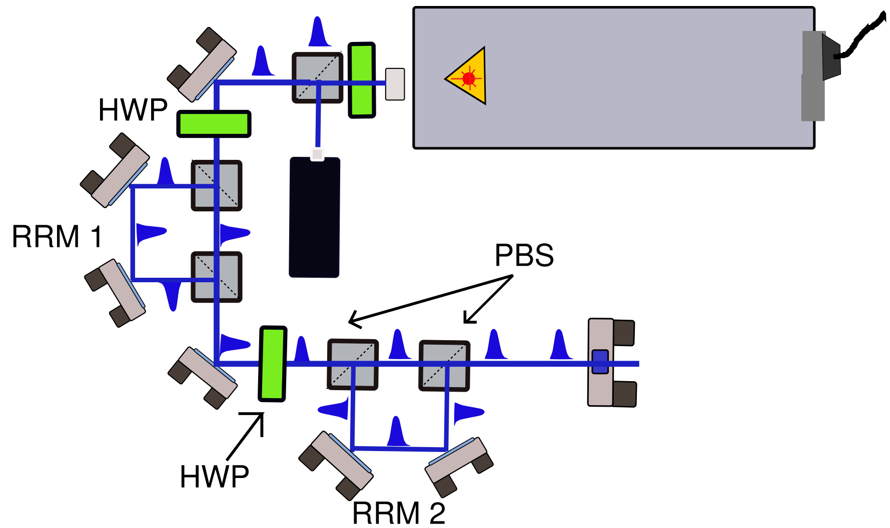

Figure A.7: **Pump rep-rate multiplying** I multiplex the pump pulse into four pulses spaced 500 ps apart with the setup shown here. The first stage consists of a HWP, two PBS (Precision Photonics/Advanced Thin Films PBS1005-TY) and two mirrors (Thorlabs NB1-K08). In this stage, I space the two pulses by 1 ns, so the total delay length is 30 cm. The second stage is identical to the first except for its delay length which is half of the first (15 cm). This creates a second and fourth pulse 500 ps after the first and third created by the first stage.

the temperature and humidity fluctuations cause small changes in the positions of the optics and each RRM needs to be realigned. The most important issue in alignment is getting the power balanced between the four pulses in order to ensure maximal entanglement and also because in order for there to be security against eavesdroppers, the probability of being in each pulse needs to be identical.

There are two equivalent methods of checking if the power in the pulses is even. Balancing power requires that each HWP before the RRM is splitting the photons equally between the two paths. The first way to measure this is by picking off a small amount of pump light after the RRMs, directing it onto a fast detector (Thorlabs SV2-FC) and measuring the output pulses from the detector on a fast oscilloscope (Agilent DSO 90804A). Because the bandwidth of this detector is 2 GHz, I have a maximum resolution of 500 ps, so even though the pump pulse is much shorter (5 ps), the pulses it produces on the scope are 500 ps in width. By looking at the relative heights of the four pulses (one from each path through the 2 RRMs) on the scope I rotate the HWPs and mirrors to balance power. This method works well for rough measurements; however, it is challenging to get the power fully balanced between the pulses using this method. The most likely reason why this is true is that the pick-off wedge plate I use (CVI LW-3-1037-UV) has a slight polarization-sensitive reflectivity. Because each pulse is alternatively H-polarized or V-polarized, this method of measurement is not the most reliable.

The more reliable method is to simply use the four pulses to pump the nonlinear crystal and collect SPDC counts. In this case both crystals in the double crystal are pumped and I use the singles count rate to determine the relative power in the pulses. I block combinations of two arms to ensure that only a single path is being taken at any given time. By successively blocking each of the paths for the four combinations and ensuring the singles counts are the same in all of them, I balance the power.

A.3.2 FBG Properties

Fiber Bragg gratings are made when the core of a single mode fiber is exposed to UV light in a periodic manner. The periodic mask is introduced over the longitudinal axis of the fiber. Because the core is photosensitive, the parts that are exposed to UV light undergo a change in refractive index. Gratings are written into a fiber with periodic masks. Although there is a periodic structure in the FBG it is still susceptible to birefringence due to stress and strain. In addition, coupling into the pigtailed end is sensitive to alignment and also to the polarization of the light entering the grating.

The FBGs in this experiment are manufactured by Teraxion. They consist of a 1 m FBG grating with pigtailed single mode fiber ends measuring $\sim$ 1.5 meters. They are terminated with a standard FC/PC ferule. The FBGs only work as reflected devices, although both ends of the FBG are pigtailed and connectorized. This allows for transmission measurement in addition to the reflection measurements. Although the light that is transmitted through the fiber does not couple into the grating, it is useful for ensuring that light is entering the pigtail fiber. I use the transmission mode for trouble-shooting whenever I do not observe reflected counts from the FBGs.

Each FBG can nominally stretch an input pulse with a $>$ 10 nm bandwidth roughly $\pm$ 830 ps. The 3 dB bandwidth of the FBG is $\sim$9-10 nm. I find that one of the + FBGs does not stretch the incoming pulse by 8.3 ns as would be expected, but only $\sim$ 4-5 ns. Plots of the reflectivity versus wavelength are given in Ch. 6. I hypothesize that the reason one of the +FBGs does not produce the total expected stretch is that part of the grating has been broken and a smaller working portion of the grating only produces a fraction of the stretch. This has been independently verified by Siddharth Ramachandran's group at Boston University.

A.3.3 FBG setup with polarization sensitivity

In the original experimental setup for securing time bins with FBGs, I rely on polarization manipulation in order to achieve separation between input light going into the FBG fiber and reflected and stretched light coming out of the FBG. Because the FBGs are reflective devices, light travels the same path backward as it does forward. To separate the incoming and outgoing light from the FBGs, I use the experimental setup shown in Fig. A.8. I send light from the 850 nm arm of the SPDC source through a PBS so that everything I obtain at either port of the PBS is linearly polarized. Then I place a QWP in the path to change the linear polarization to circular polarization. As an example, suppose I use vertically polarized light from the port in the first PBS. It then passes through the QWP and changes to left-hand-circular polarization, for example. This light enters the grating and reflects at the end of it. Upon reflection the handedness of the light changes so that it is now right-hand-circular traveling out of the grating. When it passes back through the QWP it is rotated to horizontal polarization and passes straight through the PBS instead of being reflected back the way it came in. For this case, ideally, the input and output light would be well-separated. However, physically, this is an imperfect process due to the fact that the FBGs are not polarization insensitive devices. I find that the polarization of the photons decoheres in the FBG due to birefringence in the the fiber and upon exiting the FBG is in a superposition of polarizations. This effect causes a significant portion of light to be directed into the opposite (wrong) port of the PBS and contributes to overall system loss.

Alignment of light into the FBGs is challenging, because even if light is well coupled into the pigtailed ends of the FBG, this does not translate to having the light interact with the grating. I find that FBG coupling is very sensitive to input polarization and light will not couple properly into the grating if it is not polarized in the correct direction. Although there is a QWP in the setup to control this, the long pigtailed fiber end ($\sim$ 1 m) causes polarization mixing due to birefringence. It is important then to find the correct polarization to couple into the grating and then secure the fiber pigtailed end so that bending and twisting is restricted.

Wavelength dependent polarization

Upon entering the grating, certain wavelengths of light are reflected at the beginning, middle and end of the grating. These wavelengths also experience birefringent effects due to any stress or strain of the fiber. Because the wavelengths physically reflect at different locations in the FBG, different wavelengths undergo different polarization

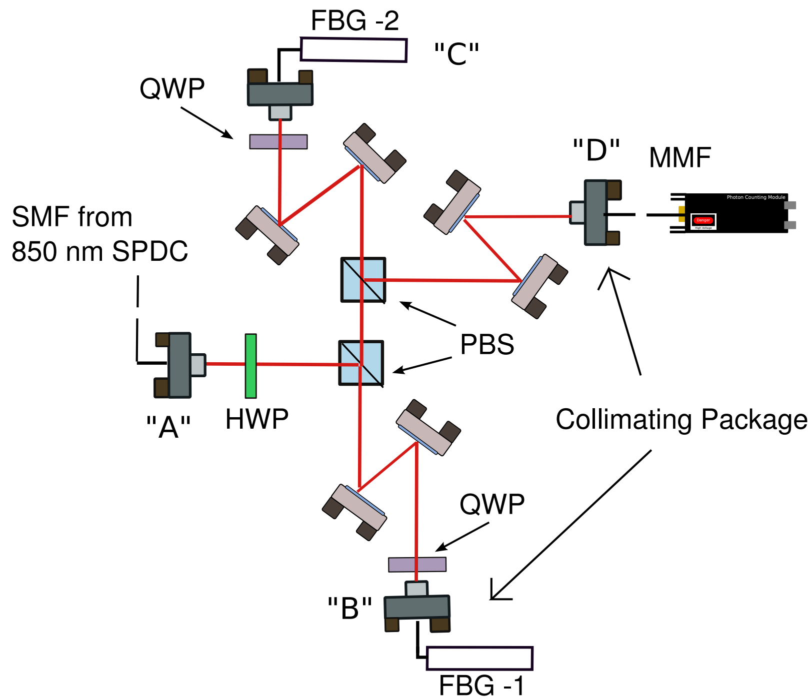

Figure A.8: **FBG polarization-based setup** 850 nm light from the SPDC source is collimated with a collimating package (Thorlabs F220FC-780) in port "A". It then travels through a HWP to rotate its polarization and then a PBS. Due to birefringence in the fiber, the polarization is not linear exiting the fiber and so some light is lost into the other port of the PBS (the straight through path). Although I could use only a single PBS in this setup, the light that leaks through the straight path in this first pass would be in the same port of the PBS that the light reflected through both gratings would be. This causes unwanted counts in the path of the signal, so I use two PBSes to take care of this. After the light is reflected in the first PBS, it travels through a QWP, two steering mirrors and into port "B" into the first FBG (FBG1). The reflected light travels this same path back and straight through the first and second PBS to port "C". It then goes through the same elements for the second FBG (FBG 2) and upon its return path, is reflected at the second PBS into a MMF (Oz Optics custom fiber) in port "D" and into the SAP500 detector. The 609 nm arm is coupled straight from the collinear SPDC setup described in Fig. A.3 to the MPD PDF detector.

changes from the different birefringence. This causes some of the reflected light to be polarized in one direction, while another portion of the reflected light is polarized in a different direction. When all the light passes back through the QWP and PBS some of the light (the light with a certain polarization), goes into the reflected port of the

PBS and does not get collected. If this process were simply an overall loss, equal for all wavelengths, this would not be an issue. However, because the light that is directed into the wrong path due to its polarization corresponds to certain wavelengths, the overall effect is that the output timing profile is not able to stretch the full 8.3 ns because the bandwidth is essentially limited.

To better illustrate this concept, suppose that in the total bandwidth (845-855 nm) 845 nm is reflected at the beginning of the grating and 855 nm travels through the whole grating and back. Now further suppose that wavelengths 845-848 nm are reflected with the correct polarization to pass through the correct port of the PBS and are collected. Wavelengths 848-852 nm undergo a different birefringent effect and are transformed into the incorrect polarization and are reflected and not collected, while wavelengths 852-855 nm are similar to the first set and are reflected correctly. In this case, although the full bandwidth entered the grating, some of it is selectively lost (the middle part of the spectrum). Because wavelength maps to time when stretched with this group velocity dispersive material, I expect to see two peaks when I look at time correlations between the twin photons – one corresponding to 845-848 nm and the other to 852-855 nm. A temporal correlation of this type might look like Fig. A.9 (a).

In Fig. A.9, I show plots of the timing correlations between one twin in the photon pair and the other, which has gone through a single FBG. I plot a histogram of the time differences between twin photon A and photon B as a function of time (in 100 ps bins). The vertical axis is in counts per bin and shows the probability of the twin photon falling in a certain time window after its twin has been detected. The delay of ~ 40 ns corresponds to the extra path a the photon that enters the FBG system travels. The width of the "pulse" seen in these plots represents the overall stretch. The difference in Fig. A.9 (a), (b) and (c) is only that the mix of polarizations changes in each due to different stresses on the fiber from being moved around. Clearly, the FBG's property

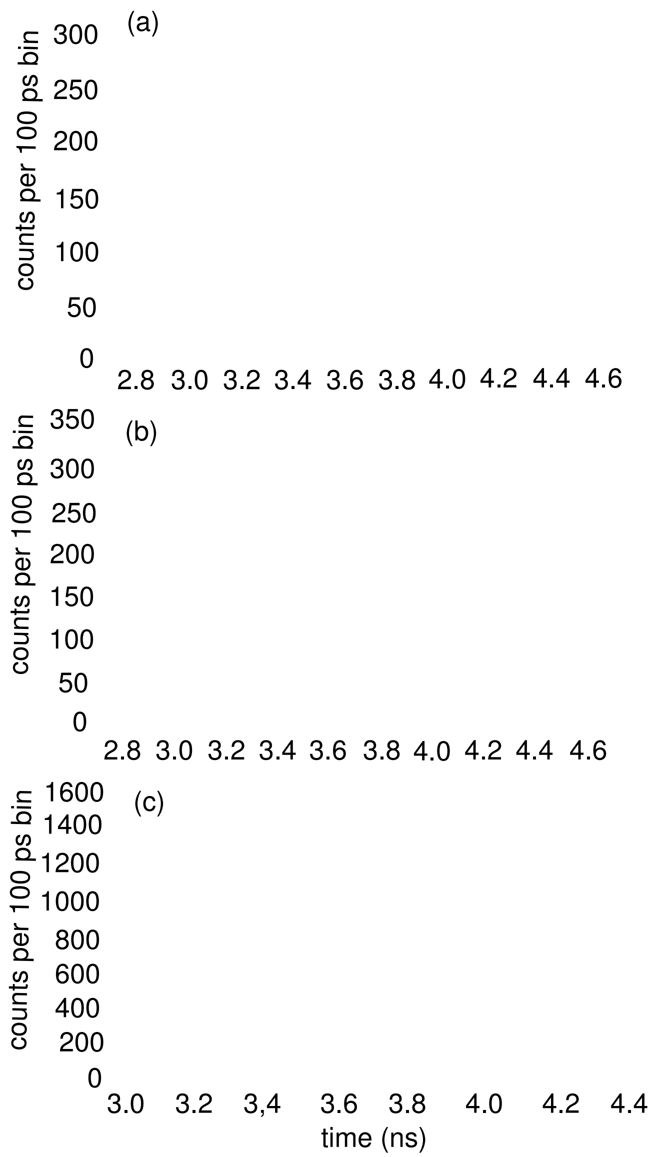

Figure A.9: **Single FBG time correlation data** I plot a histogram of the time differences between twin photons from nondegenerate SPDC. The vertical axis is number of counts in a 100 ps bin and the horizontal axis is the number of bins. (a), (b) and (c) are all taken with an identical setup. The only differences between the three plots is that I physically move the FBG between each. Moving the FBG means that there are different stresses on the fiber and thus different polarization decoherence. The scale difference in (c) is due to better coupling and possibly longer collection times.

that certain wavelengths undergo different polarization transformations explains the difference.

In each plot, different peaks can be traced back to different wavelengths being reflected with the correct polarization to be collected. Ideally, this plot should look like a flat-top pulse with 8.3 ns width. Because certain wavelengths are not collected, the pulse is either shortened, as in Fig. A.9 (c) where the collected pulse is $\sim$ 4 ns, or split in two (Fig. A.9 (a) and (b)). In these plots, if the two peaks shown were connected, they would span the 8.3 ns, but there is a dip between them. If this whole stretched pulse were used to secure 8.3 ns (or $\sim$ 16 time bins each 500 ps), there would be a large decrease in security because, for example, an eavesdropper may know one time is more probable than another time and be able to deduce, at least statistically, when the photon would arrive.

Coupling into the FBGs

The FBGs are each sensitive to the input polarization of the light, as mentioned previously in this section. For the original alignment, there is no way of knowing whether the correct polarization is entering the grating or not. To find the correct input polarization, I take a set of time-tagged data from the 609.6 nm channel (unstretched and unchirped) and 850 nm channel (stretched and chirped) and plot the coincidence window between the two channels as a function of time. If the counts couple into the grating, they reflect back and create a long window of coincidences because of their stretch. If counts do not couple into the grating, they reflect off the beginning part of of the fiber pigtail (about 1.5 m away from the grating) and back. They can also pass through the grating without coupling into it, and get reflected back (roughly 1.5 m after the grating). An example of this is shown in Fig. A.10 (a). Here, the large, sharp peak at t=200 (= 20 ns) is the light that reflects off the front of the grating. The lower broader peak in the middle is the light that gets coupled into the grating and is clearly stretched, while the last sharp peak is light reflecting from the back side of the grating. By adjusting the QWP directly in front of the grating, I control how much

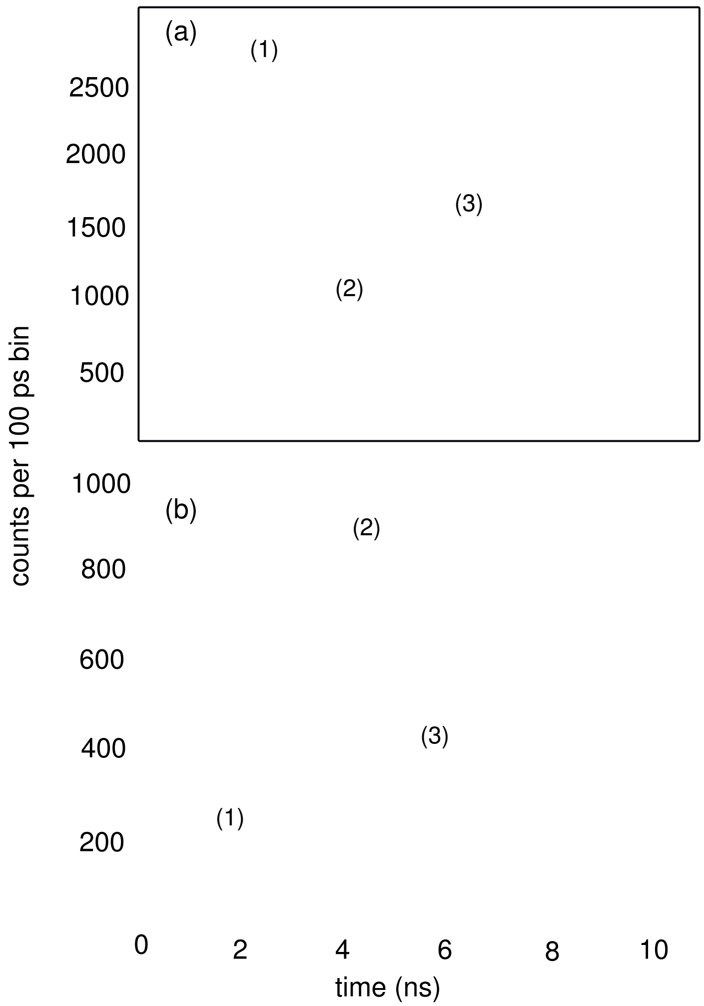

Figure A.10: **QWP control of coupling** In (a) I do not change the QWP entering the FBG at while in (b), I attempt to optimize the counts coupling into the FBG with a QWP to change the polarization. (1) and (3) are the peaks that arise from reflection off the front and back faces of the grating without coupling in while (2) is the light that gets coupled into the grating and subsequently stretched. I use the method described in the text, where I measure the coincidence counts windowed around the first reflected peak at 20 ns and minimize it.

power reflects versus couples in. In Fig. A.10 (b) I show the exact same setup with the

QWP rotated to optimally couple into the grating.

One method of measuring if the light is coupling into the grating without taking data sets for many different angles on the QWP is to use the coincidence counting FPGA program. By using a delay line in the other arm (609 nm arm) I can adjust the coincidence window to be centered around the sharp peak that occurs at 20 ns. This sharp peak is due to a reflection off of the front of the grating, so I see high coincidences when a significant portion is not coupling into the grating. To ensure more light is coupling into the grating, I minimize the coincidences with the window centered on the reflected peak at 20 nm. I want to minimize this because this reduces counts contributing to the sharp peak at 20 ns and maximizes counts into the broadened peak at around 40 ns. Using this method I am able to optimize the QWP without recording and plotting multiple data sets.

Two FBGs in polarization setup

In this section I present data taken with a combination of two gratings. There are four such combinations given that I have two gratings with each sign of chirp (+830 ps/nm and -830 ps/nm). I call the grating combinations as FBG1/FBG2, so that I have minus/minus, minus/plus, plus/minus and plus/plus. Here I present data for all four gratings, using the polarization-sensitive setup.

In Fig. A.11 I show a histogram of the times between coincidence events for the minus/plus (a) and plus/minus (c) FBG setups. In each of these plots it is clear that the pulse gets recompressed to its original temporal profile. The zoomed-in plots are Fig. A.11 (b) and (d). The FWHM of the recompressed pulse is $\sim 250 - 300$ ps, which is exactly what I expect from the detector jitter of ~ 250 ps. This indicates that the pulse gets stretched and recompressed to its original measured width, which is dictated by the joint jitter in the detectors. I verify that this is not a reflection from the front of back of the grating by its temporal location. The reflections, since they have not traveled through both gratings, appear at ~ 20 ns and ~ 55 ns while the recompressed pulses appear at 60-61 ns where the extra delay is caused by the FBG.

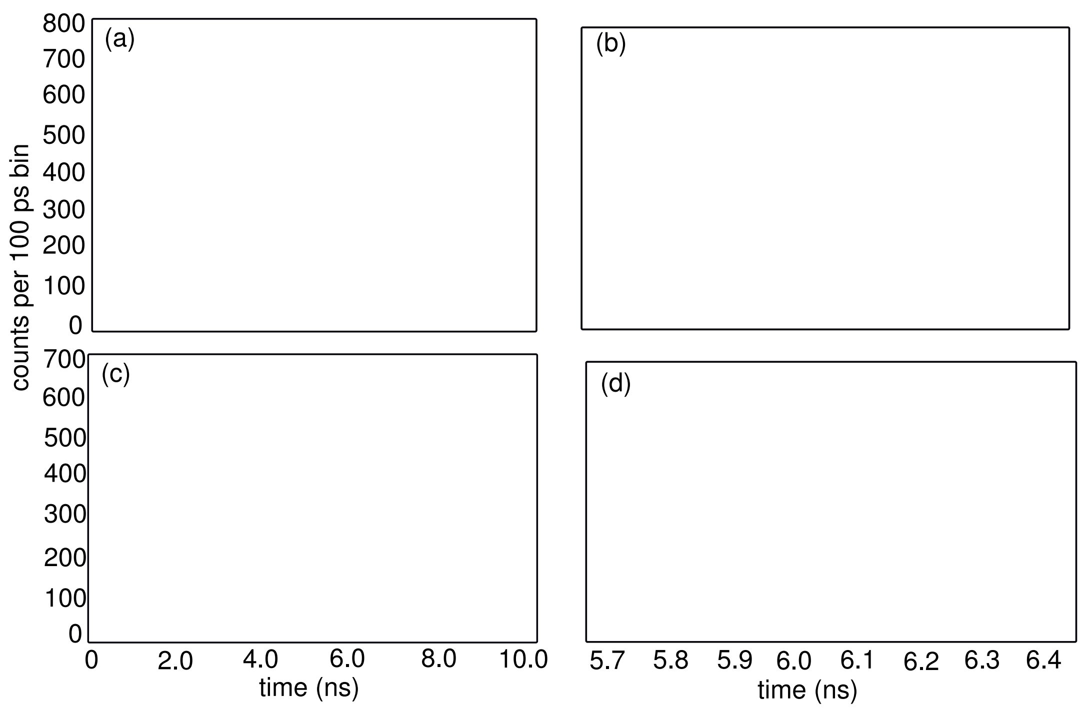

Figure A.11: **Minus/Plus and Plus/Minus Data** In (a) I plot the histogram of time differences between signal from detectors registering correlated photon pairs for the case where I use a minus FBG in port "B" and a plus FBG in port "C" of Fig. A.8. (b) is a zoomed in version of this plot to show full decompression. In (c) and (d) I show the same type of plot but for the a plus FBG in port "C" and a minus FBG in port "D".

In Fig. A.12, I show the same type of histogram, but with minus/minus and plus/plus gratings instead. The pulses here are clearly stretched, although not to the extent that I expect. For example, I expect that with 8.3 ns of stretch for each grating, that with two gratings, I would obtain 16.6 ns pulses. In my "best" data set (shown here), I see almost this amount of stretch, although the reflectivity over roughly half of it is very poor compared to the other half. In general, I never saw this amount of stretch for the plus/plus gratings, again because of the wavelength-polarization dependence discussed earlier. I frequently measured data sets with around half the stretch that I expected and sometimes less.

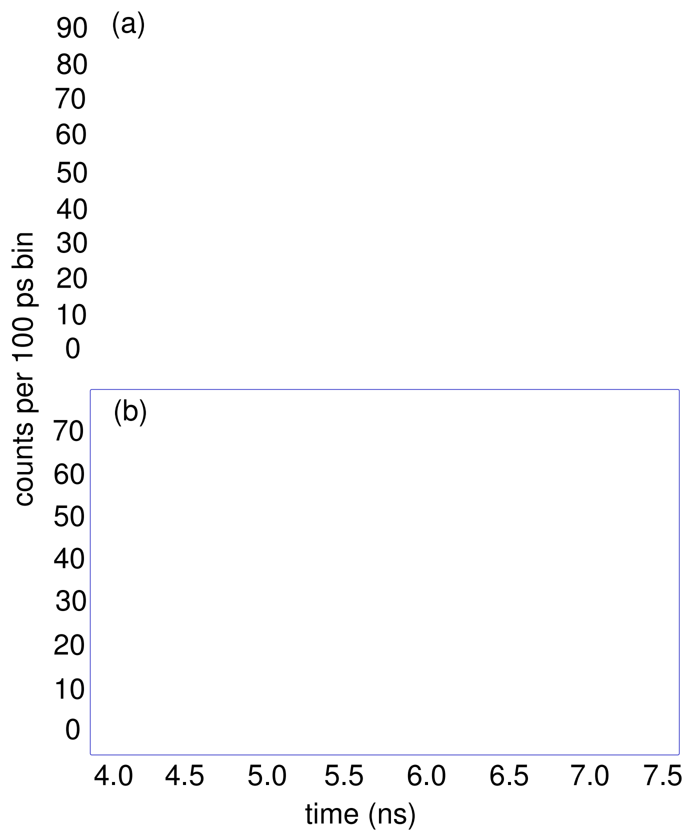

Figure A.12: **Minus/Minus and Plus/Plus Data** In (a) I plot the histogram of time differences between signals from the two detectors registering a correlated photon pair for the case of two minus FBGs in the 850 nm arm. In (b) I show the same type of plot, but with two plus FBGs in the appropriate ports.

Finally, in Fig. A.13, I show a histogram of a minus/minus FBG, but this one, the second QWP, was at the wrong angle to fully couple all the light into the grating. Some of the light coupled into the second grating, which is shown in the longer pulse that occurs around 60 ns, while some of the light only coupled into the first minus grating and then reflected off of the second one. This is the pulse at ~ 40 ns that has been stretched by roughly half of what the total stretch was.

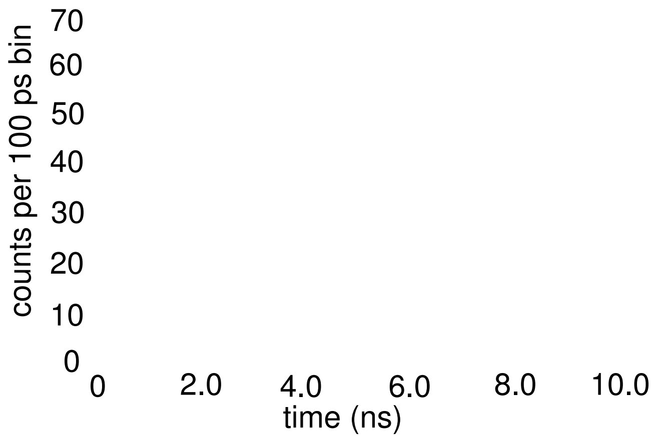

Figure A.13: **Minus/Minus versus QWP angle** I plot a histogram of time differences as in the previous two figures. Two minus FBGs were in ports "B" and "C" of Fig. A.8. The histogram shows that some light coupled into both FBGs and was stretched more, while other light was reflected off of one of the gratings and stretched less.

The amount of stretch in this case is reduced due to the polarization-dependent loss. To ensure security for a time-bin QKD system, the stretched pulse needs to be flat, meaning that the photon could be in any of the time bins equally. The polarization-sensitive case has a vastly reduced flat-topped stretch and is inferior to the system discussed in Ch. 6. However, it is important to note the birefringent properties of the FBG so as not to incorporate them into a polarization-sensitive system.

Appendix B

Detector Technologies

B.1 Introduction

In most physical systems that utilize biphoton sources, single-photon-counting detectors or single-photon avalanche photodiodes (SPADs) are necessary for detecting the biphoton pair with high probability. Single photon counting detectors are used in a wide variety of experiments from heralded-single photon systems to confocal microscopy, ultra-sensitive florescence measurements, LIDAR (Light Detection and Ranging) and quantum computing. Additionally, SPADs are crucial to the operation of most quantum communication and quantum key distribution systems. Because of their important role in these applications as well as being a crucial component for any QKD system, I devote this appendix to the discussion of details and specifications of these detectors as well as my work characterizing them.

For quantum communication systems, the detectors should have a high quantum efficiency that produces an electrical output pulse or response with just a single input photon. Commercial single-photon-counting detectors tend to have highest efficiency in the visible and NIR part of the spectrum due to the fact that silicon is highly absorptive in this range while other materials do not absorb higher or lower wavelengths as efficiently. In this chapter, I discuss how single-photon-counting detectors work. I then review the important specifications of detectors and how each affects my experiments. Finally, I discuss the detectors I chose as well as some new technologies that will potentially increase the rates and decrease the error of my current system.

B.2 Single photon avalanche detectors

Single photon avalanche detectors (SPADs) are able to detect single photons using an avalanche effect in a p-n type semiconductor material. For wavelength ranges in the visible and NIR part of the spectrum, the most commonly used semiconductor is silicon. Typically SPADs have a thin layer of silicon, and when a photon hits this layer, the photon is absorbed and creates an electron-hole pair. By itself, a single electron-hole pair drifting through a material does not produce enough current to create a signal. To rectify this, a high reverse-bias voltage is applied to the semiconductor material. If the reverse-bias voltage is above the breakdown voltage of the device (V_{br}), the SPAD is said to be in "Geiger" mode and the charges accelerate across the material. The first electron-hole pair collides with lattice atoms and ionizes them, creating other electron-hole pairs that are also accelerated due to the potential difference. Each of these pairs subsequently produce more pairs and thus the term "avalanche" process is appropriate. The gain of these devices is very high ($10^5 - 10^6$) due to the number or charge carriers produced from a single photon. The charge carriers create an electrical current that electronic circuits shape into a pulse that can be read by various electronics.

Avalanche Photodiodes (APDs), on the other hand, have a large reverse-bias voltage applied to the material, but do not operate in "Geiger" mode. They are simple linear amplifiers of an incoming signal. The Geiger-mode SPADs are therefore needed in order to detect signals that are on the single-photon level, as the avalanching is needed to amplify the signal to an "on" state.

There are some significant disadvantages to operating in Geiger-mode that I discuss in the following section. Namely, there is often an afterpulsing effect due to stray charges in the material that produce another pulse directly after the actual one. This second pulse or "afterpulse" causes errors in any system because it does not represent an actual photon event. The commonly accepted method of dealing with this issue is to shut off the reverse-bias voltage for a specific time after the photon arrive in order to allow the charges to decay. Of course, the tradeoff to this is that the detectors maximum rate is now limited by this time that they are shut off. In the next section, I discuss the interplay between all of these parameters and the various ways they affect my experiment.

The three major detectors I use in my experiments are pictured below in Fig. B.1 and are PDM and PDC modules from Micro-Photon Devices (MPD), Laser Components SAP500s mated with MPD electronics, and Perkin Elmer/Excelitas SPCMs. I discuss throughout the following sections key parameters of each that are used in the different experiments, as well as my characterization of these detectors for the experiments I discuss in this thesis.

(a) (b) (c)

Photo Credits: (a) www.advlab.org/spqm.html (b) www.lasercomponents.com (c) www.micro-photon-devices.com

Figure B.1: **SPAD models in my experiments** (a) Excelitas single photon counting module (SPCM), (b) Laser Components SAP500 detector and (c) MPD PDM module. The Laser Components SAP500 detectors that I use in the experiments were given to MPD and mated with their electronics and packaging to combine the high quantum efficiency in the SAP500 with the superior timing jitter and electronics from the MPD package.

B.3 Detector Parameters

In the following section, I discuss several different detector parameters that are important for the experiments I describe in this thesis. All detector parameters depend many parameters such as the wavelength, temperature, and the overvoltage (the voltage above V_{br}) applied to the circuit. When needed, I describe the dependence on these parameters, although detailed analysis is beyond the scope of this work and can be found in Refs. [83, 84].

B.3.1 Quantum Efficiency

The quantum efficiency (QE) or photon detection efficiency (PDE) is an important parameter that affects the overall total rate of the experiment. Formally, the quantum efficiency of a SPAD is the probability of producing an electrical output signal conditioned on having a single photon at the input. The higher the quantum efficiency, the higher the achievable rate of the system. Many companies that manufacture SPADs create devices with very high quantum efficiencies at specific wavelengths. In Fig. B.2 I show three typical curves of quantum efficiency versus wavelength from these companies.

The quantum efficiency of the detectors in these experiments have a large impact on the values of the heralding efficiencies as well as total count rates of the system. Because one of the goals of this work is to help design a high-speed QKD system, having high quantum efficiency detectors is essential. The quantum efficiency of each detector peaks at different wavelengths. The Excelitas detectors are peaked at 710 nm which is excellent for use in the noncollinear, degenerate SPDC experiment. Their quantum efficiency, although quoted at $\sim 72\%$ is closer to $63 - 64\%$. I estimated this number by using these detectors in heralding efficiency measurement at the University

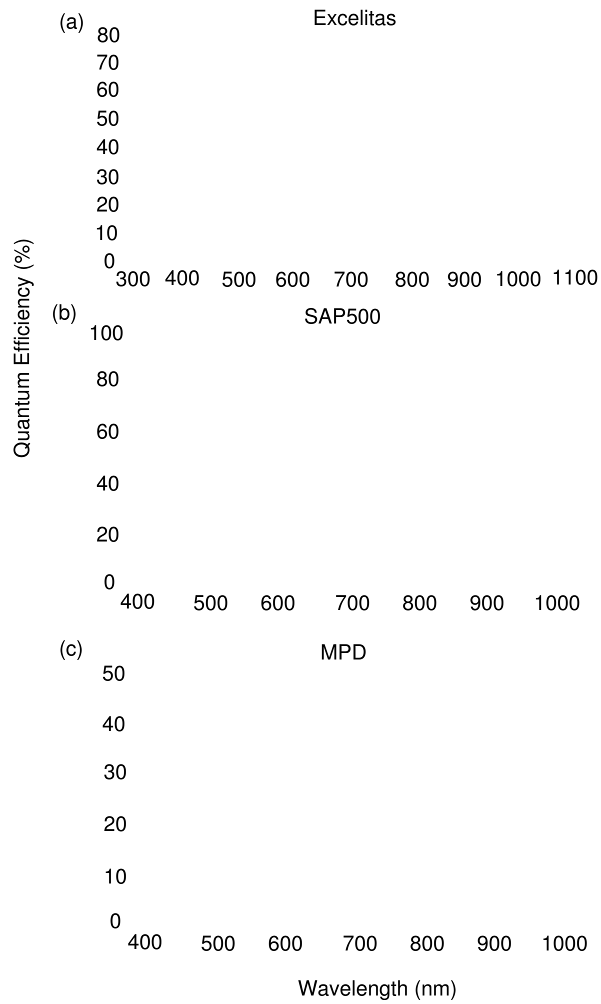

Figure B.2: **SPAD models in my experiments** Quantum efficiency versus wavelength curves for each of the three types of detectors in the experiments[1]. The Excelitas detector is peaked at the degenerate wavelength of 710 nm, while the SAP500 is peaked in the bluer part of the spectrum from 400-600 nm. Finally the MPD PDM module has a lower QE and is peaked at 550nm. For the SAP500 detector I use a different bias voltage than the one listed here as well as detectors with an AR coated window. These plots are meant to show general features of quantum efficiency for each detector for general comparisons.

of Illinois and comparing the results to the well-calibrated detectors they typically use.

For the majority of the work on testing different configurations and types of SPDC, I

[1]Excelitas–http://www.excelitas.com/Downloads/DTS_SPCM-AQRH.pdf, SAP500 – Private communication with MPD, MPD PDM –http://www.micro-photon-devices.com/Products/Photon-

use the Excelitas detectors for their high quantum efficiency. These are fiber-coupled "plug-and-play" detectors that are very straightforward to incorporate into any setup.

Although the quantum efficiency of the detectors listed here is high for standard SPAD detectors, recently, more cutting edge technologies using super-conducting nanowire single-photon detectors (SNSPD) has been demonstrated [12, 85]. These detectors were originally optimized for 1550 nm and 1310 nm and boasted higher than 90% quantum efficiency. More recently, the group in [12, 85] started fabricating these detectors in the visible and NIR parts of the spectrum and achieve high quantum efficiency of >93% [86].

In addition to high quantum efficiency, these detectors also boast no afterpulsing and low jitter. Currently, Prof. Jungsang Kim's group and collaborators at Duke University are characterizing the jitter, quantum efficiency, and dark count rate of these devices at 710 nm and 1550 nm for use in a high-speed QKD system.

B.3.2 Afterpulsing & Deadtime

Afterpulsing, like its name suggests, is a physical phenomenon that occurs after a photon has created an electrical pulse in a detector. The after pulse is a subsequent electrical pulse that occurs, not from a true photon event, but from excess carrier charges in the material that trigger another avalanche effect. In a semiconductor material, after a charge creates an electrical pulse, there are some charges that get trapped in the deeper energy levels of the material. These charges are released at later times after the initial avalanche, and, if the material still has a high reverse bias voltage applied, they create their own avalanche events. These are then "fake" pulses, as they did not originate from an actual photon event and contribute to the overall error of a system.

Counters/PDM

Charges can be trapped in multiple energy levels in the material and subsequently released at different times. It has been shown that the probability for reemission for each level follows an exponential decay law each with a different time constant, so that at longer times, there is a lower probability of causing an avalanche effect [83, 87]. Because of this exponential decay, afterpulsing is suppressed by turning off or “quenching" the circuit after an initial pulse is registered. The high reverse bias voltage is quickly shut off after the first avalanche event. Without the reverse bias, the charge carriers are released, but no avalanche event occurs. The longer the circuit is quenched for, the less probability of an afterpulse event.

However, the downfall to quenching the circuit after the intial pulse is that while the detector is shut off, it cannot “see" any other pulses, including actual photon events. For this time, the detector is essentially dead, and the name given to this off-time is the “deadtime" of the detector. This deadtime dictates the maximum count rate of the detector. For example, if the detector is turned off for 100 ns, before it can see another photon event, its maximum count rate is 10 MHz. There is then an important trade-off between the maximum count rate of a detector and the afterpulsing probability. The longer the deadtime, the lower the after pulsing probability and the lower the maximum count rate. Both of these are important parameters, especially in a QKD system because although it is important to have a high-rate system, it is also important to have a low error rate so that error correction is still possible.

For our experiments, afterpulsing probabilities of lower than 1% are considered good. The afterpulsing probability of the Excelitas detectors is quoted as 0.5% while the Laser Components detectors are 2% and the MPD detectors are 0.1-3%. I have characterized the afterpulsing probability of several of the MPD detectors, which I discuss in section B.3.5.

B.3.3 Jitter

The temporal jitter of a detector is the uncertainty in the timing of the arrival pulse triggered by a photon event. Suppose that a photon is sent to a detector every 1 μs, and suppose further that each time a photon is sent, it is registered and triggered an electronic pulse. In a perfect system, a plot of the histogram of arrival times relative to when the photon is sent results in a delta function at the delay time between when the pulse is sent and when it is registered. This corresponds to zero jitter. In reality, the photons are absorbed at different places in the semiconductor material, and each component in the electronic circuit has a finite error associate with its timing. Thus, in reality, instead of observing a delta function, I observe a Gaussian-like distribution with a long tail, whose temporal width is called the jitter.

Jitter is often quoted in the full-width-half-maximum (FWHM) of the Gaussian and is sometimes referred to as the speed of the detector (although the speed can also refer to the maximum rate). In an experimental QKD system using time bins, the jitter is a very important parameter because the jitter is essentially the resolution of the system which then dictates the size of the bins. The bin has to be at least as large as the "resolution" or uncertainty of the detector or else it would not be possible to know for sure which bin a photon was born in, introducing large errors into the final key. The jitter also dictates the resolution to the time-to-frequency measurement using the GVD material.

Decreasing jitter in SPADs is accomplished by creating faster circuit components and decreasing the length of semiconductor material used for the first photon to electron-hole pair production stage. The thinner the material, the more likely that the photon will be absorbed at the same place in the material most every time – reducing jitter. The drawback to this is that the thinner the material, the lower the quantum efficiency because the less likely a photon is to be absorbed at all. Therefore, often low-jitter detectors come at the cost of quantum efficiency. For example, the MPD detector is quoted as having a jitter of < 50 ps; however, their maximum quantum efficiency at 550 nm is barely 50%. In contrast, Excelitas detectors have quoted quantum efficiencies of up to 70%, but also larger jitter of around 500 ps due to poor electronics design. The SAP500 detectors are good balances of these two, with peak quantum efficiencies higher than that of the Excelitas at 80% at 520 nm. When the SAP500 detectors are mated with the faster MPD electronics the jitter is roughly $\sim 150-200$ ps.

Jitter is strongly dependent on the applied over-voltage and the focusing conditions of the incoming photon beam onto the detector. In general, the higher the over-voltage, the lower the jitter. This makes sense as it is more likely photons absorbed in the same place in the beginning of the material will result in avalanche events. Also important is how tightly the beam is focused onto the detector. The smaller the used area on the detector (tighter focusing), the lower the jitter. The reason for this also has to do with the probability of creating an electron from an incoming photon. Typically, detectors have less quantum efficiency at their edges and the non-uniformity causes a difference in timing across the diameter of the material. It is therefore imperative to focus tightly on most detectors in order to get the best timing jitter. In the experiments, I use short focal length lenses and XYZ translation stages to ensure a small spot size on the detectors. I typically achieve beam waists of $\sim$ 12-20 μm which is much smaller than the typical sensitive area which is 50 μm for the MPD detectors and 150 μm for the SAP500 detectors.

B.3.4 Dark Counts

Dark counts are counts that a detector registers without a photon triggering the event. These typically exclude afterpulsing and are simply due to the thermally excited electron-hole pairs in the material causing an avalanche event. Because of the thermally driven nature of dark counts, many detectors are cooled to $-10°$C, $-15°$C or $-20°$C in order to reduce the number of dark counts. Dark counts contribute to the overall error of the system because they are not true counts triggered by a photon event.

Often times dark counts and "background counts" will be used interchangeably. Background counts include dark counts, but are also made up of all the counts that are present from stray light in the setup. Although efforts of shielding are usually made, typically trace amounts of light, not due to the true signal will also result in counts. All of these make up the background noise or background counts. If shielding is done well, the background counts will be very close to the dark count levels.

Although cooling the detector to a low temperature decreases the dark counts, it has an adverse affect on the afterpulsing. When the detector is cooled, the relaxation time of the reemission of carrier charges from the deep energy levels is lengthened. Charges move more slowly and it takes longer for avalanches events to occur. Therefore, the deadtime of the detector is not long enough to stop these slower carriers and more afterpulsing occurs. This is a common theme in finding the optimal set of parameters to use for a SPAD because optimizing one parameter comes at the cost of deoptimizing another.

B.3.5 Testing single detector afterpulsing and jitter

Testing the jitter, afterpulsing and dark counts of a detector is useful because not every detector is specified by the company in the same way as it is used in an experimental setup. In order to know if the specification of the detector is valid, I perform some simple experiments to test these parameters in a methodical way. In this section I describe the experimental setup I used to calculate these parameters as well as detail my findings and explain how I calculate the parameters of interest.

The detailed experimental setup is depicted and described in Fig. B.3. Briefly, a VCSEL at 680 nm is pulsed with 1 ns pulses and focused onto a SPAD detector; here I use the MPD detector. The electrical pulse from the detector is sent to a time-tagging device (CAEN TDC V1290N and V1718). A trigger with the same clock as the source pulses for the VCSEL is also sent into the time-tagger.

I analyze the data by taking the time difference between the trigger (reference arm) and the detector hit. These time difference are then plotted on a histogram and shown in Figs. B.4 and B.5. Figure B.5 is a zoomed in version of Fig. B.4 and shows the afterpulsing tail and deadtime more clearly, while Fig. B.4 is intended to show the afterpulsing tail decaying down to the background level. In these figures, I plot the logarithm of the numbers of counts in a 1 ns bin as a function of bin number (time in ns). The reason for choosing a logarithmic plot is to better see the details of the afterpulsing.

The main peak occurs around 100 ns, meaning that relative to the trigger, there is roughly 100 ns of delay from when the BERT (Bit Error Rate Tester) sends its pulse to the laser to when the CAEN registers the same hit. This delay is caused by the path length difference arising from extra free space distance and additional cables. The width of this peak is the jitter of the detector as this is the uncertainty in the timing of the hit. The deadtime of the detector is highlighted in Fig. B.5 and is $\sim$ 65-70 ns, which is in agreement with the specifications of the MPD detector PDM module. The afterpulsing tail begins where the deadtime ends and decays down over time to the background noise or dark count level.

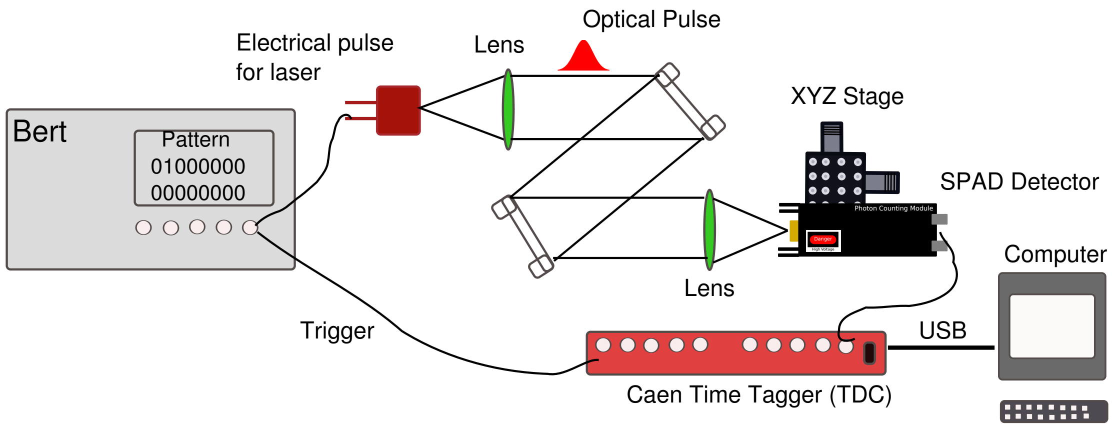

Figure B.3: **Pulsed laser experimental setup for measuring afterpulsing, jitter and dark counts** A bit error rate tester (BERT) creates a short pulse used to drive a VCSEL at 680 nm (VIXAR 10-0680S-0000-B001) and also act as a reference clock or trigger. The clock rate of the BERT is set at 1 GHz, meaning each bin (and pulse) is 1 ns in width. A bit pattern of 2^{14} zeros between each "1" ensures that there is ample time for the deadtime and afterpulsing to occur between each trigger. Each trigger pulse is $\sim$1.6 μs apart. The trigger is sent directly into a CAEN time-to-digital converter (TDC) (V1290N and V1718) and also to the VCSEL. The pulse from the BERT is attenuated by high-speed attenuators (Mini-circuits) and shaped with a Mach-Zehnder Modulator Driver (MZM) (JDSU Uniphase Optical Modulator Driver H301-2310) to achieve the desired pulse into the laser. The optical pulse from the laser is collimated by a 30 mm focal length plano-convex lens (Thorlabs LA1805-B) and sent through steering mirrors and a set of neutral density filters to attenuate it down to an average of 1 photon/pulse. It is then focused by a second lens with a larger focal length of 150 mm (Thorlabs LA1433-B) onto a SPAD detector. The electrical output of the detector is sent to the CAEN. The CAEN accepts NIM logic pulses instead of TTL. NIM is a current-based logic often used in high-energy applications. NIM pulses have a high-level of 0 V and low-level of $\sim$ -800 mV and are triggered on the fast falling edge of the pulse. The MPD detectors I characterize have NIM outputs as well as TTL outputs and so are easily compatible with the CAEN time tagger, however the Excelitas/Perkin-Elmer detectors I use only output TTL pulses. In order to convert to NIM I use high-speed inverters (Phillips Scientific Model 460) and attenuators (Minicircuits). The time-tagged data from the CAEN is sent via USB to the computer where it is written into both a binary and ASCII file. This file contains two columns: the first column is the channel number where the hit occurred, and the second channel is the time-tag. In this setup I use two channels, one for the trigger from the BERT and one from the detector.

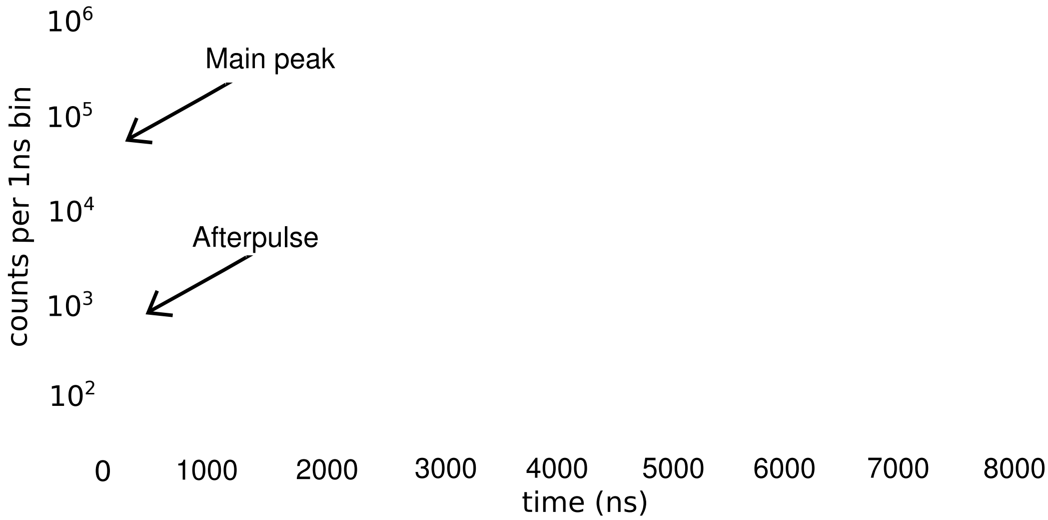

Figure B.4: **Example Data Set for MPD Detector** Full data set for an MPD PDM module with high afterpulsing probability. I subtract the raw time stamps from the trigger time stamps and histogram the resulting data. This data set is a typical example and the main features in it (main peak, deadtime, afterpulsing) are seen in all of the SPAD detectors.

Data Analysis for Afterpulsing

The afterpulsing probability is the total number of counts in the afterpulsing peak divided by the total number of counts in the main peak. For this data set the pattern length is $2^{14} = 16,384$, which is made up of 16,383 zeroes and a single one. The main pulse is at 100 ns (100 bins) and the deadtime ends at bin 170. I then use two different, but equivalent methods for calculating afterpulsing probability.

B.3.6 Method 1

1. Use Matlab to calculate total counts, counts in the main event peak (MC), total number of triggers, and total time (t)

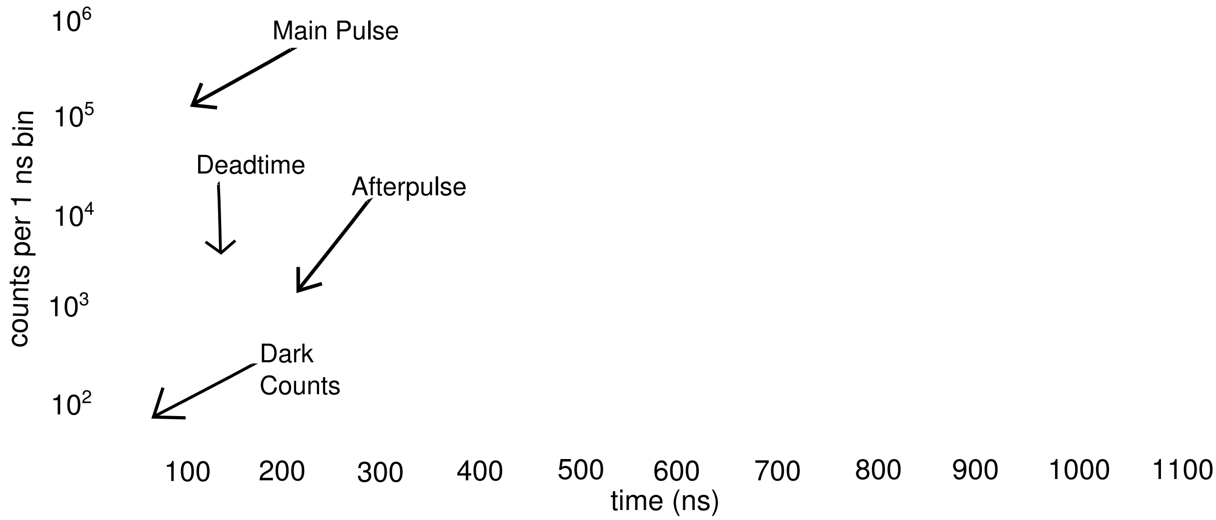

Figure B.5: **Zoom in of main peak and afterpulse** Here I show a zoomed-in version of the previous plot for the MPD detector. All parameters and data are the same. Features such as deadtime, main peak afterpulsing, and dark counts are highlighted in the figure.

2. Use Matlab to sum counts in bins 170 to 16384. This is the total number of afterpulse counts and dark counts (afterpulse counts = APC).

3. Estimate number of dark counts (DC) in those bins by multiplying dark count rate (dcps) by the total time the system was in afterpulsing $\frac{(16384-170)}{16384}t$ where t is the total time.

4. Dark count rate can either be estimated by summing up counts in the last 50 dividing that by the total time in the last 50 bins or by integrating over the offset of the curve fit and dividing by total time, which I discussed in the next method.

5. Afterpulsing probability is the number of true afterpulse counts divided by num-

ber of counts in the peak. In other words

$$\text{Afterpulsing } \% = \frac{APC - DC}{MP}. \tag{B.1}$$

B.3.7 Method 2

The second method I use to calculate afterpulsing probability is with a nonlinear least-squares fitting of the set of data.

1. Import the binary data from the output file of the TDC into a Matlab program and histogram it at 1 ns intervals.

2. Call "cftool" in the main command prompt of Matlab, which opens a fitting program. Choose parameters close to the actual values so that the fit matches the data set given.

3. The fits are sums of exponentials, power laws, or a combination of the two. I extract the fitting parameters from the program as well as the goodness of fit measures.

4. Integrate under the fitted curve from bin 170 to 16384 to obtain number of counts in afterpulsing tail.

5. In order to estimated dark count rate, take the offset parameter from the fit and integrate from 1 to 16384 and divide by total time (t) to get the dcps.

6. Taking the (afterpulse counts - dark counts)/(counts in main peak) give another estimate of the afterpulsing rate.

7. Note: Use the counts in main peak (MP) from the Matlab program because the fit only handles the afterpulsing tail.

The counts for this data set are given by Table B.1.

Table B.1: Experimental values for afterpulsing

total counts	8,631,630
triggers	435,873,752
counts in main peak	8,100,200
time	7141 s
afterpulsing counts	1,273,584
background in afterpulse	1,006,240
counts in last 50 ns	3166
dark count rate	139 cps
afterpulse prob	3.3%

I estimate the afterpulsing probability the same way as in the first method.

I fit the data with a power law and exponential fit with the function:

$$y(x) = ae^{bx} + \frac{c}{(x-d)^e} + f \tag{B.2}$$

The units of the y-axis are number of counts/bin, and the horizontal axis is bin, with every bin being 1 ns width. I test several fitting equations for afterpulsing including single, double, and triple exponentials as well as a simple power law. The combination of the two seems to produce the best fit. The reason for this being that the decay laws that govern an avalanche event are all exponential decays. However, there can be many of these levels. The main exponential tail is accounted for by the first exponential, then all subsequent exponentials are accounted for in the power law function, which is a weighted sum of exponentials.

The values for the fitting parameters are: The goodness of fit parameter R-square = 0.9936. From the fit, integrating over the afterpulse (170-16384), the equation

Table B.2: Parameters for Afterpulsing Fit

	Value [units]	95% tolerance range	Deviation
a	$1*10^7$ [counts/1 ns]	$(7.089*10^6, 1.291*10^7)$	$\pm 2.911*10^6$
b	-0.05415 [(1/ ns)]	(-0.05567,-0.05263)	$\pm$0.00152
c	$2.85*10^6$[counts/1 ns * (1ns)^e]	$(2.179*10^6, 3.47*10^6)$	$\pm 645,500$
d	-113.7 [(1 ns)]	(-118.6, -108.8)	$\pm$4.9
e	1.697 [unitless],	(1.662, 1.732)	$\pm$0.035
f	62.06 [counts/1 ns]	(61.86, 262.26)	$\pm$0.2

becomes

$$\int_{170}^{16384} [(1*10^7)e^{-0.05415x} + \frac{2.85*10^6}{(x-113.7)^{1.697}} + 62.06]*dx = 1,264,260 \pm 75,692 \text{ counts.} \tag{B.3}$$

Integrating over the dark counts I find,

$$\int_{170}^{16384} [62.06]*dx = 1,006,240 \text{ counts.} \tag{B.4}$$

This makes the dark count rate = 1,006,240*(16384-170)/(16384*t) $\simeq$ 139 dcps.

From the first method, I have (total afterpulsing counts - dark counts) = 267,344 which, when divided by the counts in the main peak is 267,344/8,100,200 = 0.033 or $\sim$ **3.3%.** From the second method, the total afterpulsing counts is 1,264,260 making the (total afterpulsing - dark counts) = 258,020. Dividing this by the counts in the main peak results in 258,020/8,100,200 = 0.0318 or $\sim$**3.18%**. To calculate the error for this fit I use error propagation analysis

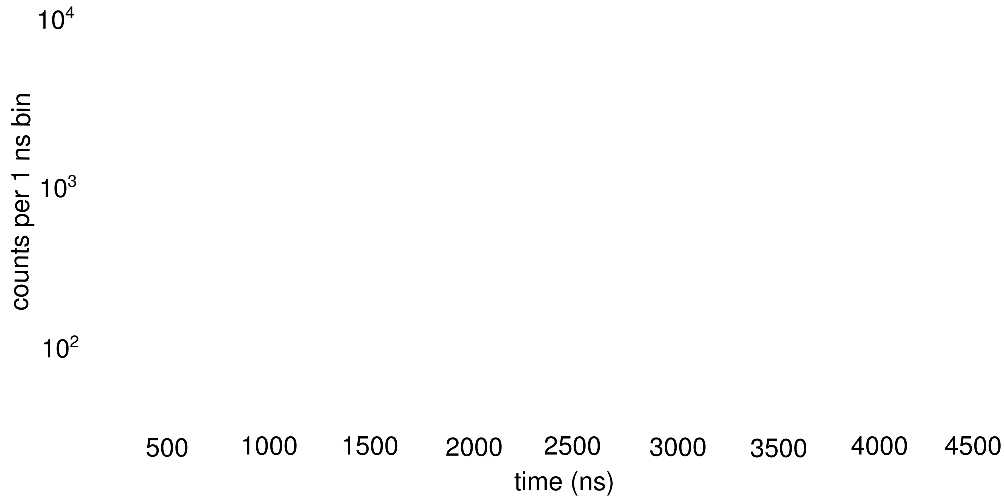

Figure B.6: **Afterpulsing Tail with Fitting** Here I plot the original data set in a range that highlights the afterpulsing tail. Again, the y-axis denotes number of counts per 1 ns bin on a logarithmic scale and the x-axis is time in 1 ns bins. The data is represented by the blue curve, while the fit is shown in red. The red fitted curve is given by Eq. B.2 with the parameters listed in Table B.2.

$$\delta y(x) = \Bigg(\left(\frac{\partial y}{\partial a}\right)^2 (\delta a)^2 + \left(\frac{\partial y}{\partial b}\right)^2 (\delta b)^2 + \left(\frac{\partial y}{\partial c}\right)^2 (\delta c)^2 \tag{B.5}$$

$$+ \left(\frac{\partial y}{\partial d}\right)^2 (\delta d)^2 + \left(\frac{\partial y}{\partial e}\right)^2 (\delta e)^2 + \left(\frac{\partial y}{\partial f}\right)^2 (\delta f)^2 \Bigg)^{1/2}. \tag{B.6}$$

Using the specific parameters, I find

$$\begin{aligned} \delta y(x) &= \Bigg((e^{bx})^2 (2.911 * 10^6)^2 + (axe^{bx})^2 (0.00152)^2 + \left(\frac{1}{(x+d)^e}\right)^2 (645.500)^2 \\ &+ \left(\frac{-ce}{(x+d)^{e+1}}\right)^2 (4.9)^2 + \left(\frac{-c * log(x+d)}{(x+d)^e}\right)^2 (0.035)^2 + (0.2)^2 \Bigg)^{1/2}. \end{aligned}$$

Integrating this from 170 to 16384 gives: **75,692** counts. Therefore the ranges of afterpulsing probability I expect are (258,017+75,692)/8,100,200 = 0.0412 or **4.12%** and (258,017+75,692)/8,100,200 = 0.0225 or **2.25%**, giving error bars of $\sim$ 1% so that the after pulse probability is 3.18$\pm$1%. I also estimate the afterpulsing probability if the deadtime were lengthened by 1 ns and find an after pulse probability of **3.14 %**.

I use this method to characterize various detectors, however, this specific MPD detector is the one with the highest afterpulsing that I observed. Because of its high afterpulsing, which I cannot tolerate in terms of the total error rate, the company sent us a replacement with much lower afterpulsing.

Analyzing the jitter

I measure the jitter in this experimental setup by measuring the width of the main peak either by estimating the width of the pulse towards the top, as it is plotted on a logarithmic scale or by separating and plotting only the pulse data and creating a fit to the main peak. I observe a jitter of $\sim$ 90-100 ps for this peak, which is the combined jitter of the BERT, laser, detector and time-tagger. The jitter of the detector will be smaller than this, although the BERT, laser and time-tagger combined should not total more than 30-50 ps of jitter. I estimate this value by company specifications for each product and by using a delay line to split the pulse and measure the jitter of the split pulse with itself. This gives an estimate of the scope jitter. This is still slightly higher jitter than expected for this detector (35-50 ps specification). In the following section, I show another method that I commonly use to test the jitter of two detectors used in my SPDC source. This is more applicable because this is closer to the actual experiments that I run throughout this thesis. Although I talk about this in a different section, the basic principles between the two ways of measuring the jitter are the same.

B.3.8 Analyzing Jitter Using SPDC

There is another useful method for characterizing jitter using an SPDC source, such as the one in my experiments. I use this pulsed source for measuring either a jitter of a single detector or the jitter in both detectors simultaneously. Because I use two detectors in most of the experiments with SPDC, it is useful to know the joint jitter from the detectors. The jitter is an independent uncertainty, so the jitter from two independent detectors add in quadrature. In this section, I first discuss the joint jitter measurement and then in the last paragraph show how I modify the setup to measure a single detector jitter.

I measure the combined jitter by sending SPDC light into each detector and analyzing the output signals from each detector (either with NIM or TTL output) on a fast oscilloscope. In more detail, I use the scope to trigger off a pulse from one detector and make sure the other detector has a small amount of delay so that both pulses do not overlap. Ideally, if the detectors are identical, the pulses overlap completely, but all detectors are slightly different and adding coaxial cable to one arm is an easy way of delaying a pulse. I histogram the pulse from the other detector over a temporal window wider than the expected jitter. The histogram function of the scope tracks where the pulse crosses this window. Typically, results in a Gaussian distribution with the width (standard deviation or FWHM) being the joint jitter. An example of this is shown in Fig. B.7. This is an actual image taken from the scope showing the two detector pulses as well as the histogram.

In order for this measurement to be accurate and the jitter minimized with this method, I ensure the following parameters are set up correctly. First, the trigger on the first pulse must be at the steepest part of the pulse slope for jitter to be minimized. Typically, for NIM pulses this is at $\sim 300 - 400$ mV and for TTL it depends on whether

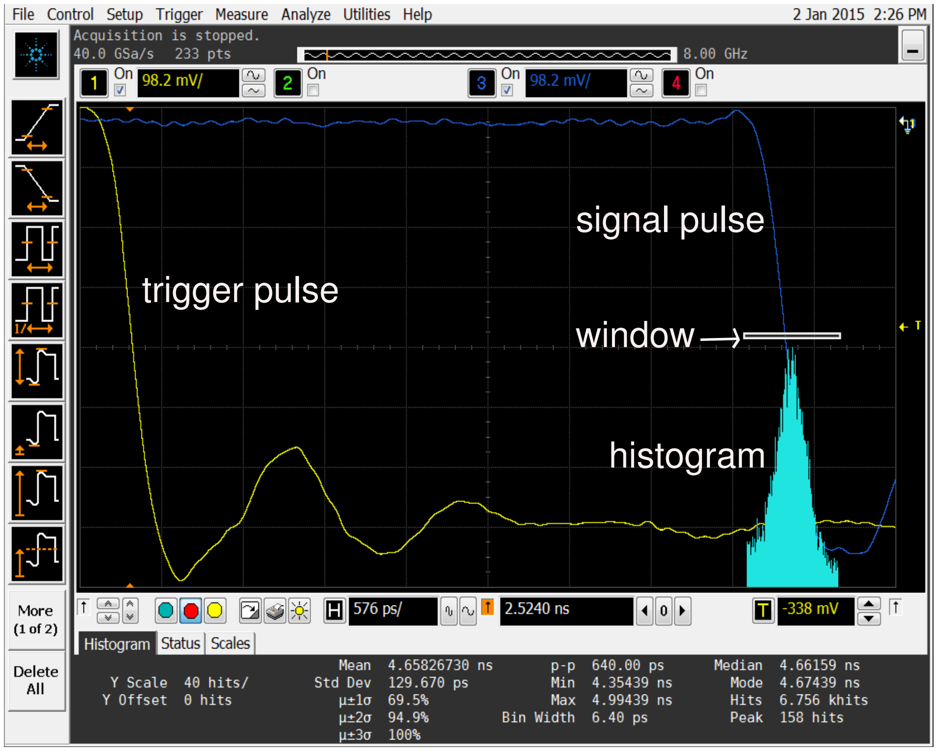

Figure B.7: **Screenshot of oscilloscope showing histogram with jitter** The yellow pulse is the trigger pulse from the first detector, while the blue pulse is from the second detector. The white rectangular box is the window over which I histogram. Every time a pulse crosses that window its position gets recorded into the histogram. I zoom in on the x-axis in this figure to make the histogram bin 6.4 ps width. I have *not* zoomed in in the horizontal axis to 10-20 mV as I state in the text.

the TTL is a 0 to 3.5 V or 0 to 5 V range. Typically for TTL the steepest part of the slope is roughly half of the total voltage. Additionally, the histogram window for the second pulse should be placed exactly at the same level the trigger is to ensure that the same parts of each pulse are being used. The vertical size of the histogram window must be set to the smallest increment to prevent the finite slope adding extra width in the temporal direction. In my case, the smallest window I set on the scope is 0.01 mV. The temporal or horizontal window must be wider than the expected jitter. This usually varies between 500 ps to 1.5 ns depending on the detectors. Finally, it is important to be "zoomed in" in each direction on the scope, making the width per division as small as possible while still keeping both pulses on the screen. In the horizontal direction this zooming directly affects the width of the histograms temporal bins. In the vertical direction, the zooming is not quite as important, and mainly decreases sampling, but can also increase resolution by small amounts. For these settings, I typically aim for a bin width of 5-10 ps which is equivalent to $\sim$ 500-600 ps per division on the horizontal axis and 2-10 mV per division in the vertical directions.

Additionally, I save the data points given on the scope and do a fit in another program (either Matlab or Mathematica). I have found that, while the scope can do a reasonably good job of finding a fit for many cases, especially the cases where the fit is very Gaussian, it can often overestimate the width of the curve by weighing too heavily on the outlying points. One way around this is to cut off the tails on the histogram by narrowing the window, while the other is to simply plot the data and fit it using a different program. In Fig. B.8, I plot the same data set as given in Fig. B.7 using Mathematica to plot and fit instead. The result for this case, which is the joint jitter between the MPD PDF (fiber-based PDM module) and an SAP500 detector is roughly 240-250 ps.

I also measure single detector jitter by using a similar experimental setup. Here instead of using one detector to trigger the other, I split the signal from a single detector and use one of the split pulses to trigger the other pulse. I use a splitter (Minicircuits part # 15542) to split the output signal of a single detector and put a long delay line in one arm. The goal is to trigger off one pulse in one arm and catch the next pulse (the one right after the deadtime of the detector) in the other arm. To do this, I need $\sim$ 60 ns of cable to compensate for the deadtime of the SAP500 and MPD detectors. The process is then the same as described above. In Fig. B.9 I plot the histogrammed data with a fit for a single fiber-based MPD detector (PDF). The jitter from the fit is

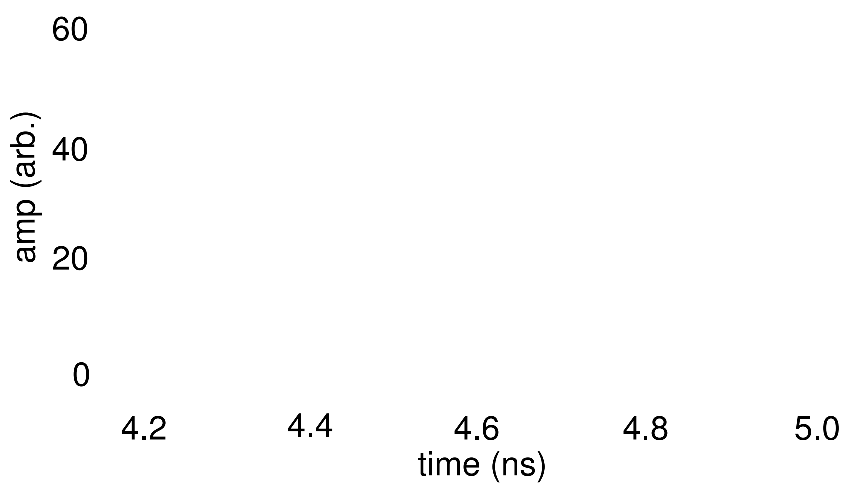

Figure B.8: **Fit to joint jitter with SAP500 and MPD PDF detector** The data set shown with blue points is the same as the histogram plotted in Fig. B.7. I extract the data and fit a Gaussian to it using a fitting program in Mathematica. The width of this curve is $\sigma = 103$ ps meaning that the FWHM = 242 ps.

$\sigma = 28 - 29$ ps and the FWHM = 66-68 ps. This result is in good agreement with the detector specifications.

B.3.9 Detector Results and Summary

Detector parameters are exceptionally relevant for any application using single-photon counting detectors such as high-speed QKD, testing heralded single photon sources, quantum computing, and many others. In this final section I give detailed results for the specifications of the detectors I use. I accumulated these both from my own measurements and also from discussions with Bradley Christensen (University of Illinois) and data sheets.

Although the idea detector with nearly 100% quantum efficiency across all wavelengths, jitter below 30 ps, negligible dark counts (1-10 cps) and next to zero after-

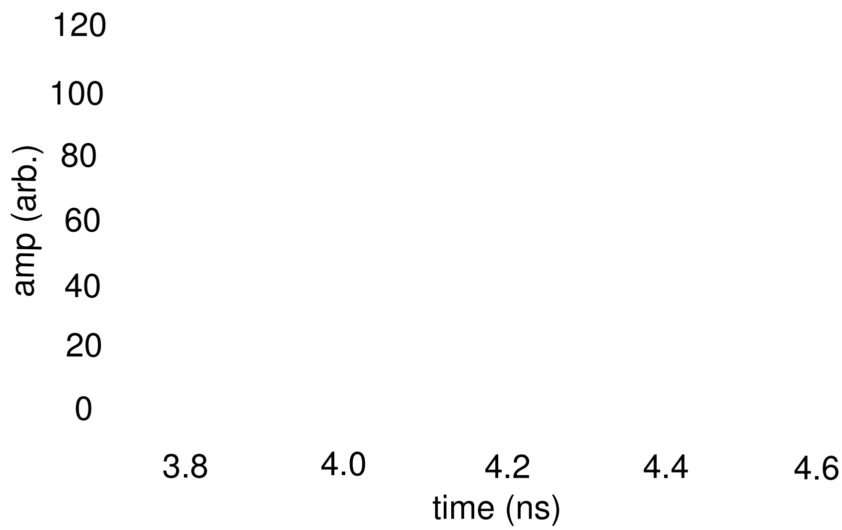

Figure B.9: **Fit to single detector jitter of MPD PDF detector** I shown the collected data from the histogram off of the scope for the jitter of a single detector, the MPD PDF detector. I fit a Gaussian to the points and find that $\sigma = 28 - 29$ ps and the FWHM jitter = 66-68 ps.

pulsing (<0.1%) is unrealistic for current SPAD detector technologies the combination of detectors I have is sufficient for successfully performing the major experiments that I present in this thesis. I take care in each set of experiments to choose the correct detector for the results I wish to obtain. For the work on SPDC theory and optimization presented in Ch 4., I use the Excelitas detectors for their high efficiency, low dark counts, and simplified implementation. In the experiments with FBGs and in implementing part of a time-bin based QKD system presented in Ch. 6, I use a combination of an MPD detector at 609 nm (PDF module) and an SAP500 detector at 850 nm. I choose these for their combined minimized jitter in order to create the smallest time bins (500 ps) and minimize overlap into adjacent time bins. Although I could have chosen two MPD detectors and reduced the jitter further, the MPD detectors have very low quantum efficiency at 850 nm.

Table B.3: Detector Specifications

	Excelitas (F)	MPD (FS)	MPD (F)	SAP500 (FS)
Quantum Efficiency	63-64% (710 nm)	27-28%	7-28%	61-63%
	40-45 % (850 nm)	9-10%	9-10%	24-25%
	60-62 % (609 nm)	42-43 %	42-43 %	75-77%
Jitter	450-500 ps	∼50-60 ps	∼50ps	170-190 ps
Afterpulsing	0.5%	1-2%	1-2%	1.5-3%
Dark Counts	300-500 cps	600-1200 cps	20-30 cps	1500-2000 cps

In Table B.3, I detail the parameters for each set of detectors I tested and used in experiments. In this table, "FS" stands for "free-space" and "F" stands for "fiber". Although each detector behaves slightly differently, the parameters presented here are representative of most of the detectors of each specific type.

Detectors and future detector technology are crucial in building high-speed QKD systems. Research in super-conducing nanowire detectors has shown much promise, although due to the large cryostat and complicated setup needed for such a detector, SPADs will most likely still be used for many years to come. It has been important, throughout my work, to be able to characterize such detectors and understand how they work so that I could make informed decisions on which detector to use for a specific experiments and also to understand better the differences between experimentally measured values and predictions.

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Biography

Hannah Elizabeth Guilbert was born on February 2nd, 1987 in Trenton, NJ. She grew up in the faculty housing of a private boarding school where her father taught (and still teaches) physics. Her love of the physical universe developed through many discussions with her father, including one memorable one about quantum mechanics at the age of 9 or 10 and the statement "One day you'll understand it". Although that statement still is not quite true, so began her love for physics and quantum mechanics. She was homeschooled by her mother until high-school where she attended The Peddie School in Hightstown, NJ where she won the Raymond Orman Biology Prize and excelled in the nature sciences.

In 2004 she matriculated to Boston College where she majored in physics and conducted undergraduate research in several different fields including metamaterials, neutrino physics, and the fractal dimensions of Jackson Pollock's paintings. She was the president of the Society of Physics Students at Boston College her senior year when the good news of acceptance into Duke's Graduate Physics program came.

In 2008 the born and raised northerner trekked to the "southern" state of NC to start her first year of graduate school in the Duke Physics program. Despite thinking that she would do her research in high-energy physics, she fell in love with optics, especially the quantum variety, and began her research in Daniel Gauthier's Quantum Electronics group in the Fall/Spring of 2009/2010. During her time in graduate school she developed a love for many new things including the city of Durham, fostering and rescuing dogs (she currently lives with three permanent fur-babies and typically one-two foster dogs), home improvement projects, and being outside almost year-round.

Publications

D.J. Gauthier, C.F. Wildfeuer, H. Guilbert, M. Stipcevic, B. Christensen, D. Kumor, P. Kwiat, K. McCusker, T. Brougham, and S.M. Barnett, ÔQuantum key distribution using hyperentangled timebin states,Õ in Proceedings of The Tenth Rochester Conference on Coherence on Quantum Optics (CQO10), N. P. Bigelow, J. H. Eberly, and C. R. Stroud, Eds. (Optical Society of America, 2014), pp. 234-239.

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